

UPSC ISS · STATISTICS PAPER-I · OBJECTIVE

Probability

Every previous-year question from 2018 to 2026, with answer, exam shortcut, tips & tricks and a step-by-step solution.

Questions in this volume **196** of 720 across the four units

Papers covered 9 — 2018 to 2026

Per paper 2018: 21 · 2019: 19 · 2020: 20 · 2021: 20 · 2022: 21 · 2023: 20 · 2024: 29 · 2025: 26 · 2026: 20

Syllabus topics 13

Syllabus concepts 32

Layout of every entry QUESTION → OPTIONS → ANSWER → EXAM SHORTCUT → TIPS & TRICKS → STEP-BY-STEP SOLUTION

Dataset version 1.0

What is authentic and what is not. The questions and their four options are reproduced word for word from the original UPSC ISS Statistics Paper-I booklets. Wording, option text, option order and question numbers are unaltered.

None of the source booklets carried an official answer key. Every ANSWER, EXAM SHORTCUT, TIPS & TRICKS note and STEP-BY-STEP SOLUTION in this volume was worked out for this project and is marked *AI-derived explanation* where it appears. They are a study aid, not an official UPSC solution.

Source defects. 3 question(s) in this volume have a defect in the printed paper. None has been silently corrected: the original text is reproduced exactly and the problem is described under the question. 1 of them has no valid option at all and is therefore left unkeyed rather than guessed.

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Questions per paper

PAPER	QUESTIONS FROM THIS UNIT	SOURCE FILE
2018	21	STATISTICS-PAPER-1-2018-2.md
2019	19	STATS_P1_ISSISS-2019.md
2020	20	STAT-1-ISSISS-2020.md
2021	20	Statistics-I-2021.md
2022	21	Stat_I_2022.md
2023	20	STATISTICS-PAPER-I-2023.md
2024	29	STATISTICS-PAPER-I-2024.md
2025	26	QP-IES-ISS-25-STATISTICS-PAPER-I-2025.md
2026	20	UPSC ISS 2026 QP Stats 1 (1).md

P1 · PROBABILITY**Classical & Axiomatic Probability, Consequences**

10 questions · 2018: 2 2019: 1 2020: 1 2021: 1 2022: 2 2023: 0 2024: 2 2025: 0 2026: 1

Weight. 10 questions (5.1% of Probability) · appeared in 7 of 9 papers · peak year 2018 (2)**Examiner's pattern.** Two flavours only. (a) Combinatorial/classical counting — round-table seating, splitting 4n people into batches, ordered vs unordered samples. (b) Geometric probability — Buffon's needle, a point dropped in a square, the condition for a quadratic to have real roots. Independence-based unions appear as the opening question of a paper.**Must-know.** $P(A_1 \cup \dots \cup A_n) = 1 - \prod(1 - P(A_i))$ for independent events; inclusion-exclusion; geometric probability = favourable area / total area; ordered samples n^r vs nPr .**1. 2018 · Q1** Classical & Axiomatic Definitions of Probability · Numerical**QUESTION**

Let A_1, A_2, A_3, A_4 be independent events with $P(A_1) = \frac{1}{2}$, $P(A_2) = \frac{1}{4}$, $P(A_3) = \frac{1}{8}$, $P(A_4) = \frac{1}{16}$. What is $P(A_1 \cup A_2 \cup A_3 \cup A_4)$?

OPTIONS

- (a) $\frac{15}{16}$
 (b) $\frac{1023}{1024}$
 (c) $\frac{709}{1024}$
 (d) $\frac{315}{1024}$

ANSWER (AI-derived — no official key exists for this paper)**(c)** $\frac{709}{1024}$ **EXAM SHORTCUT** (AI-derived explanation)

For independent events never expand inclusion-exclusion. Use $P(\text{none occurs}) = \text{product of the complements}$: $(1/2)(3/4)(7/8)(15/16) = 315/1024$, so the union is $1 - 315/1024 = 709/1024$. Ten seconds.

TIPS & TRICKS (AI-derived explanation)

- Option (d) $315/1024$ is the planted trap: it is $P(\text{no event occurs})$, i.e. the answer you get if you forget to subtract from 1.
- Denominator check: $2 \times 4 \times 8 \times 16 = 1024$, so any correct answer must have 1024 (or a divisor of it) as denominator. That instantly kills (a) $15/16$ unless it reduces, and it does not.
- $1023/1024$ in option (b) equals $1 - 1/1024$, the value you would get if all four probabilities were $1/2$ each — another distractor built from a plausible slip.
- Memorise: $P(\text{union of independent events}) = 1 - \prod(1 - p_i)$. It is the single most reused one-line identity in this part of the paper.

STEP-BY-STEP SOLUTION (AI-derived explanation)

- The events A_1, A_2, A_3, A_4 are independent, therefore their complements A'_1, A'_2, A'_3, A'_4 are also independent.
- $P(A'_1) = 1 - 1/2 = 1/2$; $P(A'_2) = 1 - 1/4 = 3/4$; $P(A'_3) = 1 - 1/8 = 7/8$; $P(A'_4) = 1 - 1/16 = 15/16$.
- $P(\text{no } A_i \text{ occurs}) = P(A'_1 \cap A'_2 \cap A'_3 \cap A'_4) = (1/2)(3/4)(7/8)(15/16) = (1 \times 3 \times 7 \times 15)/(2 \times 4 \times 8 \times 16) = 315/1024$.

4. The union is the complement of that event:

$$P(A_1 \cup A_2 \cup A_3 \cup A_4) = 1 - 315/1024 = (1024 - 315)/1024 = 709/1024.$$

5. Hence option (c) is correct.

2. **2018 · Q2** Classical & Axiomatic Definitions of Probability · Numerical

QUESTION

A lot of 15 mobiles has 4 defectives. Mobiles are examined one by one without replacement. The probability that the 9th mobile examined is the last defective is

OPTIONS

- (a) $\frac{56}{195}$
 (b) $\frac{8}{195}$
 (c) $\frac{1}{7}$
 (d) $\frac{24}{65}$

ANSWER (AI-derived — no official key exists for this paper)

(b) $\frac{8}{195}$

EXAM SHORTCUT (AI-derived explanation)

Split the event: 'exactly 3 defectives among the first 8' AND 'the 9th is the 4th defective'. Hypergeometric x conditional = $\left[\binom{4}{3} \binom{11}{5} / \binom{15}{8} \right] \times (1/7) = (56/195)(1/7) = 8/195$.

TIPS & TRICKS (AI-derived explanation)

- Option (a) $56/195$ is exactly the first factor alone - the trap for candidates who stop after the hypergeometric step and forget the 9th draw.
- After 8 draws, 7 mobiles remain and exactly 1 of them is defective, so the conditional probability of a defective on draw 9 is $1/7$ - no new combinatorics needed.
- 'The k-th item is the LAST defective' always factors as $P(\text{exactly } d-1 \text{ defectives in the first } k-1) \times P(\text{defective on draw } k \mid \text{that})$. Learn this two-step template.
- Sanity bound: the answer must be small (well under $1/7$), so (c) $1/7$ and (d) $24/65 = 0.37$ can be eliminated on size alone.

STEP-BY-STEP SOLUTION (AI-derived explanation)

- Let the lot have $N = 15$ mobiles of which $D = 4$ are defective. We need $P(\text{the 9th mobile examined is the 4th and last defective})$.
- This requires exactly 3 defectives among the first 8 mobiles, and then a defective on the 9th draw.
- $P(\text{exactly 3 defectives in the first 8}) = \frac{\binom{4}{3} \binom{11}{5}}{\binom{15}{8}} = \frac{(4)(462)}{6435} = 1848/6435 = 56/195$.
- Given that, $15 - 8 = 7$ mobiles remain and exactly $4 - 3 = 1$ of them is defective, so $P(9\text{th is defective} \mid \text{above}) = 1/7$.
- Multiplying: $(56/195)(1/7) = 8/195$.
- Hence option (b) is correct.

3. **2019 · Q11** Classical & Axiomatic Definitions of Probability · Numerical**QUESTION**

A stick of length l is tossed at random on a plane ruled with a series of parallel lines at a distance $2l$ apart. What is the probability that the stick will intersect one of the lines?

OPTIONS

- (a) $\frac{1}{2\pi}$
 (b) $\frac{3}{2\pi}$
 (c) $\frac{1}{\pi}$
 (d) $\frac{2}{3\pi}$

ANSWER (AI-derived — no official key exists for this paper)

(c) $\frac{1}{\pi}$

EXAM SHORTCUT (AI-derived explanation)

Buffon's needle with needle length L and line spacing $d (L < d)$ gives $P = 2L/(\pi d)$. With $L = l$ and $d = 2l$, $P = 2l/(2\pi l) = 1/\pi$.

TIPS & TRICKS (AI-derived explanation)

- Buffon's formula: $P = 2L/(\pi d)$, valid when the needle is shorter than the gap. Memorise it as 'two L over πd '.
- The lengths cancel whenever d is a fixed multiple of L , so the answer is a pure number - here $1/\pi$. Expect π in the denominator of any Buffon answer.
- The classical case $d = L$ gives $P = 2/\pi$, which is the historical Monte Carlo estimate of π . Doubling the spacing halves the probability to $1/\pi$ - a useful sanity check.
- Derivation sketch: the distance from the centre to the nearest line is uniform on $(0, d/2)$ and the angle is uniform on $(0, \pi/2)$; integrating the intersection condition gives the formula.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. This is Buffon's needle problem. Let the stick have length $L = l$ and let the parallel lines be $d = 2l$ apart, so $L < d$.
2. Let x be the distance from the centre of the stick to the nearest line, uniform on $(0, d/2)$, and let θ be the acute angle the stick makes with the lines, uniform on $(0, \pi/2)$, independently.
3. The stick crosses a line exactly when $x \leq (L/2) \sin \theta$.
4. $P = (\int (L/2) \sin \theta d\theta) / [(d/2)(\pi/2)] = [(L/2)(1)] / [(\pi d)/4] = 2L/(\pi d)$.
5. Substituting $L = l$ and $d = 2l$: $P = 2l/(\pi \cdot 2l) = 1/\pi$.
6. Hence option (c) is correct.

4. **2020 · Q21** Classical & Axiomatic Definitions of Probability · Conceptual**QUESTION**

Consider the following statements: 1. If ordered samples of size r are drawn from n elements, there are $n(n-1)(n-2) \cdots (n-r+1)$ samples without replacement. 2. If n distinguishable balls are randomly placed in n cells, the probability that each cell is occupied is 1.

OPTIONS

- (a) 1 only
 (b) 2 only
 (c) Both 1 and 2
 (d) Neither 1 nor 2

ANSWER (AI-derived — no official key exists for this paper)

(a) 1 only

EXAM SHORTCUT (AI-derived explanation)

Statement 1 is the standard nPr count. Statement 2 is false: the probability that n distinguishable balls fill n cells is $n!/n^n$, which is far below 1 for $n \geq 2$.

TIPS & TRICKS (AI-derived explanation)

- Ordered sampling counts: WITHOUT replacement $n(n-1)\dots(n-r+1) = nPr$; WITH replacement n^r . Learn the pair together.
- Total placements of n distinguishable balls in n cells is n^n ; favourable ones (one per cell) number $n!$, so the probability is $n!/n^n$ - for $n = 3$ that is $6/27 = 0.22$, nowhere near 1.
- Any statement asserting that a non-trivial probability equals exactly 1 deserves immediate suspicion; certainty is rare.
- This is the classical occupancy problem, and $n!/n^n$ tends to 0 rapidly - by Stirling it behaves like $\sqrt{2\pi ne}^{-n}$.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Statement 1: to draw an ordered sample of size r from n elements without replacement, the first element can be chosen in n ways, the second in $n-1$ ways, and so on down to $n-r+1$ ways for the rth.
2. The total is $n(n-1)(n-2)\dots(n-r+1) = nPr$, exactly as stated. TRUE.
3. Statement 2: placing n distinguishable balls into n cells at random gives n^n equally likely arrangements.
4. The arrangements in which every cell is occupied are exactly the ones with one ball per cell, and there are $n!$ of them.
5. Hence the probability is $n!/n^n$, which equals 1 only for $n = 1$; for $n = 3$ it is $6/27 = 0.222$. The claim that it is 1 is FALSE.
6. Only statement 1 is correct, option (a).

5. **2021 · Q29** Classical & Axiomatic Definitions of Probability · Numerical

QUESTION

16 persons, including A and B, sit at a round table. What is the probability that exactly 4 persons sit between A and B?

OPTIONS

- (a) $\frac{11}{240}$
 (b) $\frac{11}{480}$
 (c) $\frac{2}{15}$
 (d) $\frac{1}{15}$

ANSWER (AI-derived — no official key exists for this paper)

(c) $\frac{2}{15}$

EXAM SHORTCUT (AI-derived explanation)

Fix A; B is equally likely to occupy any of the remaining 15 seats. Exactly 4 people sit between them in exactly 2 of those seats (one on each side), so the probability is $2/15$.

TIPS & TRICKS (AI-derived explanation)

- Anchor one person to break the rotational symmetry - then the other person's seat is uniform over the remaining $n - 1$ positions and no factorials are needed.
- 'Exactly 4 between' must be counted on ONE arc, and there are two such arcs, hence exactly 2 favourable seats. Forgetting the second side gives $1/15$, option (d).
- Seat check: with A at seat 1, four people between means B at seat 6 (seats 2 – 5) or at seat 12 (seats 13-16 going the other way).
- The answer is free of the total 16 except through the denominator 15 - a good sign that the anchoring argument has been applied correctly.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Seat A at a fixed position; by rotational symmetry this loses no generality.
2. The remaining 15 people, including B, occupy the other 15 seats, and B is equally likely to be in any one of them, each with probability $1/15$.
3. For exactly 4 persons to sit between A and B along one arc, B must be the 5th seat away from A.
4. There are two such seats - one going clockwise and one going anticlockwise.
5. Hence the probability is $2/15$.
6. Option (c) is correct.

6. 2022 · Q4 Classical & Axiomatic Definitions of Probability · Numerical**QUESTION**

A city receives three newspapers A, B, C. A has 50% of readers, B has 30%, C has the remaining 20%. Assuming no one reads more than one paper, the probability that among 6 randomly chosen readers, 3 read A, 2 read B, and 1 reads C, is:

OPTIONS

- (a)** $\frac{1}{4}$
(b) $\frac{1}{8}$
(c) $\frac{27}{100}$
(d) $\frac{27}{200}$

ANSWER (AI-derived — no official key exists for this paper)

(d) $\frac{27}{200}$

EXAM SHORTCUT (AI-derived explanation)

Multinomial: $6!/(3!2!1!) \times (0.5)^3(0.3)^2(0.2) = 60 \times 0.125 \times 0.09 \times 0.2 = 0.135 = 27/200$.

TIPS & TRICKS (AI-derived explanation)

- Multinomial probability = $[n!/(n_1!n_2!n_3!)]p_1^{n_1}p_2^{n_2}p_3^{n_3}$. Compute the multinomial coefficient first:
 $720/(6 \times 2 \times 1) = 60$.
- 'No one reads more than one paper' is the phrase that makes the three categories mutually exclusive and exhaustive, hence multinomial rather than independent binomials.
- Keep it as a fraction: $60 \times (1/8)(9/100)(1/5) = 60 \times 9/4000 = 540/4000 = 27/200$. Fractions avoid decimal drift.
- Option (c) $27/100$ is exactly double the answer - the trap for candidates who drop a factor of 2 somewhere in the coefficient.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Each reader independently reads A, B or C with probabilities 0.5, 0.3 and 0.2, and these are mutually exclusive since no one reads more than one paper.
2. For 6 randomly chosen readers, the counts follow a multinomial distribution with $n = 6$.
3. $P(3 \text{ read A, 2 read B, 1 reads C}) = [6!/(3!2!1!)](0.5)^3(0.3)^2(0.2)^1$.
4. The multinomial coefficient is $720/(6 \times 2 \times 1) = 60$.
5. The probability factor is $(0.125)(0.09)(0.2) = 0.00225$.
6. So the probability is $60 \times 0.00225 = 0.135 = 27/200$, option (d).

7. 2022 · Q8 Classical & Axiomatic Definitions of Probability · Numerical**QUESTION**

A group of $2n$ boys and $2n$ girls is divided at random into two equal batches. What is the probability that each batch will be equally divided into boys and girls?

OPTIONS

- (a) $\frac{C(2n,n)}{C(4n,n)}$
- (b) $\frac{\{C(2n,n)\}^2}{C(4n,2n)}$
- (c) $\frac{\{C(2n,n)\}^2}{\{C(4n,n)\}^2}$
- (d) $\frac{C(2n,n)}{\{C(4n,2n)\}^2}$

ANSWER (AI-derived — no official key exists for this paper)

- (b) $\frac{\{C(2n,n)\}^2}{C(4n,2n)}$

EXAM SHORTCUT (AI-derived explanation)

Choose the first batch of $2n$ from $4n$ people in $\binom{4n}{2n}$ ways; the favourable choices take n boys from $2n$ and n girls from $2n$, giving $\left[\binom{2n}{n}\right]^2$. Hence the probability is $\left[\binom{2n}{n}\right]^2 / \binom{4n}{2n}$.

TIPS & TRICKS (AI-derived explanation)

- Count only the FIRST batch - the second is then determined, so there is no need to divide by 2 or to count both batches.
- Favourable count factorises because the boys and the girls are chosen independently: $\binom{2n}{n}$ ways for each.
- Sanity check at $n = 1$: 2 boys and 2 girls split into pairs gives $[\binom{2}{1}]^2 / \binom{4}{2} = 4/6 = 2/3$, which matches direct enumeration.
- Note that if each batch has n boys, it automatically has $2n - n = n$ girls, so one condition suffices - this is why the numerator is a simple product.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. There are $2n$ boys and $2n$ girls, i.e. $4n$ people, to be divided at random into two batches of $2n$ each.
2. It suffices to count the ways of choosing the first batch, since the second batch is then determined.
3. Total number of equally likely first batches: $\binom{4n}{2n}$.
4. For the batch to be equally divided it must contain n boys and n girls. The boys can be chosen in $\binom{2n}{n}$ ways and the girls in $\binom{2n}{n}$ ways, independently.
5. So the number of favourable choices is $[\binom{2n}{n}]^2$, and if the first batch has n boys and n girls then so does the second.
6. The probability is $[\binom{2n}{n}]^2 / \binom{4n}{2n}$, option (b).

8. 2024 · Q21 Classical & Axiomatic Definitions of Probability · Numerical**QUESTION**

A point (x, y) is to be selected from the square S containing all points (x, y) such that $0 \leq x \leq 1$ and $0 \leq y \leq 1$. What is the probability that the selected point belongs to subset of points such that $0.5 < (x + y) < 1.5$?

OPTIONS

- (a) 0.15
- (b) 0.25
- (c) 0.45
- (d) 0.75

ANSWER (AI-derived — no official key exists for this paper)

(d) 0.75

EXAM SHORTCUT (AI-derived explanation)

Work with the complement: the two corner triangles $x + y \leq 0.5$ and $x + y \geq 1.5$ each have area $(0.5)^2/2 = 0.125$, so the answer is $1 - 0.25 = 0.75$.

TIPS & TRICKS (AI-derived explanation)

- Geometric probability on the unit square = favourable AREA, since the total area is 1. Sketch the two lines $x + y = 0.5$ and $x + y = 1.5$.
- The excluded regions are right triangles at opposite corners with legs 0.5, so each has area 0.125 - far quicker than integrating the band itself.
- Symmetry check: the band is symmetric about the centre $(0.5, 0.5)$, so the two excluded triangles must be congruent - a useful confirmation.
- General rule: for the unit square, $P(x + y \leq c) = c^2/2$ for $0 \leq c \leq 1$, which gives 0.125 at $c = 0.5$ immediately.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The sample space is the unit square with area 1, so probabilities equal areas.
2. The event $0.5 < x + y < 1.5$ excludes two regions: the triangle below the line $x + y = 0.5$ and the triangle above the line $x + y = 1.5$.
3. The lower triangle has vertices $(0, 0)$, $(0.5, 0)$ and $(0, 0.5)$, so its area is $(1/2)(0.5)(0.5) = 0.125$.
4. By symmetry the upper triangle, with vertices $(1, 0.5)$, $(1, 1)$ and $(0.5, 1)$, also has area 0.125.

5. The excluded area is $0.125 + 0.125 = 0.25$.
6. Hence the required probability is $1 - 0.25 = 0.75$, option (d).

9. 2024 · Q31 Conditional Probability · Numerical

QUESTION

Two events A and B are such that $P(A) = 1/3$, $P(B) = 1/5$ and $P(A | B) + P(B | A) = 2/3$. What is the value of $12P(\bar{A} \cup \bar{B})$?

OPTIONS

- (a) $23/2$
- (b) 11
- (c) 7
- (d) 5

ANSWER (AI-derived — no official key exists for this paper)

(b) 11

EXAM SHORTCUT (AI-derived explanation)

$P(A | B) + P(B | A) = P(AB)[1/P(B) + 1/P(A)] = P(AB)(5 + 3) = 2/3$, so $P(AB) = 1/12$. Then $P(A' \cup B') = 1 - 1/12 = 11/12$ and 12 times that is 11.

TIPS & TRICKS (AI-derived explanation)

- Factor out the intersection: $P(A | B) + P(B | A) = P(AB)/P(B) + P(AB)/P(A) = P(AB)[1/P(A) + 1/P(B)]$. One line and the unknown is isolated.
- $1/P(A) + 1/P(B) = 3 + 5 = 8$, so $P(AB) = (2/3)/8 = 1/12$ immediately.
- De Morgan: $A' \cup B'$ is the complement of $A \cap B$, so $P(A' \cup B') = 1 - P(A \cap B)$. Recognising this avoids computing a union directly.
- The factor 12 in the question exists to clear the denominator, which is a strong hint that $P(AB)$ will turn out to be a twelfth.

STEP-BY-STEP SOLUTION (AI-derived explanation)

- By definition $P(A | B) = P(A \cap B)/P(B)$ and $P(B | A) = P(A \cap B)/P(A)$.
- Adding: $P(A | B) + P(B | A) = P(A \cap B)[1/P(B) + 1/P(A)] = P(A \cap B)[5 + 3] = 8P(A \cap B)$.
- Setting this equal to $2/3$ gives $P(A \cap B) = (2/3)/8 = 1/12$.
- By De Morgan's law, $A' \cup B' = (A \cap B)'$, so $P(A' \cup B') = 1 - P(A \cap B) = 1 - 1/12 = 11/12$.
- Therefore $12P(A' \cup B') = 12 \times 11/12 = 11$.
- Option (b) is correct.

10. 2026 · Q20 Classical & Axiomatic Definitions of Probability · Numerical

QUESTION

X is chosen at random between 0 and 6, and Y at random between 0 and 9. What is the probability that $Z^2 - XZ + Y = 0$ has real roots?

OPTIONS

- (a) $1/2$
- (b) $1/3$
- (c) $2/3$
- (d) $1/4$

ANSWER (AI-derived — no official key exists for this paper)

(b) $1/3$

EXAM SHORTCUT (AI-derived explanation)

Real roots need discriminant $X^2 - 4Y \geq 0$, i.e. $Y \leq X^2/4$. Since $X^2/4 \leq 9$ on $(0, 6)$, the favourable area is $\int_0^6 x^2/4 dx = 18$, out of the total area $6 \cdot 9 = 54$. So $P = 18/54 = 1/3$.

TIPS & TRICKS (AI-derived explanation)

- Geometric probability: with X and Y independent uniforms, the probability is (favourable area)/(total area) of the rectangle - no densities needed beyond the constant $1/54$.
- Check whether the curve leaves the rectangle: $y = x^2/4$ reaches 9 exactly at $x = 6$, so the parabola stays inside and no truncation ($\min$ with 9) is needed. Getting this wrong is the usual pitfall.
- The condition for real roots of $z^2 - Xz + Y = 0$ is $X^2 \geq 4Y$; note the sign convention $b^2 - 4ac$ with $a = 1, b = -X, c = Y$.
- Area under a parabola: $\int_0^6 x^2/4 dx = 6^3/12 = 18$. Memorising $\int x^2 = x^3/3$ makes this instant.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. X is uniform on $(0, 6)$ and Y is uniform on $(0, 9)$, independently, so (X, Y) is uniform on the rectangle of area $6 \cdot 9 = 54$, with joint density $1/54$.
2. The quadratic $Z^2 - XZ + Y = 0$ has real roots iff its discriminant is non-negative: $X^2 - 4Y \geq 0$, i.e. $Y \leq X^2/4$.
3. For $0 < x < 6$ the bound $x^2/4$ ranges from 0 up to $36/4 = 9$, which is exactly the top of the Y -range, so the parabola lies entirely within the rectangle and no truncation is needed.
4. Favourable area $= \int_0^6 (x^2/4) dx = (1/4)(6^3/3) = (1/4)(72) = 18$.
5. $P(\text{real roots}) = 18/54 = 1/3$.
6. Hence option (b) is correct.

P2 · PROBABILITY**Conditional Probability, Law of Total Probability, Bayes' Theorem**

13 questions · 2018: 1 2019: 2 2020: 3 2021: 0 2022: 2 2023: 1 2024: 0 2025: 2 2026: 2

Weight. 13 questions (6.6% of Probability) · appeared in 7 of 9 papers · peak year 2020 (3)**Examiner's pattern.** The urn/agency/factory Bayes template is the single most reliable item in this sub-topic — it appeared in 2018, 2019 (twice), 2023 and 2026. Two other patterns: conditioning that truncates the support of a uniform ($P(5 < X < 7 \mid X > 6)$ is a trap — the answer is 0, not a ratio of lengths), and Polya-type urns where the ball is replaced with additions.**Must-know.** Bayes with unequal priors: $P(B_k|A) = P(A|B_k)P(B_k) / \sum_j P(A|B_j)P(B_j)$. When conditioning truncates the support, intersect the events FIRST, then divide.**1. 2018 · Q4** Classical & Axiomatic Definitions of Probability · Numerical**QUESTION**

Urn 1 has 7 white, 10 black balls; urn 2 has 5 white, 12 black; urn 3 has 17 white only. An urn is chosen at random and a ball drawn is white. What are the probabilities it came from urn 1, 2, 3 respectively?

OPTIONS

- (a) $\frac{7}{29}, \frac{6}{29}, \frac{18}{29}$
 (b) $\frac{7}{29}, \frac{5}{29}, \frac{20}{29}$
 (c) $\frac{7}{29}, \frac{5}{29}, \frac{17}{29}$
 (d) None of the above

ANSWER (AI-derived — no official key exists for this paper)**(c)** $\frac{7}{29}, \frac{5}{29}, \frac{17}{29}$ **EXAM SHORTCUT (AI-derived explanation)**

All three urns hold 17 balls, so the denominators cancel. The posteriors are simply proportional to the white counts 7, 5, 17, whose total is 29. Answer: $7/29, 5/29, 17/29$ - read straight off the counts.

TIPS & TRICKS (AI-derived explanation)

- Equal urn totals + equal prior selection probability = posterior proportional to the number of favourable balls. Look for this shortcut before writing Bayes' formula.
- The three posteriors MUST sum to 1. Option (a) gives $(7 + 6 + 18)/29 = 31/29 > 1$ and option (b) gives $32/29 > 1$ - both are eliminated by inspection in about five seconds.
- Urn 3 has 17 white and 0 black, so $P(\text{white} \mid \text{urn 3}) = 1$; its posterior weight is the full 17.
- The generic form is $P(U_i \mid W) = P(U_i)P(W \mid U_i)/\sum_j P(U_j)P(W \mid U_j)$; the common prior $1/3$ cancels top and bottom.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Each urn contains 17 balls: urn 1 has 7 white and 10 black, urn 2 has 5 white and 12 black, urn 3 has 17 white and 0 black.
2. Prior: $P(\text{urn } i) = 1/3$ for each i . Likelihoods: $P(W \mid U_1) = 7/17, P(W \mid U_2) = 5/17, P(W \mid U_3) = 17/17 = 1$.
3. Total probability: $P(W) = (1/3)(7/17 + 5/17 + 17/17) = (1/3)(29/17) = 29/51$.
4. Bayes: $P(U_1 \mid W) = (1/3)(7/17)/(29/51) = (7/51)(51/29) = 7/29$.

5. Similarly $P(U_2 | W) = 5/29$ and $P(U_3 | W) = 17/29$. These sum to $29/29 = 1$, confirming the arithmetic.

6. Hence option (c) is correct.

2. 2019 · Q12 Bayes' Theorem · Numerical

QUESTION

Two cannons A_1 and A_2 fire at the same target. A_1 fires on average 9 projectiles in the time A_2 fires 10. On average 7 out of 10 projectiles from A_1 and 6 out of 10 from A_2 strike the target. If the target is struck by one projectile, the probability it was struck by a projectile from A_1 is

OPTIONS

- (a) $\frac{20}{41}$
- (b) $\frac{21}{41}$
- (c) $\frac{6}{19}$
- (d) $\frac{63}{190}$

ANSWER (AI-derived — no official key exists for this paper)

(b) $\frac{21}{41}$

EXAM SHORTCUT (AI-derived explanation)

Firing rates 9 : 10 give priors 9/19 and 10/19; the 19s cancel in Bayes' formula. So the answer is $(9)(0.7)/[(9)(0.7) + (10)(0.6)] = 6.3/12.3 = 21/41$.

TIPS & TRICKS (AI-derived explanation)

- When priors come from a ratio, keep them as raw counts (9 and 10) - the common denominator cancels top and bottom and saves all the fraction work.
- Multiply counts by hit rates to get 'effective hits': $9 \times 0.7 = 6.3$ from A_1 and $10 \times 0.6 = 6.0$ from A_2 . The posterior is just $6.3/(6.3 + 6.0)$.
- Simplify $63/123$ by dividing by 3 to reach $21/41$ - option (a) $20/41$ is the near-miss planted for arithmetic slips.
- Plausibility check: A_1 fires slightly fewer but hits slightly more often, so the posterior should be very close to $1/2$ and just above it. $21/41 = 0.512$ fits; $6/19 = 0.32$ does not.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Cannon A_1 fires 9 projectiles in the time A_2 fires 10, so a randomly chosen projectile comes from A_1 with probability 9/19 and from A_2 with probability 10/19.
2. The hit probabilities are $P(\text{strike} | A_1) = 7/10 = 0.7$ and $P(\text{strike} | A_2) = 6/10 = 0.6$.
3. Total probability of a strike: $P(\text{strike}) = (9/19)(0.7) + (10/19)(0.6) = (6.3 + 6.0)/19 = 12.3/19$.
4. By Bayes' theorem, $P(A_1 | \text{strike}) = (9/19)(0.7)/(12.3/19) = 6.3/12.3$.
5. Simplifying: $6.3/12.3 = 63/123 = 21/41$.
6. Hence option (b) is correct.

3. **2019 · Q13** Bayes' Theorem · Numerical**QUESTION**

5% of patients suffering from a certain disease are selected to undergo a new treatment believed to increase the recovery rate from 30% to 50%. A person is randomly selected from these patients after treatment and found to have recovered. What is the probability the patient received the new treatment?

OPTIONS

- (a) 0.03
- (b) 0.08
- (c) 0.80
- (d) 0.30

ANSWER (AI-derived — no official key exists for this paper)

(b) 0.08

EXAM SHORTCUT (AI-derived explanation)

Numerator $0.05 \times 0.50 = 0.025$; other branch $0.95 \times 0.30 = 0.285$. Posterior $= 0.025/0.310 = 0.0806$, about 0.08.

TIPS & TRICKS (AI-derived explanation)

- Set out the two branches before computing: treated (0.05, recovery 0.50) and untreated (0.95, recovery 0.30). Bayes is then pure arithmetic.
- A small prior is rarely overturned by a modest likelihood gain: 5% prior with only a 5/3 likelihood ratio can only reach about 8%. That intuition eliminates options (c) and (d) instantly.
- Option (a) 0.03 is the JOINT probability 0.025 rounded - the classic error of forgetting to divide by the total.
- Denominator check: $0.025 + 0.285 = 0.310$ is the overall recovery rate, 31%, which sensibly sits just above the untreated rate of 30%.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Let T be the event that the patient received the new treatment and R the event of recovery.
2. Given: $P(T) = 0.05$, $P(\text{not } T) = 0.95$, $P(R | T) = 0.50$, $P(R | \text{not } T) = 0.30$.
3. Joint probabilities: $P(T \text{ and } R) = 0.05 \times 0.50 = 0.025$ and $P(\text{not } T \text{ and } R) = 0.95 \times 0.30 = 0.285$.
4. Total probability of recovery: $P(R) = 0.025 + 0.285 = 0.310$.
5. By Bayes' theorem, $P(T | R) = 0.025/0.310 = 0.0806$.
6. Rounding, the probability is about 0.08, option (b).

4. **2020 · Q22** Conditional Probability · Numerical**QUESTION**

A die is loaded so each odd number is twice as likely as each even number. A : number > 3 . B : number is a perfect square. 1. $P(B) = 1/3$. 2. $P(B | A) = 1/2$.

OPTIONS

- (a) 1 only
- (b) 2 only

- (c) Both 1 and 2
- (d) Neither 1 nor 2

ANSWER (AI-derived — no official key exists for this paper)

(a) 1 only

EXAM SHORTCUT (AI-derived explanation)

Odd faces have probability $2/9$ each and even faces $1/9$ each. Then $P(B) = P(1) + P(4) = 2/9 + 1/9 = 1/3$ (statement 1 true), while $P(B | A) = P(4) / P(4, 5, 6) = (1/9) / (4/9) = 1/4$, not $1/2$.

TIPS & TRICKS (AI-derived explanation)

- Set up the weights first: 3 odd faces at $2p$ and 3 even faces at p give $9p = 1$, so $p = 1/9$. Everything else is a two-term sum.
- The perfect squares in 1 to 6 are 1 and 4 only - 9 is out of range. Miscounting this set is the commonest error in the item.
- For the conditional, intersect BEFORE dividing: $A = \{4, 5, 6\}$ and $B = \{1, 4\}$, so $A \cap B = \{4\}$ with probability $1/9$.
- Sanity check: $P(B) = 1/3$ while $P(B | A) = 1/4$, so knowing the number exceeds 3 makes a perfect square LESS likely - reasonable, since the square 1 is excluded by A .

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Let each even face have probability p ; then each odd face has probability $2p$. Since there are three of each, $3(2p) + 3p = 9p = 1$, so $p = 1/9$.
2. Thus $P(1) = P(3) = P(5) = 2/9$ and $P(2) = P(4) = P(6) = 1/9$.
3. Event A (number greater than 3) = $\{4, 5, 6\}$, so $P(A) = 1/9 + 2/9 + 1/9 = 4/9$.
4. Event B (perfect square) = $\{1, 4\}$, so $P(B) = 2/9 + 1/9 = 3/9 = 1/3$. Statement 1 is TRUE.
5. $A \cap B = \{4\}$, with probability $1/9$, so $P(B | A) = (1/9) / (4/9) = 1/4$.
6. Statement 2 claims $1/2$ and is therefore FALSE. Only statement 1 holds, option (a).

5. **2020 · Q30** Conditional Probability · Numerical

QUESTION

Events A, B with $P(A) = 1/4$, $P(B | A) = 1/2$, $P(A | B) = 1/4$. Let $X = 1$ if A occurs (else 0), $Y = 1$ if B occurs (else 0). 1. $P(X^2 + Y^2 = 1) = 1/4$. 2. $P(XY = X^2Y^2) = 1$.

OPTIONS

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

ANSWER (AI-derived — no official key exists for this paper)

(b) 2 only

EXAM SHORTCUT (AI-derived explanation)

$P(A \cap B) = P(A)P(B | A) = 1/8$ and $P(B) = P(A \cap B)/P(A | B) = 1/2$. Statement 1 asks for $P(\text{exactly one of } A, B) = 1/4 + 1/2 - 2(1/8) = 1/2$, not $1/4$ - false. Statement 2 is an identity for indicators - true.

TIPS & TRICKS (AI-derived explanation)

- Indicators satisfy $X^2 = X$ and $Y^2 = Y$, so $X^2 + Y^2 = 1$ means exactly one of the two events occurs, and $XY = X^2Y^2$ holds identically.
- Statement 2 needs no numbers at all - recognise it as an algebraic identity for 0/1 variables and it is true with probability 1 whatever the events.
- $P(\text{exactly one}) = P(A) + P(B) - 2P(A \cap B)$. The factor 2 is essential: it removes the intersection from BOTH marginal probabilities.
- Recover $P(B)$ from Bayes in reverse: $P(B) = P(A \cap B)/P(A | B) = (1/8)/(1/4) = 1/2$. Chaining the two conditional probabilities is the only computational step in the item.

STEP-BY-STEP SOLUTION (AI-derived explanation)

- From $P(A) = 1/4$ and $P(B | A) = 1/2$ we get $P(A \cap B) = P(A)P(B | A) = 1/8$.
- From $P(A | B) = P(A \cap B)/P(B) = 1/4$ we get $P(B) = (1/8)/(1/4) = 1/2$.
- X and Y are indicator variables, so $X^2 = X$ and $Y^2 = Y$.
- Statement 1: $X^2 + Y^2 = X + Y = 1$ means exactly one of A, B occurs.
 $P(\text{exactly one}) = P(A) + P(B) - 2P(A \cap B) = 1/4 + 1/2 - 2(1/8) = 3/4 - 1/4 = 1/2$. The claimed value $1/4$ is wrong, so statement 1 is FALSE.
- Statement 2: since $X^2 = X$ and $Y^2 = Y$ identically, $XY = X^2Y^2$ always holds, so its probability is 1. TRUE.
- Only statement 2 is correct, option (b).

6. 2020 · Q40 Bayes' Theorem · Numerical**QUESTION**

A company makes bulbs at factories A, B, C. A: 40% of output, 2% defective. B: 35%, 4% defective. C: 25%, 3% defective. Given a defective bulb, the probability it is from A is

OPTIONS

- (a) 0.20
 (b) 0.25
 (c) 0.27
 (d) 0.47

ANSWER (AI-derived — no official key exists for this paper)

(c) 0.27

EXAM SHORTCUT (AI-derived explanation)

Branch products: A gives $0.40(0.02) = 0.0080$, B gives $0.35(0.04) = 0.0140$, C gives $0.25(0.03) = 0.0075$. Total 0.0295, so $P(A | D) = 0.0080/0.0295 = 0.27$.

TIPS & TRICKS (AI-derived explanation)

- Lay out a three-row table of (share x defect rate) products before dividing; the denominator is simply their sum.
- Working in units of 1/10000 keeps the arithmetic clean: 80, 140, 75, total 295, so the answer is $80/295 = 0.271$.
- The overall defect rate 0.0295 is a useful by-product and is often the next part of such a question - compute it once and keep it.
- Plausibility: A supplies 40% of output but has the LOWEST defect rate, so its share of defectives must fall below 40% - 27% is exactly the expected direction, while option (d) 0.47 goes the wrong way.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Let D be the event that a bulb is defective. The prior shares are $P(A) = 0.40$, $P(B) = 0.35$, $P(C) = 0.25$.
2. The conditional defect rates are $P(D | A) = 0.02$, $P(D | B) = 0.04$, $P(D | C) = 0.03$.
3. Joint probabilities: $P(A \cap D) = 0.40(0.02) = 0.0080$; $P(B \cap D) = 0.35(0.04) = 0.0140$; $P(C \cap D) = 0.25(0.03) = 0.0075$.
4. Total probability of a defective bulb: $P(D) = 0.0080 + 0.0140 + 0.0075 = 0.0295$.
5. By Bayes' theorem, $P(A | D) = 0.0080/0.0295 = 0.2712$.
6. Rounding, the probability is 0.27, option (c).

7. 2022 · Q42 Bayes' Theorem · Numerical**QUESTION**

An urn contains 3 white and 4 black balls. A ball is drawn, its colour noted, returned along with two additional balls of the same colour. If a ball is drawn again, what is the probability it is white?

OPTIONS

- (a) $\frac{3}{7}$
 (b) $\frac{4}{7}$
 (c) $\frac{1}{3}$
 (d) $\frac{5}{9}$

ANSWER (AI-derived — no official key exists for this paper)

(a) $\frac{3}{7}$

EXAM SHORTCUT (AI-derived explanation)

Condition on the first draw: $(3/7)(5/9) + (4/7)(3/9) = 15/63 + 12/63 = 27/63 = 3/7$ - unchanged from the initial proportion.

TIPS & TRICKS (AI-derived explanation)

- This is a Polya urn. Its signature property is that the marginal probability of drawing white is the SAME on every draw, whatever the reinforcement - so the answer is $3/7$ without any computation.
- If you compute directly: after adding 2 balls the urn holds 9, with 5 white if the first was white and 3 white if it was black.
- The law of total probability is the standard route: $P(W_2) = P(W_1)P(W_2 | W_1) + P(B_1)P(W_2 | B_1)$.

- The invariance means the draws are exchangeable but NOT independent - a useful distinction to remember for follow-up parts.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Initially the urn has 3 white and 4 black balls, so 7 in total; after the first draw the ball is replaced together with 2 more of the same colour, leaving 9 balls.
2. If the first ball is white (probability $3/7$), the urn then holds 5 white out of 9, so $P(\text{second white} \mid \text{first white}) = 5/9$.
3. If the first ball is black (probability $4/7$), the urn holds 3 white out of 9, so $P(\text{second white} \mid \text{first black}) = 3/9$.
4. By the law of total probability, $P(\text{second white}) = (3/7)(5/9) + (4/7)(3/9) = 15/63 + 12/63 = 27/63 = 3/7$.
5. This equals the original proportion of white balls, the characteristic invariance of the Polya urn. Option (a) is correct.

8. 2022 · Q44 Classical & Axiomatic Definitions of Probability · Numerical

QUESTION

A box contains two coins, one "fair" and one "two headed". One coin is chosen at random and tossed twice. If two heads appear, what is the probability the chosen coin was "two headed"?

OPTIONS

- (a) $\frac{1}{2}$
- (b) $\frac{1}{3}$
- (c) $\frac{1}{4}$
- (d) $\frac{4}{5}$

ANSWER (AI-derived — no official key exists for this paper)

(d) $\frac{4}{5}$

EXAM SHORTCUT (AI-derived explanation)

$P(HH \mid \text{two-headed}) = 1$ and $P(HH \mid \text{fair}) = 1/4$, with equal priors. So the posterior is $1/(1 + 1/4) = 4/5$.

TIPS & TRICKS (AI-derived explanation)

- With EQUAL priors, Bayes reduces to the ratio of likelihoods: $1 : 1/4$, i.e. $4 : 1$, giving $4/5$ directly.
- Two heads is four times more likely under the two-headed coin, so the posterior must exceed the prior $1/2$ substantially - options (b) and (c) are below $1/2$ and can be dismissed.
- Extend the pattern: with k heads the posterior is $1/(1 + 2^{-k}) = 2^k/(2^k + 1)$, which tends to 1 - a useful generalisation.
- Option (a) $1/2$ is the PRIOR, the trap for candidates who forget to update on the evidence.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Let T be the event that the two-headed coin was chosen and F that the fair coin was chosen, with $P(T) = P(F) = 1/2$.
2. $P(\text{two heads} \mid T) = 1$, since both faces are heads.
3. $P(\text{two heads} \mid F) = (1/2)(1/2) = 1/4$.
4. Total probability: $P(\text{two heads}) = (1/2)(1) + (1/2)(1/4) = 1/2 + 1/8 = 5/8$.
5. By Bayes' theorem, $P(T \mid \text{two heads}) = (1/2)(1)/(5/8) = (1/2)(8/5) = 4/5$.

6. Option (d) is correct.

9. **2023 · Q1** Classical & Axiomatic Definitions of Probability · Numerical

QUESTION

The members of a firm rent cars from three rental agencies: 60% from Agency-1, 30% from Agency-2 and 10% from Agency-3. If 9% of cars from Agency-1 need an oil change, 20% of cars from Agency-2 need an oil change and 6% of cars from Agency-3 need an oil change, then what is the probability that a rental car delivered to the firm will need an oil change?

OPTIONS

- (a) 0.006
- (b) 0.054
- (c) 0.06
- (d) 0.12

ANSWER (AI-derived — no official key exists for this paper)

(d) 0.12

EXAM SHORTCUT (AI-derived explanation)

Law of total probability: $0.6(0.09) + 0.3(0.20) + 0.1(0.06) = 0.054 + 0.060 + 0.006 = 0.12$.

TIPS & TRICKS (AI-derived explanation)

- Lay the three branches out as (share x rate) products before adding: 0.054, 0.060, 0.006. The sum is the overall rate.
- Work in per-cent-of-a-per-cent to avoid decimal slips: $54 + 60 + 6 = 120$ ten-thousandths, i.e. $0.0120 \times 10 = 0.12$.
- Sanity bound: the answer must lie between the smallest rate (6%) and the largest (20%). Only 0.12 does - options (a) and (b) are individual branch products.
- Option (b) 0.054 is the Agency-1 branch alone and option (a) 0.006 the Agency-3 branch - both punish stopping before the addition.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Let A_1, A_2, A_3 be the events that a car comes from Agency-1, 2 and 3, with $P(A_1) = 0.6, P(A_2) = 0.3, P(A_3) = 0.1$.
2. Let C be the event that the car needs an oil change, with $P(C | A_1) = 0.09, P(C | A_2) = 0.20, P(C | A_3) = 0.06$.
3. By the law of total probability, $P(C) = P(A_1)P(C | A_1) + P(A_2)P(C | A_2) + P(A_3)P(C | A_3)$.
4. Branch products: $0.6(0.09) = 0.054; 0.3(0.20) = 0.060; 0.1(0.06) = 0.006$.
5. Adding: $0.054 + 0.060 + 0.006 = 0.120$.
6. Hence the probability is 0.12, option (d).

10. **2025 · Q4** Bayes' Theorem · Numerical

QUESTION

A player rolls a fair dice and simultaneously tosses a fair coin. If "head" appears on the coin, the player wins twice the value on the dice; if "tail" appears, the player wins half the value on the dice. What is the

player's expected win?

OPTIONS

- (a) 10.250
- (b) 8.625
- (c) 7.625
- (d) 4.375

ANSWER (AI-derived — no official key exists for this paper)

(d) 4.375

EXAM SHORTCUT (AI-derived explanation)

$E(D) = 3.5$. Head pays $2D$ and tail pays $D/2$, each with probability $1/2$, so the expected win is $(1/2)(2)(3.5) + (1/2)(0.5)(3.5) = 3.5 + 0.875 = 4.375$.

TIPS & TRICKS (AI-derived explanation)

- Condition on the coin: the payoff is a MULTIPLE of the die value in each case, so $E(\text{win}) = E(\text{multiplier}) \times E(D)$ with the multiplier averaging $(2 + 0.5)/2 = 1.25$.
- $1.25 \times 3.5 = 4.375$ - a one-line route once the independence of coin and die is noticed.
- $E(D) = (1 + 2 + \dots + 6)/6 = 3.5$ for a fair die; keeping it as $7/2$ makes the arithmetic exact.
- Note the payoffs are asymmetric: doubling gains more than halving loses, so the expected win must EXCEED 3.5 - but only modestly, which rules out options (a), (b) and (c).

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Let D be the value shown on the die, uniform on 1 to 6, so $E(D) = (1 + 2 + 3 + 4 + 5 + 6)/6 = 3.5$.
2. The coin is fair and independent of the die: with probability $1/2$ the player wins $2D$, and with probability $1/2$ he wins $D/2$.
3. By the law of total expectation,

$$E(\text{win}) = (1/2) E(2D) + (1/2) E(D/2) = (1/2)(2)(3.5) + (1/2)(0.5)(3.5) = 3.5 + 0.875 = 4.375.$$
4. Equivalently, the expected multiplier is $(2 + 0.5)/2 = 1.25$ and $1.25 \times 3.5 = 4.375$. Option (d) is correct.

11. 2025 · Q5 Bayes' Theorem · Numerical

QUESTION

A miner is trapped in a mine containing 3 doors. The first door leads to a tunnel that takes him to safety after 3 hours. The second returns him to the mine after 5 hours. The third returns him to the mine after 7 hours. Assuming the miner is equally likely to choose any door, what is the expected time until he reaches safety?

OPTIONS

- (a) 15 hours
- (b) 25 hours
- (c) 30 hours
- (d) 35 hours

ANSWER (AI-derived — no official key exists for this paper)**(a)** 15 hours**EXAM SHORTCUT (AI-derived explanation)**

Condition on the first door: $E = (1/3)(3) + (1/3)(5 + E) + (1/3)(7 + E) = 5 + 2E/3$, so $E/3 = 5$ and $E = 15$ hours.

TIPS & TRICKS (AI-derived explanation)

- Set up a self-referential equation: doors 2 and 3 return the miner to the SAME situation, so E reappears on the right-hand side.
- Collect the constants first: $(3 + 5 + 7)/3 = 5$, and the E -terms give $2E/3$. The equation $E = 5 + 2E/3$ then solves in one step.
- The memoryless structure is essential - after returning, the miner's expected remaining time is again E , with no memory of the failed attempts.
- Intuition check: he escapes on the first try with probability $1/3$, so he expects 3 attempts averaging 5 hours each, giving 15.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Let E be the expected time to reach safety from inside the mine.
2. Condition on which door the miner chooses first, each with probability $1/3$.
3. Door 1: he reaches safety after 3 hours, contributing $(1/3)(3)$.
4. Door 2: he wanders for 5 hours and returns to the mine, facing the same situation afresh, contributing $(1/3)(5 + E)$. Door 3 similarly contributes $(1/3)(7 + E)$.
5. So $E = (1/3)(3) + (1/3)(5 + E) + (1/3)(7 + E) = (3 + 5 + 7)/3 + (2/3)E = 5 + (2/3)E$.
6. Hence $E - (2/3)E = 5$, so $E/3 = 5$ and $E = 15$ hours, option (a).

12. **2026 · Q12** Classical & Axiomatic Definitions of Probability · Numerical**QUESTION**

Three identical urns contain white/black balls: urn 1 has 2 white, 3 black; urn 2 has 3 white, 5 black; urn 3 has 5 white, 2 black. An urn is chosen at random and a ball drawn. Given the ball is white, what is the probability the second urn was chosen?

OPTIONS

- (a)** 35/139
(b) 200/417
(c) 112/417
(d) 105/139

ANSWER (AI-derived — no official key exists for this paper)**(a)** 35/139

EXAM SHORTCUT (AI-derived explanation)

The prior $1/3$ cancels, so the posterior is just $3/8$ divided by $(2/5 + 3/8 + 5/7)$. Over LCD 280 these are 112, 105 and 200, total 417. Answer $105/417 = 35/139$.

TIPS & TRICKS (AI-derived explanation)

- When all priors are equal they cancel in Bayes' formula - the posterior is simply the likelihood divided by the sum of the likelihoods.
- Put the three white-ball probabilities over a common denominator (280 here) at once; the numerators 112, 105, 200 then make the arithmetic exact.
- Always reduce: $105/417$ has common factor 3, giving $35/139$. The unreduced form is deliberately absent, and $112/417$ is the answer for urn 1 while $200/417$ is for urn 3.
- Cross-check: the three posteriors $112/417$, $105/417$ and $200/417$ must sum to 1 - they do.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Let U_1, U_2, U_3 be the events that urn 1, 2, 3 is chosen, each with prior probability $1/3$, and let W be the event that the drawn ball is white.
2. Likelihoods: $P(W | U_1) = 2/5, P(W | U_2) = 3/8, P(W | U_3) = 5/7$.
3. Law of total probability: $P(W) = (1/3)(2/5 + 3/8 + 5/7)$.
4. Over the common denominator 280: $2/5 = 112/280, 3/8 = 105/280, 5/7 = 200/280$, so the sum is $417/280$ and $P(W) = 417/840$.
5. Bayes' theorem: $P(U_2 | W) = [(1/3)(3/8)] / [(1/3)(417/280)] = (3/8) / (417/280) = (105/280) / (417/280) = 105/417$.
6. Reducing by 3: $105/417 = 35/139$ (approximately 0.2518).
7. Hence option (a) is correct.

13. 2026 · Q16 Conditional Probability · Numerical**QUESTION**

X has distribution $f(x) = \begin{cases} \frac{1}{12}, & 0 < x < 12 \\ 0, & \text{otherwise} \end{cases}$. What is $P(5 < X < 7 | X > 6)$?

OPTIONS

- (a) $1/3$
- (b) $3/4$
- (c) $2/3$
- (d) $1/6$

ANSWER (AI-derived — no official key exists for this paper)

(d) $1/6$

EXAM SHORTCUT (AI-derived explanation)

Intersect the events first: $(5, 7)$ intersect $(6, 12) = (6, 7)$, of length 1. Divide by the length of the conditioning event $(6, 12)$, which is 6. Answer $1/6$.

TIPS & TRICKS (AI-derived explanation)

- For a uniform variable, conditional probability is just a ratio of interval lengths - no integration needed.

- Always intersect before dividing: the numerator is $P(5 < X < 7 \text{ AND } X > 6) = P(6 < X < 7)$, not $P(5 < X < 7) = 2/12$.
- The distractor $1/3$ comes from forgetting the intersection: $(2/12)/(6/12) = 1/3$. This is the single most common error on this item.
- General identity: $P(A | B) = P(A \text{ intersect } B) / P(B)$, valid whenever $P(B) > 0$.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. X is uniform on $(0, 12)$ with density $1/12$, so probabilities are proportional to interval lengths.
2. Conditioning event: $P(X > 6) = (12 - 6)/12 = 6/12 = 1/2$.
3. Intersection: $\{5 < X < 7\} \text{ intersect } \{X > 6\} = \{6 < X < 7\}$, so $P = (7 - 6)/12 = 1/12$.
4. $P(5 < X < 7 | X > 6) = (1/12)/(1/2) = 2/12 = 1/6$.
5. Hence option (d) is correct.

P3 · PROBABILITY

Random Variables & Distribution Functions

10 questions · 2018: 1 2019: 3 2020: 3 2021: 0 2022: 2 2023: 1 2024: 0 2025: 0 2026: 0

Weight. 10 questions (5.1% of Probability) · appeared in 5 of 9 papers · peak year 2019 (3)**Examiner's pattern.** Given a piecewise CDF containing an unknown alpha, solve for alpha or evaluate a probability — the exact same shell was used twice in one paper (2020 Q33 and Q36). Also: verifying a constant makes a function a valid pmf/pdf, and mixed distributions with a jump.**Must-know.** F is non-decreasing, right-continuous, $F(-\infty)=0$, $F(\infty)=1$; jump size at x equals $P(X=x)$; for a valid pdf, integral over the support = 1.1. **2018 · Q3** Discrete & Continuous Random Variables · Numerical

QUESTION

 X has p.d.f.

$$f(x) = \begin{cases} \frac{x}{2}, & 0 < x \leq 1 \\ \frac{1}{2}, & 1 < x \leq 2 \\ \frac{3-x}{3}, & 2 < x \leq 3 \end{cases}$$

What is $P(1.5 < X \leq 2.5 \mid X > 1)$?

OPTIONS

- (a) $\frac{3}{8}$
 (b) $\frac{1}{2}$
 (c) $\frac{5}{8}$
 (d) $\frac{1}{4}$

ANSWER (AI-derived — no official key exists for this paper)**Not keyed.** No option in the printed paper is correct for this item; see the source note below.

Source note. SOURCE DEFECT - stem and options are mutually inconsistent. As printed, the third branch $\frac{(3-x)}{3}$ on $2 < x \leq 3$ contributes $\frac{1}{6}$, giving a total mass of $\frac{1}{4} + \frac{1}{2} + \frac{1}{6} = \frac{11}{12}$, so the printed function is not a density. With the third branch normalised to the only value that makes f a density, $\frac{(3-x)}{2}$, the exact answer is $P(1.5 < X \leq 2.5 \mid X > 1) = \frac{(7/16)}{(3/4)} = \frac{7}{12}$, which is NOT among the four printed options; taking the branch exactly as printed gives $\frac{9}{16}$, which is also not among them. The source has therefore been preserved verbatim and this question is flagged and excluded from scoring rather than silently rewritten. No option has been invented.

EXAM SHORTCUT (AI-derived explanation)

AI-derived: with the density normalised, $P(X > 1) = \frac{3}{4}$ and $P(1.5 < X \leq 2.5) = \frac{1}{2} \times 0.5 + \text{area of the trapezium under } (3-x)/2 \text{ from } 2 \text{ to } 2.5 = 0.25 + 0.1875 = \frac{7}{16}$, giving $\frac{7}{12}$.

TIPS & TRICKS (AI-derived explanation)

- Always run the normalisation check FIRST on a piecewise density: integrate every branch and confirm the pieces sum to 1. Here they do not, which is the fastest way to spot the misprint.

- For a linear branch, the integral is just the area of a trapezium: (average height) $\times$ (width). From 2 to 2.5 the heights are 0.5 and 0.25, so the area is $0.375 \times 0.5 = 0.1875$.
- $P(A | B)$ with $B = \{X > 1\}$: the denominator is 1 minus the mass of the first branch, not 1 minus the value of the density.
- In a real exam, a stem that fails the normalisation check and whose exact answer is absent from the options is a candidate for a UPSC key challenge - mark it and move on rather than forcing a fit.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Step 0 (integrity check): $\int x/2$ over $(0,1] = 1/4$; $\int 1/2$ over $(1,2] = 1/2$; $\int (3-x)/3$ over $(2,3] = (1/3)(1/2) = 1/6$. Total = $11/12$, not 1, so the function as printed is not a probability density.
2. Reading the third branch as $(3-x)/2$ - the unique constant that normalises it - gives masses $1/4, 1/2, 1/4$ which sum to 1.
3. Denominator: $P(X > 1) = 1 - 1/4 = 3/4$.
4. Numerator: $P(1.5 < X \leq 2.5) = \int 1/2$ from 1.5 to 2 plus $\int (3-x)/2$ from 2 to 2.5 $= 0.25 + 0.1875 = 0.4375 = 7/16$.
5. Therefore $P(1.5 < X \leq 2.5 | X > 1) = (7/16)/(3/4) = 7/12 = 0.5833$.
6. $7/12$ does not equal $3/8, 1/2, 5/8$ or $1/4$, so no printed option is correct and the question is flagged as a source defect.

2. 2019 · Q14 Mathematical Expectation · Numerical**QUESTION**

A discrete random variable X takes four values $-1.5, -1, 2.5, 3$ with respective probabilities $\frac{1}{6}, \frac{1}{4}, \frac{1}{2}, \frac{1}{12}$. What is $E(|X|)$ equal to?

OPTIONS

- (a) 0
(b) $\frac{1}{4}$
(c) $\frac{3}{2}$
(d) 2

ANSWER (AI-derived — no official key exists for this paper)**(d) 2****EXAM SHORTCUT (AI-derived explanation)**

Take absolute values first: $|X| = 1.5, 1, 2.5, 3$ with the same probabilities $1/6, 1/4, 1/2, 1/12$. The products are 0.25, 0.25, 1.25, 0.25, summing to 2.

TIPS & TRICKS (AI-derived explanation)

- $E(|X|)$ is NOT $|E(X)|$ - apply the absolute value to each value before weighting. Here $E(X) = 0.75$ while $E|X| = 2$.
- Convert to twelfths at the start: the probabilities are $2/12, 3/12, 6/12, 1/12$, which also verifies that they sum to 1.
- The arithmetic is unusually friendly: three of the four products equal 0.25 and the fourth is 1.25, so the total 2 falls out at a glance.
- Option (b) $1/4$ is the value of a single term and option (c) $3/2$ comes from mishandling the largest-probability term - both punish ∂ arithmetic.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The random variable takes the values -1.5, -1, 2.5, 3 with probabilities $1/6, 1/4, 1/2, 1/12$. These sum to $2/12 + 3/12 + 6/12 + 1/12 = 1$, so the distribution is valid.
2. The absolute values are $|X| = 1.5, 1, 2.5, 3$ with the same respective probabilities.
3. $E(|X|) = 1.5(1/6) + 1(1/4) + 2.5(1/2) + 3(1/12)$.
4. Term by term: $1.5/6 = 0.25$; $1/4 = 0.25$; $2.5/2 = 1.25$; $3/12 = 0.25$.
5. Adding: $0.25 + 0.25 + 1.25 + 0.25 = 2$.
6. Hence $E(|X|) = 2$, option (d).

3. 2019 · Q37 Discrete & Continuous Random Variables · Numerical**QUESTION**

Let X be a continuous random variable with p.d.f.

$$f(x) = \begin{cases} ax, & 0 \leq x < 1 \\ a, & 1 \leq x \leq 2 \\ -ax + 3a, & 2 < x \leq 3 \\ 0, & \text{elsewhere} \end{cases}$$

What is the value of the constant a ?

OPTIONS

- (a) 1
 (b) $\frac{1}{2}$
 (c) $\frac{1}{3}$
 (d) $\frac{1}{4}$

ANSWER (AI-derived — no official key exists for this paper)

(b) $\frac{1}{2}$

EXAM SHORTCUT (AI-derived explanation)

The three pieces are a triangle, a rectangle and a triangle with areas $a/2, a$ and $a/2$, totalling $2a$. Setting $2a = 1$ gives $a = 1/2$.

TIPS & TRICKS (AI-derived explanation)

- Use AREAS rather than integrals for piecewise-linear densities: a triangle of base 1 and height a has area $a/2$, and a rectangle of width 1 and height a has area a .
- The trapezoidal shape here (rise, flat, fall) is symmetric about $x = 1.5$, which is therefore both the mean and the median - useful for follow-up parts.
- Total area $2a = 1$ gives $a = 1/2$ immediately; no calculus is needed anywhere in this item.
- Contrast with 2018Q₃, where the same trapezoidal shape appeared with an inconsistent third branch - always run this normalisation check first.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The density has three pieces: $f = ax$ on $[0, 1]$, $f = a$ on $[1, 2]$, and $f = a(3 - x)$ on $(2, 3]$.
2. Area of the first piece: the $\int ax$ from 0 to 1 = $a/2$ (a triangle of base 1 and height a).
3. Area of the second piece: the $\int a$ from 1 to 2 = a (a rectangle of width 1 and height a).

4. Area of the third piece: the $\int a(3-x)$ from 2 to 3 = $a[3x - x^2/2]$ from 2 to 3 = $a(4.5 - 4) = a/2$ (a triangle of base 1 and height a).
5. Total area = $a/2 + a + a/2 = 2a$, and this must equal 1.
6. Therefore $a = 1/2$, option (b).

4. **2019 · Q39** Mathematical Expectation · Numerical

QUESTION

The random variable X has p.d.f. $f(x) = \frac{3x^2}{2}$, $-1 < x < 1$. What are the values of $E(|X|)$ and $\text{var}(|X|)$ respectively?

OPTIONS

- (a) $\frac{1}{4}$ and $\frac{1}{80}$
- (b) $\frac{3}{4}$ and $\frac{1}{80}$
- (c) $\frac{1}{4}$ and $\frac{3}{80}$
- (d) $\frac{3}{4}$ and $\frac{3}{80}$

ANSWER (AI-derived — no official key exists for this paper)

(d) $\frac{3}{4}$ and $\frac{3}{80}$

EXAM SHORTCUT (AI-derived explanation)

By symmetry $E|X| = 2$ times the $\int x(3x^2/2)$ over $(0, 1) = 3/4$, and $E(X^2) = 3/5$. So $\text{Var}(|X|) = 3/5 - 9/16 = 3/80$.

TIPS & TRICKS (AI-derived explanation)

- Since the density is even, use $E|X| = 2$ times the integral over $(0, 1)$ - the symmetry halves the work.
- $E(|X|^2) = E(X^2)$, so you never need a separate integral for the second moment of $|X|$. That equality is the key labour-saving step.
- Arithmetic: $3/5 - 9/16 = 48/80 - 45/80 = 3/80$. Use the common denominator 80 to avoid a slip - option (b) offers $1/80$ as the near-miss.
- Sanity check: $|X|$ lies in $(0, 1)$ with mean 0.75, so its variance must be small - $3/80 = 0.0375$ is entirely plausible.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The density $f(x) = 3x^2/2$ on $(-1, 1)$ is even, and its integral is $(3/2)(2/3) = 1$, so it is valid.
2. $E(|X|) =$ the $\int |x|(3x^2/2)$ over $(-1, 1) = 2$ times the $\int_0^1 (3x^3/2) dx = 2(3/2)(1/4) = 3/4$.
3. $E(|X|^2) = E(X^2) =$ the $\int x^2(3x^2/2)$ over $(-1, 1) = 2$ times the $\int_0^1 (3x^4/2) dx = 2(3/2)(1/5) = 3/5$.
4. $\text{Var}(|X|) = E(X^2) - [E(|X|)]^2 = 3/5 - (3/4)^2 = 3/5 - 9/16$.
5. Using the common denominator 80: $48/80 - 45/80 = 3/80$.
6. Hence $E(|X|) = 3/4$ and $\text{Var}(|X|) = 3/80$, option (d).

5. **2020 · Q23** Discrete & Continuous Random Variables · Numerical

QUESTION

If $p(x) = kp^x$, $0 < p < 1$, $x = 0, 2, 4, 6, \dots$, the value of k for which $p(x)$ is a p.m.f. is

OPTIONS

- (a) $1 - p$
- (b) p
- (c) $1 - p^2$
- (d) p^2

ANSWER (AI-derived — no official key exists for this paper)**(c)** $1 - p^2$ **EXAM SHORTCUT (AI-derived explanation)**

The support is the EVEN integers, so the sum is $k(1 + p^2 + p^4 + \dots) = k/(1 - p^2) = 1$, giving $k = 1 - p^2$.

TIPS & TRICKS (AI-derived explanation)

- Read the support carefully: $x = 0, 2, 4, \dots$ means the ratio between consecutive terms is p^2 , not p . That single observation is the whole question.
- Geometric series: sum of r^j for $j \geq 0$ is $1/(1 - r)$. Here $r = p^2$, so the normaliser is $1 - p^2$.
- Option (a) $1 - p$ is the answer to the SAME question with support $0, 1, 2, 3, \dots$ - the trap for candidates who skim the support.
- Check with $p = 1/2$: $k = 3/4$, and the probabilities $3/4, 3/16, 3/64, \dots$ sum to $(3/4)/(1 - 1/4) = 1$.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The probability mass function is $p(x) = kp^x$ for $x = 0, 2, 4, 6, \dots$, i.e. only the even non-negative integers.
2. Writing $x = 2j$ for $j = 0, 1, 2, \dots$, the masses are $kp^{2j} = k(p^2)^j$.
3. For $p(x)$ to be a valid pmf, the total must be 1: $k \sum (p^2)^j = 1$.
4. Since $0 < p < 1$ we have $0 < p^2 < 1$, so the geometric series sums to $1/(1 - p^2)$.
5. Hence $k/(1 - p^2) = 1$, giving $k = 1 - p^2$.
6. Option (c) is correct.

6. 2020 · Q33 Distribution Functions & Their Properties · Theoretical**QUESTION**

The function

$$F(x) = \begin{cases} 0, & x < 0 \\ x/2, & 0 \leq x < 1 \\ (1 - \alpha)x + 2\alpha - 1, & 1 \leq x < 2 \\ 1, & x \geq 2 \end{cases}$$

is a distribution function if and only if

OPTIONS

- (a) $0 < \alpha < \frac{1}{2}$
- (b) $\frac{1}{2} \leq \alpha \leq 1$
- (c) $0 \leq \alpha \leq 1$
- (d) $\alpha = \frac{1}{2}$

ANSWER (AI-derived — no official key exists for this paper)

$$(b) \quad \frac{1}{2} \leq \alpha \leq 1$$

EXAM SHORTCUT (AI-derived explanation)

At $x = 1$ the first branch reaches $1/2$ while the second branch starts at α , so no downward jump requires $\alpha \geq 1/2$; the middle slope $1 - \alpha$ must be non-negative, so $\alpha \leq 1$. Hence $1/2 \leq \alpha \leq 1$.

TIPS & TRICKS (AI-derived explanation)

- A distribution function must be non-decreasing, right-continuous, and run from 0 to 1. Check each junction point and each branch slope separately.
- Evaluate the middle branch at both ends: at $x = 1$ it equals α , and at $x = 2$ it equals $2 - 2\alpha + 2\alpha - 1 = 1$ automatically - so the right end never constrains α .
- An UPWARD jump is permitted in a general distribution function (it just means positive mass at that point); only a downward move is forbidden. That is why $\alpha = 1$ is admissible.
- The slope condition $1 - \alpha \geq 0$ is easy to overlook; without it, $\alpha > 1$ would make F decrease on $[1, 2)$.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. F must be non-decreasing on the whole line, right-continuous, with $F(-\infty) = 0$ and $F(+\infty) = 1$.
2. On $[0, 1)$ the branch $x/2$ rises from 0 to a left-limit of $1/2$ at $x = 1$.
3. At $x = 1$ the middle branch takes the value $(1 - \alpha)(1) + 2\alpha - 1 = \alpha$. To avoid a decrease we need $\alpha \geq 1/2$ (an upward jump is permissible).
4. On $[1, 2)$ the middle branch has slope $1 - \alpha$, which must be non-negative, giving $\alpha \leq 1$.
5. At $x = 2$ the middle branch gives $(1 - \alpha)(2) + 2\alpha - 1 = 2 - 2\alpha + 2\alpha - 1 = 1$, matching the final branch, so continuity holds there for every α .
6. Combining, F is a distribution function if and only if $1/2 \leq \alpha \leq 1$, option (b).

7. 2020 · Q36 Distribution Functions & Their Properties · Theoretical**QUESTION**

The function

$$F(x) = \begin{cases} 0, & x < 0 \\ x/3, & 0 \leq x < 2 \\ (1 - \alpha)x + 3\alpha - 2, & 2 \leq x < 3 \\ 1, & x \geq 3 \end{cases}$$

is the distribution function of a continuous random variable

OPTIONS

- (a) for all $\alpha \in (0, 1)$
- (b) for all $\alpha \in (2/3, 1)$
- (c) for all $\alpha \in [2/3, 1]$
- (d) iff $\alpha = 2/3$

ANSWER (AI-derived — no official key exists for this paper)**(d)** iff $\alpha = 2/3$ **EXAM SHORTCUT (AI-derived explanation)**

For a CONTINUOUS variable F must have no jumps. At $x = 2$ the first branch gives $2/3$ and the second gives α , so continuity forces $\alpha = 2/3$ exactly.

TIPS & TRICKS (AI-derived explanation)

- 'Continuous random variable' is a strictly stronger requirement than 'distribution function': every junction must match exactly, with no jump allowed anywhere.
- Evaluate both branches AT the junction: $x/3$ at $x = 2$ is $2/3$, and $(1 - \alpha)x + 3\alpha - 2$ at $x = 2$ is $2 - 2\alpha + 3\alpha - 2 = \alpha$. Equate them.
- The junction at $x = 3$ imposes nothing: $(1 - \alpha)(3) + 3\alpha - 2 = 1$ identically, so only the $x = 2$ junction matters.
- Contrast with Q_{33} of the same paper, where an upward jump was allowed and the answer was an interval. The word 'continuous' is what collapses the interval to a single point here.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. For F to be the distribution function of a CONTINUOUS random variable, F must be continuous everywhere - no jumps are permitted.
2. As x approaches 2 from below, the branch $x/3$ tends to $2/3$.
3. At $x = 2$ the middle branch gives $(1 - \alpha)(2) + 3\alpha - 2 = 2 - 2\alpha + 3\alpha - 2 = \alpha$.
4. Continuity at $x = 2$ therefore requires $\alpha = 2/3$.
5. At $x = 3$ the middle branch gives $(1 - \alpha)(3) + 3\alpha - 2 = 3 - 3\alpha + 3\alpha - 2 = 1$, matching the final branch for every α , so this junction imposes no extra condition.
6. With $\alpha = 2/3$ the middle slope is $1 - 2/3 = 1/3 > 0$, so F is non-decreasing. Hence F is a continuous distribution function if and only if $\alpha = 2/3$, option (d).

8. **2022 · Q47** Discrete & Continuous Random Variables · Numerical**QUESTION**

The pdf of X is $f(x) = \frac{kx^3}{(1+2x)^6}$, $x > 0$ (0 otherwise). What is the value of k ?

OPTIONS

- (a)** 80
(b) 160
(c) 240
(d) 320

ANSWER (AI-derived — no official key exists for this paper)**(d)** 320**EXAM SHORTCUT (AI-derived explanation)**

Substitute $u = 2x$ to get $(1/16)$ times the $\int u^3/(1+u)^6 du = (1/16)B(4, 2) = (1/16)(1/20) = 1/320$. Hence $k = 320$.

TIPS & TRICKS (AI-derived explanation)

- Beta second-kind integral: the $\int_0^\infty u^{m-1}/(1+u)^{m+n} du = B(m, n)$. Here $m-1=3$ and $m+n=6$, so $m=4$ and $n=2$.
- The substitution $u=2x$ is what converts $(1+2x)$ into $(1+u)$; the factor picked up is $(1/2)^3$ from x^3 and $(1/2)$ from dx , giving $1/16$.
- $B(4, 2) = \Gamma(4)\Gamma(2)/\Gamma(6) = (6)(1)/120 = 1/20$ - compute with factorials, not integrals.
- Cross-check with 2018Q₂₀, which used the same density and showed that $2X/(1+2X) \sim \text{Beta}(4, 2)$ - the same parameters appear, confirming $m=4$ and $n=2$.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The density must integrate to 1: k times the $\int_0^\infty x^3/(1+2x)^6 dx = 1$.
2. Substitute $u=2x$, so $x=u/2$ and $dx=du/2$. Then $x^3=u^3/8$ and the integral becomes $(1/8)(1/2)$ times the $\int u^3/(1+u)^6 du = (1/16)$ times that integral.
3. The standard Beta $\int$ the second kind gives the $\int_0^\infty u^{m-1}/(1+u)^{m+n} du = B(m, n)$; here $m-1=3$ and $m+n=6$, so $m=4$ and $n=2$.
4. $B(4, 2) = \Gamma(4)\Gamma(2)/\Gamma(6) = (3!)(1!)/5! = 6/120 = 1/20$.
5. So the integral equals $(1/16)(1/20) = 1/320$.
6. Therefore $k/320 = 1$ and $k = 320$, option (d).

9. 2022 · Q50 Distribution Functions & Their Properties · Numerical**QUESTION**

The distribution function of X is

$$F(x) = \begin{cases} 0, & x < 0 \\ \frac{x}{5}, & 0 \leq x \leq 5 \\ 1, & x > 5 \end{cases}$$

What is $P[1.5 \leq X \leq 2.5]$ equal to?

OPTIONS

- (a) 1
(b) $\frac{3}{5}$
(c) $\frac{2}{5}$
(d) $\frac{1}{5}$

ANSWER (AI-derived — no official key exists for this paper)

(d) $\frac{1}{5}$

EXAM SHORTCUT (AI-derived explanation)

$F(x) = x/5$ on $[0, 5]$ is the $U(0, 5)$ distribution, so the probability of any sub-interval is its length divided by 5: $(2.5 - 1.5)/5 = 1/5$.

TIPS & TRICKS (AI-derived explanation)

- A linear CDF means a UNIFORM distribution; probabilities are then proportional to interval length, with no integration needed.
- $P(a \leq X \leq b) = F(b) - F(a) = 2.5/5 - 1.5/5 = 0.5 - 0.3 = 0.2$ - the direct route if you prefer the CDF.

- For a continuous variable the endpoints carry no mass, so $\leq$ and $<$ give the same answer - the closed interval in the stem is a distractor.
- Option (c) $2/5$ is the trap for candidates who use the interval length 1 over the half-range, or who mis-subtract $0.5 - 0.1$.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The distribution function $F(x) = x/5$ for $0 \leq x \leq 5$ is that of a $\text{Uniform}(0, 5)$ random variable, with density $1/5$ on that range.
2. For any interval, $P(a \leq X \leq b) = F(b) - F(a)$.
3. $F(2.5) = 2.5/5 = 0.5$ and $F(1.5) = 1.5/5 = 0.3$.
4. Hence $P(1.5 \leq X \leq 2.5) = 0.5 - 0.3 = 0.2$.
5. Equivalently, the interval has length 1 out of a total range of 5, giving $1/5$.
6. Option (d) is correct.

10. 2023 · Q6 Discrete & Continuous Random Variables · Numerical**QUESTION**

The pdf of a random variable X is given by

$$f(x) = \begin{cases} 0.2, & 0 \leq x \leq 5 \\ 0, & \text{otherwise} \end{cases}$$

Consider the following probabilities: 1. $P(2 \leq X \leq 3) = 0.2$ 2. $P(X > 3.5) = 0.3$ 3. $P(|x| \geq 2) = 0.6$ 4. $P(X < 4) = 0.8$ How many of the probabilities given above are correct?

OPTIONS

- (a) Only one
- (b) Only two
- (c) Only three
- (d) All four

ANSWER (AI-derived — no official key exists for this paper)

(d) All four

EXAM SHORTCUT (AI-derived explanation)

The density is $\text{Uniform}(0, 5)$, so every probability is (interval length)/5. All four values - 0.2, 0.3, 0.6 and 0.8 - check out.

TIPS & TRICKS (AI-derived explanation)

- A constant density 0.2 over a range of length 5 is $U(0, 5)$; probabilities are then just lengths divided by 5, with no integration at all.
- Statement 3 has the trap: $|X| \geq 2$ looks two-sided, but X is confined to $[0, 5]$, so it reduces to $X \geq 2$ and gives $3/5 = 0.6$.
- Check each in turn: $(3 - 2)/5 = 0.2$; $(5 - 3.5)/5 = 0.3$; $(5 - 2)/5 = 0.6$; $(4 - 0)/5 = 0.8$. Four quick divisions.
- When every statement in a set is a straightforward computation from the same simple density, expect 'all four' - but verify the one with the absolute value.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The density is $f(x) = 0.2$ on $[0, 5]$, which is the $\text{Uniform}(0, 5)$ distribution; probabilities equal interval length divided by 5.
2. Statement 1: $P(2 \leq X \leq 3) = (3 - 2)/5 = 0.2$. CORRECT.
3. Statement 2: $P(X > 3.5) = (5 - 3.5)/5 = 1.5/5 = 0.3$. CORRECT.
4. Statement 3: since X takes only non-negative values, $|X| \geq 2$ is the same as $X \geq 2$, giving $(5 - 2)/5 = 0.6$. CORRECT.
5. Statement 4: $P(X < 4) = (4 - 0)/5 = 0.8$. CORRECT.
6. All four probabilities are correct, option (d).

P4 · PROBABILITY

Standard Discrete Distributions

20 questions · 2018: 1 2019: 5 2020: 1 2021: 1 2022: 3 2023: 1 2024: 4 2025: 2 2026: 2

Weight. 20 questions (10.2% of Probability) · appeared in 9 of 9 papers · peak year 2019 (5)**Examiner's pattern.** Poisson approximation to the binomial, binomial mode and skewness, and waiting-time counting are the workhorses. Negative binomial appears disguised ('the seventh child is the second daughter', 'the first win is on trial N'). Hypergeometric shows up as sampling without replacement with a variance ask. Multinomial appears once as a covariance ask.**Must-know.** NB: $P(X=n) = C(n-1, r-1) p^r q^{n-r}$; hypergeometric $\text{Var} = n(K/N)(1-K/N)(N-n)/(N-1)$; multinomial $\text{Cov}(X_i, X_j) = -np_i p_j$; binomial mode = $\text{floor}((n+1)p)$.1. **2018 · Q7** Binomial Distribution · Conceptual

QUESTION

 $X \sim \text{Binomial}(n = 13, p = 0.3)$. 1. The mode of X is 3. 2. The distribution is negatively skewed.

OPTIONS

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

ANSWER (AI-derived — no official key exists for this paper)**(d)** Neither 1 nor 2**EXAM SHORTCUT** (AI-derived explanation)

Mode of $\text{Bin}(n, p) = \text{floor}((n+1)p) = \text{floor}(14 \times 0.3) = \text{floor}(4.2) = 4$, not 3. Skewness sign = sign of $(1-2p) = \text{sign of } 0.4 > 0$, so positively skewed, not negatively. Both statements fail.

TIPS & TRICKS (AI-derived explanation)

- Mode formula: $\text{floor}((n+1)p)$; if $(n+1)p$ is an integer there are two modes, $(n+1)p$ and $(n+1)p - 1$. Here 4.2 is not an integer, so the mode is unique at 4.
- Binomial skewness = $(1-2p)/\sqrt{npq}$. Its sign depends only on p : $p < 0.5$ gives positive skew, $p > 0.5$ gives negative skew, $p = 0.5$ is symmetric. You never need to compute the value.
- Common trap: using $np = 3.9$ and rounding to 3. The mean is 3.9 but the mode is 4 - mean and mode are not interchangeable for a skewed discrete law.
- When both statements in a 1/2/both/neither item look 'nearly right', suspect (d). Examiners build these from two near-misses.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Here $X \sim \text{Binomial}(n = 13, p = 0.3)$, so $q = 0.7$.
2. Statement 1 (mode): the mode is $\text{floor}((n+1)p) = \text{floor}(14 \times 0.3) = \text{floor}(4.2) = 4$. Confirm by ratio:
 $P(X=4)/P(X=3) = [(n-3)/4](p/q) = (10/4)(3/7) = 30/28 > 1$, so $P(X=4) > P(X=3)$ and 3 is not the mode. Statement 1 is FALSE.
3. Statement 2 (skewness): for the binomial, the coefficient of skewness is
 $(1-2p)/\sqrt{npq} = (1-0.6)/\sqrt{13 \times 0.3 \times 0.7} = 0.4/\sqrt{2.73} = +0.242$.

4. The skewness is positive, so the distribution is positively skewed, not negatively skewed. Statement 2 is FALSE.
5. Neither statement is correct, so option (d) is right.

2. **2019 · Q15** Classical & Axiomatic Definitions of Probability · Numerical

QUESTION

If the probability of having a male or a female child is 0.50, what is the probability that a family's seventh child is their second daughter?

OPTIONS

- (a) $\frac{3}{64}$
- (b) $\frac{15}{64}$
- (c) $\frac{21}{64}$
- (d) $\frac{1}{64}$

ANSWER (AI-derived — no official key exists for this paper)

(a) $\frac{3}{64}$

EXAM SHORTCUT (AI-derived explanation)

The 7th child is the 2nd daughter means exactly 1 daughter among the first 6, then a daughter.

$$P = {}^6C_1 (1/2)^6 \times (1/2) = 6/128 = 3/64.$$

TIPS & TRICKS (AI-derived explanation)

- This is the negative binomial pattern: $P(\text{the } r\text{th success occurs on trial } n) = {}^{n-1}C_{r-1} p^r q^{n-r}$. Here $n = 7, r = 2, p = 1/2$, giving ${}^6C_1 (1/2)^7 = 6/128$.
- Do not use the plain binomial ${}^7C_2 (1/2)^7 = 21/128$ - that answers 'exactly 2 daughters in 7 children', which is planted as option (c) in the equivalent $21/64$ form.
- The last trial is FIXED as a success, so the combination is chosen from $n - 1 = 6$ trials, not from 7. That single distinction is what the item tests.
- $6/128$ must be reduced to $3/64$ to match the options; always reduce before comparing.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Let a 'success' be the birth of a daughter, with $p = 0.5$ for each child independently.
2. The event is that the 2nd daughter arrives exactly on the 7th child. This requires exactly 1 daughter among the first 6 children, and then a daughter on the 7th.
3. $P(\text{exactly 1 daughter in 6 children}) = {}^6C_1 (0.5)^1 (0.5)^5 = 6(0.5)^6 = 6/64$.
4. $P(7\text{th child is a daughter}) = 0.5$, independently of the first six.
5. Multiplying: $(6/64)(1/2) = 6/128 = 3/64$.
6. Hence option (a) is correct.

3. 2019 · Q16 Poisson Distribution · Numerical**QUESTION**

Let X be a Poisson random variable with p.m.f. $P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$, $k = 0, 1, 2, \dots$, $\lambda > 0$. If $Y = X^2 + 3$, what is $P(Y = y)$ equal to?

OPTIONS

- (a) $\frac{e^{-\lambda} \lambda^{\sqrt{y-3}}}{\sqrt{(y-3)!}}$, for $y = \{3, 4, 7, 12, \dots\}$
- (b) $\frac{e^{-\lambda} \lambda^{-\sqrt{y-3}}}{\sqrt{(3-y)!}}$, for $y = \{3, 4, 7, 12, \dots\}$
- (c) $\frac{e^{-\lambda} \lambda^{\sqrt{3-y}}}{\sqrt{(3-y)!}}$, for $y = \{4, 7, 12, \dots\}$
- (d) $\frac{e^{-\lambda} \lambda^{-\sqrt{3-y}}}{\sqrt{(3-y)!}}$, for $y = \{4, 7, 12, \dots\}$ (exponent signs/arguments differ across (a)–(d) exactly as printed in source — not a transcription error)

ANSWER (AI-derived — no official key exists for this paper)

(a) $\frac{e^{-\lambda} \lambda^{\sqrt{y-3}}}{\sqrt{(y-3)!}}$, for $y = \{3, 4, 7, 12, \dots\}$

EXAM SHORTCUT (AI-derived explanation)

$Y = X^2 + 3$ is one-to-one on the non-negative integers, so $X = \sqrt{y-3}$ and the support is $y = 3, 4, 7, 12, 19, \dots$. Substituting $k = \sqrt{y-3}$ into the Poisson pmf gives option (a).

TIPS & TRICKS (AI-derived explanation)

- For a strictly monotone transformation of a DISCRETE variable, simply invert: $P(Y = y) = P(X = g^{-1}(y))$. No Jacobian is ever involved for discrete variables.
- Build the support explicitly from $k = 0, 1, 2, 3$: $y = 3, 4, 7, 12$. Options (c) and (d) omit $y = 3$ (the image of $k = 0$) and are eliminated at once.
- Options (b), (c) and (d) also contain $\sqrt{3-y}$ or a negative exponent, both of which are undefined or wrong for $y > 3$ - a sign check alone finishes the item.
- The source prints the factorial as $\sqrt{(y-3)!}$ where $(\sqrt{y-3})!$ is meant; the intended reading is $k!$ with $k = \sqrt{y-3}$. Read past the typography and match on the support and the exponent.

STEP-BY-STEP SOLUTION (AI-derived explanation)

- X is Poisson with $P(X = k) = e^{-\lambda} \lambda^k / k!$ for $k = 0, 1, 2, \dots$
- The transformation $Y = X^2 + 3$ is strictly increasing on the non-negative integers, so it is one-to-one there and can be inverted: $X = \sqrt{Y-3}$.
- The support of Y is obtained from $k = 0, 1, 2, 3, \dots$: $y = 0 + 3 = 3, 1 + 3 = 4, 4 + 3 = 7, 9 + 3 = 12$, and so on, i.e. y in $\{3, 4, 7, 12, 19, \dots\}$.
- For each such y , $P(Y = y) = P(X = \sqrt{y-3}) = e^{-\lambda} \lambda^{\sqrt{y-3}} / (\sqrt{y-3})!$.
- Options (c) and (d) exclude $y = 3$ from the support and use $\sqrt{3-y}$, which is not real for $y > 3$; option (b) uses a negative exponent on λ . All three are inadmissible.
- Option (a) has the correct support and the correct exponent $\sqrt{y-3}$, so it is the intended key.

4. **2019 · Q17** Standard Discrete Distributions · Conceptual**QUESTION**

A person purchases a weekly lottery ticket until he wins. If p is the probability of winning, the expected number of weeks required to win is

OPTIONS

- (a) $\frac{1}{p}$
 (b) $\frac{1}{p} - 1$
 (c) $\frac{1}{1-p}$
 (d) $\frac{1}{1-p} - 1$

ANSWER (AI-derived — no official key exists for this paper)

(a) $\frac{1}{p}$

EXAM SHORTCUT (AI-derived explanation)

The number of weekly tickets until the first win is geometric on $\{1, 2, 3, \dots\}$, whose mean is $1/p$.

TIPS & TRICKS (AI-derived explanation)

- Two geometric means to keep apart: counting TRIALS until the first success gives $1/p$; counting FAILURES before it gives $q/p = 1/p - 1$ (option b).
- The phrase 'until he wins' includes the winning week itself, so the trial-counting version applies.
- Intuition check: if $p = 1/50$ you expect about 50 weeks, which is $1/p$. If the answer were $1/(1-p)$ it would be barely more than 1 week - clearly wrong.
- Options (c) and (d) use $1-p$ in place of p and correspond to counting until the first LOSS, a different experiment altogether.

STEP-BY-STEP SOLUTION (AI-derived explanation)

- Each week the person buys one ticket, winning with probability p independently of other weeks.
- Let X be the number of weeks up to and including the first win. Then $P(X = k) = (1-p)^{k-1}p$ for $k = 1, 2, 3, \dots$
- This is the geometric distribution on the positive integers.
- Its expectation is $E(X) = \sum k(1-p)^{k-1}p = p \cdot 1/p^2 = 1/p$.
- Hence the expected number of weeks required is $1/p$.
- Option (a) is correct.

5. **2019 · Q18** Binomial Distribution · Theoretical**QUESTION**

Consider the following statements: 1. Binomial distribution with mean $a > 0$ and variance $b > 0$ is such that $\frac{a^2}{(a-b)^2}$ is an integer. 2. Binomial distribution with mean 3.5 and variance 1.75 is bimodal. 3. For a binomial distribution with mean and variance equal to 4 and $\frac{4}{3}$ respectively, $P(X \geq 1) > 0.99$. 4. Sum of two independent binomial variates is a binomial variate. Which of the above statements are correct?

OPTIONS

- (a) 1 and 3
 (b) 1, 2 and 4

(c) 2 and 3 only**(d)** 2, 3 and 4**ANSWER (AI-derived — no official key exists for this paper)****(c)** 2 and 3 only**EXAM SHORTCUT (AI-derived explanation)**

Statement 4 fails at once (sums of binomials are binomial only when p is common), and statement 1 gives $a^2/(a-b)^2 = 1/p^2$, not generally an integer. Statements 2 and 3 both check out, so the answer is 2 and 3 only.

TIPS & TRICKS (AI-derived explanation)

- Recover n and p from mean and variance: $p = 1 - b/a$ and $n = a/p$. For statement 2, $a = 3.5$ and $b = 1.75$ give $p = 0.5$ and $n = 7$; for statement 3, $a = 4$ and $b = 4/3$ give $p = 2/3$ and $n = 6$.
- Bimodality test: the binomial is bimodal exactly when $(n+1)p$ is an INTEGER, the modes being $(n+1)p$ and $(n+1)p - 1$. Here $(7+1)(0.5) = 4$, so modes 3 and 4. Statement 2 is true.
- For statement 1, note $a - b = np - npq = np^2$, so $a^2/(a-b)^2 = (np)^2/(np^2)^2 = 1/p^2$. It is $a^2/(a-b)$ that equals n and is an integer - the squared denominator as printed makes the claim false.
- Sums of independent binomials add only when the success probability is shared: $B(m, p) + B(n, p) = B(m+n, p)$. With different p the sum is not binomial, which kills statement 4 and options (b) and (d).

STEP-BY-STEP SOLUTION (AI-derived explanation)

- Write $a = np$ and $b = npq$ for the binomial mean and variance, so $a - b = np - npq = np(1 - q) = np^2$.
- Statement 1: $a^2/(a-b)^2 = (np)^2/(np^2)^2 = n^2p^2/(n^2p^4) = 1/p^2$, which is not an integer for general p (it is $a^2/(a-b)$ that equals n). FALSE.
- Statement 2: $a = 3.5, b = 1.75$ give $q = b/a = 0.5$, so $p = 0.5$ and $n = a/p = 7$. Since $(n+1)p = 8(0.5) = 4$ is an integer, the distribution has two modes, 3 and 4 - it is bimodal. TRUE.
- Statement 3: $a = 4, b = 4/3$ give $q = (4/3)/4 = 1/3$, so $p = 2/3$ and $n = 4/(2/3) = 6$. Then $P(X \geq 1) = 1 - (1/3)^6 = 1 - 1/729 = 0.99863 > 0.99$. TRUE.
- Statement 4: if $X \sim B(m, p_1)$ and $Y \sim B(n, p_2)$ are independent, $X + Y$ is binomial only when $p_1 = p_2$. In general it is not. FALSE.
- Only statements 2 and 3 are correct, option (c).

6. 2019 · Q32 Mathematical Expectation · Theoretical**QUESTION**

r balls are drawn one at a time without replacement from a bag containing n white and m black balls. What is the variance of the number of black balls drawn?

OPTIONS

- (a)** $\frac{mr(m+n-2)}{(m+n)(m+n+1)}$
- (b)** $\frac{mnr(m+n-r)}{(m+n)^2(m+n+1)}$
- (c)** $\frac{mr(m+n)^2}{(m+n-1)(m+n-r)}$
- (d)** $\frac{(m+n)(m+n-r)}{(m+n)^2}$

ANSWER (AI-derived — no official key exists for this paper)

(b) $\frac{mnr(m+n-r)}{(m+n)^2(m+n+1)}$

Source note. SOURCE NOTE: sampling is without replacement, so the number of black balls is HYPERGEOMETRIC with exact variance $r(m/N)(n/N)(N-r)/(N-1)$ where $N = m+n$, i.e. $mnr(m+n-r)/[(m+n)^2(m+n-1)]$. Option (b) as printed carries $(m+n+1)$ in place of $(m+n-1)$ in the denominator. Every other option is structurally wrong (wrong symmetry in m and n , or not even a variance), so (b) is the only defensible key; the sign in its last denominator factor is preserved exactly as printed and flagged here rather than silently corrected.

EXAM SHORTCUT (AI-derived explanation)

Without replacement means hypergeometric: $\text{Var} = rpq(N-r)/(N-1)$ with $p = m/N, q = n/N$ and $N = m+n$, i.e. $mnr(m+n-r)/[(m+n)^2(m+n-1)]$. Option (b) is the only choice with that structure.

TIPS & TRICKS (AI-derived explanation)

- Sampling WITHOUT replacement always brings the finite population correction $(N-r)/(N-1)$; with replacement the count would be $\text{Binomial}(r, m/N)$ with variance rpq and no correction factor.
- Symmetry check: swapping the roles of m and n must leave the variance unchanged, because black and white are interchangeable labels. Options (a), (c) and (d) are not symmetric in m and n and can be struck out instantly.
- Boundary check: putting $r = N$ must give variance 0 (every ball is drawn, so the count is certain). Option (b)'s factor $(m+n-r)$ vanishes at $r = N$, confirming its structure.
- The hypergeometric mean is rm/N ; pairing mean and variance together makes this family quick to recognise in the exam.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Let $N = m+n$ be the total number of balls, of which m are black. Drawing r balls one at a time WITHOUT replacement makes X , the number of black balls drawn, hypergeometric.
2. Write X as a sum of indicators $I_1 + \dots + I_r$, where $I_k = 1$ if the k th ball drawn is black. Each I_k has mean m/N , so $E(X) = rm/N$.
3. Each $\text{Var}(I_k) = (m/N)(n/N)$, and for k not equal to j , $\text{Cov}(I_k, I_j) = -(m/N)(n/N)/(N-1)$.
4. Summing:

$$\text{Var}(X) = r(m/N)(n/N) - r(r-1)(m/N)(n/N)/(N-1) = r(m/N)(n/N)[1 - (r-1)/(N-1)] = r(mn/N^2)(N-r)/(N-1)$$
5. That is $\text{Var}(X) = mnr(m+n-r)/[(m+n)^2(m+n-1)]$. It is symmetric in m and n and vanishes at $r = N$, as it must.
6. Option (b) is the only choice with this structure (it prints $(m+n+1)$ where the exact result has $(m+n-1)$); the remaining options are not even symmetric in m and n . Hence (b).

7. 2020 · Q25 Geometric Distribution · Factual**QUESTION**

Distributions possessing the lack-of-memory property are

OPTIONS

- (a)** Gamma and Beta distributions
- (b)** Beta and Geometric distributions

(c) Geometric and Exponential distributions

(d) Exponential and Cauchy distributions

ANSWER (AI-derived — no official key exists for this paper)

(c) Geometric and Exponential distributions

EXAM SHORTCUT (AI-derived explanation)

The lack-of-memory (memoryless) property characterises exactly two distributions: the GEOMETRIC in the discrete case and the EXPONENTIAL in the continuous case.

TIPS & TRICKS (AI-derived explanation)

- Memorise the pairing: geometric (discrete) and exponential (continuous) are the ONLY memoryless distributions, and each is uniquely characterised by that property.
- Statement of the property: $P(X > s + t \mid X > s) = P(X > t)$. Waiting time already spent gives no information about remaining waiting time.
- The exponential is a special Γ (shape 1), but a general Γ is NOT memoryless - option (a) is eliminated by that distinction.
- The Cauchy has no mean at all and is defined on the whole real line, so it cannot describe a waiting time and is not memoryless.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. A distribution is memoryless if $P(X > s + t \mid X > s) = P(X > t)$ for all $s, t > 0$ - the past duration gives no information about the future.
2. For the exponential with rate λ , $P(X > t) = e^{-\lambda t}$, so $P(X > s + t) / P(X > s) = e^{-\lambda t} = P(X > t)$. Memoryless, and it is the ONLY continuous distribution with this property.
3. For the geometric with success probability p , $P(X > k) = (1 - p)^k$, so the same cancellation occurs. It is the only discrete distribution with the property.
4. The Beta distribution has bounded support and the general Γ has an increasing or decreasing hazard rate, so neither is memoryless.
5. The Cauchy is symmetric on the whole real line and is not a waiting-time distribution at all.
6. Hence the memoryless distributions are the geometric and the exponential, option (c).

8. **2021 · Q24** Poisson Distribution · Numerical

QUESTION

To find the prevalence of a virus in a city of population 1,00,000, a blood test on 200 randomly selected citizens gave 8 positive results. The number of affected persons in a random sample of size 500 can approximately be taken as

OPTIONS

- (a) Poisson(40)
- (b) Poisson(20)
- (c) Poisson(8)
- (d) Poisson(16/5)

ANSWER (AI-derived — no official key exists for this paper)**(b)** Poisson(20)**EXAM SHORTCUT (AI-derived explanation)**

Estimated prevalence = $8/200 = 0.04$. In a sample of 500 the expected number affected is $500 \times 0.04 = 20$, so the count is approximately Poisson(20).

TIPS & TRICKS (AI-derived explanation)

- Poisson approximation to the binomial: with n large and p small, $\text{Binomial}(n, p)$ is approximately $\text{Poisson}(np)$. Here $np = 20$.
- Estimate p from the pilot sample first - 8 out of 200 gives 4%. The city population of 100000 is a distractor and never enters the calculation.
- Option (c) $\text{Poisson}(8)$ reuses the pilot COUNT instead of rescaling to the new sample size - the commonest error.
- Option (d) $\text{Poisson}(16/5) = \text{Poisson}(3.2)$ comes from dividing rather than multiplying by the size ratio $500/200 = 2.5$.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The pilot test of 200 citizens gave 8 positives, so the estimated prevalence is $p = 8/200 = 0.04$.
2. In a fresh random sample of size $n = 500$, the number affected is $\text{Binomial}(500, 0.04)$.
3. Its mean is $np = 500 \times 0.04 = 20$.
4. Since n is large and p is small, the binomial is well approximated by a Poisson distribution with the same mean.
5. Hence the number affected is approximately $\text{Poisson}(20)$.
6. Option (b) is correct.

9. 2022 · Q1 Binomial Distribution · Conceptual**QUESTION**

Suppose X_1, X_2, X_3, X_4 are independent where X_1, X_2, X_3 have Bernoulli $(\frac{1}{3})$ distribution and X_4 follows $B(3, \frac{2}{3})$. The distribution of $X_1 + X_2 + X_3 + 3 - X_4$ is:

OPTIONS

- (a) $B(5, \frac{1}{3})$
- (b) $B(6, \frac{1}{3})$
- (c) $B(6, \frac{2}{3})$
- (d) not a Binomial

ANSWER (AI-derived — no official key exists for this paper)**(b)** $B(6, \frac{1}{3})$ **EXAM SHORTCUT (AI-derived explanation)**

$X_1 + X_2 + X_3 \sim B(3, 1/3)$, and $3 - X_4 \sim B(3, 1/3)$ because reflecting a $B(3, 2/3)$ swaps p and q . Two independent $B(3, 1/3)$ variables add to $B(6, 1/3)$.

TIPS & TRICKS (AI-derived explanation)

- Reflection rule: if $X \sim B(n, p)$ then $n - X \sim B(n, 1 - p)$. Spotting this converts the awkward ' $3 - X_4$ ' into a friendly binomial.
- Additive property: independent $B(m, p) + B(n, p) = B(m + n, p)$ - but ONLY when p is common, which is exactly what the reflection arranges here.
- Check the mean: $3(1/3) + (3 - 3(2/3)) = 1 + 1 = 2$, and $B(6, 1/3)$ has mean 2. A quick confirmation.
- Option (c) $B(6, 2/3)$ is the trap for candidates who forget to reflect X_4 and simply add the two success probabilities.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. X_1, X_2, X_3 are i.i.d. Bernoulli($1/3$), so their sum $S = X_1 + X_2 + X_3 \sim \text{Binomial}(3, 1/3)$.
2. $X_4 \sim \text{Binomial}(3, 2/3)$. If $X \sim B(n, p)$ then $n - X \sim B(n, 1 - p)$, since counting failures instead of successes swaps p and q .
3. Hence $3 - X_4 \sim \text{Binomial}(3, 1/3)$.
4. S and $3 - X_4$ are independent (they involve disjoint sets of the original variables) and share the same success probability $1/3$.
5. By the additive property of the binomial, their sum is $\text{Binomial}(3 + 3, 1/3) = \text{Binomial}(6, 1/3)$.
6. Option (b) is correct.

10. 2022 · Q2 Standard Discrete Distributions · Numerical**QUESTION**

A large number of candidates from three States A, B, C appear at a selection test. 10 candidates are selected independently, of which X, Y, Z are respectively from A, B, C. If a candidate from A had twice as much chance as one from B or C to be selected, $\text{cov}(Y, Z)$ equals:

OPTIONS

- (a) $-\frac{5}{4}$
 (b) $-\frac{5}{8}$
 (c) 0
 (d) $\frac{15}{8}$

ANSWER (AI-derived — no official key exists for this paper)

(b) $-\frac{5}{8}$

EXAM SHORTCUT (AI-derived explanation)

The selection probabilities are $1/2, 1/4, 1/4$ for A, B, C. For a multinomial,
 $\text{Cov}(Y, Z) = -np_B p_C = -10(1/4)(1/4) = -5/8$.

TIPS & TRICKS (AI-derived explanation)

- Multinomial covariance: $\text{Cov}(X_i, X_j) = -np_i p_j$ for i not equal to j , always NEGATIVE because the counts must sum to n .
- Set the weights first: A is twice as likely as each of B and C, so the weights are $2 : 1 : 1$ and the probabilities are $1/2, 1/4, 1/4$.

- The sign alone eliminates options (c) and (d): more from B necessarily means fewer from C, so the covariance must be negative.
- Companion formulas: $E(X_i) = np_i$ and $\text{Var}(X_i) = np_i(1 - p_i)$ - the whole multinomial toolkit is three lines long.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Let the selection probability of a candidate from B be k . A candidate from A is twice as likely, so $p_A = 2k$, and C matches B, so $p_C = k$.
2. Since the probabilities sum to 1: $2k + k + k = 4k = 1$, giving $k = 1/4$.
3. Hence $p_A = 1/2, p_B = 1/4, p_C = 1/4$.
4. With 10 independent selections, (X, Y, Z) follows a multinomial distribution with $n = 10$ and these probabilities.
5. For the multinomial, $\text{Cov}(Y, Z) = -np_B p_C = -10(1/4)(1/4) = -10/16 = -5/8$, option (b).

11. 2022 · Q6 Binomial Distribution · Numerical**QUESTION**

If X has binomial distribution with $n = 3$ and $p = \frac{1}{3}$, what is $E((X - 1)^3)$?

OPTIONS

- (a) $\frac{1}{9}$
 (b) $\frac{2}{9}$
 (c) $\frac{4}{9}$
 (d) $\frac{5}{9}$

ANSWER (AI-derived — no official key exists for this paper)

(b) $\frac{2}{9}$

EXAM SHORTCUT (AI-derived explanation)

With $p = 1/3$ the masses are $8/27, 12/27, 6/27, 1/27$ and $(X - 1)^3$ takes the values $-1, 0, 1, 8$. So $E = (-8 + 0 + 6 + 8)/27 = 6/27 = 2/9$.

TIPS & TRICKS (AI-derived explanation)

- Tabulate the function values before weighting: $(X - 1)^3 = -1, 0, 1, 8$ for $X = 0, 1, 2, 3$. The cube makes the last value dominate.
- Keep everything over 27: the numerator is $-1(8) + 0(12) + 1(6) + 8(1) = 6$, so the answer is $6/27 = 2/9$ with no decimals.
- The $X = 1$ term contributes nothing, which removes the largest probability from the calculation - a useful simplification to notice early.
- Alternative route: expand $E[(X - 1)^3] = E(X^3) - 3E(X^2) + 3E(X) - 1$ using $E(X) = 1, E(X^2) = 5/3, E(X^3) = 3$ - but direct summation is faster here.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. $X \sim \text{Binomial}(3, 1/3)$, so
 $P(X = 0) = (2/3)^3 = 8/27, P(X = 1) = 3(1/3)(2/3)^2 = 12/27, P(X = 2) = 3(1/3)^2(2/3) = 6/27, P(X = 3) = (1/3)^3 = 1/27$
2. The function values are $(X - 1)^3 = (-1)^3 = -1, 0^3 = 0, 1^3 = 1$ and $2^3 = 8$ for $X = 0, 1, 2, 3$.

$$3. E[(X - 1)^3] = (-1)(8/27) + 0(12/27) + 1(6/27) + 8(1/27) = (-8 + 0 + 6 + 8)/27 = 6/27 = 2/9.$$

4. Option (b) is correct.

12. **2023 · Q4** Mathematical Expectation · Numerical

QUESTION

A random variable X has probability mass function given by

$$P(X = x) = \frac{1}{5^{10}} \binom{10}{x} 2^x 3^{10-x}, \quad x = 0, 1, 2, \dots, 10$$

The mean and variance of X are respectively:

OPTIONS

- (a) 2, 4
- (b) 2.4, 4
- (c) 4, 2.4
- (d) 4, 2

ANSWER (AI-derived — no official key exists for this paper)

(c) 4, 2.4

EXAM SHORTCUT (AI-derived explanation)

Rewrite the pmf as $\binom{10}{x} (2/5)^x (3/5)^{10-x}$: a Binomial(10, 0.4). Mean = 4 and variance = $10(0.4)(0.6) = 2.4$.

TIPS & TRICKS (AI-derived explanation)

- Divide the 5^{10} into the two factors: $2^x/5^x = (2/5)^x$ and $3^{10-x}/5^{10-x} = (3/5)^{10-x}$. The binomial form then appears immediately.
- Read off $n = 10$ and $p = 2/5 = 0.4$, giving $q = 0.6$. Everything else is two multiplications.
- For a binomial the variance npq is always LESS than the mean np , since $q < 1$ - that ordering alone eliminates options (a) and (b).
- Mean 4 and variance 2.4 in that order; option (b) 2.4, 4 reverses them, the classic careless-reading trap.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The pmf is $P(X = x) = \binom{10}{x} 2^x 3^{10-x} / 5^{10}$ for $x = 0, 1, \dots, 10$.
2. Split the denominator: $5^{10} = 5^x \cdot 5^{10-x}$, so $P(X = x) = \binom{10}{x} (2/5)^x (3/5)^{10-x}$.
3. This is exactly the Binomial distribution with $n = 10$ and $p = 2/5 = 0.4$, so $q = 0.6$.
4. Mean = $np = 10(0.4) = 4$.
5. Variance = $npq = 10(0.4)(0.6) = 2.4$.
6. Hence the mean and variance are 4 and 2.4, option (c).

13. **2024 · Q27** Standard Discrete Distributions · Conceptual

COMMON DATA FOR THIS GROUP

Questions 27–28 are based on: gambles are independent and each one results in the player being equally likely to win or lose 1 unit. Let W denote the net winnings of a gambler whose strategy is to stop gambling

immediately after his first win.

QUESTION

If the first win is on trial N , then the winnings is:

OPTIONS

- (a) $2/N$
- (b) $2 - N$
- (c) $2N$
- (d) $(2 - N)/N$

ANSWER (AI-derived — no official key exists for this paper)

(b) $2 - N$

EXAM SHORTCUT (AI-derived explanation)

If the first win comes on trial N , the gambler has lost the first $N - 1$ trials and won the N th, so the net is $+1 - (N - 1) = 2 - N$.

TIPS & TRICKS (AI-derived explanation)

- Count the outcomes: $N - 1$ losses at -1 each and one win at $+1$. Net $= 1 - (N - 1) = 2 - N$.
- Check small cases: $N = 1$ gives $+1$ (win at once) and $N = 2$ gives 0 (one loss then one win). The formula $2 - N$ reproduces both.
- The winnings are a LINEAR function of N and can be negative, so options (a), (c) and (d) - which are ratios or always positive - are structurally wrong.
- Note the strategy guarantees stopping with probability 1, but the net winnings are unbounded below - the basis of the next part.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Each gamble results in a gain of $+1$ with probability $1/2$ or a loss of 1 with probability $1/2$, and the gambler stops immediately after his first win.
2. If the first win occurs on trial N , then trials 1 through $N - 1$ were all losses and trial N was a win.
3. Total from the losses: $-(N - 1)$.
4. Total from the win: $+1$.
5. Net winnings $W = 1 - (N - 1) = 2 - N$.
6. Check: $N = 1$ gives $W = +1$ and $N = 2$ gives $W = 0$. Option (b) is correct.

14. 2024 · Q28 Standard Discrete Distributions · Numerical

COMMON DATA FOR THIS GROUP

Questions 27–28 are based on: gambles are independent and each one results in the player being equally likely to win or lose 1 unit. Let W denote the net winnings of a gambler whose strategy is to stop gambling immediately after his first win.

QUESTION

What is $P(W < 0)$ equal to?

OPTIONS

- (a) $1/2$
- (b) $1/4$
- (c) $3/4$
- (d) 0

ANSWER (AI-derived — no official key exists for this paper)**(b)** $1/4$ **EXAM SHORTCUT** (AI-derived explanation)
$$W = 2 - N \text{ is negative exactly when } N \geq 3, \text{ and } P(N \geq 3) = 1 - 1/2 - 1/4 = 1/4.$$
TIPS & TRICKS (AI-derived explanation)

- Translate the event first: $W < 0$ means $2 - N < 0$, i.e. $N > 2$, i.e. $N \geq 3$ since N is an integer.
- N is geometric with $p = 1/2$, so $P(N = n) = (1/2)^n$ and $P(N \geq 3) = (1/2)^2 = 1/4$ by the tail formula $P(N \geq k) = q^{k-1}$.
- The boundary case $N = 2$ gives $W = 0$, which is NOT negative - excluding it correctly is the crux of the item.
- Sanity check: $P(W > 0) = P(N = 1) = 1/2$ and $P(W = 0) = P(N = 2) = 1/4$, so the three probabilities sum to 1.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. From the previous part, $W = 2 - N$ where N is the trial of the first win.
2. $W < 0$ requires $2 - N < 0$, that is $N > 2$; since N is a positive integer this means $N \geq 3$.
3. N follows a geometric distribution on $\{1, 2, 3, \dots\}$ with success probability $1/2$, so $P(N = n) = (1/2)^n$.
4. $P(N \geq 3) = 1 - P(N = 1) - P(N = 2) = 1 - 1/2 - 1/4 = 1/4$.
5. Equivalently, $P(N \geq 3) = (1/2)^2 = 1/4$, the probability that the first two trials are both losses.
6. Hence $P(W < 0) = 1/4$, option (b).

15. **2024 · Q29** Standard Discrete Distributions · Numerical**COMMON DATA FOR THIS GROUP**

Questions 29–30 are based on: suppose that a balanced, six-faced dice is rolled repeatedly until the same number appears on two successive rolls and let X denote the number of rolls that are required.

QUESTIONWhat is the value of $P(X = 10)$?**OPTIONS**

- (a) $\frac{5^8}{6^9}$
- (b) $\frac{5^9}{6^{10}}$
- (c) $\frac{5^{10}}{6^{10}}$
- (d) $\frac{5^{11}}{6^{12}}$

ANSWER (AI-derived — no official key exists for this paper)

$$(a) \frac{5^8}{6^9}$$

EXAM SHORTCUT (AI-derived explanation)

After the first roll, each subsequent roll differs from its predecessor with probability $5/6$ and matches with probability $1/6$. So $P(X = n) = (5/6)^{n-2}(1/6)$, and $P(X = 10) = 5^8/6^9$.

TIPS & TRICKS (AI-derived explanation)

- The first roll is free - it can be anything. From the second roll onwards each comparison is a Bernoulli trial with match probability $1/6$.
- For $X = n$ we need $n - 2$ consecutive 'different' outcomes followed by one 'same', giving $(5/6)^{n-2}(1/6)$.
- Combine the powers: $(5/6)^8(1/6) = 5^8/6^8 \times 1/6 = 5^8/6^9$ - watch that the denominator exponent is one MORE than the numerator's.
- Check the smallest case: $P(X = 2) = (5/6)^0(1/6) = 1/6$, which is obviously right - the second roll simply matches the first.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The first roll can show any face, so it imposes no condition. From the second roll onwards, each roll either matches its immediate predecessor (probability $1/6$) or differs from it (probability $5/6$), independently.
2. The event $X = n$ means that rolls $2, 3, \dots, n - 1$ each DIFFER from the preceding roll, and roll n MATCHES roll $n - 1$.
3. That is $n - 2$ 'different' outcomes followed by one 'same'.
4. Hence $P(X = n) = (5/6)^{n-2}(1/6)$ for $n = 2, 3, 4, \dots$
5. For $n = 10$: $P(X = 10) = (5/6)^8(1/6) = 5^8/6^8 \times 1/6 = 5^8/6^9$, option (a).

16. 2024 · Q30 Mathematical Expectation · Numerical**COMMON DATA FOR THIS GROUP**

Questions 29–30 are based on: suppose that a balanced, six-faced dice is rolled repeatedly until the same number appears on two successive rolls and let X denote the number of rolls that are required.

QUESTION

What is the value of $E(X)$?

OPTIONS

- (a) 17
(b) 7
(c) $18/35$
(d) $17/25$

ANSWER (AI-derived — no official key exists for this paper)

$$(b) 7$$

EXAM SHORTCUT (AI-derived explanation)

$X - 1$ is geometric on $\{1, 2, \dots\}$ with success probability $1/6$, so $E(X - 1) = 6$ and $E(X) = 7$.

TIPS & TRICKS (AI-derived explanation)

- Shift to a standard family: $X - 1$ counts the comparisons made from the second roll onwards, each succeeding with probability $1/6$.
- Geometric mean on $\{1, 2, \dots\}$ is $1/p = 6$, so add back the free first roll to get 7.
- Direct check with the pmf: $E(X) = \sum n(5/6)^{n-2}(1/6)$, which evaluates to 7.
- Intuition: you expect to wait 6 comparisons for a repeat, plus the initial roll - so 7 rolls in all.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. From the previous part, $P(X = n) = (5/6)^{n-2}(1/6)$ for $n = 2, 3, 4, \dots$
2. Set $Y = X - 1$, so $P(Y = k) = (5/6)^{k-1}(1/6)$ for $k = 1, 2, 3, \dots$
3. This is the geometric distribution on the positive integers with success probability $p = 1/6$.
4. Its mean is $E(Y) = 1/p = 6$.
5. Therefore $E(X) = E(Y) + 1 = 6 + 1 = 7$.
6. Option (b) is correct.

17. 2025 · Q17 Geometric Distribution · Theoretical**QUESTION**

Let $(X_1, \dots, X_k)$ be a random sample of size k from a geometric distribution with pmf $f(x) = pq^{x-1}$; $x = 1, 2, 3, \dots$. Then $\min(X_1, \dots, X_k)$ is distributed geometrically with parameter:

OPTIONS

- (a) $1 - p^k$
- (b) $1 - (1 - p)^k$
- (c) $(1 - p)^k$
- (d) None of the above

ANSWER (AI-derived — no official key exists for this paper)

(b) $1 - (1 - p)^k$

EXAM SHORTCUT (AI-derived explanation)

$P(X > m) = q^m$, so $P(\min > m) = q^{km} = (q^k)^m$. The minimum is geometric with failure probability q^k , i.e. success probability $1 - (1 - p)^k$.

TIPS & TRICKS (AI-derived explanation)

- Work with the SURVIVAL function: $P(X > m) = q^m$ for the geometric on $\{1, 2, \dots\}$, and independence turns the k -fold minimum into the k th power.
- Recognise $(q^k)^m$ as the survival function of a geometric with new failure probability $q' = q^k$, hence $p' = 1 - q^k$.
- Substituting $q = 1 - p$ gives $p' = 1 - (1 - p)^k$ - exactly the probability that at least one of the k variables equals its first trial.
- Sanity check at $k = 1$: the formula returns p , as it must. Option (c) would give $(1 - p)$, which fails this test.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. For the geometric distribution with pmf $f(x) = pq^{x-1}$ on $x = 1, 2, 3, \dots$, the survival function is $P(X > m) = q^m$.
2. Let $M = \min(X_1, \dots, X_k)$ with the X_i independent copies.
3. Then $P(M > m) = P(\text{every } X_i > m) = [q^m]^k = (q^k)^m$.
4. This is the survival function of a geometric distribution with failure probability q^k , hence success probability $1 - q^k$.
5. Since $q = 1 - p$, the parameter is $1 - (1 - p)^k$.
6. Check at $k = 1$: the formula gives p , as required. Option (b) is correct.

18. 2025 · Q48 Poisson Distribution · Numerical

QUESTION

Use the Poisson distribution to compute the probability of getting the ace of spades at least twice in 104 trials of drawing a single card (with replacement) from a well-shuffled pack (given $e^{-2} = 0.1353$). What is the probability?

OPTIONS

- (a) 0.6235
- (b) 0.5941
- (c) 0.5419
- (d) None of the above

ANSWER (AI-derived — no official key exists for this paper)

(b) 0.5941

EXAM SHORTCUT (AI-derived explanation)

$\lambda = np = 104(1/52) = 2$, so $P(\text{at least } 2) = 1 - e^{-2} - 2e^{-2} = 1 - 3(0.1353) = 0.5941$.

TIPS & TRICKS (AI-derived explanation)

- Poisson approximation parameter: $\lambda = np$. Here $104/52 = 2$ exactly, which is why e^{-2} is supplied.
- 'At least twice' means $1 - P(0) - P(1)$; computing $P(2)$ alone is the classic error.
- $P(0) + P(1) = e^{-2}(1 + 2) = 3e^{-2}$ - factor out e^{-2} before multiplying.
- $3 \times 0.1353 = 0.4059$, so the answer is $1 - 0.4059 = 0.5941$. Option (c) 0.5419 is a digit transposition of the correct value.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The probability of drawing the ace of spades on a single trial is $p = 1/52$, and there are $n = 104$ independent trials with replacement.
2. The Poisson approximation uses $\lambda = np = 104/52 = 2$.
3. $P(X \geq 2) = 1 - P(X = 0) - P(X = 1)$.
4. $P(X = 0) = e^{-2}$ and $P(X = 1) = 2e^{-2}$, so $P(X = 0) + P(X = 1) = 3e^{-2}$.
5. With $e^{-2} = 0.1353$: $3(0.1353) = 0.4059$.
6. Hence $P(X \geq 2) = 1 - 0.4059 = 0.5941$, option (b).

19. 2026 · Q10 Negative Binomial Distribution · Numerical**QUESTION**

X, Y, Z are independent. X and Y each have a geometric distribution with parameter $1/2$, and Z has a negative binomial distribution with parameters 3 and p . $E(X + Y - Z) = 0$ if and only if p equals:

OPTIONS

- (a) $1/3$
- (b) $3/4$
- (c) $2/5$
- (d) $3/5$

ANSWER (AI-derived — no official key exists for this paper)

(d) $3/5$

Source note. The answer depends on the counting convention. Using the standard Indian-textbook convention in which both the geometric and the negative binomial count FAILURES (geometric mean q/p , negative binomial mean r/p), $E(X) = 1$ and $E(Y) = 1$ and $2 = 3(1-p)/p$ gives $p = 3/5$, option (d). If instead both are taken as counting TRIALS (means $1/p$ and r/p), then $4 = 3/p$ gives $p = 3/4$, option (b), which is why both values appear among the choices. Option (d) is keyed here on the failure-counting convention; a candidate should state the convention used.

EXAM SHORTCUT (AI-derived explanation)

With the failure-counting convention, geometric($1/2$) has mean $q/p = 1$, so $E(X + Y) = 2$, and NB(3, p) has mean $3q/p$. Set $3(1 - p)/p = 2$ to get $5p = 3, p = 3/5$.

TIPS & TRICKS (AI-derived explanation)

- State your convention before computing: geometric as number of failures has mean q/p ; as number of trials it has mean $1/p$. The same split applies to the negative binomial (r/p versus r/p).
- Whichever convention you adopt, use it for BOTH distributions - the geometric is the negative binomial with $r = 1$, so mixing conventions is internally inconsistent.
- Expectation is linear whether or not the variables are independent, so $E(X + Y - Z) = E(X) + E(Y) - E(Z)$; the stated independence is needed only if a variance were asked.
- The two consistent conventions produce exactly the two 'nice' options $3/5$ and $3/4$ - recognising that tells you which pair of distractors are decoys.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Adopt the standard convention in which the geometric variable counts the number of failures before the first success: $P(X = x) = q^x p, x = 0, 1, 2, \dots$, with $E(X) = q/p$.
2. With $p = 1/2$: $E(X) = (1/2)/(1/2) = 1$, and likewise $E(Y) = 1$, so $E(X + Y) = 2$.
3. Under the matching convention, the negative binomial with parameters $r = 3$ and p counts the failures before the 3rd success: $P(Z = z) = \binom{z+2}{z} p^3 q^z$, with $E(Z) = 3q/p = 3(1 - p)/p$.
4. By linearity, $E(X + Y - Z) = 0$ requires $E(Z) = E(X) + E(Y) = 2$.
5. So $3(1 - p)/p = 2$, giving $3 - 3p = 2p$, hence $5p = 3$ and $p = 3/5$.
6. (Consistency check: with $p = 3/5$, $E(Z) = 3(2/5)/(3/5) = 2$, matching $E(X) + E(Y) = 2$.)

7. Hence option (d) is correct under the failure-counting convention; see the note above for the trials convention.

20. **2026 · Q17** Binomial Distribution · Theoretical

QUESTION

X has a binomial distribution with parameters n , p , and $q = 1 - p$. What is $P[X = 0 \text{ or even}]$?

OPTIONS

- (a) $2(q - p)^n$
 (b) $[1 + (q - p)^n]/2$
 (c) $[1 - (q - p)^n]/2$
 (d) $1 + (q - p)^n$

ANSWER (AI-derived — no official key exists for this paper)

(b) $[1 + (q - p)^n]/2$

EXAM SHORTCUT (AI-derived explanation)

Add the expansions of $(q + p)^n$ and $(q - p)^n$: odd terms cancel and even terms double. So the sum of even-index terms is $[1 + (q - p)^n]/2$. Answer (b).

TIPS & TRICKS (AI-derived explanation)

- The even/odd splitting trick: sum of even terms = $[f(1) + f(-1)]/2$ and sum of odd terms = $[f(1) - f(-1)]/2$, applied to the pgf.
- Here $f(s) = (q + ps)^n$, so $f(1) = 1$ and $f(-1) = (q - p)^n$. That is the entire derivation.
- Sanity check with $n = 1$: $P(X = 0) = q$, and the formula gives $[1 + (q - p)]/2 = (1 + q - p)/2 = (q + q)/2 = q$ since $p + q = 1$. Correct.
- Options (a) and (d) can exceed 1 (take $p = 0$, giving 2 and 2 respectively), so they are impossible probabilities; option (c) is the odd-count probability.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. $P(X = x) = \binom{n}{x} p^x q^{n-x}$. We want $P(X \text{ is even, including } 0) = \text{sum over even } x \text{ of } \binom{n}{x} p^x q^{n-x}$.
2. Consider the binomial expansions $(q + p)^n = \sum \binom{n}{x} p^x q^{n-x} = 1$, and $(q - p)^n = \sum \binom{n}{x} (-p)^x q^{n-x} = \sum (-1)^x \binom{n}{x} p^x q^{n-x}$.
3. Adding the two: the terms with odd x cancel and the terms with even x double.
4. So $(q + p)^n + (q - p)^n = 2 * \text{sum over even } x \text{ of } \binom{n}{x} p^x q^{n-x}$.
5. Therefore $P(X \text{ even}) = [1 + (q - p)^n]/2$.
6. Verification at $n = 1$: $[1 + q - p]/2 = q = P(X = 0)$, correct. (Correspondingly $P(X \text{ odd}) = [1 - (q - p)^n]/2$, which is option (c) - a deliberate distractor.)
7. Hence option (b) is correct.

P5 · PROBABILITY

Standard Continuous Distributions

21 questions · 2018: 1 2019: 2 2020: 1 2021: 0 2022: 2 2023: 3 2024: 2 2025: 7 2026: 3

Weight. 21 questions (10.7% of Probability) · appeared in 8 of 9 papers · peak year 2025 (7)**Examiner's pattern.** Beta of the first kind is the recent favourite (mode, and recovering parameters from mean and variance). Also standard: exponential mean-vs-variance comparison, Laplace CDF, Cauchy having no mean, and writing $P(a \leq X \leq b)$ for a normal in terms of Φ .**Must-know.** Beta-I mode = $(a-1)/(a+b-2)$ for $a, b > 1$; Laplace $F(x) = 0.5\exp(x)$ for $x < 0$; Cauchy has no moments; exponential: $\text{Var} > \text{mean}$ iff $\theta < 1$.**1. 2018 · Q10** Standard Continuous Distributions · Conceptual**QUESTION**

For $f(x) = \theta e^{-\theta x}$, $x > 0$: 1. Variance $>$ mean if $0 < \theta < 1$. 2. Variance = mean if $\theta = 1$. 3. Variance $<$ mean if $\theta > 1$.

OPTIONS

- (a) 1 and 3 only
- (b) 2 and 3 only
- (c) 1 and 2 only
- (d) 1, 2 and 3

ANSWER (AI-derived — no official key exists for this paper)**(d)** 1, 2 and 3**EXAM SHORTCUT** (AI-derived explanation)

For $\text{Exp}(\theta)$: mean = $1/\theta$, variance = $1/\theta^2$. Comparing $1/\theta^2$ with $1/\theta$ reduces to comparing $1/\theta$ with 1, i.e. comparing 1 with θ . So all three statements are just $\theta < 1, \theta = 1, \theta > 1$ - all correct.

TIPS & TRICKS (AI-derived explanation)

- $\text{Var}/\text{mean} = (1/\theta^2)/(1/\theta) = 1/\theta$. The whole item collapses to 'is $1/\theta$ bigger, equal to, or smaller than 1?'.
- Note that θ here is the RATE (the parameter in $\theta e^{-\theta x}$), not the mean. Mixing up rate and mean parameterisation inverts every inequality.
- When three statements exhaustively cover $\theta < 1, \theta = 1$ and $\theta > 1$ and each is internally consistent, (d) 'all of them' is the natural answer - check one and the pattern usually confirms.
- For any distribution, comparing mean and variance is fastest via the ratio (the index of dispersion), not by subtracting.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. For the exponential density $f(x) = \theta e^{-\theta x}$, $x > 0$, the mean is $E(X) = 1/\theta$ and the variance is $\text{Var}(X) = 1/\theta^2$.
2. Form the ratio $\text{Var}(X)/E(X) = (1/\theta^2)/(1/\theta) = 1/\theta$.
3. Statement 1: variance $>$ mean means $1/\theta > 1$, i.e. $\theta < 1$. Combined with $\theta > 0$ this is exactly $0 < \theta < 1$. TRUE.
4. Statement 2: variance = mean means $1/\theta = 1$, i.e. $\theta = 1$ (both equal 1). TRUE.
5. Statement 3: variance $<$ mean means $1/\theta < 1$, i.e. $\theta > 1$. TRUE.

6. All three statements are correct, so option (d) is right.

2. 2019 · Q35 Uniform / Rectangular Distribution · Numerical

QUESTION

The round-off error to the second decimal place has uniform distribution on $(-0.05, 0.05)$. What is the probability that the absolute error in the sum of 1000 numbers is less than 2? [Given $\Phi(2.19) = 0.985$]

OPTIONS

- (a) 0.92
- (b) 0.95
- (c) 0.97
- (d) 0.99

ANSWER (AI-derived — no official key exists for this paper)

(c) 0.97

EXAM SHORTCUT (AI-derived explanation)

Each error has variance $(0.1)^2/12 = 1/1200$, so the sum of 1000 has variance $5/6$ and SD 0.9129. Then $P(|S| < 2) = P(|Z| < 2.19) = 2(0.985) - 1 = 0.97$.

TIPS & TRICKS (AI-derived explanation)

- Uniform(a, b) variance is $(b - a)^2/12$. Here $b - a = 0.1$, so the variance is $0.01/12$ - watch the square on the WIDTH, not on the endpoint.
- Variances of independent errors add: $1000 \times (1/1200) = 5/6 = 0.8333$, giving SD = 0.9129. The CLT then makes the sum approximately normal.
- $2/0.9129 = 2.19$, exactly the argument the examiner supplies - a strong confirmation that the set-up is right.
- For a two-sided probability use $2\Phi(z) - 1$, not $\Phi(z)$. Forgetting this gives 0.985 and no matching option.

STEP-BY-STEP SOLUTION (AI-derived explanation)

- Each round-off error is Uniform $(-0.05, 0.05)$, with mean 0 and variance $(0.1)^2/12 = 0.01/12 = 1/1200$.
- The total error S is the sum of 1000 independent such errors, so $E(S) = 0$ and $\text{Var}(S) = 1000/1200 = 5/6 = 0.8333$.
- The standard deviation is $\sqrt{0.8333} = 0.9129$, and by the Central Limit Theorem S is approximately normal.
- $P(|S| < 2) = P(|Z| < 2/0.9129) = P(|Z| < 2.19)$.
- Using the supplied value $\Phi(2.19) = 0.985$: $P(|Z| < 2.19) = 2(0.985) - 1 = 0.97$.
- Hence option (c) is correct.

3. 2019 · Q38 Normal Distribution · Numerical

QUESTION

If the p.d.f. of the normal distribution of X is $f(x) = k \exp\left(-\frac{1}{8}x^2 + 2x\right)$, $-\infty < x < \infty$, the mean and variance of the distribution are respectively

OPTIONS

- (a) 8 and 2
 (b) 4 and 2
 (c) 8 and 4
 (d) 4 and 4

ANSWER (AI-derived — no official key exists for this paper)**(c)** 8 and 4**EXAM SHORTCUT (AI-derived explanation)**

Match the exponent to $-(x - \mu)^2/(2\sigma^2)$. The x^2 coefficient $-1/8$ gives $\sigma^2 = 4$; the x coefficient $\mu/\sigma^2 = 2$ then gives $\mu = 8$.

TIPS & TRICKS (AI-derived explanation)

- Standard expansion: $-(x - \mu)^2/(2\sigma^2) = -x^2/(2\sigma^2) + (\mu/\sigma^2)x - \mu^2/(2\sigma^2)$. Read σ^2 off the x^2 term FIRST, then μ from the x term.
- From $-1/(2\sigma^2) = -1/8$ we get $2\sigma^2 = 8$, so $\sigma^2 = 4$ (not 8) - halving at the right moment is the whole trick.
- With $\sigma^2 = 4$, the x -coefficient $\mu/4 = 2$ gives $\mu = 8$. Options (a) and (b) mix up which parameter is which.
- The constant k absorbs $\exp(-\mu^2/(2\sigma^2))$ together with the usual $1/(\sigma\sqrt{2\pi})$; it carries no information about μ or σ and can be ignored.

STEP-BY-STEP SOLUTION (AI-derived explanation)

- The normal density is $f(x) = [1/(\sigma\sqrt{2\pi})] \exp(-(x - \mu)^2/(2\sigma^2))$.
- Expanding the exponent: $-(x - \mu)^2/(2\sigma^2) = -x^2/(2\sigma^2) + (\mu/\sigma^2)x - \mu^2/(2\sigma^2)$.
- The given exponent is $-x^2/8 + 2x$. Comparing coefficients of x^2 : $-1/(2\sigma^2) = -1/8$, so $2\sigma^2 = 8$ and $\sigma^2 = 4$.
- Comparing coefficients of x : $\mu/\sigma^2 = 2$, so $\mu = 2\sigma^2 = 2(4) = 8$.
- The constant term $-\mu^2/(2\sigma^2)$ is absorbed into k , which is consistent.
- Hence the mean is 8 and the variance is 4, option (c).

4. 2020 · Q34 Exponential Distribution · Numerical**QUESTION**

The maximum possible probability an exponential distribution (any parameter) can give to the interval $[3, 6]$ is

OPTIONS

- (a) $\frac{1}{2}$
 (b) $\frac{1}{3}$
 (c) $\frac{1}{4}$
 (d) $\frac{2}{3}$

ANSWER (AI-derived — no official key exists for this paper)**(c)** $\frac{1}{4}$

EXAM SHORTCUT (AI-derived explanation)

$P(3 \leq X \leq 6) = e^{-3L} - e^{-6L}$. Put $u = e^{-3L}$ in $(0, 1)$, so the probability is $u - u^2$, maximised at $u = 1/2$ with value $1/4$.

TIPS & TRICKS (AI-derived explanation)

- Substituting $u = e^{-3\lambda}$ turns a transcendental maximisation into the elementary quadratic $u - u^2$ - the key move in this item.
- Note that $6 = 2 \times 3$, so $e^{-6\lambda} = u^2$ exactly. The doubling of the endpoints is what makes the substitution work.
- $u - u^2$ has its maximum at $u = 1/2$, i.e. $e^{-3\lambda} = 1/2$ and $\lambda = (\ln 2)/3 = 0.231$. The optimal mean is therefore about 4.33, comfortably inside $[3, 6]$ - a good plausibility check.
- The maximum value $1/4$ is a bound over ALL exponential distributions, so no single choice of parameter can do better - the reason option (a) $1/2$ is unattainable.

STEP-BY-STEP SOLUTION (AI-derived explanation)

- For an exponential distribution with rate λ , $P(3 \leq X \leq 6) = e^{-3\lambda} - e^{-6\lambda}$.
- Let $u = e^{-3\lambda}$, which ranges over $(0, 1)$ as λ ranges over $(0, \infty)$. Then $e^{-6\lambda} = u^2$.
- So the probability is $g(u) = u - u^2$.
- Differentiating, $g'(u) = 1 - 2u = 0$ at $u = 1/2$, and $g''(u) = -2 < 0$, so this is a maximum.
- The maximum value is $g(1/2) = 1/2 - 1/4 = 1/4$, attained when $e^{-3\lambda} = 1/2$, i.e. $\lambda = (\ln 2)/3$.
- Hence the maximum possible probability is $1/4$, option (c).

5. 2022 · Q43 Standard Continuous Distributions · Numerical**QUESTION**

Let X be continuous with density $f(x) = \frac{1}{2}e^{-|x|}$, $-\infty < x < \infty$. What is $P(-1 < X < 2)$ equal to?

OPTIONS

- (a) $\frac{1}{2}(2 - e^{-1} - e^{-2})$
 (b) $\frac{1}{2}(2 + e^{-1} - e^{-2})$
 (c) $\frac{1}{2}(2 - e^{-1} + e^{-2})$
 (d) $\frac{1}{2}(2 + e^{-1} + e^{-2})$

ANSWER (AI-derived — no official key exists for this paper)

(a) $\frac{1}{2}(2 - e^{-1} - e^{-2})$

EXAM SHORTCUT (AI-derived explanation)

Split at 0: the integral from -1 to 0 gives $(1/2)(1 - e^{-1})$ and from 0 to 2 gives $(1/2)(1 - e^{-2})$. Adding: $(1/2)(2 - e^{-1} - e^{-2})$.

TIPS & TRICKS (AI-derived explanation)

- The Laplace density has $|x|$ in the exponent, so ALWAYS split the range at 0 - on the left the density is $(1/2)e^x$ and on the right $(1/2)e^{-x}$.
- Each half integrates to $(1/2)(1 - e^{-a})$ where a is the distance from 0, so the two pieces are structurally identical and both SUBTRACT an exponential.

- Both exponential terms must therefore carry a minus sign, which identifies option (a) immediately; options (b), (c) and (d) have at least one wrong sign.
- Numerical check: $(1/2)(2 - 0.3679 - 0.1353) = 0.7484$, a sensible probability for an interval covering most of the density's mass.

STEP-BY-STEP SOLUTION (AI-derived explanation)

- The density is the Laplace (double exponential) $f(x) = (1/2)e^{-|x|}$, symmetric about 0.
- Split the interval at 0. For $-1 < x < 0$, $|x| = -x$ so $f(x) = (1/2)e^x$.
- The $\int_{-1}^0 (1/2)e^x dx = (1/2)[e^x]$ from -1 to 0 $= (1/2)(1 - e^{-1})$.
- For $0 < x < 2$, $|x| = x$ so $f(x) = (1/2)e^{-x}$, and the integral from 0 to 2 is $(1/2)[-e^{-x}]$ from 0 to 2 $= (1/2)(1 - e^{-2})$.
- Adding: $P(-1 < X < 2) = (1/2)(1 - e^{-1}) + (1/2)(1 - e^{-2}) = (1/2)(2 - e^{-1} - e^{-2})$.
- Numerically this is 0.748. Option (a) is correct.

6. 2022 · Q48 Beta Distribution · Numerical**QUESTION**

A beta distribution of first type with parameters m, n has mean $\frac{1}{3}$ and variance $\frac{2}{63}$. What are m and n respectively?

OPTIONS

- (a) 2, 4
(b) 4, 2
(c) 2, 2
(d) 3, 6

ANSWER (AI-derived — no official key exists for this paper)

(a) 2, 4

EXAM SHORTCUT (AI-derived explanation)

Mean $m/(m+n) = 1/3$ gives $n = 2m$. Substituting into the variance $2m^2/[9m^2(3m+1)] = 2/(9(3m+1)) = 2/63$ gives $3m+1 = 7$, so $m = 2$ and $n = 4$.

TIPS & TRICKS (AI-derived explanation)

- Beta(m, n) first kind: mean $= m/(m+n)$, variance $= mn/[(m+n)^2(m+n+1)]$. Write both down before substituting.
- Use the mean FIRST to express n in terms of m ; the variance equation then has a single unknown and the m^2 factors cancel.
- Variance shortcut: variance $= \text{mean}(1 - \text{mean})/(m+n+1) = (1/3)(2/3)/(m+n+1) = 2/(9(m+n+1))$. Setting this to $2/63$ gives $m+n+1 = 7$ directly, so $m+n = 6$.
- With $m+n = 6$ and $m/(m+n) = 1/3$, we get $m = 2$ and $n = 4$ in one step - the fastest route of all.

STEP-BY-STEP SOLUTION (AI-derived explanation)

- For a Beta distribution of the first kind with parameters m and n , the mean is $m/(m+n)$ and the variance is $mn/[(m+n)^2(m+n+1)]$.
- From the mean: $m/(m+n) = 1/3$, so $3m = m+n$ and $n = 2m$.

3. A convenient form of the variance is $\text{mean} \times (1 - \text{mean}) / (m + n + 1) = (1/3)(2/3) / (m + n + 1) = 2 / (9(m + n + 1))$.
4. Setting this equal to $2/63$ gives $9(m + n + 1) = 63$, so $m + n + 1 = 7$ and $m + n = 6$.
5. Combining with $n = 2m$: $3m = 6$, so $m = 2$ and $n = 4$.
6. Check: $\text{mean} = 2/6 = 1/3$ and $\text{variance} = (2)(4) / [36 \times 7] = 8/252 = 2/63$. Option (a) is correct.

7. 2023 · Q3 Standard Continuous Distributions · Conceptual

QUESTION

If $X \sim \exp(\theta)$, then consider the following statements: 1. Variance > Mean if $0 < \theta < 1$ 2. Minimum order statistic is chi-square distribution 3. Lack of memory property exists only if $\theta > 1$ Which of the statements given above is/are correct?

OPTIONS

- (a) 1 only
- (b) 3 only
- (c) 1 and 2
- (d) 1 and 3

ANSWER (AI-derived — no official key exists for this paper)

(a) 1 only

EXAM SHORTCUT (AI-derived explanation)

Only statement 1 holds: $\text{Var}/\text{Mean} = 1/\theta$, so the variance exceeds the mean exactly when $\theta < 1$. The minimum of exponentials is exponential, not χ^2 , and memorylessness holds for every θ .

TIPS & TRICKS (AI-derived explanation)

- Exponential(θ) with θ as RATE: mean $1/\theta$, variance $1/\theta^2$, so the ratio is $1/\theta$ and the comparison reduces to θ versus 1.
- Minimum of n i.i.d. $\text{Exp}(\theta)$ is $\text{Exp}(n\theta)$ - the exponential family is closed under minima, which is the standard result statement 2 misquotes.
- Memorylessness is a property of the exponential FAMILY, holding for every rate parameter; no condition on θ is needed, so statement 3 is false.
- The χ^2 link that does exist is $\text{Exp}(1/2) = \chi^2_2$ - a special case, not a statement about minima.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. For $X \sim \text{Exponential}$ with rate θ , the mean is $1/\theta$ and the variance is $1/\theta^2$.
2. Statement 1: $\text{Var}/\text{Mean} = (1/\theta^2)/(1/\theta) = 1/\theta$, which exceeds 1 exactly when $\theta < 1$. So the variance exceeds the mean for $0 < \theta < 1$. TRUE.
3. Statement 2: for a sample of n i.i.d. $\text{Exponential}(\theta)$ variables, $P(\min > x) = e^{-n\theta x}$, so the minimum is $\text{Exponential}(n\theta)$, not χ^2 . FALSE.
4. Statement 3: the lack-of-memory property $P(X > s + t | X > s) = P(X > t)$ follows from $e^{-\theta(s+t)}/e^{-\theta s} = e^{-\theta t}$, which holds for every $\theta > 0$. No restriction to $\theta > 1$ exists. FALSE.
5. Only statement 1 is correct.

6. Hence option (a) is correct.

8. **2023 · Q7** Normal Distribution · Numerical

QUESTION

If $\varphi(\cdot)$ is pdf of $N(0, 1)$ distribution and A represents a circle of unit area with centre at origin, then what is the value of $\iint_A \varphi(x)\varphi(y) dx dy$?

OPTIONS

- (a) $\frac{1}{2} + e^{-\pi}$
- (b) $1 - e^{-\frac{1}{2\pi}}$
- (c) $1 - e^{-2\pi}$
- (d) $1 - e^{-\frac{1}{\sqrt{\pi}}}$

ANSWER (AI-derived — no official key exists for this paper)

(b) $1 - e^{-\frac{1}{2\pi}}$

EXAM SHORTCUT (AI-derived explanation)

Unit area means $\pi r^2 = 1$, so $r^2 = 1/\pi$. The integral is $P(X^2 + Y^2 \leq r^2)$ with $X^2 + Y^2 \sim \chi_2^2 = \text{Exp}(1/2)$, giving $1 - e^{-r^2/2} = 1 - e^{-1/(2\pi)}$.

TIPS & TRICKS (AI-derived explanation)

- 'Unit AREA' not unit radius - solve $\pi r^2 = 1$ first to get $r^2 = 1/\pi$. Reading it as $r = 1$ leads straight to option (c).
- For independent standard normals, $X^2 + Y^2 \sim \chi^2$ with 2 d.f., which is Exponential with rate $1/2$ - a fact worth memorising.
- Hence $P(X^2 + Y^2 \leq c) = 1 - e^{-c/2}$ for any $c > 0$; substituting $c = 1/\pi$ gives the answer in one step.
- The bivariate standard normal density is circularly symmetric, which is precisely why a circular region turns a double integral into a one-dimensional χ^2 probability.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The integrand $\phi(x)\phi(y)$ is the joint density of two independent standard normal variables, so the integral is $P((X,Y) \text{ lies in } A)$.
2. A is a circle centred at the origin with unit AREA, so $\pi r^2 = 1$ and $r^2 = 1/\pi$.
3. The event is therefore $X^2 + Y^2 \leq r^2 = 1/\pi$.
4. For independent standard normals, $X^2 + Y^2$ follows a χ^2 distribution with 2 degrees of freedom, which is exponential with rate $1/2$.
5. Hence $P(X^2 + Y^2 \leq c) = 1 - e^{-c/2}$, and with $c = 1/\pi$ this is $1 - e^{-1/(2\pi)}$.
6. Option (b) is correct.

9. **2023 · Q18** Beta Distribution · Numerical

COMMON DATA FOR THIS GROUP

Questions 18–20 are based on the following: let the joint pdf of X and Y be

$$f(x, y) = \begin{cases} \frac{\lambda x^{12}}{(1 + xy)^{20}}, & x > 0, y > 1 \\ 0, & \text{otherwise} \end{cases}$$

QUESTION

What is the value of λ ? (where $B(m, n) = \int_0^1 x^{m-1}(1-x)^{n-1} dx$)

OPTIONS

- (a) $\frac{1}{B(12, 7)}$
 (b) $\frac{19}{B(12, 7)}$
 (c) $\frac{7}{B(12, 7)}$
 (d) $\frac{84}{B(12, 7)}$

ANSWER (AI-derived — no official key exists for this paper)

(b) $\frac{19}{B(12, 7)}$

EXAM SHORTCUT (AI-derived explanation)

Integrating x first (substituting $u = xy$) gives $B(13, 7)/y^{13}$; integrating that over $y > 1$ gives $B(13, 7)/12$. So $\lambda = 12/B(13, 7) = 19/B(12, 7)$, since $B(13, 7) = (12/19)B(12, 7)$.

TIPS & TRICKS (AI-derived explanation)

- Beta second-kind integral: the $\int_0^\infty u^{m-1}/(1+u)^{m+n} du = B(m, n)$. With numerator power 12 and denominator power 20, $m = 13$ and $n = 7$.
- The substitution $u = xy$ turns $(1 + xy)$ into $(1 + u)$ and produces the factor y^{-13} , which then integrates cleanly over $(1, \infty)$ to $1/12$.
- Convert $B(13, 7)$ to $B(12, 7)$ using $B(m, n) = \Gamma(m)\Gamma(n)/\Gamma(m+n)$: the ratio $B(13, 7)/B(12, 7) = 12/19$, so $\lambda = 12 \times 19/12$ divided by $B(12, 7) = 19/B(12, 7)$.
- The options are all expressed in $B(12, 7)$, which is the signal that a Gamma-ratio conversion is expected - do it at the very end.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Require the double $\int \lambda x^{12}/(1 + xy)^{20}$ over $x > 0, y > 1$ to equal 1.
2. Inner integral over x : substitute $u = xy$, so $x = u/y$ and $dx = du/y$, giving the $\int (u/y)^{12}/(1 + u)^{20} du/y = (1/y^{13})$ times the $\int u^{12}/(1 + u)^{20} du$.
3. That standard integral is $B(m, n)$ with $m - 1 = 12$ and $m + n = 20$, i.e. $B(13, 7)$. So the inner integral equals $B(13, 7)/y^{13}$.
4. Outer integral over y from 1 to ∞ : the $\int y^{-13} dy = [y^{-12}/(-12)]$ from 1 to $\infty = 1/12$.
5. Hence $\lambda B(13, 7)/12 = 1$, giving $\lambda = 12/B(13, 7)$.
6. Since $B(13, 7) = \Gamma(13)\Gamma(7)/\Gamma(20)$ and $B(12, 7) = \Gamma(12)\Gamma(7)/\Gamma(19)$, their ratio is $(12!/11!)(18!/19!) = 12/19$, so $B(13, 7) = (12/19)B(12, 7)$ and $\lambda = 12 \times 19/(12B(12, 7)) = 19/B(12, 7)$, option (b).

10. 2024 · Q10 Cauchy Distribution · Numerical

COMMON DATA FOR THIS GROUP

Questions 7–10 are based on: let $X_1, X_2, X_3, \dots, X_7$ be iid random variables, each having the $N(0, 1)$ distribution.

QUESTION

If $\lambda \left(\frac{X_3 + X_4 + X_5 + X_6 + X_7}{X_1 + X_2} \right) \sim \text{Cauchy}(0, 1)$ distribution, then what is the value of $5\lambda^2$?

OPTIONS

- (a) 1
- (b) 2
- (c) 4
- (d) 5

ANSWER (AI-derived — no official key exists for this paper)

(b) 2

EXAM SHORTCUT (AI-derived explanation)

The ratio of $N(0, 5)$ to $N(0, 2)$ is $\sqrt{5/2}$ times a standard Cauchy, so λ must equal $\sqrt{2/5}$ and $5\lambda^2 = 5(2/5) = 2$.

TIPS & TRICKS (AI-derived explanation)

- Key fact: for independent $N(0, a^2)$ and $N(0, b^2)$, the ratio is (a/b) times a standard Cauchy - the scale parameter is the RATIO of the standard deviations.
- Variances: the five-term sum has variance 5 and the two-term sum has variance 2, so the scale is $\sqrt{5}/\sqrt{2} = \sqrt{5/2}$.
- To standardise, λ must be the reciprocal $\sqrt{2/5}$, and squaring gives $\lambda^2 = 2/5$ - hence $5\lambda^2 = 2$.
- The question asks for $5\lambda^2$ precisely so that the answer is an integer; computing λ itself and then squaring is the safest order.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. $X_3 + X_4 + X_5 + X_6 + X_7$ is normal with mean 0 and variance 5; $X_1 + X_2$ is normal with mean 0 and variance 2, and the two are independent.
2. For independent $U \sim N(0, a^2)$ and $V \sim N(0, b^2)$, the ratio U/V has a Cauchy distribution centred at 0 with scale a/b .
3. Here $a = \sqrt{5}$ and $b = \sqrt{2}$, so the ratio is $\text{Cauchy}(0, \sqrt{5/2})$.
4. Multiplying a $\text{Cauchy}(0, s)$ variable by λ gives $\text{Cauchy}(0, \lambda s)$, so for the result to be $\text{Cauchy}(0, 1)$ we need $\lambda\sqrt{5/2} = 1$.
5. Hence $\lambda = \sqrt{2/5}$ and $\lambda^2 = 2/5$.
6. Therefore $5\lambda^2 = 5(2/5) = 2$, option (b).

11. **2024 · Q38** Exponential Distribution · Numerical

QUESTION

Let $x_1, x_2, \dots, x_{10}$ be a sample of size 10 from an exponential distribution with mean θ . The lower bound to the probability $P(0 < \bar{X} < 2\theta)$, where $\bar{X} = \frac{1}{10} \sum_{i=1}^{10} X_i$, is:

OPTIONS

- (a) 0.01
 (b) 0.09
 (c) 0.1
 (d) 0.9

ANSWER (AI-derived — no official key exists for this paper)**(d)** 0.9**EXAM SHORTCUT** (AI-derived explanation)

$\bar{X}$ has mean θ and variance $\theta^2/10$, and the interval $(0, 2\theta)$ is $\theta \pm \theta$. Chebyshev gives at least $1 - (\theta^2/10)/\theta^2 = 0.9$.

TIPS & TRICKS (AI-derived explanation)

- The words 'lower bound to the probability' signal Chebyshev - no distributional detail beyond mean and variance is needed.
- For exponential with mean θ the variance is θ^2 , so $\text{Var}(\bar{X}) = \theta^2/n = \theta^2/10$.
- The interval $(0, 2\theta)$ is exactly symmetric about the mean θ with half-width θ , so $k\sigma = \theta$ with $\sigma = \theta/\sqrt{10}$, giving $k = \sqrt{10}$ and the bound $1 - 1/10$.
- θ cancels completely, which is why the answer is a pure number - a good check that the set-up is right.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. For an exponential distribution with mean θ , the variance is also θ^2 .
2. The sample mean of 10 observations has $E(\bar{X}) = \theta$ and $\text{Var}(\bar{X}) = \theta^2/10$.
3. The event $0 < \bar{X} < 2\theta$ is $|\bar{X} - \theta| < \theta$, an interval symmetric about the mean with half-width θ .
4. Chebyshev's inequality gives $P(|\bar{X} - \theta| < e) \geq 1 - \text{Var}(\bar{X})/e^2$ with $e = \theta = 1 - (\theta^2/10)/\theta^2 = 1 - 1/10 = 0.9$.
5. Hence the lower bound is 0.9, option (d).

12. **2025 · Q1** Beta Distribution · Factual**QUESTION**

X follows Beta distribution of first kind $\text{Beta}(\alpha, \beta)$ with $\alpha > 1$ and $\beta > 1$. What is the mode?

OPTIONS

- (a) $\frac{\alpha-1}{\alpha+\beta-2}$
 (b) $\frac{\alpha}{\alpha+\beta}$
 (c) $\frac{\alpha+1}{\alpha+\beta+1}$
 (d) $\frac{\alpha+\beta}{\alpha+\beta-1}$

ANSWER (AI-derived — no official key exists for this paper)**(a)** $\frac{\alpha-1}{\alpha+\beta-2}$

EXAM SHORTCUT (AI-derived explanation)

Differentiating the Beta kernel $x^{a-1}(1-x)^{b-1}$ and setting the derivative to zero gives the mode $(a-1)/(a+b-2)$, valid when both parameters exceed 1.

TIPS & TRICKS (AI-derived explanation)

- Mode of $\text{Beta}(a, b) = (a-1)/(a+b-2)$; mean $= a/(a+b)$. The '-1' and '-2' pattern distinguishes the mode from the mean at a glance.
- The condition $a > 1$ and $b > 1$ is exactly what makes the density unimodal with an interior maximum, so the formula applies.
- Symmetry check: for $a = b$ the formula gives $1/2$, as it must for a symmetric Beta - option (d) fails this at once.
- Boundary check: as a grows with b fixed the mode tends to 1, which the formula reproduces; option (b) is the MEAN, the standard near-miss.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The Beta density of the first kind is $f(x) = x^{a-1}(1-x)^{b-1}/B(a, b)$ on $(0, 1)$.
2. Take logarithms: $\log f = (a-1)\log x + (b-1)\log(1-x) + \text{constant}$.
3. Differentiating and setting to zero: $(a-1)/x - (b-1)/(1-x) = 0$.
4. Cross-multiplying: $(a-1)(1-x) = (b-1)x$, so $a-1 = x(a-1+b-1) = x(a+b-2)$.
5. Hence the mode is $x = (a-1)/(a+b-2)$, and since $a > 1$ and $b > 1$ this lies strictly inside $(0, 1)$ and is a maximum.
6. Option (a) is correct.

13. 2025 · Q2 Beta Distribution · Conceptual**QUESTION**

For Beta distribution of first kind $\text{Beta}(\alpha, \beta)$, consider the following statements: I. For $\alpha > 1, \beta > 1$; density has unique mode. II. For $\alpha + \beta = 2$ such that $\alpha = \beta = 1$, all points in $[0, 2]$ are modes. Which of the statements given above is/are correct?

OPTIONS

- (a) I only
- (b) II only
- (c) Both I and II
- (d) Neither I nor II

ANSWER (AI-derived — no official key exists for this paper)

(a) I only

EXAM SHORTCUT (AI-derived explanation)

Statement I is true - with both parameters above 1 the Beta density is strictly concave in log scale and has a unique interior mode. Statement II fails because $\text{Beta}(1, 1)$ is Uniform on $[0, 1]$, not $[0, 2]$.

TIPS & TRICKS (AI-derived explanation)

- Beta support is ALWAYS $[0, 1]$, whatever the parameters - so any claim about $[0, 2]$ is wrong by definition.

- Beta(1, 1) is exactly the Uniform(0, 1) distribution: the density is the constant 1, so every point of $[0, 1]$ is a mode.
- For $a > 1$ and $b > 1$ the density vanishes at both endpoints and is positive inside, so a unique interior maximum must exist - statement I follows.
- Watch the support in every Beta question; the first-kind Beta lives on $(0, 1)$ and the second-kind Beta on $(0, \infty)$.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Statement I: for $a > 1$ and $b > 1$ the Beta density $x^{a-1}(1-x)^{b-1}$ vanishes at both $x = 0$ and $x = 1$ and is positive in between, and $\log f$ is strictly concave, so there is exactly one interior maximum at $(a-1)/(a+b-2)$. TRUE.
2. Statement II: with $a = b = 1$ the density becomes $x^0(1-x)^0/B(1, 1) = 1$ on $(0, 1)$, which is the Uniform(0, 1) distribution.
3. Every point of $[0, 1]$ is then a mode, since the density is constant there.
4. However the statement asserts that all points of $[0, 2]$ are modes, whereas the Beta distribution of the first kind has support $[0, 1]$ only, and the density is zero outside it. FALSE.
5. Only statement I is correct.
6. Hence option (a) is correct.

14. 2025 · Q3 Poisson Distribution · Conceptual**QUESTION**

Let X be a continuous random variable with the following pdf:

$$f(x) = \begin{cases} \frac{1}{8}, & 2 \leq x \leq 10 \\ 0, & \text{otherwise} \end{cases}$$

What is the distribution of $Y = aX + b$, provided $a > 0$?

OPTIONS

- (a) $U(8a + b, 12a + b)$
- (b) $U(2a + b, 10a + b)$
- (c) $\text{Poisson}(ba^{1/8})$
- (d) None of the above

ANSWER (AI-derived — no official key exists for this paper)

(b) $U(2a + b, 10a + b)$

EXAM SHORTCUT (AI-derived explanation)

A constant density $1/8$ on $[2, 10]$ is $U(2, 10)$; a positive linear map sends the endpoints to $2a + b$ and $10a + b$, so $Y \sim U(2a + b, 10a + b)$.

TIPS & TRICKS (AI-derived explanation)

- A linear transformation of a uniform variable is uniform - only the endpoints move, and they move by the same map as the variable.
- Since $a > 0$ the order of the endpoints is preserved; a negative a would swap them, which is why the sign condition is stated.

- Check the density: the range widens by a factor a , so the height falls to $1/(8a)$ - consistent with $U(2a + b, 10a + b)$.
- Option (a) uses the wrong endpoints (8 and 12 rather than 2 and 10) - read the support from the stem, not from the constant $1/8$.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The density is the constant $1/8$ on $[2, 10]$, an interval of length 8, so $X \sim \text{Uniform}(2, 10)$.
2. Let $Y = aX + b$ with $a > 0$, a strictly increasing linear map.
3. The endpoints map to $2a + b$ and $10a + b$, and the order is preserved because a is positive.
4. For y between these values, $F_Y(y) = P(X \leq (y - b)/a) = [(y - b)/a - 2]/8$, which is linear in y - so Y has a constant density.
5. That density is $1/(8a)$ over an interval of length $8a$, confirming $Y \sim \text{Uniform}(2a + b, 10a + b)$.
6. Option (b) is correct.

15. 2025 · Q6 Laplace Distribution · Numerical**QUESTION**

The distribution function of the standard Laplace distribution with pdf $f(x) = \frac{1}{2}e^{-|x|}$, $-\infty < x < \infty$ is:

OPTIONS

- (a) $F_X(x) = \begin{cases} \frac{e^x}{2}, & x \leq 0 \\ 1 - \frac{e^{-x}}{2}, & x > 0 \end{cases}$
- (b) $F_X(x) = \begin{cases} \frac{e^x}{2}, & x \leq 0 \\ 1 - e^{-x}, & x > 0 \end{cases}$
- (c) $F_X(x) = \begin{cases} \frac{e^{-x}}{2}, & x \leq 0 \\ \frac{e^x}{2}, & x > 0 \end{cases}$
- (d) $F_X(x) = \begin{cases} 1 - \frac{e^{-x}}{2}, & x \leq 0 \\ 1, & x > 0 \end{cases}$

ANSWER (AI-derived — no official key exists for this paper)

$$(a) F_X(x) = \begin{cases} \frac{e^x}{2}, & x \leq 0 \\ 1 - \frac{e^{-x}}{2}, & x > 0 \end{cases}$$

EXAM SHORTCUT (AI-derived explanation)

Integrating $(1/2)e^x$ up to $x \leq 0$ gives $e^x/2$; for $x > 0$ add the remaining area to get $1 - e^{-x}/2$.

TIPS & TRICKS (AI-derived explanation)

- Split at 0 because of the $|x|$: the density is $(1/2)e^x$ on the left and $(1/2)e^{-x}$ on the right.
- $F(0)$ must equal $1/2$ by symmetry. Check each option at $x = 0$: only options (a) and (b) give $1/2$ from the left branch, and only (a) gives $1/2$ from the right.
- The right branch is $1/2 + \int_0^x (1/2)e^{-t} dt = 1/2 + (1/2)(1 - e^{-x}) = 1 - e^{-x}/2$.
- Limits check: $F(-\infty) = 0$ and $F(+\infty) = 1$. Option (c) tends to 0 at $+\infty$ and option (d) is discontinuous - both invalid.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The standard Laplace density is $f(x) = (1/2)e^{-|x|}$, i.e. $(1/2)e^x$ for $x \leq 0$ and $(1/2)e^{-x}$ for $x > 0$.
2. For $x \leq 0$: $F(x) = \int_{-\infty}^x (1/2)e^t dt = (1/2)e^x$.

3. In particular $F(0) = 1/2$, as symmetry requires.
4. For $x > 0$: $F(x) = 1/2 + \int_0^x (1/2)e^{-t} dt = 1/2 + (1/2)(1 - e^{-x}) = 1 - (1/2)e^{-x}$.
5. So $F(x) = e^x/2$ for $x \leq 0$ and $1 - e^{-x}/2$ for $x > 0$, option (a).

16. **2025 · Q54** Uniform / Rectangular Distribution · Numerical

QUESTION

If X is uniformly distributed with mean 1 and variance $4/3$, then $P[X < 0]$ equals:

OPTIONS

- (a) $1/2$
- (b) $1/3$
- (c) $1/4$
- (d) $1/8$

ANSWER (AI-derived — no official key exists for this paper)

(c) $1/4$

EXAM SHORTCUT (AI-derived explanation)

$(a + b)/2 = 1$ and $(b - a)^2/12 = 4/3$ give $b - a = 4$ and $a + b = 2$, so $X \sim U(-1, 3)$. Then $P(X < 0) = 1/4$.

TIPS & TRICKS (AI-derived explanation)

- Uniform parameters from moments: the mean gives $a + b = 2(\text{mean})$ and the variance gives $(b - a)^2 = 12$ (variance).
- $12 \times 4/3 = 16$, so $b - a = 4$. Solving with $a + b = 2$ gives $a = -1$ and $b = 3$.
- Probability = length of the sub-interval divided by the total range: $(0 - (-1))/4 = 1/4$.
- Sanity check: the mean 1 sits three-quarters of the way along $[-1, 3]$, so the mass below 0 should be well under a half - $1/4$ is right.

STEP-BY-STEP SOLUTION (AI-derived explanation)

- Let $X \sim U(a, b)$. The mean is $(a + b)/2 = 1$, so $a + b = 2$.
- The variance is $(b - a)^2/12 = 4/3$, so $(b - a)^2 = 16$ and $b - a = 4$ (taking $b > a$).
- Solving the pair: adding gives $2b = 6$ so $b = 3$, and then $a = -1$.
- Hence X is uniform on $(-1, 3)$, a range of length 4.
- $P(X < 0) = (0 - (-1))/4 = 1/4$.
- Option (c) is correct.

17. **2025 · Q55** Normal Distribution · Factual

QUESTION

If $X \sim N(\mu, \sigma^2)$, then $P[a \leq X \leq b]$ in terms of the standard normal cdf $\Phi(\cdot)$ is:

OPTIONS

- (a) $\Phi\left(\frac{a-\mu}{\sigma}\right) - \Phi\left(\frac{b-\mu}{\sigma}\right)$
- (b) $\Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right)$

(c) $\Phi\left(\frac{a+\mu}{\sigma}\right) - \Phi\left(\frac{b+\mu}{\sigma}\right)$

(d) $\Phi\left(\frac{b+\mu}{\sigma}\right) - \Phi\left(\frac{a+\mu}{\sigma}\right)$

ANSWER (AI-derived — no official key exists for this paper)

(b) $\Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right)$

EXAM SHORTCUT (AI-derived explanation)

Standardise both limits and subtract the smaller from the larger: $P(a \leq X \leq b) = \Phi((b - \mu)/\sigma) - \Phi((a - \mu)/\sigma)$.

TIPS & TRICKS (AI-derived explanation)

- Standardise by SUBTRACTING the mean and dividing by the standard deviation: $z = (x - \mu)/\sigma$. Options (c) and (d) add μ instead and are wrong.
- The upper limit's Φ comes FIRST in the subtraction, since the CDF is increasing and the probability must be non-negative. Option (a) reverses the order.
- Structural check: if $b = a$ the expression must give 0, and if b tends to ∞ while a tends to minus ∞ it must give 1. Only option (b) satisfies both.
- Generic rule: $P(a \leq X \leq b) = F(b) - F(a)$ for any continuous variable, and standardising simply expresses F through Φ .

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. For $X \sim N(\mu, \sigma^2)$, the standardised variable $Z = (X - \mu)/\sigma$ follows $N(0, 1)$.
2. The event $a \leq X \leq b$ is equivalent to $(a - \mu)/\sigma \leq Z \leq (b - \mu)/\sigma$.
3. For any continuous variable, $P(a \leq X \leq b) = F(b) - F(a)$.
4. Expressing this through the standard normal CDF: $P = \Phi((b - \mu)/\sigma) - \Phi((a - \mu)/\sigma)$.
5. The upper limit appears first because Φ is increasing, so the difference is non-negative.
6. Option (b) is correct.

18. 2025 · Q58 Normal Distribution · Numerical

QUESTION

Let $X \sim N(\mu, \sigma^2)$ with $\sigma^2 = \mu^2$, $\mu > 0$. The expression for $P[(X < -\mu) \mid (X < \mu)]$ in terms of the $N(0, 1)$ cdf $\Phi(\cdot)$ is:

OPTIONS

(a) $\Phi(-2)$

(b) $\Phi(2)$

(c) $1 - \Phi(2)$

(d) $2[1 - \Phi(2)]$

ANSWER (AI-derived — no official key exists for this paper)

(d) $2[1 - \Phi(2)]$

EXAM SHORTCUT (AI-derived explanation)

With $\sigma = \mu$, $P(X < -\mu) = \Phi(-2) = 1 - \Phi(2)$ and $P(X < \mu) = \Phi(0) = 1/2$, so the conditional probability is $2[1 - \Phi(2)]$.

TIPS & TRICKS (AI-derived explanation)

- $\sigma^2 = \mu^2$ with $\mu > 0$ means $\sigma = \mu$, so standardising gives $z = (x - \mu)/\mu$ - both limits become clean integers.
- $x = -\mu$ gives $z = -2$ and $x = \mu$ gives $z = 0$. The denominator is therefore exactly $1/2$, which is what produces the factor 2.
- Because $\{X < -\mu\}$ is contained in $\{X < \mu\}$, the intersection is just the numerator event - no extra work is needed.
- $\Phi(-2) = 1 - \Phi(2)$ by symmetry; dividing by $1/2$ doubles it, giving $2[1 - \Phi(2)] = 0.0455$.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Since $\sigma^2 = \mu^2$ and $\mu > 0$, we have $\sigma = \mu$, so $Z = (X - \mu)/\mu$ is standard normal.
2. The event $X < -\mu$ corresponds to $Z < (-\mu - \mu)/\mu = -2$, so $P(X < -\mu) = \Phi(-2) = 1 - \Phi(2)$.
3. The event $X < \mu$ corresponds to $Z < (\mu - \mu)/\mu = 0$, so $P(X < \mu) = \Phi(0) = 1/2$.
4. Since $\{X < -\mu\}$ is a subset of $\{X < \mu\}$, the intersection is $\{X < -\mu\}$ itself.
5. Hence $P(X < -\mu | X < \mu) = [1 - \Phi(2)]/(1/2) = 2[1 - \Phi(2)]$.
6. Numerically this is $2(0.0228) = 0.0455$. Option (d) is correct.

19. 2026 · Q4 Normal Distribution · Conceptual**COMMON DATA FOR THIS GROUP**

Shared data for Q4–Q5. Let $f_X(x)$ be the pdf of $N(0, 1)$ and $\Phi(x) = \int_{-\infty}^x f_X(\theta) d\theta$.

QUESTION

What is $\Phi(-100)$ equal to?

OPTIONS

- (a) $-\Phi(100)$
- (b) $1 - \Phi(100)$
- (c) $\frac{1}{2}\Phi(100)$
- (d) $\frac{1}{2} - \Phi(100)$

ANSWER (AI-derived — no official key exists for this paper)

(b) $1 - \Phi(100)$

EXAM SHORTCUT (AI-derived explanation)

The standard normal density is symmetric about 0, so $\Phi(-z) = 1 - \Phi(z)$ for every z . Put $z = 100$. Answer (b).

TIPS & TRICKS (AI-derived explanation)

- Memorise the symmetry relation $\Phi(-z) + \Phi(z) = 1$; it is the single most-used property of the standard normal CDF.
- A CDF can never be negative, so option (a) is impossible on sight.
- Option (d) would give a negative number since $\Phi(100)$ is essentially 1; another instant rejection.

- Numerically $\Phi(-100)$ is about 10^{-2174} , i.e. effectively 0, consistent with $1 - \Phi(100)$.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. By definition $\Phi(x) = \int_{-\infty}^x f_X(\theta) d\theta$, where f_X is the $N(0, 1)$ density.
2. The density is symmetric: $f_X(-t) = f_X(t)$.
3. Substituting $\theta = -u$ in the integral for $\Phi(-z)$ gives $\Phi(-z) = \int_z^{\infty} f_X(u) du = 1 - \Phi(z)$.
4. With $z = 100$: $\Phi(-100) = 1 - \Phi(100)$.
5. Options (a) and (d) would be negative (since $\Phi(100)$ is essentially 1), which is impossible for a CDF; option (c) contradicts the symmetry relation.
6. Hence option (b) is correct.

20. 2026 · Q5 Normal Distribution · Conceptual

COMMON DATA FOR THIS GROUP

Shared data for Q4-Q5. Let $f_X(x)$ be the pdf of $N(0, 1)$ and $\Phi(x) = \int_{-\infty}^x f_X(\theta) d\theta$.

QUESTION

What is $\int_{-10}^{10} \int_{-y}^y f_X(x) dx dy$ equal to?

OPTIONS

- (a) $1/2$
- (b) $1/4$
- (c) 0
- (d) $1/3$

ANSWER (AI-derived — no official key exists for this paper)

(c) 0

EXAM SHORTCUT (AI-derived explanation)

The inner integral $g(y) = \int_{-y}^y f_X(x) dx = \Phi(y) - \Phi(-y)$ is an odd function of y , and the outer range $[-10, 10]$ is symmetric about 0. An odd function integrated over a symmetric interval gives 0.

TIPS & TRICKS (AI-derived explanation)

- Do the inner integral symbolically first - here it collapses to $2\Phi(y) - 1$, whose oddness is the whole point.
- Check oddness explicitly: $g(-y) = \Phi(-y) - \Phi(y) = -g(y)$. Do not assume the inner integral is automatically non-negative just because f_X is.
- When $y < 0$ the limits $-y$ to y run backwards, so the inner integral is negative - this is the step most candidates miss, which is why $1/2$ is the popular wrong answer.
- General principle: $\int_{-a}^a$ an odd function is 0; $\int_{-a}^a$ an even function is $2 \times$ integral from 0 to a .

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Let $g(y) = \int_{-y}^y f_X(x) dx = \Phi(y) - \Phi(-y)$.
2. Using $\Phi(-y) = 1 - \Phi(y)$, we get $g(y) = 2\Phi(y) - 1$.
3. Then $g(-y) = 2\Phi(-y) - 1 = 2(1 - \Phi(y)) - 1 = 1 - 2\Phi(y) = -g(y)$, so g is an odd function of y .
4. The required double integral is $\int_{-10}^{10} g(y) dy$.

5. An odd integrable function integrated over an interval symmetric about the origin gives 0: the contribution from y in $(0, 10)$ exactly cancels that from y in $(-10, 0)$.
6. Therefore the value is 0.
7. Hence option (c) is correct.

21. 2026 · Q14 Discrete & Continuous Random Variables · Factual

QUESTION

X is a continuous random variable whose density is constant on $a \leq x \leq b$ and zero elsewhere. What is the standard deviation of X ?

OPTIONS

- (a) $(2a + b)/(2\sqrt{3})$
- (b) $(b - a)/(2\sqrt{3})$
- (c) $(a + b)/2$
- (d) $(b - a)/2$

ANSWER (AI-derived — no official key exists for this paper)

(b) $(b - a)/(2\sqrt{3})$

EXAM SHORTCUT (AI-derived explanation)

Constant density on $[a, b]$ means X is uniform, whose variance is $(b - a)^2/12$, so the SD is $(b - a)/(2\sqrt{3})$. Answer (b).

TIPS & TRICKS (AI-derived explanation)

- 'Density constant on an interval and zero elsewhere' is the definition of the continuous uniform distribution - recognise the phrasing.
- Uniform facts: mean $(a + b)/2$, variance $(b - a)^2/12$, and $\sqrt{12} = 2\sqrt{3}$.
- Options (c) and (d) are the mean and the half-range respectively - both are spread or location quantities, but neither is the SD.
- A standard deviation must be translation invariant, i.e. it can depend only on $b - a$, not on $a + b$. That instantly kills options (a) and (c).

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. If the density is constant on $a \leq x \leq b$ and zero elsewhere, then $f(x) = 1/(b - a)$ on $[a, b]$, i.e. X follows the uniform distribution $U(a, b)$.
2. Mean: $E(X) = (a + b)/2$.
3. $E(X^2) = \int_a^b x^2/(b - a) dx = (b^3 - a^3)/[3(b - a)] = (a^2 + ab + b^2)/3$.
4. $\text{Var}(X) = (a^2 + ab + b^2)/3 - (a + b)^2/4 = [4(a^2 + ab + b^2) - 3(a^2 + 2ab + b^2)]/12 = (a^2 - 2ab + b^2)/12 = (b - a)^2/12$.
5. Standard deviation = $\sqrt{(b - a)^2/12} = (b - a)/\sqrt{12} = (b - a)/(2\sqrt{3})$.
6. Hence option (b) is correct.

P6 · PROBABILITY

Random Vectors: Joint, Marginal & Conditional Distributions

40 questions · 2018: 4 2019: 2 2020: 2 2021: 5 2022: 2 2023: 6 2024: 3 2025: 7 2026: 9

Weight. 40 questions (20.4% of Probability) · appeared in 9 of 9 papers · peak year 2026 (9)**Examiner's pattern.** THE highest-yield block in Probability (40 items). Modern papers bundle 2-3 questions onto one joint density — 2021, 2023, 2025 and 2026 all did this, so one setup can be worth 3 marks. The drill is always the same: marginal, then conditional, then a probability over a region, then $E(XY)$ or a conditional moment. Watch the region: triangular supports ($0 < x < y < 1$) and curved supports ($x^2 \leq y$) are deliberately chosen to punish careless limits.**Must-know.** $f_X(x) = \int f(x,y) dy$; $f(y|x) = f(x,y)/f_X(x)$; independence iff $f(x,y)$ factorises AND the support is a rectangle; always sketch the region before setting limits.**1.** **2018 · Q5** Random Vectors & Joint Distributions · Conceptual

QUESTION

 X, Y have joint p.d.f.

$$f(x,y) = \begin{cases} \frac{1+xy}{4}, & |x| < 1, |y| < 1 \\ 0, & \text{otherwise} \end{cases}$$

1. X and Y are independent. 2. X^2 and Y^2 are independent.

OPTIONS

- (a) 1 only
 (b) 2 only
 (c) Both 1 and 2
 (d) Neither 1 nor 2

ANSWER (AI-derived — no official key exists for this paper)**(b)** 2 only**EXAM SHORTCUT** (AI-derived explanation)

Both marginals are $\text{Uniform}(-1, 1)$ with density $1/2$, so the product $1/4$ does not equal $(1+xy)/4$ - statement 1 fails. But integrating the xy term over any symmetric rectangle kills it, so squares are independent - statement 2 holds. Answer: 2 only.

TIPS & TRICKS (AI-derived explanation)

- The 'copula-style' density $(1+xy)/4$ on the square is a classic: dependent variables whose squares (and whose signs) behave independently.
- Marginal shortcut: integrating $(1+xy)/4$ in y over $(-1, 1)$ kills the odd term xy , leaving $2/4 = 1/2$. Any odd-in- y term always vanishes on a symmetric interval.
- $\text{Cov}(X, Y)$ here is NOT zero ($E(XY) = 1/9$), so this is not the usual 'uncorrelated but dependent' example - do not answer from pattern memory.
- For 1-only / 2-only / both / neither items, test the CHEAPER statement first. Here statement 1 takes one line, so start there and use it to eliminate two options immediately.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Marginal of X : $f_X(x) = \text{integral over } y \text{ in } (-1, 1) \text{ of } (1 + xy)/4 dy = (1/4)[y + xy^2/2] \text{ from } -1 \text{ to } 1 = (1/4)(2) = 1/2$ for $|x| < 1$. By symmetry $f_Y(y) = 1/2$ for $|y| < 1$.
2. Statement 1: $f_X(x)f_Y(y) = 1/4$, but the joint density is $(1 + xy)/4$. These differ whenever xy is not 0, so X and Y are NOT independent. Statement 1 is false.
3. Statement 2: take any $0 < a, b \leq 1$. $P(X^2 \leq a^2, Y^2 \leq b^2) = P(|X| \leq a, |Y| \leq b) = \text{double } \int (1 + xy)/4 \text{ over } [-a, a] \times [-b, b]$.
4. The xy term is odd in x on a symmetric interval, so it integrates to 0; the remaining term gives $(1/4)(2a)(2b) = ab$.
5. Also $P(X^2 \leq a^2) = P(|X| \leq a) = a$ and $P(Y^2 \leq b^2) = b$. Since $ab = axb$, the joint CDF of (X^2, Y^2) factorises, so X^2 and Y^2 ARE independent. Statement 2 is true.
6. Only statement 2 is correct, so option (b) is right.

2. 2018 · Q12 Random Vectors & Joint Distributions · Numerical**QUESTION**

The joint distribution function of X, Y is $F(x, y) = (1 - e^{-x})(1 - e^{-y})$ for $x, y > 0$ (else 0). What is $P(|X - Y| \leq 10)$?

OPTIONS

- (a) $1 - e^{-10}$
- (b) $\frac{1}{2}(1 - e^{-10})$
- (c) $1 - \frac{e^{-10}}{2}$
- (d) None of the above

ANSWER (AI-derived — no official key exists for this paper)

(a) $1 - e^{-10}$

EXAM SHORTCUT (AI-derived explanation)

The joint CDF factorises into $(1 - e^{-x})(1 - e^{-y})$, so X and Y are i.i.d. $\text{Exp}(1)$. For i.i.d. exponentials, $|X - Y|$ is again $\text{Exp}(1)$ by memorylessness, so $P(|X - Y| \leq 10) = 1 - e^{-10}$.

TIPS & TRICKS (AI-derived explanation)

- A factorised joint CDF is the fastest possible independence certificate - no integration, no marginals.
- Memorise: for i.i.d. $\text{Exp}(\lambda)$, the absolute difference $|X - Y| \sim \text{Exp}(\lambda)$. This turns a two-dimensional integral into a one-line lookup.
- Intuition for that fact: condition on which variable is smaller; by the lack-of-memory property the excess of the larger over the smaller is again $\text{Exp}(\lambda)$.
- Option (b) is what you get if you forget to double the two symmetric regions $X > Y$ and $Y > X$; option (c) mishandles the complement.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The joint CDF $F(x, y) = (1 - e^{-x})(1 - e^{-y})$ for $x, y > 0$ factorises as $F_X(x)F_Y(y)$, so X and Y are independent, each Exponential with mean 1.

2. Let $D = |X - Y|$. For $d > 0$, $P(D > d) = P(X - Y > d) + P(Y - X > d)$, and by symmetry this equals $2P(X - Y > d)$.
3. $P(X - Y > d) = \int e^{-y} P(X > y + d) dy = \int e^{-y} e^{-(y+d)} dy = e^{-d} \int e^{-2y} dy = e^{-d}/2$.
4. Therefore $P(D > d) = 2 \times e^{-d}/2 = e^{-d}$, i.e. $|X - Y| \sim \text{Exponential}(1)$.
5. Putting $d = 10$ gives $P(|X - Y| > 10) = e^{-10}$, so $P(|X - Y| \leq 10) = 1 - e^{-10}$.
6. Hence option (a) is correct.

3. **2018 · Q13** Random Vectors & Joint Distributions · Numerical

QUESTION

X, Y have joint p.d.f. $f(x, y) = 2e^{-x}e^{-2y}$, $0 < x, y < \infty$. What is $P(X < Y)$?

OPTIONS

- (a) $\frac{1}{3}$
- (b) $\frac{1}{2}$
- (c) $\frac{1}{4}$
- (d) $\frac{1}{6}$

ANSWER (AI-derived — no official key exists for this paper)

(a) $\frac{1}{3}$

EXAM SHORTCUT (AI-derived explanation)

The density factorises: $X \sim \text{Exp}(\text{rate } 1)$, $Y \sim \text{Exp}(\text{rate } 2)$, independent. For independent exponentials, $P(X < Y) = \text{rate}_X / (\text{rate}_X + \text{rate}_Y) = 1/(1 + 2) = 1/3$.

TIPS & TRICKS (AI-derived explanation)

- Memorise the competing-risks formula $P(X < Y) = \lambda_X / (\lambda_X + \lambda_Y)$ for independent exponentials. It converts this whole question into one division.
- Read the rates off the exponents, not off the constants: e^{-x} gives rate 1 and e^{-2y} gives rate 2; the leading 2 is only the normalising constant.
- Sanity check: Y has the larger rate, so Y is stochastically smaller and $X < Y$ should have probability less than $1/2$. That alone eliminates (b) $1/2$.
- The same formula answers 'which component fails first' reliability questions, a recurring ISS theme.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The joint density $f(x, y) = 2e^{-x}e^{-2y}$ on $x, y > 0$ factorises into $g(x) = e^{-x}$ and $h(y) = 2e^{-2y}$, so $X \sim \text{Exp}(1)$ and $Y \sim \text{Exp}(2)$ independently.
2. $P(X < Y) = \int f_Y(y) P(X < y) dy$.
3. Now $P(X < y) = 1 - e^{-y}$, so the integral is the $\int_0^\infty 2e^{-2y}(1 - e^{-y}) dy = \int 2e^{-2y} dy - \int 2e^{-3y} dy = 1 - 2/3 = 1/3$.
4. This agrees with the general rule $P(X < Y) = \lambda_X / (\lambda_X + \lambda_Y) = 1/(1 + 2) = 1/3$.
5. Hence option (a) is correct.

4. 2018 · Q15 Random Vectors & Joint Distributions · Numerical**QUESTION**Joint pmf of X, Y :

$Y \backslash X$	-1	0	1
-1	1/6	1/3	1/6
0	0	0	0
1	1/6	0	1/6

1. X, Y are independent. 2. $\text{Cov}(X, Y) = 0$.**OPTIONS**

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

ANSWER (AI-derived — no official key exists for this paper)**(b)** 2 only**EXAM SHORTCUT (AI-derived explanation)**

A single zero cell settles independence: $P(X = 0, Y = 1) = 0$ but $P(X = 0)P(Y = 1) = (1/3)(1/3)$ is non-zero, so they are dependent. $E(X) = 0$ by symmetry and $E(XY) = 1/6 - 1/6 - 1/6 + 1/6 = 0$, so $\text{Cov} = 0$. Only statement 2 holds.

TIPS & TRICKS (AI-derived explanation)

- To DISPROVE independence, hunt for a zero entry whose marginal product is non-zero. It is a one-cell check and never requires the full table.
- This is the textbook 'uncorrelated but dependent' table - the standard counterexample to 'zero covariance implies independence'.
- Because X is symmetric about 0 with $E(X) = 0$, $\text{Cov}(X, Y) = E(XY)$ alone; you can skip computing $E(Y)$ entirely.
- Marginals here are X : 1/3, 1/3, 1/3 and Y : 2/3, 0, 1/3. Compute marginals once and reuse them for both statements.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Row sums give the marginal of Y :
 $P(Y = -1) = 1/6 + 1/3 + 1/6 = 2/3$, $P(Y = 0) = 0$, $P(Y = 1) = 1/6 + 0 + 1/6 = 1/3$.
2. Column sums give the marginal of X :
 $P(X = -1) = 1/6 + 0 + 1/6 = 1/3$, $P(X = 0) = 1/3 + 0 + 0 = 1/3$, $P(X = 1) = 1/6 + 0 + 1/6 = 1/3$. The grand total is 1, so the table is consistent.
3. Statement 1: $P(X = 0, Y = 1) = 0$ from the table, but $P(X = 0)P(Y = 1) = (1/3)(1/3) = 1/9$, which is not 0. Hence X and Y are NOT independent; statement 1 is false.
4. Statement 2: $E(X) = (-1)(1/3) + 0(1/3) + 1(1/3) = 0$.
5. $E(XY) = (-1)(-1)(1/6) + (1)(-1)(1/6) + (-1)(1)(1/6) + (1)(1)(1/6) = 1/6 - 1/6 - 1/6 + 1/6 = 0$ (the $X = 0$ column and the $Y = 0$ row contribute nothing).

6. So $\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = 0 - 0 = 0$; statement 2 is true. Only statement 2 holds, option (b).

5. 2019 · Q2 Random Vectors & Joint Distributions · Theoretical

QUESTION

The joint distribution of two random variables X and Y is given below:

$X \backslash Y$	-1	1
0	α_1	α_2
1	α_3	α_4

where $\alpha_i \geq 0$ for all i and $\sum_1^4 \alpha_i = 1$. Which of the following are correct? 1. $E(X^r) = \alpha_3 + \alpha_4$, for all r 2. $E(XY) = \alpha_4 - \alpha_3$ 3. $E(XY)^{2n} = \alpha_3 + \alpha_4$, for all n

OPTIONS

- (a) 1 and 2 only
- (b) 1 and 3 only
- (c) 2 and 3 only
- (d) 1, 2 and 3

ANSWER (AI-derived — no official key exists for this paper)

(d) 1, 2 and 3

EXAM SHORTCUT (AI-derived explanation)

X takes only 0 and 1, so $X^r = X$ for every r and $E(X^r) = P(X = 1) = \alpha_3 + \alpha_4$. XY takes the values 0, -1, +1, so $E(XY) = \alpha_4 - \alpha_3$ and $(XY)^{2n}$ is 1 exactly when $X = 1$, giving $\alpha_3 + \alpha_4$. All three hold.

TIPS & TRICKS (AI-derived explanation)

- Indicator trick: a variable taking only 0 and 1 satisfies $X^r = X$ for all $r \geq 1$, so all its moments coincide with its mean. That settles statement 1 with no algebra.
- Build the value table of XY before touching statement 3: $XY = 0$ on the whole $X = 0$ row, $XY = -1$ with probability α_3 and $XY = +1$ with probability α_4 .
- An EVEN power kills the sign, so $(XY)^{2n}$ equals 1 whenever XY is non-zero. Hence its expectation is $P(XY \text{ not } 0) = \alpha_3 + \alpha_4$, the same as statement 1's answer - a nice internal consistency check.
- When several statements about the same tiny table all reduce to the row/column sums, expect 'all of them' - but verify the one with the sign, here statement 2.

STEP-BY-STEP SOLUTION (AI-derived explanation)

- The table gives $P(X = 0, Y = -1) = \alpha_1$, $P(X = 0, Y = 1) = \alpha_2$, $P(X = 1, Y = -1) = \alpha_3$, $P(X = 1, Y = 1) = \alpha_4$.
- Statement 1: X takes only the values 0 and 1, so $X^r = X$ for every $r \geq 1$. Hence $E(X^r) = E(X) = 0(\alpha_1 + \alpha_2) + 1(\alpha_3 + \alpha_4) = \alpha_3 + \alpha_4$. TRUE.
- Statement 2: $E(XY) = (0)(-1)\alpha_1 + (0)(1)\alpha_2 + (1)(-1)\alpha_3 + (1)(1)\alpha_4 = -\alpha_3 + \alpha_4 = \alpha_4 - \alpha_3$. TRUE.
- Statement 3: the product XY takes the value 0 with probability $\alpha_1 + \alpha_2$, the value -1 with probability α_3 , and +1 with probability α_4 .
- Raising to the even power $2n$ maps 0 to 0 and both -1 and +1 to 1, so $E[(XY)^{2n}] = \alpha_3 + \alpha_4$ for every n . TRUE.

6. All three statements are correct, option (d).

6. **2019 · Q20** Marginal Distributions · Numerical

QUESTION

Let $f(x, y) = 4xy$, $0 \leq x \leq 1$, $0 \leq y \leq 1$ be the joint p.d.f. of (X, Y) . The marginal distribution of X is

OPTIONS

- (a) $f_X(x) = 4x$; $0 \leq x \leq 1$
- (b) $f_X(x) = 2x$; $0 \leq x \leq 1$
- (c) $f_X(x) = \frac{x}{2}$; $0 \leq x \leq 1$
- (d) $f_X(x) = x$; $0 \leq x \leq 1$

ANSWER (AI-derived — no official key exists for this paper)

(b) $f_X(x) = 2x$; $0 \leq x \leq 1$

EXAM SHORTCUT (AI-derived explanation)

Integrate out y : the $\int 4xy \, dy$ from 0 to 1 is $4 \times (1/2) = 2x$. So $f_X(x) = 2x$ on $[0, 1]$.

TIPS & TRICKS (AI-derived explanation)

- Marginal of X = integrate the joint density over ALL values of y . Here the $y - \int y$ over $(0, 1)$ is $1/2$, so the factor 4 halves to 2.
- Normalisation check: the $\int 2x$ over $(0, 1)$ is 1, confirming a valid density. Option (a) integrates to 2 and option (c) to $1/4$, so both are invalid densities and can be struck out immediately.
- The joint density factorises as $(2x)(2y)$, so X and Y are independent Beta(2, 1) variables - recognising the product form gives both marginals at once.
- Whenever the options are proportional to one another, use the 'integrates to 1' test - it decides the question faster than doing the integral.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The joint density is $f(x, y) = 4xy$ on the unit square $0 \leq x \leq 1, 0 \leq y \leq 1$.
2. The marginal density of X is $f_X(x) = \int 4xy \, dy = 4x[y^2/2]$ from 0 to 1 $= 4 \times (1/2) = 2x$.
3. So $f_X(x) = 2x$ for $0 \leq x \leq 1$.
4. Check: the $\int 2x$ from 0 to 1 is $[x^2]$ from 0 to 1 $= 1$, a valid density.
5. Hence option (b) is correct.

7. **2020 · Q26** Random Vectors & Joint Distributions · Numerical

QUESTION

X, Y have joint p.d.f. $f(x, y) = 6(1 - x - y)$ for $x > 0, y > 0, x + y < 1$ (else 0). What is $P(X \geq 0.4)$?

OPTIONS

- (a) 0.21
- (b) 0.25
- (c) 0.31

(d) 0.35**ANSWER (AI-derived — no official key exists for this paper)****(a)** 0.21**EXAM SHORTCUT (AI-derived explanation)**

Integrate out y to get the marginal $f_X(x) = 3(1-x)^2$ on $(0, 1)$. Then $P(X \geq 0.4) =$ the integral from 0.4 to 1 $= (0.6)^3 = 0.216$, i.e. about 0.21.

TIPS & TRICKS (AI-derived explanation)

- Get the marginal first: for a fixed x the y -range is $(0, 1-x)$, and the inner $\int 6(1-x-y) dy$ is $3(1-x)^2$ - a clean, memorable result.
- Recognise $f_X(x) = 3(1-x)^2$ as the Beta(1, 3) density, so $P(X > c) = (1-c)^3$ directly. No second integral is needed.
- $(0.6)^3 = 0.216$ rounds to 0.21 at two decimals; option (b) 0.25 would correspond to $(0.63)^3$, so keep three decimals before rounding.
- Sanity check: X is concentrated near 0 (its mean is $1/4$), so $P(X \geq 0.4)$ should be well under a half - which rules out (c) and (d) as too large only marginally, so do the computation.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The joint density is $f(x, y) = 6(1-x-y)$ on the triangle $x > 0, y > 0, x+y < 1$.
2. For fixed x in $(0, 1)$, y ranges over $(0, 1-x)$, so $f_X(x) =$ the $\int_0^{1-x} 6(1-x-y) dy$.
3. Substituting $u = 1-x-y$, the integral is 6 times the $\int u du$ from 0 to $1-x = 6(1-x)^2/2 = 3(1-x)^2$.
4. Thus X has the Beta(1, 3) density $3(1-x)^2$ on $(0, 1)$.
5. $P(X \geq 0.4) =$ the $\int_{0.4}^1 3(1-x)^2 dx = [-(1-x)^3]$ from 0.4 to 1 $= 0 + (0.6)^3 = 0.216$.
6. Rounding, the probability is about 0.21, option (a).

8. 2020 · Q37 Random Vectors & Joint Distributions · Theoretical**QUESTION**

The function

$$F(x, y) = \begin{cases} \alpha, & x, y > 0; x + y < 1 \\ 1, & x, y > 0; x + y \geq 1 \\ 0, & \text{otherwise} \end{cases}$$

is the c.d.f. of a random vector (X, Y) **OPTIONS**

- (a)** for no real value of α
- (b)** iff $0 < \alpha < 1$
- (c)** iff $0 \leq \alpha < 1$
- (d)** iff $0 < \alpha \leq 1$

ANSWER (AI-derived — no official key exists for this paper)**(a)** for no real value of α

EXAM SHORTCUT (AI-derived explanation)

The rectangle $(0.2, 0.8] \times (0.2, 0.8]$ has F-measure $1 - 1 - 1 + \alpha = \alpha - 1$, which is negative for every $\alpha < 1$; and $\alpha = 1$ destroys right-continuity along the axes. So no value of α works.

TIPS & TRICKS (AI-derived explanation)

- A bivariate CDF must satisfy the RECTANGLE inequality $F(b, d) - F(b, c) - F(a, d) + F(a, c) \geq 0$ for all $a < b, c < d$. Marginal monotonicity alone is not enough - this is the test the examiner is probing.
- Choose the rectangle so that three corners lie in the region $x + y \geq 1$ and one corner in $x + y < 1$; then the expression collapses to $\alpha - 1$.
- Even $\alpha = 1$ fails: F would be 1 for all $x, y > 0$ but 0 on the axes, so $\lim$ as x decreases to 0 of $F(x, y) = 1$ while $F(0, y) = 0$ - right-continuity is violated.
- Lesson: a function that is non-decreasing in each argument separately can still fail to be a joint CDF. Always run the rectangle test on two-dimensional candidates.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. A function F is a bivariate distribution function only if, for all $a < b$ and $c < d$, the rectangle measure $F(b, d) - F(b, c) - F(a, d) + F(a, c)$ is non-negative.
2. Take $a = c = 0.2$ and $b = d = 0.8$. Then $F(0.8, 0.8) = 1$ (since $1.6 \geq 1$), $F(0.8, 0.2) = 1$ (since $1.0 \geq 1$), $F(0.2, 0.8) = 1$ (since $1.0 \geq 1$), and $F(0.2, 0.2) = \alpha$ (since $0.4 < 1$).
3. The rectangle measure is $1 - 1 - 1 + \alpha = \alpha - 1$, which would be the probability that (X, Y) lies in that rectangle.
4. This is negative whenever $\alpha < 1$, so no α below 1 can work. Since F is bounded by 1, α cannot exceed 1 either.
5. For $\alpha = 1$, F equals 1 on the whole open quadrant and 0 on the axes, so $\lim$ as x decreases to 0 of $F(x, y) = 1$ while $F(0, y) = 0$ - right-continuity fails and F is still not a distribution function.
6. Hence F is a distribution function for no real value of α , option (a).

9. 2021 · Q1 Random Vectors & Joint Distributions · Numerical**COMMON DATA FOR THIS GROUP**

Q1-Q3 refer to the following: Let the joint p.d.f. of X and Y be

$$f(x, y) = \begin{cases} 6(1 - x - y), & x > 0, y > 0, x + y < 1 \\ 0, & \text{otherwise} \end{cases}$$

QUESTION

What is the value of $E(XY)$?

OPTIONS

- (a) 10
 (b) 20
 (c) $\frac{1}{10}$
 (d) $\frac{1}{20}$

ANSWER (AI-derived — no official key exists for this paper)**(d)** $\frac{1}{20}$ **EXAM SHORTCUT (AI-derived explanation)**

Use the Dirichlet $\int x^{a-1}y^{b-1}(1-x-y)^{c-1}$ equals $\Gamma(a)\Gamma(b)\Gamma(c)/\Gamma(a+b+c)$. For $xy(1-x-y)$ that is $(1)(1)(1)/120$, so $E(XY) = 6/120 = 1/20$.

TIPS & TRICKS (AI-derived explanation)

- Memorise the Dirichlet formula for the triangle $x, y > 0, x + y < 1$ - every moment of this density is one substitution away from it, with no iterated integration at all.
- Read the parameters straight off the powers: x^1 gives $a = 2$, y^1 gives $b = 2$, $(1-x-y)^1$ gives $c = 2$, so the denominator is $\Gamma(6) = 120$.
- Check normalisation the same way: $a = b = 1, c = 2$ gives $1/\Gamma(4) = 1/6$, and $6 \times 1/6 = 1$, confirming the density is valid.
- Options (a) 10 and (b) 20 are impossible on size grounds - X and Y both lie in $(0, 1)$, so $E(XY)$ must be well below 1.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The joint density is $f(x, y) = 6(1 - x - y)$ on the triangle $x > 0, y > 0, x + y < 1$.
2. The Dirichlet $\int x^{a-1}y^{b-1}(1-x-y)^{c-1} dx dy = \Gamma(a)\Gamma(b)\Gamma(c)/\Gamma(a+b+c)$.
3. For $E(XY)$ we need the $\int xy(1-x-y)$, i.e. $a-1 = 1, b-1 = 1, c-1 = 1$, so $a = b = c = 2$.
4. The integral equals $\Gamma(2)\Gamma(2)\Gamma(2)/\Gamma(6) = (1)(1)(1)/120 = 1/120$.
5. Multiplying by the constant 6: $E(XY) = 6/120 = 1/20$.
6. Hence option (d) is correct.

10. **2021 · Q2** Random Vectors & Joint Distributions · Numerical**COMMON DATA FOR THIS GROUP**

Q1-Q3 refer to the following: Let the joint p.d.f. of X and Y be

$$f(x, y) = \begin{cases} 6(1-x-y), & x > 0, y > 0, x+y < 1 \\ 0, & \text{otherwise} \end{cases}$$

QUESTION

What is the value of $\text{Cov}(X, Y)$?

OPTIONS

- (a)** $\frac{1}{40}$
(b) $\frac{1}{80}$
(c) $-\frac{1}{40}$
(d) $-\frac{1}{80}$

ANSWER (AI-derived — no official key exists for this paper)**(d)** $-\frac{1}{80}$

EXAM SHORTCUT (AI-derived explanation)

$E(X) = E(Y) = 1/4$ by symmetry (Dirichlet with $a = 2, b = 1, c = 2$ gives $1/24$, times 6). With $E(XY) = 1/20$ from the previous part, $\text{Cov} = 1/20 - 1/16 = -1/80$.

TIPS & TRICKS (AI-derived explanation)

- Exploit symmetry: the density is symmetric in x and y , so $E(X) = E(Y)$ and only one marginal moment need be computed.
- The sign must be NEGATIVE: the constraint $x + y < 1$ means a large X forces a small Y . That alone eliminates options (a) and (b).
- Common denominator 80: $1/20 = 4/80$ and $1/16 = 5/80$, so the difference is $-1/80$ exactly. Convert before subtracting.
- Reuse the previous part's $E(XY) = 1/20$ rather than recomputing - shared-stem questions are designed so that each answer feeds the next.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. By the Dirichlet formula, $E(X) = 6$ times the $\int x(1-x-y)$, which has $a = 2, b = 1, c = 2$, giving $6\Gamma(2)\Gamma(1)\Gamma(2)/\Gamma(5) = 6/24 = 1/4$.
2. The density is symmetric in x and y , so $E(Y) = 1/4$ as well.
3. From the previous part, $E(XY) = 1/20$.
4. $\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = 1/20 - (1/4)(1/4) = 1/20 - 1/16$.
5. Using the common denominator 80: $4/80 - 5/80 = -1/80$.
6. The covariance is negative, as expected from the constraint $x + y < 1$. Option (d) is correct.

11. **2021 · Q4** Conditional Expectation · Numerical**COMMON DATA FOR THIS GROUP**

Q4-Q5 refer to the following: Let the joint density function of X and Y be

$$f(x, y) = \begin{cases} 8xy, & 0 < x < y < 1 \\ 0, & \text{otherwise} \end{cases}$$

QUESTION

What is $E[Y \mid X = x]$ equal to?

OPTIONS

- (a) $\frac{1}{5} \left(\frac{1-x^3}{1+x^2} \right)$
- (b) $\frac{2}{5} \left(\frac{1+x^3}{1-x^2} \right)$
- (c) $\frac{2}{3} \left(\frac{1-x^3}{1-x^2} \right)$
- (d) $\frac{1+x^3}{1+x^2}$

ANSWER (AI-derived — no official key exists for this paper)

(c) $\frac{2}{3} \left(\frac{1-x^3}{1-x^2} \right)$

EXAM SHORTCUT (AI-derived explanation)

$f_X(x) = \text{the } \int 8xy \, dy \text{ from } x \text{ to } 1 = 4 \times (1 - x^2), \text{ so } f(y | x) = 2y/(1 - x^2) \text{ on } (x, 1). \text{ Then}$
 $E(Y | X = x) = 2(1 - x^3)/(3(1 - x^2)).$

TIPS & TRICKS (AI-derived explanation)

- Mind the region: $0 < x < y < 1$ means y runs from x to 1 , not from 0 to 1 . Getting the limits right is the whole difficulty here.
- The conditional density $2y/(1 - x^2)$ on $(x, 1)$ integrates to 1 - always verify this before taking moments.
- The factor $2/3$ in front is the signature of integrating y times y : the $y^3/3$ term produces the 3 in the denominator and the 2 comes from the conditional density.
- Boundary check at $x \rightarrow 1$: $(1 - x^3)/(1 - x^2) \rightarrow 3/2$, so $E(Y | X = x) \rightarrow 1$, which is right since Y is squeezed against 1 . Option (c) passes; options (a) and (b) do not.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The joint density is $f(x, y) = 8xy$ on the region $0 < x < y < 1$.
2. Marginal of X : $f_X(x) = \text{the integral from } y = x \text{ to } 1 \text{ of } 8xy \, dy = 8x[y^2/2] \text{ from } x \text{ to } 1 = 4 \times (1 - x^2), \text{ for } 0 < x < 1.$
3. Conditional density: $f(y | x) = 8xy/[4 \times (1 - x^2)] = 2y/(1 - x^2)$ for $x < y < 1$.
4. $E(Y | X = x) = \text{the } \int_x^1 y \cdot 2y/(1 - x^2) \, dy = [2/(1 - x^2)][y^3/3] \text{ from } x \text{ to } 1 = [2/(1 - x^2)](1 - x^3)/3 = (2/3)(1 - x^3)/(1 - x^2).$
5. Hence option (c) is correct.

12. 2021 · Q5 Random Vectors & Joint Distributions · Numerical**COMMON DATA FOR THIS GROUP**

Q4–Q5 refer to the following: Let the joint density function of X and Y be

$$f(x, y) = \begin{cases} 8xy, & 0 < x < y < 1 \\ 0, & \text{otherwise} \end{cases}$$

QUESTION

What is the value of $[Y^2]_{X=0.5}$?

OPTIONS

- (a) 0.5
 (b) 0.525
 (c) 0.6
 (d) 0.625

ANSWER (AI-derived — no official key exists for this paper)

(d) 0.625

EXAM SHORTCUT (AI-derived explanation)

$E(Y^2 | X = x) = [2/(1 - x^2)](1 - x^4)/4 = (1 + x^2)/2$. At $x = 0.5$ that is $(1 + 0.25)/2 = 0.625$.

TIPS & TRICKS (AI-derived explanation)

- The algebra collapses beautifully: $(1 - x^4)/(1 - x^2) = 1 + x^2$, so the conditional second moment is simply $(1 + x^2)/2$.
- Reuse the conditional density $2y/(1 - x^2)$ from the previous part - shared-stem items are built to be solved in sequence.
- Factorising differences of powers $(1 - x^4 = (1 - x^2)(1 + x^2))$ is the standard simplification in these conditional-moment questions.
- Consistency check: $E(Y | X = 0.5) = (2/3)(0.875/0.75) = 0.778$, and $E(Y^2 | X = 0.5) = 0.625$ exceeds $0.778^2 = 0.605$, so the conditional variance is positive as it must be.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. From the previous part, the conditional density of Y given $X = x$ is $f(y | x) = 2y/(1 - x^2)$ for $x < y < 1$.
2. $E(Y^2 | X = x) = \int_x^1 y^2 \cdot 2y/(1 - x^2) dy = [2/(1 - x^2)][y^4/4]$ from x to $1 = [2/(1 - x^2)](1 - x^4)/4 = (1 - x^4)/(2(1 - x^2))$.
3. Since $1 - x^4 = (1 - x^2)(1 + x^2)$, this simplifies to $(1 + x^2)/2$.
4. At $x = 0.5$: $(1 + 0.25)/2 = 1.25/2 = 0.625$.
5. Hence option (d) is correct.

13. 2021 · Q16 Random Vectors & Joint Distributions · Numerical**QUESTION**

X, Y have joint p.d.f. $f(x, y) = \frac{1}{3}(x + y)$, $0 \leq x \leq 1$, $0 \leq y \leq 2$. What is $E(XY)$?

OPTIONS

- (a) $\frac{2}{3}$
 (b) $\frac{5}{3}$
 (c) $\frac{2}{9}$
 (d) $\frac{5}{9}$

ANSWER (AI-derived — no official key exists for this paper)

(a) $\frac{2}{3}$

EXAM SHORTCUT (AI-derived explanation)

$E(XY) = (1/3)$ times the double $\int xy(x + y)$. Integrating in y from 0 to 2 gives $2x^2 + 8x/3$; integrating that in x from 0 to 1 gives 2, so $E(XY) = 2/3$.

TIPS & TRICKS (AI-derived explanation)

- Integrate over the WIDER range first (y from 0 to 2 here) - the inner integral $2x^2 + 8x/3$ is then simple to carry forward.
- Validity check: the double $\int (x + y)/3$ over the rectangle is $(1/3)(1 + 2) = 1$, so the density is correct and the limits are right.
- This is the same density as 2018Q14, where $6E(XY)$ contributed 4 to the answer - recognising the repeat saves the whole computation.
- Magnitude check: X averages about $5/9$ and Y about $10/9$, so a product expectation near $2/3$ is entirely plausible; options (c) and (d) are too small.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The density is $f(x, y) = (x + y)/3$ on $0 \leq x \leq 1, 0 \leq y \leq 2$. Its total integral is $(1/3)(1 + 2) = 1$, so it is valid.
2. $E(XY) = (1/3)$ times the double $\int xy(x + y)$ over the rectangle.
3. Inner $\int (x^2y + xy^2) dy = [x^2y^2/2 + xy^3/3]$ from 0 to 2 $= 2x^2 + 8x/3$.
4. Outer $\int (2x^2 + 8x/3) dx = 2/3 + 4/3 = 2$.
5. Therefore $E(XY) = (1/3)(2) = 2/3$.
6. Hence option (a) is correct.

14. 2022 · Q7 Random Vectors & Joint Distributions · Numerical**QUESTION**

X and Y have joint p.m.f. $p(x, y) = \frac{x+y}{21}$ for $x = 1, 2, 3; y = 1, 2$ (0 otherwise). What is $P(X - Y \leq 0.5)$ equal to?

OPTIONS

- (a) $\frac{2}{3}$
 (b) $\frac{4}{7}$
 (c) $\frac{3}{7}$
 (d) $\frac{17}{21}$

ANSWER (AI-derived — no official key exists for this paper)

(c) $\frac{3}{7}$

EXAM SHORTCUT (AI-derived explanation)

$X - Y$ is an integer, so $X - Y \leq 0.5$ is the same as $X \leq Y$. The qualifying cells are $(1, 1), (1, 2), (2, 2)$ with masses $2/21, 3/21, 4/21$, totalling $9/21 = 3/7$.

TIPS & TRICKS (AI-derived explanation)

- Convert a fractional bound on a DISCRETE difference into an integer one: $X - Y \leq 0.5$ means $X - Y \leq 0$, i.e. $X \leq Y$. This removes all ambiguity.
- List the qualifying (x, y) pairs explicitly rather than summing blindly - with only six cells the enumeration is quick and safe.
- Verify the pmf first: the six masses $(x + y)/21$ are 2, 3, 3, 4, 4, 5 over 21, which sum to $21/21 = 1$.
- Option (d) $17/21$ is the complement of the wrong event ($X \geq Y$ or similar) - always double-check which inequality you are summing over.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The joint pmf is $p(x, y) = (x + y)/21$ for $x = 1, 2, 3$ and $y = 1, 2$. The six masses are $2/21, 3/21, 3/21, 4/21, 4/21, 5/21$, summing to 1.
2. Since X and Y take integer values, $X - Y$ is an integer, so the event $X - Y \leq 0.5$ is identical to $X - Y \leq 0$, i.e. $X \leq Y$.
3. The pairs with $x \leq y$ are $(1, 1), (1, 2)$ and $(2, 2)$.
4. Their probabilities are $p(1, 1) = 2/21, p(1, 2) = 3/21$ and $p(2, 2) = 4/21$.
5. Adding: $(2 + 3 + 4)/21 = 9/21 = 3/7$.

6. Option (c) is correct.

15. **2022 · Q49** Marginal Distributions · Numerical

QUESTION

Let (X, Y) be jointly distributed with pdf $f(x, y) = 2, 0 < x < y < 1$. The marginal pdf of Y is:

OPTIONS

- (a) $f(y) = 1 - y, 0 < y < 1$
- (b) $f(y) = 2y, 0 < y < 1$
- (c) $f(y) = 1 - 2y, 0 < y < 1$
- (d) $f(y) = 2 - 3y, 0 < y < 1$

ANSWER (AI-derived — no official key exists for this paper)

(b) $f(y) = 2y, 0 < y < 1$

EXAM SHORTCUT (AI-derived explanation)

For fixed y , x ranges over $(0, y)$, so $f_Y(y) = \int_0^y 2 dx$ from 0 to $y = 2y$ on $(0, 1)$.

TIPS & TRICKS (AI-derived explanation)

- The region $0 < x < y < 1$ is the triangle below the line $y = x$ reflected - for a given y , x runs from 0 up to y . Sketching the triangle prevents limit errors.
- Integrating a constant density over a variable range gives a LINEAR marginal - so the answer must be proportional to y , which only option (b) is.
- Validity check: the $\int 2y$ over $(0, 1)$ is 1 , whereas options (a), (c) and (d) integrate to $1/2, 0$ and $1/2$ respectively - all invalid.
- The area of the triangle is $1/2$, so the constant density must be 2 for the total probability to be 1 - a quick confirmation that the set-up is right.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The joint density is $f(x, y) = 2$ on the triangular region $0 < x < y < 1$, whose area is $1/2$, so the total probability is $2 \times 1/2 = 1$ as required.
2. The marginal density of Y is obtained by integrating out x over the range permitted for that y .
3. For a fixed y in $(0, 1)$, the constraint $0 < x < y$ means x ranges from 0 to y .
4. $f_Y(y) = \int_0^y 2 dx = 2y$.
5. Check: the $\int 2y$ over $(0, 1)$ is $[y^2]$ from 0 to $1 = 1$, a valid density.
6. Hence $f_Y(y) = 2y$ for $0 < y < 1$, option (b).

16. **2023 · Q8** Random Vectors & Joint Distributions · Numerical

COMMON DATA FOR THIS GROUP

Questions 8–10 are based on the following: let the joint pdf of two random variables X and Y be

$$f(x, y) = \frac{1}{\sqrt{2\pi y}} e^{-\frac{1}{2y}(x-12y)^2}, \quad -\infty < x < \infty, 0 < y < 1$$

QUESTION

What is the value of $E(X)$?

OPTIONS

- (a) 1
- (b) 2
- (c) 4
- (d) 6

ANSWER (AI-derived — no official key exists for this paper)

(d) 6

EXAM SHORTCUT (AI-derived explanation)

The density says $X | Y = y \sim N(12y, y)$, and integrating out x leaves $f_Y(y) = 1$ on $(0, 1)$, so $Y \sim U(0, 1)$. Then $E(X) = E(12Y) = 12(1/2) = 6$.

TIPS & TRICKS (AI-derived explanation)

- Recognise the normal kernel in x : $(1/\sqrt{2\pi y}) \exp(-(x - 12y)^2/(2y))$ is exactly the $N(12y, y)$ density, so the conditional law is read off directly.
- Because that kernel integrates to 1 for each y , the marginal of Y is the constant 1 on $(0, 1)$ - i.e. Y is $\text{Uniform}(0, 1)$. This identification drives all three parts.
- Tower property: $E(X) = E[E(X | Y)] = E(12Y)$. Never attempt the double integral directly.
- $E(Y) = 1/2$, $E(Y^2) = 1/3$ and $\text{Var}(Y) = 1/12$ for $U(0, 1)$ - keep all three ready, since the next two parts need them.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Write the joint density as $f(x, y) = [1/\sqrt{2\pi y}] \exp(-(x - 12y)^2/(2y))$ for $-\infty < x < \infty$ and $0 < y < 1$.
2. For fixed y this is exactly the $N(12y, y)$ density in x , so $X | Y = y \sim N(12y, y)$.
3. Integrating that normal density over x gives 1 for every y , so the marginal density of Y is 1 on $(0, 1)$, i.e. $Y \sim \text{Uniform}(0, 1)$.
4. By the tower property, $E(X) = E[E(X | Y)] = E(12Y)$.
5. Since $E(Y) = 1/2$ for $U(0, 1)$, $E(X) = 12(1/2) = 6$.
6. Hence option (d) is correct.

17. 2023 · Q9 Random Vectors & Joint Distributions · Numerical
COMMON DATA FOR THIS GROUP

Questions 8–10 are based on the following: let the joint pdf of two random variables X and Y be

$$f(x, y) = \frac{1}{\sqrt{2\pi y}} e^{-\frac{1}{2y}(x-12y)^2}, \quad -\infty < x < \infty, \quad 0 < y < 1$$

QUESTION

What is the value of $\text{Var}(X)$?

OPTIONS

- (a) 1
(b) 2
(c) 12
(d) 12.5

ANSWER (AI-derived — no official key exists for this paper)**(d)** 12.5**EXAM SHORTCUT** (AI-derived explanation)

$$\text{Var}(X) = E[\text{Var}(X | Y)] + \text{Var}[E(X | Y)] = E(Y) + \text{Var}(12Y) = 1/2 + 144(1/12) = 0.5 + 12 = 12.5.$$

TIPS & TRICKS (AI-derived explanation)

- Law of total variance: $\text{Var}(X) = E[\text{Var}(X | Y)] + \text{Var}[E(X | Y)]$. Both pieces are needed - dropping either gives 12 or 0.5, and option (c) 12 is planted for exactly that slip.
- From $X | Y \sim N(12Y, Y)$: the conditional variance is Y itself, whose expectation is $1/2$; the conditional mean is $12Y$, whose variance is $144 \text{Var}(Y)$.
- $\text{Var}(Y) = 1/12$ for $U(0, 1)$, so $144(1/12) = 12$ - the clean cancellation confirms the set-up.
- The two contributions are of very different sizes here (0.5 versus 12), which is a useful reminder that the between-group term usually dominates.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. From the joint density, $X | Y = y \sim N(12y, y)$ and $Y \sim \text{Uniform}(0, 1)$.
2. The law of total variance gives $\text{Var}(X) = E[\text{Var}(X | Y)] + \text{Var}[E(X | Y)]$.
3. The conditional variance is $\text{Var}(X | Y) = Y$, so $E[\text{Var}(X | Y)] = E(Y) = 1/2$.
4. The conditional mean is $E(X | Y) = 12Y$, so $\text{Var}[E(X | Y)] = 144 \text{Var}(Y) = 144(1/12) = 12$.
5. Adding: $\text{Var}(X) = 0.5 + 12 = 12.5$.
6. Hence option (d) is correct.

18. **2023 · Q16** Random Vectors & Joint Distributions · Numerical**COMMON DATA FOR THIS GROUP**

Questions 15–17 are based on the following joint pmf of X and Y ($0 < \theta < 1$):

$X \backslash Y$	-2	-1	0	1	2
-1	θ	2θ	3θ	θ	θ
0	2θ	3θ	θ	0	0
1	θ	θ	θ	θ	2θ

QUESTION

What is the value of $\text{Var}(X | Y = 2)$?

OPTIONS

- (a) $1/3$

- (b) $2/3$
- (c) $8/9$
- (d) $2/9$

ANSWER (AI-derived — no official key exists for this paper)

(c) $8/9$

EXAM SHORTCUT (AI-derived explanation)

With conditional probabilities $1/3$ on -1 and $2/3$ on $+1$: $E(X | Y = 2) = 1/3$ and $E(X^2 | Y = 2) = 1$, so the variance is $1 - 1/9 = 8/9$.

TIPS & TRICKS (AI-derived explanation)

- Since X takes only the values -1 and $+1$ here, $X^2 = 1$ with certainty, so $E(X^2 | Y = 2) = 1$ without any computation.
- $E(X | Y = 2) = -1(1/3) + 1(2/3) = 1/3$ - the same conditional distribution as the previous part, so reuse it.
- Variance $= 1 - (1/3)^2 = 8/9$. Option (a) $1/3$ is the conditional MEAN, the classic mis-selection.
- A two-point variable on $\{-1, 1\}$ with probability p on $+1$ has variance $4p(1 - p) = 4(2/3)(1/3) = 8/9$ - a fast cross-check.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. From the previous part, conditionally on $Y = 2$ the variable X takes the value -1 with probability $1/3$ and $+1$ with probability $2/3$.
2. $E(X | Y = 2) = (-1)(1/3) + (1)(2/3) = 1/3$.
3. Since X takes only the values -1 and $+1$ in this conditional distribution, $X^2 = 1$ always, so $E(X^2 | Y = 2) = 1$.
4. $\text{Var}(X | Y = 2) = E(X^2 | Y = 2) - [E(X | Y = 2)]^2 = 1 - 1/9 = 8/9$.
5. Equivalently, for a two-point variable on $\{-1, 1\}$ with $P(+1) = p$, the variance is $4p(1 - p) = 4(2/3)(1/3) = 8/9$. Option (c) is correct.

19. 2023 · Q17 Conditional Expectation · Numerical

COMMON DATA FOR THIS GROUP

Questions 15–17 are based on the following joint pmf of X and Y ($0 < \theta < 1$):

$X \backslash Y$	-2	-1	0	1	2
-1	θ	2θ	3θ	θ	θ
0	2θ	3θ	θ	0	0
1	θ	θ	θ	θ	2θ

QUESTION

What is the value of $E(XY | Y = 2)$?

OPTIONS

- (a) 1
- (b) $1/3$

(c) $2/3$ (d) $8/9$ **ANSWER (AI-derived — no official key exists for this paper)****(c)** $2/3$ **EXAM SHORTCUT (AI-derived explanation)**

Given $Y = 2$, the factor Y is the constant 2, so $E(XY | Y = 2) = 2E(X | Y = 2) = 2(1/3) = 2/3$.

TIPS & TRICKS (AI-derived explanation)

- Conditioning on $Y = 2$ makes Y a CONSTANT inside the expectation, so it can be pulled straight out - no joint computation is needed.
- Reuse $E(X | Y = 2) = 1/3$ from the previous part; these three items are deliberately chained.
- Option (b) $1/3$ is $E(X | Y = 2)$ itself, the trap for candidates who forget to multiply by 2.
- General rule: $E(g(Y)h(X) | Y = y) = g(y)E(h(X) | Y = y)$ - worth stating as a formula, since it appears constantly.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. We condition on the event $Y = 2$, so within this conditional distribution Y takes the constant value 2.
2. Hence $E(XY | Y = 2) = E(2X | Y = 2) = 2E(X | Y = 2)$.
3. From the earlier parts, the conditional distribution of X given $Y = 2$ puts mass $1/3$ on -1 and $2/3$ on $+1$.
4. So $E(X | Y = 2) = (-1)(1/3) + (1)(2/3) = 1/3$.
5. Therefore $E(XY | Y = 2) = 2(1/3) = 2/3$.
6. Option (c) is correct.

20. **2023 · Q19** Marginal Distributions · Numerical**COMMON DATA FOR THIS GROUP**

Questions 18–20 are based on the following: let the joint pdf of X and Y be

$$f(x, y) = \begin{cases} \frac{\lambda x^{12}}{(1 + xy)^{20}}, & x > 0, y > 1 \\ 0, & \text{otherwise} \end{cases}$$

QUESTION

What is the marginal distribution of X ?

OPTIONS

- (a) Beta Type-II distribution with parameters (12, 7)
- (b) Beta Type-II distribution with parameters (13, 7)
- (c) Beta Type-I distribution with parameters (12, 7)
- (d) Beta Type-I distribution with parameters (13, 7)

ANSWER (AI-derived — no official key exists for this paper)**(a)** Beta Type-II distribution with parameters (12, 7)

EXAM SHORTCUT (AI-derived explanation)

Integrating out y gives $f_X(x) = \lambda x^{11} / [19(1+x)^{19}] = x^{11} / [B(12, 7)(1+x)^{19}]$, the Beta second-kind density with $m = 12$ and $n = 7$.

TIPS & TRICKS (AI-derived explanation)

- Beta type-II (second kind) density: $f(x) = x^{m-1} / [B(m, n)(1+x)^{m+n}]$ on $x > 0$. Match the two exponents to read off m and n .
- Here the numerator power is 11 so $m = 12$, and the denominator power is 19 so $m + n = 19$, giving $n = 7$.
- Support decides the type: X ranges over $(0, \infty)$, which is Beta type-II; Beta type-I lives on $(0, 1)$. That eliminates options (c) and (d) at once.
- The substitution $t = 1 + xy$ with y from 1 to ∞ makes t run from $1 + x$ to ∞ , which is what produces the $(1+x)^{-19}$ factor.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The marginal of X is $f_X(x) = \int \lambda x^{12} / (1+xy)^{20} dy$.
2. Substitute $t = 1 + xy$, so $dt = xdy$ and $y = 1$ corresponds to $t = 1 + x$, with t running to ∞ .
3. $f_X(x) = \lambda x^{12}$ times the $\int_{1+x}^{\infty} t^{-20} dt / x = \lambda x^{11} [t^{-19} / 19]$ evaluated at $t = 1 + x = \lambda x^{11} / [19(1+x)^{19}]$.
4. With $\lambda = 19/B(12, 7)$, this becomes $f_X(x) = x^{11} / [B(12, 7)(1+x)^{19}]$ for $x > 0$.
5. The Beta distribution of the second kind with parameters (m, n) has density $x^{m-1} / [B(m, n)(1+x)^{m+n}]$; matching gives $m - 1 = 11$ and $m + n = 19$, so $m = 12$ and $n = 7$.
6. Hence X follows a Beta Type-II distribution with parameters $(12, 7)$, option (a).

21. 2023 · Q20 Marginal Distributions · Numerical**COMMON DATA FOR THIS GROUP**

Questions 18–20 are based on the following: let the joint pdf of X and Y be

$$f(x, y) = \begin{cases} \frac{\lambda x^{12}}{(1+xy)^{20}}, & x > 0, y > 1 \\ 0, & \text{otherwise} \end{cases}$$

QUESTION

Marginal distribution of Y has pdf:

OPTIONS

- (a) $\frac{7}{y^8}; y > 1$ and 0 otherwise
- (b) $\frac{11}{y^{12}}; y > 1$ and 0 otherwise
- (c) $\frac{18}{y^{19}}; y > 1$ and 0 otherwise
- (d) $\frac{12}{y^{13}}; y > 1$ and 0 otherwise

ANSWER (AI-derived — no official key exists for this paper)

- (d) $\frac{12}{y^{13}}; y > 1$ and 0 otherwise

EXAM SHORTCUT (AI-derived explanation)

The x-integral computed earlier gives $B(13, 7)/y^{13}$, so $f_Y(y) = \lambda B(13, 7)/y^{13} = 12/y^{13}$ on $y > 1$.

TIPS & TRICKS (AI-derived explanation)

- Reuse the inner integral from the normalisation step: it already equals $B(13, 7)/y^{13}$, so only the constant λ needs attaching.
- $\lambda B(13, 7) = 12$ exactly, because λ was chosen to make the whole double integral 1 and the $y - \int y^{-13}$ is $1/12$.
- Validity check: the $\int 12y^{-13}$ over $(1, \infty)$ is $12/12 = 1$. Options (a), (b) and (c) integrate to 1 as well, so use the derivation rather than this test.
- This is a Pareto density with shape 12 on $(1, \infty)$ - recognising the family makes the exponent pairing (12, 13) easy to remember.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The marginal of Y is $f_Y(y) = \int \lambda x^{12}/(1 + xy)^{20} dx$.
2. As computed in the normalisation step, substituting $u = xy$ gives this integral as $\lambda B(13, 7)/y^{13}$ for $y > 1$.
3. From the normalisation, $\lambda B(13, 7)/12 = 1$, so $\lambda B(13, 7) = 12$.
4. Hence $f_Y(y) = 12/y^{13}$ for $y > 1$, and 0 otherwise.
5. Check: the $\int 12y^{-13} dy$ from 1 to $\infty = 12[y^{-12}/(-12)]$ from 1 to $\infty = 1$, a valid density.
6. Option (d) is correct.

22. 2024 · Q32 Random Vectors & Joint Distributions · Numerical**QUESTION**

Let the joint distribution of X and Y be given by

$$f(x, y) = \begin{cases} cxye^{-(x^2+y^2)}, & x > 0, y > 0 \\ 0, & \text{otherwise} \end{cases}$$

With reference to the above, consider the following statements: I. The value of c for $f(x, y)$ to be a proper pdf is 4. II. The probability $P(X^2 > 2Y^2)$ is $1/3$. III. X and Y are independently distributed. Which of the statements given above are correct?

OPTIONS

- (a) I and II only
- (b) II and III only
- (c) I and III only
- (d) I, II and III

ANSWER (AI-derived — no official key exists for this paper)

(d) I, II and III

EXAM SHORTCUT (AI-derived explanation)

The density factorises as $(2xe^{-x^2})(2ye^{-y^2})$ once $c = 4$, so X and Y are i.i.d. Rayleigh, X^2 and Y^2 are i.i.d. $\text{Exp}(1)$, and $P(X^2 > 2Y^2) = E(e^{-2V}) = 1/3$. All three statements hold.

TIPS & TRICKS (AI-derived explanation)

- Normalisation: the $\int x e^{-x^2}$ over $(0, \infty)$ is $1/2$, so $c(1/2)^2 = 1$ gives $c = 4$ at once.
- Once $c = 4$ the density splits into a product of identical factors, which settles independence (statement III) with no extra work.
- Transform to squares: $P(X^2 > t) = P(X > \sqrt{t}) = e^{-t}$, so $X^2 \sim \text{Exp}(1)$ - this turns statement II into a simple exponential comparison.
- $P(U > 2V)$ for i.i.d. $\text{Exp}(1)$ is $E[e^{-2V}] = \int e^{-3v} dv = 1/3$ - the general form is $1/(1+k)$ for $P(U > kV)$.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Statement I: the double $\int cxy e^{-(x^2+y^2)}$ over $x, y > 0$ factorises as c times [the $\int x e^{-x^2} dx]^2 = c(1/2)^2 = c/4$. Setting this equal to 1 gives $c = 4$. TRUE.
2. Statement III: with $c = 4$ the density is $4xy e^{-x^2} e^{-y^2} = (2x e^{-x^2})(2y e^{-y^2})$, a product of a function of x alone and a function of y alone, so X and Y are independent (each Rayleigh). TRUE.
3. For statement II, note $P(X > a) = \int_a^\infty 2x e^{-x^2} dx = e^{-a^2}$, so $P(X^2 > t) = e^{-t}$: $X^2 \sim \text{Exponential}(1)$, and likewise Y^2 .
4. Write $U = X^2$ and $V = Y^2$, independent $\text{Exp}(1)$. Then $P(X^2 > 2Y^2) = P(U > 2V) = E[P(U > 2V | V)] = E[e^{-2V}]$.
5. $E[e^{-2V}] = \int_0^\infty e^{-2v} e^{-v} dv = \int_0^\infty e^{-3v} dv = 1/3$. TRUE.
6. All three statements are correct, option (d).

23. 2024 · Q33 Poisson Distribution · Numerical**QUESTION**

Let X and Y have joint probability mass function

$$P(X = x, Y = y) = \frac{e^{-2}}{x!(y-x)!}, \quad x = 0, 1, 2, \dots, y \text{ and } y = 1, 2, 3, \dots$$

With reference to the above, consider the following statements: I. Y follows Poisson distribution with parameter $1/2$. II. $E(X) = 1$ Which of the statements given above is/are correct?

OPTIONS

- (a) I only
- (b) II only
- (c) Both I and II
- (d) Neither I nor II

ANSWER (AI-derived — no official key exists for this paper)

(b) II only

EXAM SHORTCUT (AI-derived explanation)

Summing over x for fixed y gives $P(Y = y) = e^{-2} 2^y / y!$, i.e. $\text{Poisson}(2)$ - not $1/2$, so statement I fails. Conditionally $X | Y = y \sim \text{Binomial}(y, 1/2)$, so $E(X) = E(Y)/2 = 1$ and statement II holds.

TIPS & TRICKS (AI-derived explanation)

- Multiply and divide by $y!$: $e^{-2}/(x!(y-x)!) = [e^{-2}/y!]\binom{y}{x}$, and the binomial coefficients sum to 2^y .
- That immediately gives the marginal $P(Y = y) = e^{-2} 2^y / y!$, which is Poisson with parameter 2 - so 'parameter $1/2$ ' is wrong.

- The same manipulation shows $P(X = x | Y = y) = \binom{y}{x}/2^y = \text{Binomial}(y, 1/2)$, whence $E(X | Y) = Y/2$.
- Tower property: $E(X) = E(Y)/2 = 2/2 = 1$. Statement II is confirmed without touching the joint sum again.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Write the joint pmf as $P(X = x, Y = y) = e^{-2}/(x!(y-x)!) = [e^{-2}/y!] * y!/(x!(y-x)!) = [e^{-2}/y!] \binom{y}{x}$.
2. Summing over $x = 0, 1, \dots, y$ and using the binomial identity $\sum \binom{y}{x} = 2^y$ gives $P(Y = y) = e^{-2}2^y/y!$.
3. This is the Poisson distribution with parameter 2, not 1/2. Statement I is FALSE.
4. Dividing the joint pmf by this marginal gives $P(X = x | Y = y) = \binom{y}{x}/2^y$, i.e. $X | Y = y \sim \text{Binomial}(y, 1/2)$.
5. Hence $E(X | Y) = Y/2$, and by the tower property $E(X) = E(Y)/2 = 2/2 = 1$. Statement II is TRUE.
6. Only statement II is correct, option (b).

24. 2024 · Q35 Geometric Distribution · Numerical**QUESTION**

Let X and Y be jointly distributed with joint probability mass function

$$P(x, y) = \frac{1}{2^{y+2}(y+1)} \left(\frac{2y+1}{2y+2} \right)^x, \quad x, y = 0, 1, 2, \dots$$

With reference to the above, consider the following statements: I. The marginal distribution of Y is geometric distribution. II. $\text{Var}(Y) = 1$ Which of the statements given above is/are correct?

OPTIONS

- (a) I only
- (b) II only
- (c) Both I and II
- (d) Neither I nor II

ANSWER (AI-derived — no official key exists for this paper)

(a) I only

EXAM SHORTCUT (AI-derived explanation)

Summing the geometric series over x gives $P(Y = y) = (1/2)^{y+1}$, a geometric distribution with $p = 1/2$ on $\{0, 1, 2, \dots\}$, whose variance is $q/p^2 = 2$, not 1. So statement I holds and II fails.

TIPS & TRICKS (AI-derived explanation)

- For fixed y the x -part is a geometric series with ratio $r = (2y+1)/(2y+2)$, so its sum is $1/(1-r) = 2y+2$ - the $(y+1)$ in the denominator is placed to cancel with it.
- After cancelling, $P(Y = y) = (2y+2)/[2^{y+2}(y+1)] = 2/2^{y+2} = (1/2)^{y+1}$ - a clean geometric law.
- Geometric on $\{0, 1, 2, \dots\}$ with parameter p : mean q/p and variance q/p^2 . With $p = 1/2$ these are 1 and 2 respectively.
- Statement II quotes the MEAN (1) as the variance - the classic swap, and the only error in the pair.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Fix y and sum the joint pmf over $x = 0, 1, 2, \dots$. The x -dependence is r^x with $r = (2y+1)/(2y+2)$, and $0 < r < 1$.
2. The geometric series sums to $1/(1-r) = 1/[1 - (2y+1)/(2y+2)] = 2y+2$.

3. Hence $P(Y = y) = [1/(2^{y+2}(y+1))](2y+2) = 2(y+1)/[2^{y+2}(y+1)] = 2/2^{y+2} = (1/2)^{y+1}$.

4. This is the geometric distribution on $\{0, 1, 2, \dots\}$ with success probability $p = 1/2$, so statement I is TRUE.

5. For that distribution, mean $= q/p = 1$ and variance $= q/p^2 = (1/2)/(1/4) = 2$.

6. Statement II claims $\text{Var}(Y) = 1$, which is in fact the mean. FALSE. Only statement I is correct, option (a).

25. 2025 · Q7 Marginal Distributions · Numerical

QUESTION

Let X_1 and X_2 have joint pdf $f(x_1, x_2) = \begin{cases} 2, & 0 < x_1 < x_2 < 1 \\ 0, & \text{otherwise} \end{cases}$. What is the marginal distribution of X_1 ?

OPTIONS

(a) $f_1(x_1) = 1, 0 < x_1 < 1$

(b) $f_1(x_1) = 2 - 2x_1, 0 < x_1 < 1$

(c) $f_1(x_1) = 2x_1, 0 < x_1 < 1$

(d) $f_1(x_1) = 3x_1^2, 0 < x_1 < 1$

ANSWER (AI-derived — no official key exists for this paper)

(b) $f_1(x_1) = 2 - 2x_1, 0 < x_1 < 1$

EXAM SHORTCUT (AI-derived explanation)

For fixed x_1 , the variable x_2 ranges over $(x_1, 1)$, so $f_1(x_1) = 2(1 - x_1)$ on $(0, 1)$.

TIPS & TRICKS (AI-derived explanation)

- Get the limits from the region: $0 < x_1 < x_2 < 1$ means x_2 runs from x_1 up to 1, a range of length $1 - x_1$.
- Integrating the constant 2 over that range gives $2(1 - x_1)$ - a DECREASING density, since small x_1 leaves more room for x_2 .
- Validity check: the $\int 2(1 - x_1)$ over $(0, 1)$ is $2(1 - 1/2) = 1$. Option (c) 2×1 also integrates to 1, so use the limits, not this test, to choose.
- Contrast with 2022Q49, where the same density gave $f_Y(y) = 2y$ for the OTHER variable - the two marginals are mirror images.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The joint density is $f(x_1, x_2) = 2$ on the triangle $0 < x_1 < x_2 < 1$, whose area is $1/2$, so the total probability is $2(1/2) = 1$.
2. The marginal of X_1 is obtained by integrating out x_2 over the values it may take for that x_1 .
3. The constraint $x_1 < x_2 < 1$ means x_2 ranges from x_1 to 1.
4. $f_1(x_1) = \int_{x_1}^1 2 dx_2 = 2(1 - x_1)$, for $0 < x_1 < 1$.
5. Check: the $\int 2(1 - x_1)$ over $(0, 1) = 2[x_1 - x_1^2/2]$ from 0 to 1 $= 2(1/2) = 1$, a valid density.
6. Option (b) is correct.

26. 2025 · Q11 Random Vectors & Joint Distributions · Numerical

QUESTION

The joint probability of two events A and B is given by:

	B	$\bar{B}$
A	0.4	0.1
$\bar{A}$	0.2	0.3

I. A and B are independent events. II. $P(A | \bar{B}) + P(\bar{A} | B) > \frac{1}{2}$. Which is/are correct?

OPTIONS

- (a) I only
 (b) II only
 (c) Both I and II
 (d) Neither I nor II

ANSWER (AI-derived — no official key exists for this paper)

(b) II only

EXAM SHORTCUT (AI-derived explanation)

$P(A) = 0.5$ and $P(B) = 0.6$, so $P(A)P(B) = 0.30$ but $P(A \cap B) = 0.40$ - not independent. And $P(A | B') + P(A' | B) = 0.1/0.4 + 0.2/0.6 = 0.25 + 0.333 = 0.583 > 0.5$.

TIPS & TRICKS (AI-derived explanation)

- Get the marginals by summing rows and columns: $P(A) = 0.4 + 0.1 = 0.5$ and $P(B) = 0.4 + 0.2 = 0.6$.
- Independence test: compare $P(A \cap B)$ with $P(A)P(B)$. Here 0.40 against 0.30, so the events are positively associated, not independent.
- For the conditionals, divide the relevant cell by its column or row total: $P(A | B') = 0.1/(0.1 + 0.3)$ and $P(A' | B) = 0.2/(0.4 + 0.2)$.
- $0.25 + 0.3333 = 0.5833$ exceeds 0.5, so statement II holds - compute both fractions exactly rather than estimating.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. From the table: $P(A \cap B) = 0.4$, $P(A \cap B') = 0.1$, $P(A' \cap B) = 0.2$, $P(A' \cap B') = 0.3$, summing to 1.
2. Marginals: $P(A) = 0.4 + 0.1 = 0.5$ and $P(B) = 0.4 + 0.2 = 0.6$.
3. Statement I: $P(A)P(B) = 0.5 \times 0.6 = 0.30$, but $P(A \cap B) = 0.40$. Since these differ, A and B are NOT independent. FALSE.
4. Statement II: $P(A | B') = P(A \cap B')/P(B') = 0.1/0.4 = 0.25$, and $P(A' | B) = P(A' \cap B)/P(B) = 0.2/0.6 = 0.3333$.
5. Their sum is 0.5833, which exceeds 1/2. TRUE.
6. Only statement II is correct, option (b).

27. **2025 · Q16** Random Vectors & Joint Distributions · Numerical

QUESTION

The joint pmf of X and Y is:

Y \ X	0	1	2
0	3/28	9/28	3/28
1	3/14	3/14	0

$Y \setminus X$	0	1	2
2	1/28	0	0

I. $\text{Cov}(X, Y) = 3/4$ II. $E(X) = E(Y)$ Which is/are correct?

OPTIONS

- (a) I only
 (b) II only
 (c) Both I and II
 (d) Neither I nor II

ANSWER (AI-derived — no official key exists for this paper)

(d) Neither I nor II

EXAM SHORTCUT (AI-derived explanation)

Marginals give $E(X) = 3/4$ and $E(Y) = 1/2$, so they are unequal and statement II fails. $E(XY) = 6/28 = 3/14$, so $\text{Cov} = 3/14 - 3/8 = -9/56$, not $3/4$, and statement I fails too.

TIPS & TRICKS (AI-derived explanation)

- Convert every entry to 28ths first: the table becomes 3, 9, 3/6, 6, 0/1, 0, 0, which sums to 28 - confirming the pmf is valid.
- Column sums give the X-marginal (10, 15, 3)/28 and row sums the Y-marginal (15, 12, 1)/28. Compute both before evaluating either statement.
- Only the cell $(x = 1, y = 1)$ contributes to $E(XY)$, since every other cell has $x = 0, y = 0$ or probability 0 - so $E(XY) = 6/28$ directly.
- The covariance must be NEGATIVE here (large X goes with small Y in this table), so a positive value like $3/4$ is implausible on inspection.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Express all entries in 28ths: row $Y = 0$ is 3, 9, 3; row $Y = 1$ is 6, 6, 0; row $Y = 2$ is 1, 0, 0. These sum to 28, so the pmf is valid.
2. Column sums give the marginal of X: $P(X = 0) = 10/28, P(X = 1) = 15/28, P(X = 2) = 3/28$, so $E(X) = (0 + 15 + 6)/28 = 21/28 = 3/4$.
3. Row sums give the marginal of Y: $P(Y = 0) = 15/28, P(Y = 1) = 12/28, P(Y = 2) = 1/28$, so $E(Y) = (0 + 12 + 2)/28 = 14/28 = 1/2$.
4. Statement II: $E(X) = 3/4$ is not equal to $E(Y) = 1/2$. FALSE.
5. For the covariance, only the cell $x = 1, y = 1$ contributes to $E(XY)$ (all others have a zero factor or zero probability), so $E(XY) = (1)(1)(6/28) = 3/14$.
6. $\text{Cov}(X, Y) = 3/14 - (3/4)(1/2) = 12/56 - 21/56 = -9/56$, not $3/4$. Statement I is FALSE. Neither statement holds, option (d).

28. **2025 · Q41** Binomial Distribution · Conceptual

COMMON DATA FOR THIS GROUP

Shared data for Q41–Q44. Let (X_1, X_2, X_3) be a random vector with joint pmf:

$$P(X_1 = x_1, X_2 = x_2, X_3 = x_3) = \begin{cases} \frac{50!}{x_1! x_2! x_3! (50 - x_1 - x_2 - x_3)!} \left(\frac{1}{4}\right)^{x_1} \left(\frac{1}{4}\right)^{x_2} \left(\frac{1}{4}\right)^{x_3} \left(\frac{1}{4}\right)^{50 - x_1 - x_2 - x_3} & \text{if } x_1 + x_2 + x_3 \leq 50 \\ 0 & \text{otherwise} \end{cases}$$

QUESTION

The distribution of $(50 - X_1)$ is:

OPTIONS

- (a) Binomial(50, 1/4)
- (b) Binomial(50, 3/4)
- (c) Poisson(3/4)
- (d) Poisson(1/4)

ANSWER (AI-derived — no official key exists for this paper)

(b) Binomial(50, 3/4)

EXAM SHORTCUT (AI-derived explanation)

The joint pmf is Multinomial(50; 1/4, 1/4, 1/4, 1/4), so $X_1 \sim \text{Binomial}(50, 1/4)$ and therefore $50 - X_1 \sim \text{Binomial}(50, 3/4)$.

TIPS & TRICKS (AI-derived explanation)

- Any single component of a multinomial is BINOMIAL with the same n and its own p - collapse the other categories into 'not category 1'.
- Reflection rule: $X \sim B(n, p)$ implies $n - X \sim B(n, 1 - p)$. Here $1 - 1/4 = 3/4$.
- Mean check: $E(50 - X_1) = 50 - 12.5 = 37.5$, and $B(50, 3/4)$ has mean 37.5. Consistent.
- The Poisson options are implausible because $50 - X_1$ is bounded above by 50, whereas a Poisson variable is unbounded.

STEP-BY-STEP SOLUTION (AI-derived explanation)

- The joint pmf is the multinomial with $n = 50$ trials and four categories, each of probability 1/4 (the fourth category being $50 - x_1 - x_2 - x_3$).
- For a multinomial, each individual count is binomial: grouping all categories other than the first together, $X_1 \sim \text{Binomial}(50, 1/4)$.
- Hence $50 - X_1$ counts the trials NOT falling in category 1, and each trial fails to fall in category 1 with probability 3/4.
- Therefore $50 - X_1 \sim \text{Binomial}(50, 3/4)$.
- Check: $E(50 - X_1) = 50 - 50(1/4) = 37.5 = 50(3/4)$, as required.
- Option (b) is correct.

29. **2025 · Q43** Conditional Expectation · Numerical

COMMON DATA FOR THIS GROUP

Shared data for Q41–Q44. Let (X_1, X_2, X_3) be a random vector with joint pmf:

$$P(X_1 = x_1, X_2 = x_2, X_3 = x_3) = \begin{cases} \frac{50!}{x_1! x_2! x_3! (50 - x_1 - x_2 - x_3)!} \left(\frac{1}{4}\right)^{x_1} \left(\frac{1}{4}\right)^{x_2} \left(\frac{1}{4}\right)^{x_3} \left(\frac{1}{4}\right)^{50 - x_1 - x_2 - x_3} & \text{if } x_1 + x_2 + x_3 \leq 50 \\ 0 & \text{otherwise} \end{cases}$$

QUESTION

What is $E(X_1 \mid X_2 = 5)$?

OPTIONS

- (a) 8
(b) 10
(c) 12
(d) 15

ANSWER (AI-derived — no official key exists for this paper)

(d) 15

EXAM SHORTCUT (AI-derived explanation)

Given $X_2 = 5$, the remaining 45 trials fall into the other three categories with conditional probability $(1/4)/(3/4) = 1/3$ each, so $X_1 \mid X_2 = 5 \sim \text{Binomial}(45, 1/3)$ with mean 15.

TIPS & TRICKS (AI-derived explanation)

- Conditioning on one multinomial count leaves a multinomial on the REMAINING trials, with probabilities renormalised by dividing by $1 - p_2$.
- Remaining trials: $50 - 5 = 45$. Renormalised probability: $(1/4)/(1 - 1/4) = 1/3$.
- Mean = $45(1/3) = 15$. Note this is LARGER than the unconditional mean 12.5, which seems odd until you recall that $X_2 = 5$ is below its own mean of 12.5, freeing extra trials.
- The same conditional distribution answers the next part, so record $\text{Binomial}(45, 1/3)$ before moving on.

STEP-BY-STEP SOLUTION (AI-derived explanation)

- The vector follows a multinomial with $n = 50$ and four equally likely categories.
- Conditioning on $X_2 = 5$ leaves $50 - 5 = 45$ trials to be distributed among the remaining three categories.
- Each of those trials falls in category 1 with the renormalised probability $p_1/(1 - p_2) = (1/4)/(3/4) = 1/3$.
- Hence $X_1 \mid X_2 = 5 \sim \text{Binomial}(45, 1/3)$.
- Its mean is $45(1/3) = 15$.
- Option (d) is correct.

30. **2025 · Q44** Random Vectors & Joint Distributions · Numerical

COMMON DATA FOR THIS GROUP

Shared data for Q41–Q44. Let (X_1, X_2, X_3) be a random vector with joint pmf:

$$P(X_1 = x_1, X_2 = x_2, X_3 = x_3) = \begin{cases} \frac{50!}{x_1! x_2! x_3! (50 - x_1 - x_2 - x_3)!} \left(\frac{1}{4}\right)^{x_1} \left(\frac{1}{4}\right)^{x_2} \left(\frac{1}{4}\right)^{x_3} \left(\frac{1}{4}\right)^{50 - x_1 - x_2 - x_3} & \text{if } x_1 + x_2 + x_3 \leq 50 \\ 0 & \text{otherwise} \end{cases}$$

QUESTION

What is $V(X_1 | X_2 = 5)$?

OPTIONS

- (a) 8
- (b) 10
- (c) 12
- (d) 15

ANSWER (AI-derived — no official key exists for this paper)

(b) 10

EXAM SHORTCUT (AI-derived explanation)

From the previous part $X_1 | X_2 = 5 \sim \text{Binomial}(45, 1/3)$, whose variance is $45(1/3)(2/3) = 10$.

TIPS & TRICKS (AI-derived explanation)

- Reuse the conditional distribution rather than rederiving it - shared-stem items are designed to chain.
- Binomial variance $npq = 45(1/3)(2/3)$. Compute $45/3 = 15$ first, then $15 \times 2/3 = 10$.
- The conditional variance 10 is below the conditional mean 15, as it must be for a binomial (since $q < 1$).
- Option (d) 15 is the conditional MEAN, offered as the standard mean-variance confusion.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. From the previous part, X_1 given $X_2 = 5$ follows a Binomial distribution with 45 trials and success probability $1/3$.
2. For a binomial the variance is npq .
3. Here $n = 45, p = 1/3$ and $q = 2/3$.
4. $V(X_1 | X_2 = 5) = 45(1/3)(2/3) = 15(2/3) = 10$.
5. Option (b) is correct.

31. **2025 · Q52** Random Vectors & Joint Distributions · Numerical

QUESTION

A non-negative continuous random variable (X, Y) has joint density $f(x, y) = \begin{cases} 4xye^{-(x^2+y^2)}, & x \geq 0, y \geq 0 \\ 0, & \text{otherwise} \end{cases}$.

Let $T^2 = X^2 + Y^2$. The density of T is:

OPTIONS

- (a) $2t^3e^{-t^2}$
- (b) $2t^{3/2}e^{-t^3}$
- (c) $3t^2e^{-t^2}$
- (d) $3te^{-t^3}$

ANSWER (AI-derived — no official key exists for this paper)

(a) $2t^3e^{-t^2}$

EXAM SHORTCUT (AI-derived explanation)

X^2 and Y^2 are i.i.d. $\text{Exp}(1)$, so $T^2 = X^2 + Y^2 \sim \Gamma(2, 1)$ with density we^{-w} . Transforming to $T = \sqrt{W}$ gives $2t^3e^{-t^2}$.

TIPS & TRICKS (AI-derived explanation)

- First identify the marginals: $4xye^{-(x^2+y^2)}$ factorises as $(2xe^{-x^2})(2ye^{-y^2})$, so X and Y are i.i.d. Rayleigh.
- $P(X^2 > u) = P(X > \sqrt{u}) = e^{-u}$, so $X^2 \sim \text{Exp}(1)$; the sum of two such is $\Gamma(\text{shape } 2, \text{rate } 1)$ with density we^{-w} .
- Change variable $W = T^2$: $dw/dt = 2t$, so $f_T(t) = f_W(t^2)(2t) = t^2e^{-t^2}(2t) = 2t^3e^{-t^2}$.
- Validity check: the $\int 2t^3e^{-t^2}$ over $(0, \infty)$ is 1 (substitute $u = t^2$), confirming a proper density.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The joint density factorises as $f(x, y) = (2xe^{-x^2})(2ye^{-y^2})$, so X and Y are independent, each with the Rayleigh density $2xe^{-x^2}$ on $x > 0$.
2. For such a variable, $P(X > a) = e^{-a^2}$, so $P(X^2 > u) = e^{-u}$: $X^2 \sim \text{Exponential}(1)$, and likewise Y^2 .
3. Hence $W = T^2 = X^2 + Y^2$ is the sum of two independent $\text{Exponential}(1)$ variables, i.e. $\Gamma(\text{shape } 2, \text{rate } 1)$, with density $f_W(w) = we^{-w}$ for $w > 0$.
4. Now $T = \sqrt{W}$, so $w = t^2$ and $dw/dt = 2t$.
5. $f_T(t) = f_W(t^2) |dw/dt| = t^2e^{-t^2}(2t) = 2t^3e^{-t^2}$ for $t > 0$.
6. Its integral over $(0, \infty)$ is 1, confirming validity. Option (a) is correct.

32. 2026 · Q1 Random Vectors & Joint Distributions · Numerical**COMMON DATA FOR THIS GROUP**

Shared data for Q1–Q3. X and Y have discrete joint distribution

$$P(X = x, Y = y) = \begin{cases} c|x + y| & \text{for } x, y \in \{-2, -1, 0, 1, 2\} \\ 0 & \text{otherwise} \end{cases}$$

QUESTION

What is $\text{Var}(X)$ equal to?

OPTIONS

- (a) 2.7
(b) 2.4
(c) 2.35
(d) 1.17

ANSWER (AI-derived — no official key exists for this paper)

(c) 2.35

EXAM SHORTCUT (AI-derived explanation)

Row sums of $|x + y|$ for $x = -2, -1, 0, 1, 2$ are 10, 7, 6, 7, 10, total 40, so $c = 1/40$ and $P(X) = (0.25, 0.175, 0.15, 0.175, 0.25)$. Symmetry gives $E(X) = 0$, so $\text{Var}(X) = E(X^2) = 2(4)(0.25) + 2(1)(0.175) = 2 + 0.35 = 2.35$.

TIPS & TRICKS (AI-derived explanation)

- Get c first: the number of pairs summing to s is $5 - |s|$ for $|s| \leq 4$, so $\sum |x + y| = 2(4 \cdot 1 + 3 \cdot 2 + 2 \cdot 3 + 1 \cdot 4) = 40$ and $c = 1/40$.
- The marginal row sums 10, 7, 6, 7, 10 are symmetric about $x = 0$, which forces $E(X) = 0$ and turns the variance into a pure $E(X^2)$ computation.
- Only $x = +/2$ and $x = +/1$ contribute to $E(X^2)$; $x = 0$ contributes nothing, so it is a two-term sum.
- The same marginal serves for Y by symmetry, so $\text{Var}(Y) = 2.35$ as well - handy if a companion question asks for it.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Normalising constant: sum over x, y in $\{-2, -1, 0, 1, 2\}$ of $|x + y|$. The number of pairs with $x + y = s$ is $5 - |s|$, so the total is $4(1) + 3(2) + 2(3) + 1(4) + 0(5) + 1(4) + 2(3) + 3(2) + 4(1) = 40$. Hence $c = 1/40$.
2. Marginal of X : $P(X = x) = (1/40) \cdot \sum |x + y|$.
3. $x = -2$: $|x + y|$ values are 4, 3, 2, 1, 0, sum 10, so $P = 10/40 = 0.25$.
4. $x = -1$: 3, 2, 1, 0, 1, sum 7, so $P = 7/40 = 0.175$.
5. $x = 0$: 2, 1, 0, 1, 2, sum 6, so $P = 6/40 = 0.15$.
6. $x = 1$: 1, 0, 1, 2, 3, sum 7, so $P = 0.175$. $x = 2$: 0, 1, 2, 3, 4, sum 10, so $P = 0.25$. (Total $40/40 = 1$, check.)
7. $E(X) = (-2)(0.25) + (-1)(0.175) + 0(0.15) + 1(0.175) + 2(0.25) = 0$ by symmetry.
8. $E(X^2) = 4(0.25) + 1(0.175) + 0 + 1(0.175) + 4(0.25) = 1 + 0.175 + 0.175 + 1 = 2.35$.
9. $\text{Var}(X) = E(X^2) - [E(X)]^2 = 2.35 - 0 = 2.35$.
10. Hence option (c) is correct.

33. 2026 · Q2 Conditional Expectation · Numerical**COMMON DATA FOR THIS GROUP**

Shared data for Q1–Q3. X and Y have discrete joint distribution

$$P(X = x, Y = y) = \begin{cases} c|x + y| & \text{for } x, y \in \{-2, -1, 0, 1, 2\} \\ 0 & \text{otherwise} \end{cases}$$

QUESTION

What is $E(Y \mid X = 1)$ equal to?

OPTIONS

- (a) $1/7$
 (b) $3/7$
 (c) $4/7$
 (d) $6/7$

ANSWER (AI-derived — no official key exists for this paper)

(d) $6/7$

EXAM SHORTCUT (AI-derived explanation)

Along the row $X = 1$ the weights $|1 + y|$ for $y = -2, -1, 0, 1, 2$ are 1, 0, 1, 2, 3 with total 7. So

$$E(Y | X = 1) = [(-2)(1) + (-1)(0) + 0(1) + 1(2) + 2(3)]/7 = 6/7.$$

TIPS & TRICKS (AI-derived explanation)

- For a conditional expectation on a discrete grid, work with the unnormalised row weights and divide by the row total once at the end - the constant c cancels.
- The row total 7 is exactly the marginal weight $P(X = 1) = 7/40$ found in the previous part; reuse it rather than recomputing.
- Note that $|1 + y| = 0$ at $y = -1$, so that term drops out entirely.
- Sanity check the sign: the row weights are heavier on the positive y side (2 and 3 versus 1 and 0), so $E(Y | X = 1)$ must be positive, ruling out any negative answer.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The joint weights along the row $X = 1$ are $P(X = 1, Y = y) = |1 + y|/40$.
2. $y = -2$: $|-1| = 1$; $y = -1$: $|0| = 0$; $y = 0$: $|1| = 1$; $y = 1$: $|2| = 2$; $y = 2$: $|3| = 3$.
3. Row total = $(1 + 0 + 1 + 2 + 3)/40 = 7/40 = P(X = 1)$.
4. Conditional pmf: $P(Y = y | X = 1) = |1 + y|/7$, giving $1/7, 0, 1/7, 2/7, 3/7$ for $y = -2, -1, 0, 1, 2$.
5. $E(Y | X = 1) = (-2)(1/7) + (-1)(0) + (0)(1/7) + (1)(2/7) + (2)(3/7) = (-2 + 0 + 0 + 2 + 6)/7 = 6/7$.
6. Hence option (d) is correct.

34. 2026 · Q3 Random Vectors & Joint Distributions · Numerical**COMMON DATA FOR THIS GROUP**

Shared data for Q1–Q3. X and Y have discrete joint distribution

$$P(X = x, Y = y) = \begin{cases} c|x + y| & \text{for } x, y \in \{-2, -1, 0, 1, 2\} \\ 0 & \text{otherwise} \end{cases}$$

QUESTION

What is $P(|X - Y| \leq 1)$ equal to?

OPTIONS

- (a) 0.06
(b) 0.7
(c) 0.12
(d) 0.41

ANSWER (AI-derived — no official key exists for this paper)

(b) 0.7

EXAM SHORTCUT (AI-derived explanation)

$|x - y| \leq 1$ means the main diagonal plus the two adjacent diagonals. Weights: diagonal $|2x|$ gives $4 + 2 + 0 + 2 + 4 = 12$; each off-diagonal gives $3 + 1 + 1 + 3 = 8$. Total $12 + 8 + 8 = 28$, so $P = 28/40 = 0.7$.

TIPS & TRICKS (AI-derived explanation)

- Organise the 25 cells by diagonals: on $y = x$ the weight is $|2x|$, on $y = x + 1$ it is $|2x + 1|$, on $y = x - 1$ it is $|2x - 1|$. Each diagonal is then a four- or five-term sum.

- The two off-diagonals give the same total by the symmetry of the weight in x and y - compute one and double it.
- Always divide by 40, not by 25: the distribution is not uniform on the grid.
- 0.7 and 0.3 are complements, and 0.06/0.12 are far too small to be a 13-cell event - a quick magnitude check rules them out.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The event $|X - Y| \leq 1$ covers the cells with $y = x, y = x + 1$ and $y = x - 1$, i.e. $5 + 4 + 4 = 13$ of the 25 cells.
2. Cells with $y = x$: weight $|x + y| = |2x| = 4, 2, 0, 2, 4$ for $x = -2, -1, 0, 1, 2$, total 12.
3. Cells with $y = x + 1$ ($x = -2, -1, 0, 1$): weight $|2x + 1| = 3, 1, 1, 3$, total 8.
4. Cells with $y = x - 1$ ($x = -1, 0, 1, 2$): weight $|2x - 1| = 3, 1, 1, 3$, total 8.
5. Total weight = $12 + 8 + 8 = 28$.
6. $P(|X - Y| \leq 1) = 28/40 = 0.7$.
7. Hence option (b) is correct.

35. 2026 · Q6 Conditional Expectation · Numerical**COMMON DATA FOR THIS GROUP**

Shared data for Q6–Q7. The joint pdf of X and Y is

$$f(x, y) = \begin{cases} \frac{2(x + 2y)}{3}, & 0 < x < 1, 0 < y < 1 \\ 0, & \text{elsewhere} \end{cases}$$

QUESTION

What is $E[X | Y = 0.5]$?

OPTIONS

- (a) $1/9$
- (b) $1/3$
- (c) $5/9$
- (d) $7/9$

ANSWER (AI-derived — no official key exists for this paper)

(c) $5/9$

EXAM SHORTCUT (AI-derived explanation)

$f_Y(y) = \int_0^1 2(x + 2y)/3 \, dx = (1 + 4y)/3$, so $f_Y(0.5) = 1$ exactly. Hence $f(x | y = 0.5) = 2(x + 1)/3$ and $E(X | Y = 0.5) = (2/3)(1/3 + 1/2) = 5/9$.

TIPS & TRICKS (AI-derived explanation)

- The marginal at $y = 0.5$ evaluating to exactly 1 is the designed simplification - the conditional density is then just the joint density with y substituted.
- Substitute $y = 0.5$ before integrating, not after; carrying a symbolic y through doubles the work.
- Verify the conditional integrates to 1: $\int_0^1 2(x + 1)/3 \, dx = (2/3)(1/2 + 1) = 1$. A ten-second check that catches algebra slips.

- Since the conditional density increases in x , $E(X | Y = 0.5)$ must exceed $1/2$ - that alone eliminates options (a) and (b).

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Marginal of Y : $f_Y(y) = \int_0^1 2(x + 2y)/3 \, dx = (2/3)[x^2/2 + 2xy]$ from 0 to 1 $= (2/3)(1/2 + 2y) = (1 + 4y)/3$, for $0 < y < 1$.
2. At $y = 0.5$: $f_Y(0.5) = (1 + 2)/3 = 1$.
3. Conditional density: $f(x | y = 0.5) = f(x, 0.5)/f_Y(0.5) = [2(x + 1)/3]/1 = 2(x + 1)/3$, for $0 < x < 1$.
4. Check: $\int_0^1 2(x + 1)/3 \, dx = (2/3)(1/2 + 1) = 1$, as required.
5. $E(X | Y = 0.5) = \int_0^1 x \cdot 2(x + 1)/3 \, dx = (2/3) \int_0^1 (x^2 + x) \, dx = (2/3)(1/3 + 1/2) = (2/3)(5/6) = 5/9$.
6. Hence option (c) is correct.

36. 2026 · Q7 Conditional Expectation · Numerical**COMMON DATA FOR THIS GROUP**

Shared data for Q6–Q7. The joint pdf of X and Y is

$$f(x, y) = \begin{cases} \frac{2(x + 2y)}{3}, & 0 < x < 1, 0 < y < 1 \\ 0, & \text{elsewhere} \end{cases}$$

QUESTION

What is the conditional variance of X given $Y = 0.5$?

OPTIONS

- (a) 11/162
- (b) 13/162
- (c) 5/54
- (d) 17/162

ANSWER (AI-derived — no official key exists for this paper)

(b) 13/162

EXAM SHORTCUT (AI-derived explanation)

With $f(x | y = 0.5) = 2(x + 1)/3$: $E(X^2 | \cdot) = (2/3)(1/4 + 1/3) = 7/18$ and $E(X | \cdot) = 5/9$.
 $\text{Var} = 7/18 - 25/81 = (63 - 50)/162 = 13/162$.

TIPS & TRICKS (AI-derived explanation)

- Reuse the conditional density and the conditional mean from the previous part; only $E(X^2 | Y = 0.5)$ is new work.
- Convert to the common denominator 162 immediately: $7/18 = 63/162$ and $25/81 = 50/162$, so the subtraction is a single step.
- Sanity bound: X lies in $(0, 1)$, so its conditional variance cannot exceed $1/4 = 40.5/162$; $13/162 = 0.080$ is comfortably inside, and even the uniform value $1/12 = 13.5/162$ is close, as expected for a gently sloping density.

- The distractors $11/162$, $5/54 = 15/162$ and $17/162$ all come from small slips in $E(X^2)$ or in squaring $5/9$ - keep fractions exact throughout.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. From the previous part, $f(x | Y = 0.5) = 2(x + 1)/3$ on $0 < x < 1$, with $E(X | Y = 0.5) = 5/9$.
2. $E(X^2 | Y = 0.5) = \int_0^1 x^2 \cdot 2(x + 1)/3 dx = (2/3) \int_0^1 (x^3 + x^2) dx = (2/3)(1/4 + 1/3) = (2/3)(7/12) = 7/18$.
3. $\text{Var}(X | Y = 0.5) = E(X^2 | Y = 0.5) - [E(X | Y = 0.5)]^2 = 7/18 - (5/9)^2 = 7/18 - 25/81$.
4. Common denominator 162: $7/18 = 63/162$ and $25/81 = 50/162$.
5. $\text{Var} = (63 - 50)/162 = 13/162$ (approximately 0.0802).
6. Hence option (b) is correct.

37. 2026 · Q8 Conditional Distributions · Numerical**COMMON DATA FOR THIS GROUP**

Shared data for Q8–Q9. The joint pdf of (X, Y) is

$$f(x, y) = \begin{cases} x^2 + \frac{xy}{3}, & 0 < x < 1, 0 < y < 2 \\ 0, & \text{otherwise} \end{cases}$$

QUESTION

The conditional density of Y given X is:

OPTIONS

- (a) $\frac{3x + y}{2(3x + 1)}$; $0 < y < 2, 0 < x < 1$
- (b) $\frac{x(3x + y)}{2x(3 + x)}$; $0 < y < 2, 0 < x < 1$
- (c) $\frac{x(3 + y)}{3x + 1}$; $0 < y < 2, 0 < x < 1$
- (d) $\frac{3x + y}{x(6 + y)}$; $0 < y < 2, 0 < x < 1$

ANSWER (AI-derived — no official key exists for this paper)

(a) $\frac{3x + y}{2(3x + 1)}$; $0 < y < 2, 0 < x < 1$

EXAM SHORTCUT (AI-derived explanation)

Integrate out y over $(0, 2)$: $f_X(x) = 2x^2 + 2x/3 = (2x/3)(3x + 1)$. Dividing the joint by this cancels the x , leaving $(3x + y)/[2(3x + 1)]$. Option (a).

TIPS & TRICKS (AI-derived explanation)

- Conditional of Y given X means divide by the marginal of X , so integrate the joint over the y -range - keep track of which variable is being eliminated.
- Factor the joint as $x(3x + y)/3$ before dividing; the common factor x then cancels visibly and the answer's shape becomes obvious.
- Elimination test: for a valid conditional density of Y given X , integrating over $0 < y < 2$ must give 1. Option (a): $\int (3x + y)/[2(3x + 1)] dy = (6x + 2)/[2(3x + 1)] = 1$. It passes; the others do not.
- The conditional pdf of Y given X must not have a stray free x that fails to normalise - options (b), (c), (d) all break the normalisation check.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Marginal of X: $f_X(x) = \int_0^2 (x^2 + xy/3) dy = [x^2y + xy^2/6]$ from 0 to 2 = $2x^2 + 4x/6 = 2x^2 + 2x/3$, for $0 < x < 1$.
2. Factor: $f_X(x) = (2x/3)(3x + 1)$.
3. Check normalisation: $\int_0^1 (2x^2 + 2x/3) dx = 2/3 + 1/3 = 1$, as required.
4. Joint density factored: $f(x, y) = x^2 + xy/3 = (x/3)(3x + y)$.
5. Conditional density: $f(y | x) = f(x, y)/f_X(x) = [(x/3)(3x + y)]/[(2x/3)(3x + 1)] = (3x + y)/[2(3x + 1)]$, for $0 < y < 2, 0 < x < 1$.
6. Verification: $\int_0^2 (3x + y)/[2(3x + 1)] dy = [6x + 2]/[2(3x + 1)] = 1$.
7. Hence option (a) is correct.

38. 2026 · Q9 Conditional Distributions · Numerical**COMMON DATA FOR THIS GROUP**

Shared data for Q8–Q9. The joint pdf of (X, Y) is

$$f(x, y) = \begin{cases} x^2 + \frac{xy}{3}, & 0 < x < 1, 0 < y < 2 \\ 0, & \text{otherwise} \end{cases}$$

QUESTION

The conditional density of X given Y is:

OPTIONS

- (a) $\frac{x(3x + y)}{2x(3 + x)}$; $0 < x < 1, 0 < y < 2$
- (b) $\frac{3x + y}{x(6 + y)}$; $0 < x < 1, 0 < y < 2$
- (c) $\frac{2x(3x + y)}{2 + y}$; $0 < x < 1, 0 < y < 2$
- (d) $\frac{x(6x + y)}{2 + xy}$; $0 < x < 1, 0 < y < 2$

ANSWER (AI-derived — no official key exists for this paper)

(c) $\frac{2x(3x + y)}{2 + y}$; $0 < x < 1, 0 < y < 2$

EXAM SHORTCUT (AI-derived explanation)

$f_Y(y) = \int_0^1 (x^2 + xy/3) dx = 1/3 + y/6 = (2 + y)/6$. Dividing the joint by this gives $6(x^2 + xy/3)/(2 + y) = 2 \times (3x + y)/(2 + y)$. Option (c).

TIPS & TRICKS (AI-derived explanation)

- Now integrate out x over (0, 1) - the opposite variable from the previous part; mixing the two up is the classic error.
- Write the marginal as a single fraction $(2 + y)/6$ before dividing, so the factor 6 comes up cleanly in the numerator.
- Normalisation check: $\int_0^1 2 \times (3x + y)/(2 + y) dx = (2 + y)/(2 + y) = 1$. Option (c) passes; test the others the same way and they fail.
- Compare with the previous answer: the conditional of Y given X kept x in the denominator's shape, whereas this one keeps y - a useful structural cross-check.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Marginal of Y: $f_Y(y) = \int_0^1 (x^2 + xy/3) dx = [x^3/3 + x^2y/6]$ from 0 to 1 = $1/3 + y/6 = (2+y)/6$, for $0 < y < 2$.
2. Check: $\int_0^2 (2+y)/6 dy = [2y + y^2/2]/6$ from 0 to 2 = $(4+2)/6 = 1$, as required.
3. Conditional density: $f(x|y) = f(x,y)/f_Y(y) = (x^2 + xy/3) \cdot 6/(2+y) = (6x^2 + 2xy)/(2+y) = 2 \times (3x+y)/(2+y)$, for $0 < x < 1, 0 < y < 2$.
4. Verification: $\int_0^1 2 \times (3x+y)/(2+y) dx = [2x^3 + x^2y]/(2+y)$ from 0 to 1 = $(2+y)/(2+y) = 1$.
5. Hence option (c) is correct.

39. 2026 · Q19 Random Vectors & Joint Distributions · Numerical

QUESTION

The joint pdf of X and Y is $f(x,y) = \begin{cases} \frac{21}{2}x^2y, & x^2 \leq y, 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$. What is $P(X \geq Y)$?

OPTIONS

- (a) 0.7
(b) 0.3
(c) 0.12
(d) 0.88

ANSWER (AI-derived — no official key exists for this paper)

(b) 0.3

EXAM SHORTCUT (AI-derived explanation)

The support is $x^2 \leq y \leq 1$, so $X \geq Y$ means $x^2 \leq y \leq x$. Integrating:
 $(21/2) \int x^2(x^2 - x^4)/2 dx = (21/4)(1/5 - 1/7) = (21/4)(2/35) = 0.3$.

TIPS & TRICKS (AI-derived explanation)

- Sketch the region first: $y = x^2$ is the lower boundary and $y = 1$ the upper; the line $y = x$ cuts it, and on $(0, 1)$ we have $x^2 \leq x$, so the y-limits are x^2 to x .
- Do the y-integral first (it is a plain power of y), leaving a single polynomial integral in x.
- Verify the constant: integral over the full support gives $(21/4)(1/3 - 1/7) = (21/4)(4/21) = 1$, so 21/2 is right and your limits are set up correctly.
- Note 0.7 is the complement $P(X < Y)$; read the inequality direction carefully before choosing.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The density $f(x,y) = (21/2)x^2y$ is supported on $\{0 < x < 1, x^2 \leq y \leq 1\}$.
2. Normalisation check: $\int_0^1 \int_{x^2}^1 (21/2)x^2y dy dx = \int_0^1 (21/4)x^2(1 - x^4) dx = (21/4)(1/3 - 1/7) = (21/4)(4/21) = 1$. Correct.
3. For $P(X \geq Y)$ we need $y \leq x$, and combined with the support $y \geq x^2$ the y-range is $x^2 \leq y \leq x$ (valid since $x^2 \leq x$ for $0 < x < 1$).
4. Inner integral: $\int_{x^2}^x (21/2)x^2y dy = (21/2)x^2 \cdot (x^2 - x^4)/2 = (21/4)(x^4 - x^6)$.
5. Outer integral: $(21/4) \int_0^1 (x^4 - x^6) dx = (21/4)(1/5 - 1/7) = (21/4)(2/35) = 42/140 = 0.3$.
6. Hence option (b) is correct.

40. **2026 · Q21** Conditional Expectation · Numerical**QUESTION**

X and Y have joint pdf $f(x, y) = \frac{1}{3}(x + y)$; $0 \leq x \leq 1$, $0 \leq y \leq 2$. What is $E[Y \mid X = \frac{1}{2}]$? (corrected from "X = 2" — see note above)

OPTIONS

- (a) 8/9
(b) 10/9
(c) 11/9
(d) 13/9

ANSWER (AI-derived — no official key exists for this paper)

(c) 11/9

EXAM SHORTCUT (AI-derived explanation)

$f_X(x) = (1/3)(2x + 2)$, so $f_X(1/2) = 1$ and the conditional density is just $(0.5 + y)/3$ on $(0, 2)$. Then $E(Y \mid X = 1/2) = (1/3)[0.5(2) + 8/3] = (1/3)(11/3) = 11/9$.

TIPS & TRICKS (AI-derived explanation)

- Integrate y over its full range 0 to 2 (not 0 to 1) when finding f_X - mismatched limits are the commonest slip on rectangular but non-square supports.
- $f_X(1/2) = (2/3)(1/2 + 1) = 1$ exactly, so the conditional density equals the joint density with $x = 1/2$ substituted - a designed simplification.
- Check the conditional integrates to 1: $\int_0^2 (0.5 + y)/3 dy = (1 + 2)/3 = 1$.
- Bound check: Y lies in $(0, 2)$ and the conditional density increases in y, so the answer must exceed 1; 8/9 is therefore impossible.

STEP-BY-STEP SOLUTION (AI-derived explanation)

- Verify the joint density: $\int_0^1 \int_0^2 (x + y)/3 dy dx = \int_0^1 (2x + 2)/3 dx = (1 + 2)/3 = 1$. Valid.
- Marginal of X: $f_X(x) = \int_0^2 (x + y)/3 dy = (1/3)[xy + y^2/2]$ from 0 to 2 = $(1/3)(2x + 2) = (2/3)(x + 1)$, for $0 \leq x \leq 1$.
- At $x = 1/2$: $f_X(1/2) = (2/3)(3/2) = 1$.
- Conditional density: $f(y \mid X = 1/2) = [(1/2 + y)/3]/1 = (0.5 + y)/3$, for $0 \leq y \leq 2$.
- Check: $\int_0^2 (0.5 + y)/3 dy = (1/3)[0.5(2) + 4/2] = (1/3)(1 + 2) = 1$. Valid.
- $E(Y \mid X = 1/2) = \int_0^2 y(0.5 + y)/3 dy = (1/3)[0.25y^2 + y^3/3]$ from 0 to 2 = $(1/3)(1 + 8/3) = (1/3)(11/3) = 11/9$.
- Hence option (c) is correct.

P7 · PROBABILITY

Distributions of Functions of Random Variables

15 questions · 2018: 3 2019: 2 2020: 3 2021: 1 2022: 1 2023: 2 2024: 2 2025: 0 2026: 1

Weight. 15 questions (7.7% of Probability) · appeared in 8 of 9 papers · peak year 2018 (3)**Examiner's pattern.** Pure transformation drills, and the same half-dozen results recycle: X/Y of two independent standard normals $\rightarrow$ Cauchy; $X \cdot Y$ of two iid exponentials $\rightarrow$ Laplace; $-\ln X$ for $U(0,1)$ $\rightarrow$ exponential; the probability integral transform $-\log F(X)$ $\rightarrow$ exponential(1); $Y = X^2$ on $(0,1)$; sum of iid exponentials $\rightarrow$ Gamma.**Must-know.** Jacobian method: $f_Y(y) = f_X(g^{-1}(y)) |d/dy g^{-1}(y)|$. For non-monotone maps sum over all branches. Memorise the six standard results above.**1. 2018 · Q8** Gamma Distribution · Conceptual

QUESTION

If $X \sim N(4, 100)$, the distribution of $Y = \frac{1}{2} \left(\frac{X - 4}{10} \right)^2$ is

OPTIONS

- (a) Exponential with mean 4
- (b) Chi-square with 1 degree of freedom
- (c) Gamma distribution with parameter $1/2$
- (d) Normal with mean 4 and variance 10

ANSWER (AI-derived — no official key exists for this paper)**(c)** Gamma distribution with parameter $1/2$ **EXAM SHORTCUT (AI-derived explanation)** $(X - 4)/10 = Z \sim N(0, 1)$, so $Z^2 \sim \chi_1^2 = \Gamma(\text{shape } 1/2, \text{scale } 2)$. Halving a Γ scales the scale parameter to 1, leaving Γ with parameter $1/2$ - option (c), not the χ^2 in option (b).**TIPS & TRICKS (AI-derived explanation)**

- The factor $1/2$ is the whole question. Z^2 is χ_1^2 ; $Z^2/2$ is NOT, because χ^2 has a fixed scale of 2. Read the coefficient before answering.
- Standard identity: $\chi_v^2 = \Gamma(\text{shape } v/2, \text{scale } 2)$. Dividing by 2 gives $\Gamma(\text{shape } v/2, \text{scale } 1)$, the ' Γ ' with parameter $v/2$ of Indian textbook notation.
- Variance 100 means $\sigma = 10$, so the standardising divisor is 10 and not 100. Mis-standardising is the second trap here.
- Option (b) is the answer to the question the examiner did NOT ask - it is correct for $Y = Z^2$. Distinguishing (b) from (c) is precisely the discrimination being tested.

STEP-BY-STEP SOLUTION (AI-derived explanation)

- $X \sim N(4, 100)$ means mean 4 and variance 100, so the standard deviation is 10 and $Z = (X - 4)/10 \sim N(0, 1)$.
- The statistic is $Y = (1/2)Z^2$.
- It is standard that $Z^2 \sim \chi^2$ with 1 degree of freedom, and χ_1^2 is the same as $\Gamma(\text{shape } 1/2, \text{scale } 2)$.
- If $W \sim \Gamma(\text{shape } a, \text{scale } b)$ then $cW \sim \Gamma(\text{shape } a, \text{scale } cb)$. With $c = 1/2$ and $b = 2$, $Y = Z^2/2 \sim \Gamma(\text{shape } 1/2, \text{scale } 1)$.

5. Direct check: the density of Y is $f(y) = y^{-1/2}e^{-y}/\Gamma(1/2)$ for $y > 0$, which is exactly the standard Γ density with parameter $1/2$.
6. Hence Y follows a Γ distribution with parameter $1/2$, option (c).

2. 2018 · Q9 Uniform / Rectangular Distribution · Theoretical

QUESTION

$X \sim G(\alpha_1, \beta)$, $Y \sim G(\alpha_2, \beta)$ independent. $\frac{X}{X+Y}$ is distributed as

OPTIONS

- (a) $G(\alpha_1 + \alpha_2, \beta)$
- (b) $\beta_1(\alpha_1, \alpha_2)$
- (c) $U(0, 1)$
- (d) $G(\alpha_1, \alpha_2)$

ANSWER (AI-derived — no official key exists for this paper)

(b) $\beta_1(\alpha_1, \alpha_2)$

EXAM SHORTCUT (AI-derived explanation)

The Gamma-Beta connection: for independent $G(a_1, b)$ and $G(a_2, b)$ with the SAME scale, $X/(X+Y) \sim \text{Beta first kind } (a_1, a_2)$, independent of the total $X+Y$. Read it straight off.

TIPS & TRICKS (AI-derived explanation)

- Both Gammas must share the same scale parameter β for this to hold - that is why the question states it. If the scales differed the ratio would not be Beta.
- Companion results worth memorising together: $X+Y \sim G(a_1+a_2, b)$; $X/Y \sim \text{Beta second kind } (a_1, a_2)$; and $X/(X+Y)$ is independent of $X+Y$.
- Range check: $X/(X+Y)$ lies in $(0, 1)$, which immediately rules out (a) $G(a_1+a_2, b)$ and (d) $G(a_1, a_2)$ since Γ variables are unbounded above.
- Option (c) $U(0, 1)$ is only the special case $a_1 = a_2 = 1$, so it is too narrow to be the general answer.

STEP-BY-STEP SOLUTION (AI-derived explanation)

- Let $X \sim \Gamma(a_1, b)$ and $Y \sim \Gamma(a_2, b)$ be independent, sharing the common scale parameter b .
- Apply the transformation $U = X/(X+Y)$, $V = X+Y$, so $X = UV$ and $Y = V(1-U)$. The Jacobian of the transformation is V .
- The joint density becomes proportional to $[(uv)^{a_1-1}(v(1-u))^{a_2-1}e^{-v/b}]v = [u^{a_1-1}(1-u)^{a_2-1}][v^{a_1+a_2-1}e^{-v/b}]$.
- This factorises into a function of u alone and a function of v alone, so U and V are independent.
- The u -factor $u^{a_1-1}(1-u)^{a_2-1}$ on $(0, 1)$ is exactly the Beta distribution of the first kind with parameters a_1 and a_2 .
- Hence $X/(X+Y) \sim \text{Beta first kind } (a_1, a_2)$, option (b).

3. 2018 · Q20 Beta Distribution · Numerical**QUESTION**

X has p.d.f. $f(x) = \frac{kx^3}{(1+2x)^6}$, $x > 0$. $Y = \frac{2X}{1+2X}$ is a Beta distribution of the first kind with parameters:

OPTIONS

- (a) $m = 4, n = 2$
 (b) $m = 5, n = 1$
 (c) $m = 3, n = 3$
 (d) $m = 5, n = 2$

ANSWER (AI-derived — no official key exists for this paper)

(a) $m = 4, n = 2$

EXAM SHORTCUT (AI-derived explanation)

Substitute $y = 2x/(1+2x)$, so $1+2x = 1/(1-y)$ and $x = y/(2(1-y))$. The density picks up $x^3(1+2x)^{-6} |dx/dy| \sim y^3(1-y)^1$, i.e. Beta(4, 2). So $m = 4, n = 2$.

TIPS & TRICKS (AI-derived explanation)

- Match exponents at the end: $y^{m-1}(1-y)^{n-1} = y^3(1-y)^1$ gives $m = 4$ and $n = 2$ by inspection. Never integrate to identify a Beta.
- The transformation $y = 2x/(1+2x)$ is the standard Beta-second-kind to Beta-first-kind map; its inverse relation $1-y = 1/(1+2x)$ is the piece that does all the work.
- Check the parameter total: a density of the form $x^{m-1}/(1+2x)^{m+n}$ transforms to Beta(m, n). Here the exponent $6 = m+n$ and the numerator power $3 = m-1$, giving $m = 4, n = 2$ in one line.
- Do not forget the Jacobian $dx/dy = 1/(2(1-y)^2)$ - omitting it changes n from 2 to 4 and lands you on a wrong option.

STEP-BY-STEP SOLUTION (AI-derived explanation)

- Let $Y = 2X/(1+2X)$. Then $1-Y = 1/(1+2X)$, so $1+2X = 1/(1-Y)$ and $X = Y/(2(1-Y))$, with Y in $(0, 1)$ as X ranges over $(0, \infty)$.
- Differentiating, $dx/dy = 1/(2(1-y)^2)$.
- The density of X is $f(x) = kx^3/(1+2x)^6$. Substituting: $x^3 = y^3/(8(1-y)^3)$ and $(1+2x)^{-6} = (1-y)^6$.
- So $f(x) = k[y^3/(8(1-y)^3)](1-y)^6 = (k/8)y^3(1-y)^3$.
- The density of Y is $g(y) = f(x) |dx/dy| = (k/8)y^3(1-y)^3 \cdot 1/(2(1-y)^2) = (k/16)y^3(1-y)$.
- Comparing with the Beta first kind density proportional to $y^{m-1}(1-y)^{n-1}$: $m-1 = 3$ and $n-1 = 1$, so $m = 4$ and $n = 2$, option (a).

4. 2019 · Q31 Discrete & Continuous Random Variables · Numerical**QUESTION**

Let X have p.d.f. $f(x) = \frac{2x}{\pi^2}$, $0 < x < \pi$. Let $Y = \sin X$. For $0 < y < 1$, the p.d.f. of Y is given by

OPTIONS

- (a) $\frac{2\pi}{\sqrt{1-y^2}}$
 (b) $\frac{\pi}{2}\sqrt{1-y^2}$

(c) $\frac{2}{\pi} \sqrt{1-y^2}$

(d) $\frac{2}{\pi \sqrt{1-y^2}}$

ANSWER (AI-derived — no official key exists for this paper)

(d) $\frac{2}{\pi \sqrt{1-y^2}}$

EXAM SHORTCUT (AI-derived explanation)

$\sin x = y$ has TWO roots in $(0, \pi)$, namely x and $\pi - x$, and they sum to π . Since f is proportional to x , the two contributions add to $2\pi/\pi^2$, and dividing by $|\cos x| = \sqrt{1-y^2}$ gives $2/(\pi \sqrt{1-y^2})$.

TIPS & TRICKS (AI-derived explanation)

- Always count the pre-images before transforming: on $(0, \pi)$ the sine is two-to-one, so the density of Y is a SUM of two terms.
- The elegance here is that $x_1 + x_2 = \pi$ exactly, so the linear density $2x/\pi^2$ sums to $2\pi/\pi^2 = 2/\pi$ independently of y - the answer needs no arcsine at all.
- $|dy/dx| = |\cos x| = \sqrt{1-\sin^2 x} = \sqrt{1-y^2}$ at BOTH roots, so the Jacobian factor is common and can be taken outside.
- Validity check: the $\int 2/(\pi \sqrt{1-y^2})$ over $(0, 1)$ is $(2/\pi)(\pi/2) = 1$. Options (a), (b) and (c) fail this test.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. X has density $f(x) = 2x/\pi^2$ on $(0, \pi)$, and $Y = \sin X$ takes values in $(0, 1)$.
2. For a given y in $(0, 1)$, the equation $\sin x = y$ has two solutions in $(0, \pi)$: $x_1 = \arcsin(y)$ in $(0, \pi/2)$ and $x_2 = \pi - \arcsin(y)$ in $(\pi/2, \pi)$.
3. At both roots $|dy/dx| = |\cos x| = \sqrt{1-y^2}$.
4. By the change-of-variable formula for a two-to-one map,
 $g(y) = f(x_1)/\sqrt{1-y^2} + f(x_2)/\sqrt{1-y^2} = [f(x_1) + f(x_2)]/\sqrt{1-y^2}$.
5. Now $f(x_1) + f(x_2) = (2/\pi^2)(x_1 + x_2) = (2/\pi^2)(\pi) = 2/\pi$, since $x_1 + x_2 = \pi$.
6. Therefore $g(y) = 2/(\pi \sqrt{1-y^2})$ for $0 < y < 1$, option (d). (Check: its integral over $(0, 1)$ is $(2/\pi)(\pi/2) = 1$.)

5. **2019 · Q40** Cauchy Distribution · Theoretical**QUESTION**

Let X and Y be two i.i.d. random variables with p.d.f. $f(x) = 2e^{-2x}$, $x \geq 0$. The distribution of $Z = X - Y$ is

OPTIONS

- (a) exponential
- (b) beta
- (c) Laplace
- (d) Cauchy

ANSWER (AI-derived — no official key exists for this paper)

(c) Laplace

EXAM SHORTCUT (AI-derived explanation)

The difference of two i.i.d. exponentials is symmetric about 0 with an exponential decay on each side - that is exactly the Laplace (double exponential) distribution.

TIPS & TRICKS (AI-derived explanation)

- Memorise: $X - Y$ for i.i.d. $\text{Exp}(\lambda)$ is Laplace with density $(\lambda/2)e^{-\lambda|z|}$ on the whole line. Here $\lambda = 2$, so $f(z) = e^{-2|z|}$.
- Support test alone settles it: Z can be negative, but an exponential and a β variable cannot. Options (a) and (b) are eliminated in one glance.
- MGF route: $M_Z(t) = M_X(t)M_Y(-t) = [2/(2-t)][2/(2+t)] = 4/(4-t^2)$, which is the Laplace MGF. A Cauchy variable has no MGF at all, ruling out option (d).
- Note the companion result from 2018Q₁₂: $|X - Y|$ for i.i.d. exponentials is again exponential. Taking the absolute value of a Laplace variable gives an exponential - the two facts fit together.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. X and Y are i.i.d. with density $2e^{-2x}$ for $x \geq 0$, i.e. Exponential with rate 2.
2. $Z = X - Y$ is symmetric about 0, since X and Y are exchangeable, and it can take any real value.
3. For $z > 0$, the density of Z is the $\int f(z+y)f(y)dy = \text{the } \int 4e^{-2(z+y)}e^{-2y}dy = 4e^{-2z}(1/4) = e^{-2z}$.
4. By symmetry the density is e^{2z} for $z < 0$, so $f_Z(z) = e^{-2|z|}$ for all real z .
5. This is the Laplace (double exponential) density with scale 1/2, confirmed by the MGF
 $M_Z(t) = [2/(2-t)][2/(2+t)] = 4/(4-t^2)$.
6. Hence option (c) is correct.

6. 2020 · Q17 Normal Distribution · Theoretical**QUESTION**

X, Y are independent standard normal. The distribution of $U = \left(\frac{X-Y}{X+Y}\right)^2$ is

OPTIONS

- (a) Chi-square with 2 df
- (b) Chi-square with 1 df
- (c) F with (2, 2) df
- (d) F with (1, 1) df

ANSWER (AI-derived — no official key exists for this paper)

(d) F with (1, 1) df

EXAM SHORTCUT (AI-derived explanation)

$X - Y$ and $X + Y$ are independent $N(0, 2)$ variables (their covariance is $\text{Var}(X) - \text{Var}(Y) = 0$). So U is the ratio of two independent $\chi_1^2/1$ quantities, i.e. $F(1, 1)$.

TIPS & TRICKS (AI-derived explanation)

- For i.i.d. normals, the sum and the difference are INDEPENDENT - a special property of the normal family that this question is built on.

- $\text{Cov}(X - Y, X + Y) = \text{Var}(X) - \text{Var}(Y) = 0$, and joint normality then upgrades zero correlation to full independence.
- The common variance 2 cancels in the ratio, so no scaling constant survives - the answer is a pure F with 1 and 1 degrees of freedom.
- Equivalently $(X - Y)/(X + Y)$ is a standard Cauchy, and the square of a Cauchy is $F(1, 1)$. Both routes give option (d).

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. X and Y are independent $N(0, 1)$, so $X - Y \sim N(0, 2)$ and $X + Y \sim N(0, 2)$.
2. $\text{Cov}(X - Y, X + Y) = \text{Var}(X) - \text{Var}(Y) = 1 - 1 = 0$, and since the pair is jointly normal, $X - Y$ and $X + Y$ are INDEPENDENT.
3. Set $A = (X - Y)/\sqrt{2}$ and $B = (X + Y)/\sqrt{2}$; these are independent $N(0, 1)$.
4. Then $U = ((X - Y)/(X + Y))^2 = A^2/B^2$, and A^2 and B^2 are independent χ^2 variables each with 1 degree of freedom.
5. By definition, the ratio $(\chi_1^2/1)/(\chi_1^2/1)$ of independent chi-squares follows $F(1, 1)$.
6. Hence $U \sim F(1, 1)$, option (d).

7. 2020 · Q24 Discrete & Continuous Random Variables · Numerical

QUESTION

X, Y are independent normal with means 1, 2 and SDs 3, 4. $Z = X - Y$. 1. Median and SD of Z are -1 and 5 respectively. 2. $P(Z + 1 < 0) = 0.5$.

OPTIONS

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

ANSWER (AI-derived — no official key exists for this paper)

(c) Both 1 and 2

EXAM SHORTCUT (AI-derived explanation)

$Z = X - Y \sim N(1 - 2, 9 + 16) = N(-1, 25)$, so its median equals its mean -1 and its SD is 5 - statement 1 holds. And $P(Z + 1 < 0) = P(Z < -1) = P(Z < \text{median}) = 0.5$ - statement 2 holds.

TIPS & TRICKS (AI-derived explanation)

- Convert SDs to variances before adding: 3 and 4 give 9 and 16, whose sum is 25 and whose square root is 5. Adding the SDs directly would give 7 - the classic error.
- Variances ADD for a difference of independent variables; the minus sign disappears on squaring.
- The normal distribution is symmetric, so mean = median = mode. Statement 2 is then just 'the probability of being below the median is $1/2$ '.
- Note how statements 1 and 2 reinforce each other: once the median is identified as -1 , statement 2 follows without any table look-up.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. $X \sim N(1, 3^2) = N(1, 9)$ and $Y \sim N(2, 4^2) = N(2, 16)$, independent.
2. $Z = X - Y$ is normal with mean $1 - 2 = -1$ and variance $9 + 16 = 25$, so $SD(Z) = 5$.
3. Because the normal distribution is symmetric about its mean, the median of Z equals its mean, -1 . Statement 1 is TRUE.
4. $P(Z + 1 < 0) = P(Z < -1)$.
5. Since -1 is the median of Z , this probability is exactly 0.5 . Statement 2 is TRUE.
6. Both statements are correct, option (c).

8. 2020 · Q38 Random Vectors & Joint Distributions · Numerical**QUESTION**

The random vector (X, Y, Z) has density f on $\mathbb{R}^3$. The random vector $\left(\frac{X+Y}{2}, \frac{Y+Z}{2}, \frac{Z+X}{2}\right)$ has density

OPTIONS

- (a) $g(u, v, w) = f(u - v + w, u + v - w, -u + v + w)$
- (b) $g(u, v, w) = 4f(u - v + w, u + v - w, -u + v + w)$
- (c) $g(u, v, w) = 4f\left(\frac{u - v + w}{2}, \frac{u + v - w}{2}, \frac{-u + v + w}{2}\right)$
- (d) $g(u, v, w) = 8f\left(\frac{u - v + w}{2}, \frac{u + v - w}{2}, \frac{-u + v + w}{2}\right)$

ANSWER (AI-derived — no official key exists for this paper)

(b) $g(u, v, w) = 4f(u - v + w, u + v - w, -u + v + w)$

EXAM SHORTCUT (AI-derived explanation)

The linear map has matrix $(1/2)[[1, 1, 0], [0, 1, 1], [1, 0, 1]]$ with determinant $(1/8)(2) = 1/4$, so the density picks up the factor $1/|J| = 4$. Inverting gives $x = u - v + w, y = u + v - w, z = -u + v + w$.

TIPS & TRICKS (AI-derived explanation)

- For a linear transformation, $g(u) = f(A^{-1}u)/|\det A|$. Compute $\det A$ once; here $\det$ of $[[1, 1, 0], [0, 1, 1], [1, 0, 1]]$ is 2 , and the factor $(1/2)^3$ gives $\det A = 1/4$.
- Invert cleverly: adding all three equations gives $x + y + z = u + v + w$, and then each variable follows by subtracting twice one of the new coordinates.
- Check the inverse by substitution: $(x + y)/2 = ((u - v + w) + (u + v - w))/2 = u$. Do this once and the arguments are confirmed.
- Options (c) and (d) halve the arguments as well, which would correspond to a different transformation entirely - the inverse here has NO factor of $1/2$ inside f .

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Write $U = (X + Y)/2, V = (Y + Z)/2, W = (Z + X)/2$, a linear transformation with matrix $A = (1/2)[[1, 1, 0], [0, 1, 1], [1, 0, 1]]$.
2. The determinant of $[[1, 1, 0], [0, 1, 1], [1, 0, 1]]$ is $1(1) - 1(-1) + 0 = 2$, so $\det A = (1/2)^3(2) = 1/4$.
3. Invert: from $x + y = 2u, y + z = 2v, z + x = 2w$, adding gives $2(x + y + z) = 2(u + v + w)$, so $x + y + z = u + v + w$.

4. Then $x = (x + y + z) - (y + z) = (u + v + w) - 2v = u - v + w$; $y = (x + y + z) - (z + x) = u + v - w$;
 $z = (x + y + z) - (x + y) = -u + v + w$.
5. The density transforms as $g(u, v, w) = f(x, y, z)/|\det A| = 4f(u - v + w, u + v - w, -u + v + w)$.
6. Hence option (b) is correct.

9. 2021 · Q27 Exponential Distribution · Theoretical

QUESTION

Let X have continuous c.d.f. F . Statement-1: $Y = -\log F(X)$ is exponentially distributed. Statement-2: $F(X)$ is uniformly distributed on $[0, 1]$.

OPTIONS

- (a) Both true, Statement-2 is the correct explanation of Statement-1
- (b) Both true, Statement-2 is *not* the correct explanation of Statement-1
- (c) Statement-1 true, Statement-2 false
- (d) Statement-1 false, Statement-2 true

ANSWER (AI-derived — no official key exists for this paper)

(a) Both true, Statement-2 is the correct explanation of Statement-1

EXAM SHORTCUT (AI-derived explanation)

$F(X) \sim U(0, 1)$ by the probability integral transform, and $-\log U \sim \text{Exp}(1)$ for U uniform. So both statements are true and Statement-2 is precisely why Statement-1 holds.

TIPS & TRICKS (AI-derived explanation)

- Probability integral transform: for continuous F , $U = F(X)$ is exactly $\text{Uniform}(0, 1)$. This is the workhorse of simulation and of distribution-free methods.
- Companion fact: if $U \sim U(0, 1)$ then $-\log U \sim \text{Exponential}(1)$, since $P(-\log U > t) = P(U < e^{-t}) = e^{-t}$.
- For 'assertion and reason' items, first check both statements independently, THEN ask whether the second is the mechanism behind the first. Here it plainly is.
- The chain $U(0, 1) \rightarrow -\log U \rightarrow \text{Exp}(1)$ is also the inverse-transform method for generating exponential random numbers - a useful anchor.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Statement-2: for a continuous CDF F , $P(F(X) \leq u) = P(X \leq F^{-1}(u)) = u$ for $0 < u < 1$, so $F(X) \sim \text{Uniform}(0, 1)$. TRUE.
2. Statement-1: put $U = F(X)$, which is $\text{Uniform}(0, 1)$ by Statement-2, and let $Y = -\log U$.
3. For $t > 0$, $P(Y > t) = P(-\log U > t) = P(U < e^{-t}) = e^{-t}$.
4. That is the survival function of an $\text{Exponential}(1)$ variable, so Y is exponentially distributed. TRUE.
5. The derivation of Statement-1 rests entirely on the uniformity of $F(X)$, so Statement-2 is the correct explanation of Statement-1.
6. Option (a) is correct.

10. **2022 · Q60** Exponential Distribution · Numerical**QUESTION**

If $X \sim U[0, 1]$, the pdf of $(-2 \log X)$ is:

OPTIONS

- (a) Uniform
- (b) $\beta\left(\frac{1}{2}, \frac{1}{2}\right)$
- (c) $\chi^2_{n/2}$ degrees of freedom
- (d) Exponential with $\theta = 1/2$

ANSWER (AI-derived — no official key exists for this paper)

(d) Exponential with $\theta = 1/2$

EXAM SHORTCUT (AI-derived explanation)

$P(-2 \log X > y) = P(X < e^{-y/2}) = e^{-y/2}$, the survival function of an Exponential with rate 1/2 (equivalently χ^2 with 2 d.f.).

TIPS & TRICKS (AI-derived explanation)

- Work with the SURVIVAL function for monotone transformations - it avoids Jacobians entirely and the algebra is one line.
- $-\log U \sim \text{Exp}(1)$ for U uniform, so $-2 \log U \sim \text{Exp}(\text{rate } 1/2)$ with mean 2. Scaling an exponential by c divides the rate by c .
- $\text{Exp}(\text{rate } 1/2)$ is exactly χ^2 with 2 degrees of freedom - a fact worth carrying, and the basis of the Box-Muller and inverse-transform simulation methods.
- Option (c) names a χ^2 with $n/2$ degrees of freedom, but no n appears in the problem - a structural giveaway that it is a distractor.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Let $X \sim U[0, 1]$ and $Y = -2 \log X$, which takes values in $(0, \infty)$.
2. For $y > 0$, $P(Y > y) = P(-2 \log X > y) = P(\log X < -y/2) = P(X < e^{-y/2})$.
3. Since X is uniform on $[0, 1]$, this probability is simply $e^{-y/2}$.
4. So the survival function of Y is $e^{-y/2}$, which is that of an exponential distribution with rate parameter 1/2 (mean 2).
5. Differentiating, the density is $f(y) = (1/2)e^{-y/2}$ for $y > 0$. Equivalently $Y \sim \chi^2$ with 2 degrees of freedom.
6. Hence the distribution is exponential with $\theta = 1/2$, option (d).

11. **2023 · Q5** Discrete & Continuous Random Variables · Numerical**QUESTION**

If $X \sim \text{log-normal}(\mu, \sigma^2)$ and $Y = \sigma X^\mu$; $\sigma > 0$, $-\infty < \mu < \infty$, then what is the value of

$$\text{Var} \left(\left(\log \frac{Y}{\sigma} - \mu^2 \right)^2 \right)?$$

OPTIONS

- (a) $105\mu^4\sigma^4$

- (b) $15\mu^4\sigma^4$
- (c) $3\mu^4\sigma^4$
- (d) $2\mu^4\sigma^4$

ANSWER (AI-derived — no official key exists for this paper)

(d) $2\mu^4\sigma^4$

EXAM SHORTCUT (AI-derived explanation)

$\log(Y/\sigma) = \mu \log X = \mu W$ with $W \sim N(\mu, \sigma^2)$, so the bracket is $\mu(W - \mu) = \mu\sigma Z$. Its square is $\mu^2\sigma^2 Z^2$ and $\text{Var}(Z^2) = 2$, giving $2\mu^4\sigma^4$.

TIPS & TRICKS (AI-derived explanation)

- Peel the transformation one layer at a time: $Y = \sigma X^\mu$ gives $Y/\sigma = X^\mu$, so $\log(Y/\sigma) = \mu \log X$ - the σ cancels by design.
- X lognormal means $W = \log X \sim N(\mu, \sigma^2)$, so $\mu W - \mu^2 = \mu(W - \mu)$, a centred normal with SD $\mu\sigma$.
- The square of a centred normal is $(\mu\sigma)^2$ times a χ_1^2 , and $\text{Var}(\chi_1^2) = 2$ - the source of the factor 2.
- The distractors encode wrong χ^2 variances: 3 (option c) and 15 or 105 (options b, a) come from higher moments of the normal, so pinning $\text{Var}(Z^2) = 2$ is decisive.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. X is lognormal(μ, σ^2), so $W = \log X$ follows $N(\mu, \sigma^2)$.
2. $Y = \sigma X^\mu$ gives $Y/\sigma = X^\mu$, hence $\log(Y/\sigma) = \mu \log X = \mu W$.
3. Therefore $\log(Y/\sigma) - \mu^2 = \mu W - \mu^2 = \mu(W - \mu)$.
4. Write $W - \mu = \sigma Z$ with $Z \sim N(0, 1)$. Then the bracket is $\mu\sigma Z$ and its square is $\mu^2\sigma^2 Z^2$.
5. Since $Z^2 \sim \chi^2$ with 1 degree of freedom, $\text{Var}(Z^2) = 2$.
6. Hence $\text{Var}(\mu^2\sigma^2 Z^2) = \mu^4\sigma^4 \text{Var}(Z^2) = 2\mu^4\sigma^4$, option (d).

12. 2023 · Q12 Exponential Distribution · Conceptual

QUESTION

If X, Y and Z are iid Exponential(1) variates, then the distribution of $\frac{X + Y + Z}{3}$ is:

OPTIONS

- (a) Exponential(1/3)
- (b) Exponential(3)
- (c) Gamma($n = 3, \lambda = 1/3$)
- (d) Gamma($n = 3, \lambda = 3$)

ANSWER (AI-derived — no official key exists for this paper)

(d) Gamma($n = 3, \lambda = 3$)

EXAM SHORTCUT (AI-derived explanation)

The sum of three i.i.d. Exp(1) variables is $\Gamma(\text{shape } 3, \text{rate } 1)$; dividing by 3 multiplies the rate by 3, giving

$$\Gamma(n = 3, \lambda = 3).$$

TIPS & TRICKS (AI-derived explanation)

- Additive property: the sum of n i.i.d. $\text{Exp}(\lambda)$ variables is $\Gamma(\text{shape } n, \text{rate } \lambda)$. Here the sum is $\Gamma(3, 1)$.
- Scaling rule: if $W \sim \Gamma(\text{shape } a, \text{rate } b)$ then $W/c \sim \Gamma(\text{shape } a, \text{rate } bc)$. Dividing by 3 raises the rate from 1 to 3, leaving the shape unchanged.
- The SHAPE never changes under scaling, so any answer with shape other than 3 - options (a) and (b), which are exponential, i.e. shape 1 - is wrong.
- Check the mean: E of the average of three $\text{Exp}(1)$ variables is 1, and $\Gamma(3, \text{rate } 3)$ has mean $3/3 = 1$. Confirmed.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. X, Y, Z are i.i.d. $\text{Exponential}(1)$, which is Γ with shape 1 and rate 1.
2. By the additive property of the Γ family with a common rate, $X + Y + Z \sim \Gamma(\text{shape } 3, \text{rate } 1)$.
3. Let $W = X + Y + Z$ and consider $W/3$.
4. If $W \sim \Gamma(\text{shape } a, \text{rate } b)$, then $W/c \sim \Gamma(\text{shape } a, \text{rate } bc)$; with $c = 3$ and $b = 1$ the rate becomes 3.
5. Hence $(X + Y + Z)/3 \sim \Gamma$ with $n = 3$ and $\lambda = 3$, whose mean is $3/3 = 1$, agreeing with the mean of an average of $\text{Exp}(1)$ variables.
6. Option (d) is correct.

13. 2024 · Q22 Gamma Distribution · Theoretical**QUESTION**

Let $(X_1, X_2, X_3, \dots, X_n)$ be a random sample from $U(0, 1)$ distribution. $-\ln X$ follows:

OPTIONS

- (a) gamma distribution
- (b) negative exponential distribution
- (c) half-normal distribution
- (d) None of the above

ANSWER (AI-derived — no official key exists for this paper)

(b) negative exponential distribution

EXAM SHORTCUT (AI-derived explanation)

$P(-\ln X > t) = P(X < e^{-t}) = e^{-t}$ for X uniform on $(0, 1)$, which is the survival function of the negative exponential with mean 1.

TIPS & TRICKS (AI-derived explanation)

- Use the survival function for a monotone transformation: it avoids Jacobians and gives the answer in one line.
- $\text{Exp}(1)$ is also $\Gamma(1, 1)$, so option (a) is not strictly false - but 'negative exponential' is the precise and standard description asked for.
- This is the inverse-transform method for simulating exponential variates, so the identification is worth knowing cold.
- The half-normal arises from $|N(0, 1)|$, an entirely different construction with no logarithm involved.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Let $X \sim U(0, 1)$ and set $W = -\ln X$, which takes values in $(0, \infty)$.
2. For $t > 0$, $P(W > t) = P(-\ln X > t) = P(\ln X < -t) = P(X < e^{-t})$.
3. Since X is uniform on $(0, 1)$, this probability is simply e^{-t} .
4. The survival function e^{-t} identifies the negative exponential distribution with parameter 1, whose density is e^{-w} for $w > 0$.
5. (Equivalently this is $\Gamma(1, 1)$, but the exponential is the exact and standard description.)
6. Option (b) is correct.

14. 2024 · Q39 Discrete & Continuous Random Variables · Numerical**QUESTION**

Suppose $Y = X^2$ and $f(x) = 1$, $0 < x < 1$. What is the pdf of Y ?

OPTIONS

- (a) $\sqrt{y}$, $0 < y < 1$
- (b) $\frac{\sqrt{y}}{2}$, $0 < y < 1$
- (c) $\frac{1}{2\sqrt{y}}$, $0 < y < 1$
- (d) y , $0 < y < 1$

ANSWER (AI-derived — no official key exists for this paper)

(c) $\frac{1}{2\sqrt{y}}$, $0 < y < 1$

EXAM SHORTCUT (AI-derived explanation)

For $X \sim U(0, 1)$ and $Y = X^2$, $x = \sqrt{y}$ and $dx/dy = 1/(2\sqrt{y})$, so $f_Y(y) = 1 \cdot 1/(2\sqrt{y})$ on $(0, 1)$.

TIPS & TRICKS (AI-derived explanation)

- One-to-one on $(0, 1)$: the square is increasing there, so no doubling is needed - just invert and multiply by the Jacobian.
- $|dx/dy| = 1/(2\sqrt{y})$ is the whole content of the transformation, since $f_X = 1$ contributes nothing.
- Validity check: the $\int 1/(2\sqrt{y})$ over $(0, 1)$ is $[\sqrt{y}]$ from 0 to 1 = 1. Options (a), (b) and (d) integrate to $2/3$, $1/3$ and $1/2$, so all are invalid.
- The density blows up at $y = 0$, which is correct: squaring crowds the small values of X towards 0.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. X has density $f(x) = 1$ on $(0, 1)$, and $Y = X^2$ maps $(0, 1)$ one-to-one onto $(0, 1)$ increasingly.
2. Inverting, $x = \sqrt{y}$ for $0 < y < 1$.
3. The Jacobian is $dx/dy = 1/(2\sqrt{y})$.
4. By the change-of-variable formula, $f_Y(y) = f_X(\sqrt{y}) |dx/dy| = 1 \cdot 1/(2\sqrt{y})$.
5. So $f_Y(y) = 1/(2\sqrt{y})$ for $0 < y < 1$, whose integral is $[\sqrt{y}]$ from 0 to 1 = 1, confirming it is a valid density.
6. Option (c) is correct.

15. **2026 · Q11** Cauchy Distribution · Theoretical**QUESTION**

If $X \sim N(0, 1)$ and $Y \sim N(0, 1)$ are independent, the distribution of X/Y is:

OPTIONS

- (a) Chi-square with 2 df
- (b) Cauchy(0,1)
- (c) Cauchy(1,1)
- (d) F-distribution with df (1,1)

ANSWER (AI-derived — no official key exists for this paper)

(b) Cauchy(0,1)

EXAM SHORTCUT (AI-derived explanation)

The ratio of two independent standard normals is the standard Cauchy - a memorised standard result.
Answer (b).

TIPS & TRICKS (AI-derived explanation)

- Memorise the family: X/Y is Cauchy(0, 1); X^2/Y^2 is $F(1, 1)$; $X^2 + Y^2$ is χ^2 with 2 df; $X/|Y|$ is also Cauchy.
- Option (d) is the trap: $F(1, 1)$ is the distribution of the ratio of SQUARES, not of the variables themselves.
- The Cauchy has no mean and no variance, which is why the ratio of two well-behaved normals is famously pathological - a favourite examiner point.
- Cauchy(0, 1) has density $1/[\pi(1 + z^2)]$ on the whole real line; the location 0 and scale 1 follow from both normals having mean 0 and variance 1, so (c) with location 1 is wrong.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Let $Z = X/Y$ with X, Y independent $N(0, 1)$.
2. Using the transformation $Z = X/Y, W = Y$ with Jacobian $|w|$, the joint density of (Z, W) is $|w| \cdot (1/(2\pi)) \exp(-(z^2 w^2 + w^2)/2)$.
3. Integrating out w : $f_Z(z) = (1/(2\pi)) \cdot \int |w| \exp(-w^2(1 + z^2)/2) dw = (1/(2\pi)) \cdot 2/(1 + z^2) = 1/[\pi(1 + z^2)]$.
4. This is exactly the standard Cauchy density, i.e. Cauchy with location 0 and scale 1.
5. χ^2 with 2 df (option a) is the distribution of $X^2 + Y^2$, not X/Y ; $F(1, 1)$ (option d) is the distribution of X^2/Y^2 ; Cauchy(1, 1) (option c) has the wrong location parameter.
6. Hence option (b) is correct.

P8 · PROBABILITY

Mathematical Expectation & Conditional Expectation

24 questions · 2018: 4 2019: 0 2020: 2 2021: 3 2022: 2 2023: 3 2024: 8 2025: 1 2026: 1

Weight. 24 questions (12.2% of Probability) · appeared in 8 of 9 papers · peak year 2024 (8)**Examiner's pattern.** Mostly computational: $E(XY)$ and Var of linear combinations from a joint law, conditional expectation from a table or density, and the tower property. A recurring trap is a series whose sum is only conditionally convergent — the question is whether the mean exists at all, not what it equals.**Must-know.** $\text{Var}(aX+bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X,Y)$; $E(X) = E[E(X|Y)]$; $\text{Var}(X) = E[\text{Var}(X|Y)] + \text{Var}[E(X|Y)]$; a mean exists only if $E|X| < \infty$.**1. 2018 · Q6** Discrete & Continuous Random Variables · Conceptual

QUESTION

 X has p.d.f. $f(x) = 2/x^3$, $x \geq 1$ (and 0 for $x < 1$). Which is correct?

OPTIONS

- (a) Both mean and variance exist.
- (b) Mean exists but variance does not.
- (c) Neither mean nor variance exists.
- (d) Variance exists but mean does not.

ANSWER (AI-derived — no official key exists for this paper)**(b)** Mean exists but variance does not.**EXAM SHORTCUT** (AI-derived explanation)A Pareto-type tail $f(x) \sim 2/x^3$ has $E(X^k)$ finite only when $k < 2$. So $k = 1$ (mean) converges, $k = 2$ (second moment) diverges: mean exists, variance does not.**TIPS & TRICKS** (AI-derived explanation)

- Tail rule to memorise: if $f(x)$ behaves like c/x^{a+1} for large x , then $E(X^k)$ exists exactly when $k < a$. Here $a = 2$, so only the first moment survives.
- The integrand for the mean is $2/x^2$, which converges; for $E(X^2)$ it is $2/x$, whose integral is $\log x$ and diverges. Reduce the power first, integrate second.
- Distributions with heavy polynomial tails (Pareto, Cauchy, t with low d.f.) are the standard vehicle for 'which moments exist' questions - expect one every couple of papers.
- Variance cannot exist while the mean fails to, so option (d) is logically impossible and can be struck out before any integration.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The density is $f(x) = 2/x^3$ for $x \geq 1$ and 0 otherwise. Check: $\int_1^\infty 2x^{-3} dx = [-x^{-2}]$ from 1 to $\infty = 1$, so f is a valid density.
2. Mean: $E(X) = \int_1^\infty x(2/x^3) dx = \int_1^\infty 2x^{-2} dx = [-2/x]$ from 1 to $\infty = 0 - (-2) = 2$. Finite, so the mean exists and equals 2.
3. Second moment: $E(X^2) = \int_1^\infty x^2(2/x^3) dx = \int_1^\infty 2/x dx = [2 \log x]$ from 1 to ∞ , which diverges.
4. Since $E(X^2)$ is infinite, $\text{Var}(X) = E(X^2) - [E(X)]^2$ does not exist.

5. Mean exists but variance does not, so option (b) is correct.

2. **2018 · Q11** Mathematical Expectation · Numerical

QUESTION

A rod of length $2l$ is broken at random into two pieces. The mean and variance of the length of the shorter piece are

OPTIONS

- (a) $\frac{l}{2}, \frac{l^2}{3}$
 (b) $l, \frac{l^2}{3}$
 (c) $\frac{l}{2}, \frac{l^2}{12}$
 (d) $l, \frac{l^2}{12}$

ANSWER (AI-derived — no official key exists for this paper)

(c) $\frac{l}{2}, \frac{l^2}{12}$

EXAM SHORTCUT (AI-derived explanation)

If the break point is $U \sim \text{Uniform}(0, 2l)$, the shorter piece is $\min(U, 2l - U)$, which is $\text{Uniform}(0, l)$. So mean = $l/2$ and variance = $l^2/12$ straight from the uniform formulas.

TIPS & TRICKS (AI-derived explanation)

- Key insight: the minimum of U and $2l - U$ for U uniform on $(0, 2l)$ is again uniform, on $(0, l)$. Prove it once, then reuse it - this 'broken rod' set-up recurs constantly.
- $\text{Uniform}(0, a)$ has mean $a/2$ and variance $a^2/12$. With $a = l$ that is $l/2$ and $l^2/12$ - no integration required.
- The mean must be less than l (the shorter piece can never exceed half the rod), which kills options (b) and (d) immediately.
- Variance $l^2/3$ in option (a) is the variance of $\text{Uniform}(0, 2l)$ (namely $4l^2/12$), the value you get if you forget to take the shorter piece.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Let the rod occupy the interval $(0, 2l)$ and let U , the break point, be uniformly distributed on $(0, 2l)$, so $f_U(u) = 1/(2l)$.
2. The two pieces have lengths U and $2l - U$. The shorter piece is $S = \min(U, 2l - U)$, which takes values in $(0, l)$.
3. For $0 < s < l$, $P(S \leq s) = P(U \leq s) + P(U \geq 2l - s) = s/(2l) + s/(2l) = s/l$.
4. That is the CDF of a $\text{Uniform}(0, l)$ variable, so $S \sim U(0, l)$ with density $1/l$ on $(0, l)$.
5. Hence $E(S) = l/2$ and $\text{Var}(S) = l^2/12$.
6. So the mean and variance are $l/2$ and $l^2/12$, option (c).

3. **2018 · Q14** Random Vectors & Joint Distributions · Numerical

QUESTION

X, Y have joint p.d.f. $f(x, y) = \frac{1}{3}(x + y)$, $0 < x < 1$, $0 < y < 2$. What is $E[3X(X + 2Y)]$?

OPTIONS

- (a) $\frac{9}{2}$
 (b) $\frac{29}{6}$
 (c) $\frac{31}{6}$
 (d) $\frac{11}{2}$

ANSWER (AI-derived — no official key exists for this paper)**(c)** $\frac{31}{6}$ **EXAM SHORTCUT (AI-derived explanation)**

Expand first: $E[3X(X + 2Y)] = 3E(X^2) + 6E(XY)$. With $f = (x + y)/3$ on $(0, 1) \times (0, 2)$, $E(X^2) = 7/18$ and $E(XY) = 2/3$, so the value is $7/6 + 4 = 31/6$.

TIPS & TRICKS (AI-derived explanation)

- Always expand the bracket by linearity before integrating. Integrating $3 \times (x + 2y)$ directly is the same work but far more error-prone.
- Normalisation check takes five seconds here: the double $\int (x + y)/3$ over the rectangle is $(1/3)(1 + 2) = 1$, so the density is valid and your set-up is right.
- Integrate over y first (its range 0 to 2 is the wider one); the inner $\int (x + y) dy$ from 0 to 2 is $2x + 2$, which is clean to carry forward.
- The answer exceeds 4 because the $6E(XY)$ term alone contributes 4 - use that to eliminate any option below 4 at a glance.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The density is $f(x, y) = (x + y)/3$ on $0 < x < 1, 0 < y < 2$. Check: $\int (x + y)/3 dy$ from 0 to 2 = $(2x + 2)/3$, and integrating that over x from 0 to 1 gives $(1 + 2)/3 = 1$. Valid density.
2. By linearity, $E[3X(X + 2Y)] = 3E(X^2) + 6E(XY)$.
3. $E(X^2) = \int_0^1 x^2(2x + 2)/3 dx = (1/3) \int (2x^3 + 2x^2) dx = (1/3)(1/2 + 2/3) = (1/3)(7/6) = 7/18$.
4. $E(XY) = (1/3) \int_0^1 x[\int_0^2 y(x + y) dy] dx$. The inner integral is $[xy^2/2 + y^3/3]$ from 0 to 2 = $2x + 8/3$.
5. So $E(XY) = (1/3) \int_0^1 (2x^2 + 8x/3) dx = (1/3)(2/3 + 4/3) = (1/3)(2) = 2/3$.
6. Therefore $E[3X(X + 2Y)] = 3(7/18) + 6(2/3) = 7/6 + 4 = 7/6 + 24/6 = 31/6$, option (c).

4. 2018 · Q16 Discrete & Continuous Random Variables · Conceptual**QUESTION**

X has p.d.f. $f(x) = 1 + x$ for $-1 < x \leq 0$; $1 - x$ for $0 < x < 1$ (else 0). $U = X$, $V = X^2$. 1. $\text{Cov}(U, V) = 0$. 2. U, V are independent.

OPTIONS

- (a) 1 only
 (b) 2 only
 (c) Both 1 and 2
 (d) Neither 1 nor 2

ANSWER (AI-derived — no official key exists for this paper)**(a)** 1 only**EXAM SHORTCUT (AI-derived explanation)**

The density is the symmetric triangular law on $(-1, 1)$, so all odd moments vanish:

$\text{Cov}(X, X^2) = E(X^3) - E(X)E(X^2) = 0 - 0 = 0$. But $V = U^2$ is a function of U , so they cannot be independent. Statement 1 only.

TIPS & TRICKS (AI-derived explanation)

- $V = X^2$ is a deterministic function of $U = X$. A non-degenerate variable is never independent of a non-constant function of itself - answer statement 2 without any computation.
- Symmetry about 0 makes every odd moment zero. Spot the symmetry of f (it satisfies $f(-x) = f(x)$) and $E(X) = E(X^3) = 0$ follow instantly.
- This is the mirror image of the previous item: there, zero covariance with dependence in a table; here, zero covariance with functional dependence. Both illustrate that $\text{Cov} = 0$ does not imply independence.
- If you are ever tempted by 'both 1 and 2', remember that independence is strictly stronger than uncorrelatedness - never the other way round.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The density $f(x) = 1 + x$ on $(-1, 0]$ and $1 - x$ on $(0, 1)$ is the symmetric triangular density; it satisfies $f(-x) = f(x)$.
2. By symmetry every odd moment vanishes: $E(X) = 0$ and $E(X^3) = 0$.
3. Statement 1: $\text{Cov}(U, V) = \text{Cov}(X, X^2) = E(X \cdot X^2) - E(X)E(X^2) = E(X^3) - 0 \cdot E(X^2) = 0 - 0 = 0$. Statement 1 is TRUE.
4. Statement 2: $V = X^2$ is a (non-constant) deterministic function of $U = X$. For instance $P(|U| < 1/2, V > 1/2) = 0$ while both marginal probabilities are strictly positive, so the joint law does not factorise.
5. Hence U and V are dependent and statement 2 is FALSE.
6. Only statement 1 is correct, so option (a) is right.

5. 2020 · Q27 Conditional Expectation · Numerical**QUESTION**

The mean of $f(x) = 12x^2(1 - x)$, $0 \leq x \leq 1$ (else 0), is

OPTIONS

- (a)** 0.5
(b) 0.6
(c) 0.7
(d) 0.8

ANSWER (AI-derived — no official key exists for this paper)**(b)** 0.6

EXAM SHORTCUT (AI-derived explanation)

$12x^2(1-x)$ is the Beta(3, 2) density, whose mean is $a/(a+b) = 3/5 = 0.6$.

TIPS & TRICKS (AI-derived explanation)

- Recognise the Beta kernel $x^{a-1}(1-x)^{b-1}$: here $a-1=2$ and $b-1=1$, so $a=3$ and $b=2$.
- Beta mean = $a/(a+b)$ and variance = $ab/[(a+b)^2(a+b+1)]$. Recognition beats integration every time.
- Verify the constant: $1/B(3,2) = \Gamma(5)/(\Gamma(3)\Gamma(2)) = 24/(2 \times 1) = 12$, matching the printed density - a good check that a and b are right.
- Direct integration also works: the $\int 12x^3(1-x) dx$ over $(0,1) = 12(1/4 - 1/5) = 12/20 = 0.6$.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The density is $f(x) = 12x^2(1-x)$ on $[0,1]$.
2. Comparing with the Beta density $x^{a-1}(1-x)^{b-1}/B(a,b)$, we read $a-1=2$ and $b-1=1$, so $a=3$ and $b=2$.
3. The normalising constant is $1/B(3,2) = \Gamma(5)/(\Gamma(3)\Gamma(2)) = 24/(2 \times 1) = 12$, confirming the identification.
4. For a Beta(a,b) variable the mean is $a/(a+b)$.
5. Hence the mean is $3/(3+2) = 3/5 = 0.6$.
6. Option (b) is correct.

6. 2020 · Q39 Binomial Distribution · Numerical**QUESTION**

If $X \sim \text{Binomial}(n=3, p=1/3)$, what is $E\left(\frac{X}{X+1}\right)$?

OPTIONS

- (a) $\frac{43}{108}$
 (b) $\frac{5}{12}$
 (c) $\frac{47}{108}$
 (d) $\frac{49}{108}$

ANSWER (AI-derived — no official key exists for this paper)

(a) $\frac{43}{108}$

EXAM SHORTCUT (AI-derived explanation)

With $p=1/3$ the masses are $8/27, 12/27, 6/27, 1/27$. Then
 $E[X/(X+1)] = 0 + (1/2)(12/27) + (2/3)(6/27) + (3/4)(1/27) = (6 + 4 + 0.75)/27 = 43/108$.

TIPS & TRICKS (AI-derived explanation)

- $X/(X+1)$ is a non-linear function, so $E[X/(X+1)]$ is NOT $E(X)/(E(X)+1)$. Compute the sum term by term.
- Keep every mass over the common denominator 27; the weighted terms $6/27, 4/27$ and $(3/4)/27$ then add trivially.
- The final fraction has denominator $108 = 4 \times 27$, which is why all four options share that denominator - a hint that a quarter appears in the last term.
- Check by bounds: $X/(X+1)$ takes the values $0, 1/2, 2/3, 3/4$ with the mass concentrated on the small values, so the answer must be well below $1/2$. $43/108 = 0.398$ fits.

STEP-BY-STEP SOLUTION (AI-derived explanation)

- $X \sim \text{Binomial}(3, 1/3)$, so
 $P(X = 0) = (2/3)^3 = 8/27$, $P(X = 1) = 3(1/3)(2/3)^2 = 12/27$, $P(X = 2) = 3(1/3)^2(2/3) = 6/27$, $P(X = 3) = (1/3)^3 = 1/27$.
 These sum to 1.
- The function values are $X/(X + 1) = 0, 1/2, 2/3, 3/4$ for $X = 0, 1, 2, 3$.
- $E[X/(X + 1)] = 0(8/27) + (1/2)(12/27) + (2/3)(6/27) + (3/4)(1/27)$.
- Term by term: $(1/2)(12/27) = 6/27$; $(2/3)(6/27) = 4/27$; $(3/4)(1/27) = 3/108$.
- Converting to 108ths: $24/108 + 16/108 + 3/108 = 43/108$.
- Hence $E[X/(X + 1)] = 43/108$, option (a).

7. 2021 · Q22 Conditional Expectation · Numerical**QUESTION**

Let n be a natural number and Y be distributed on $\{-n, \dots, -1, 0, 1, \dots, n\}$ with $P(Y = k) = 2P(Y = -k) > 0$ for every $k \in \{1, \dots, n\}$. For $y \in \{1, 4, 9, \dots, n^2\}$, $E(Y \mid Y^2 = y)$ equals

OPTIONS

- (a) $-\sqrt{y/3}$
 (b) $-\sqrt{y/2}$
 (c) $\sqrt{y}/3$
 (d) $y/2$

ANSWER (AI-derived — no official key exists for this paper)

(c) $\sqrt{y}/3$

EXAM SHORTCUT (AI-derived explanation)

Given $Y^2 = k^2$ the only possibilities are $+k$ and $-k$, with probabilities in the ratio 2:1. So
 $E(Y \mid Y^2 = k^2) = k(2/3) - k(1/3) = k/3 = \sqrt{y}/3$.

TIPS & TRICKS (AI-derived explanation)

- Conditioning on $Y^2 = y$ collapses the sample space to just two points, $+\sqrt{y}$ and $-\sqrt{y}$ - the whole problem is a two-point weighted average.
- The condition $P(Y = k) = 2P(Y = -k)$ makes the conditional probabilities $2/3$ and $1/3$ regardless of k , so the answer is a fixed multiple of $\sqrt{y}$.
- Sign check: the positive value is twice as likely, so the conditional mean must be POSITIVE - options (a) and (b) are eliminated at once.
- Dimension check: $E(Y \mid Y^2 = y)$ must scale like $\sqrt{y}$, not like y . That rules out option (d).

STEP-BY-STEP SOLUTION (AI-derived explanation)

- Fix $y = k^2$ with k in $\{1, \dots, n\}$. The event $\{Y^2 = y\}$ is $\{Y = k\}$ union $\{Y = -k\}$.
- Write $P(Y = -k) = p$, so $P(Y = k) = 2p$ by hypothesis, and $P(Y^2 = y) = 3p$.
- Conditional probabilities: $P(Y = k \mid Y^2 = y) = 2p/(3p) = 2/3$ and $P(Y = -k \mid Y^2 = y) = p/(3p) = 1/3$.
- Hence $E(Y \mid Y^2 = y) = k(2/3) + (-k)(1/3) = k/3$.
- Since $k = \sqrt{y}$ is positive, $E(Y \mid Y^2 = y) = \sqrt{y}/3$.

6. Option (c) is correct.

8. **2021 · Q28** Conditional Expectation · Numerical

QUESTION

$Y_1 = 2X_1 + 3X_2$, $Y_2 = X_1 + 2X_2$, with X_1, X_2 independent, $\text{Var}(Y_1) = 72$, $\text{Var}(Y_2) = 25$. What is $\text{Var}\left(\frac{2}{3}X_1 - 4X_2\right)$?

OPTIONS

- (a) 10
- (b) 22
- (c) 60
- (d) 68

ANSWER (AI-derived — no official key exists for this paper)

(d) 68

EXAM SHORTCUT (AI-derived explanation)

Independence gives $4V_1 + 9V_2 = 72$ and $V_1 + 4V_2 = 25$. Solving, $V_2 = 4$ and $V_1 = 9$. Then $\text{Var}((2/3)X_1 - 4X_2) = (4/9)(9) + 16(4) = 4 + 64 = 68$.

TIPS & TRICKS (AI-derived explanation)

- Set up two linear equations in V_1 and V_2 from the two given variances - independence kills all covariance terms, so the coefficients are simply squared.
- Solve by elimination: from $V_1 = 25 - 4V_2$, substitution gives $100 - 16V_2 + 9V_2 = 72$, so $7V_2 = 28$ and $V_2 = 4$.
- In the final variance the minus sign vanishes on squaring: $(-4)^2 = 16$. Options (a) and (b) come from subtracting instead of adding.
- The fraction $2/3$ is squared to $4/9$, which cancels neatly against $V_1 = 9$ to give 4 - a sign the intermediate values are right.

STEP-BY-STEP SOLUTION (AI-derived explanation)

- X_1 and X_2 are independent, so $\text{Var}(aX_1 + bX_2) = a^2 \text{Var}(X_1) + b^2 \text{Var}(X_2)$. Write $V_1 = \text{Var}(X_1)$ and $V_2 = \text{Var}(X_2)$.
- From $Y_1 = 2X_1 + 3X_2$: $4V_1 + 9V_2 = 72$.
- From $Y_2 = X_1 + 2X_2$: $V_1 + 4V_2 = 25$, so $V_1 = 25 - 4V_2$.
- Substituting: $4(25 - 4V_2) + 9V_2 = 72$, i.e. $100 - 16V_2 + 9V_2 = 72$, so $7V_2 = 28$ and $V_2 = 4$.
- Then $V_1 = 25 - 16 = 9$.
- $\text{Var}((2/3)X_1 - 4X_2) = (4/9)(9) + 16(4) = 4 + 64 = 68$, option (d).

9. **2021 · Q38** Mathematical Expectation · Numerical

QUESTION

X takes value $(-1)^{i+1} \frac{2^i}{i!}$ with probability $\left(\frac{1}{2}\right)^i$, for $i = 1, 2, 3, \dots$ (e.g. $X = 2$ w.p. $1/2$, $X = -2$ w.p. $1/4$, ...). What is $E(X)$?

OPTIONS

- (a) 1
- (b) e^{-1}
- (c) $e - 1$
- (d) $1 - e^{-1}$

ANSWER (AI-derived — no official key exists for this paper)**(d)** $1 - e^{-1}$ **EXAM SHORTCUT (AI-derived explanation)**

The 2^i and $(1/2)^i$ cancel, leaving $E(X) = \text{sum of } (-1)^{i+1}/i! \text{ for } i \geq 1 = -[e^{-1} - 1] = 1 - e^{-1}$.

TIPS & TRICKS (AI-derived explanation)

- Multiply value by probability FIRST: $[(-1)^{i+1}2^i/i!] \times (1/2)^i = (-1)^{i+1}/i!$. The cancellation is the entire design of the question.
- Recognise the exponential series: $\sum (-1)^i/i! = e^{-1}$. The required sum starts at $i = 1$ and has the opposite sign, giving $1 - e^{-1}$.
- Check the probabilities sum to 1: sum of $(1/2)^i$ for $i \geq 1 = 1$, so the distribution is valid.
- Numerically $1 - e^{-1} = 0.632$, and the first two terms of the sum are $1 - 1/2 = 0.5$ with the next $+1/6 = 0.667$ - the ∂ sums bracket 0.632, confirming the answer.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. X takes the value $(-1)^{i+1}2^i/i!$ with probability $(1/2)^i$ for $i = 1, 2, 3, \dots$. The probabilities sum to 1, so the distribution is valid.
2. The contribution of the i th term to the expectation is $[(-1)^{i+1}2^i/i!](1/2)^i$.
3. The factors 2^i and $(1/2)^i$ cancel, leaving $(-1)^{i+1}/i!$.
4. Hence $E(X) = \sum (-1)^{i+1}/i! = -\sum (-1)^i/i!$.
5. Since $\sum (-1)^i/i! = e^{-1}$, the sum from $i = 1$ equals $e^{-1} - 1$.
6. Therefore $E(X) = -(e^{-1} - 1) = 1 - e^{-1}$, option (d).

10. **2022 · Q5** Poisson Distribution · Numerical**QUESTION**

If X is a Poisson random variable with $P(X = 1) = P(X = 2)$, what is $V(X)$ equal to?

OPTIONS

- (a) 0
- (b) 1
- (c) 2
- (d) 3

ANSWER (AI-derived — no official key exists for this paper)**(c)** 2

EXAM SHORTCUT (AI-derived explanation)

$P(X = 1) = P(X = 2)$ gives $e^{-L}L = e^{-L}L^2/2$, so $L = 2$. For the Poisson, variance = mean = 2.

TIPS & TRICKS (AI-derived explanation)

- Take the RATIO of consecutive Poisson probabilities: $P(X = k + 1)/P(X = k) = \lambda/(k + 1)$. Setting it to 1 with $k = 1$ gives $\lambda = 2$ immediately.
- For the Poisson, mean = variance = λ , so once λ is found the answer needs no further work.
- The exponential factor $e^{-\lambda}$ cancels on both sides - never expand it.
- General fact worth carrying: $P(X = k) = P(X = k + 1)$ exactly when $\lambda = k + 1$, so equal consecutive probabilities pinpoint λ at once.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. For a Poisson variable with parameter λ , $P(X = k) = e^{-\lambda}\lambda^k/k!$.
2. The condition $P(X = 1) = P(X = 2)$ gives $e^{-\lambda}\lambda = e^{-\lambda}\lambda^2/2$.
3. Cancelling $e^{-\lambda}$ and one factor of λ (which is non-zero): $1 = \lambda/2$.
4. Hence $\lambda = 2$.
5. For the Poisson distribution the variance equals the mean, so $V(X) = \lambda = 2$.
6. Option (c) is correct.

11. **2022 · Q9** Conditional Expectation · Numerical**COMMON DATA FOR THIS GROUP**

Q9-Q10 share this setup. Let the joint density function of X and Y be

$$f(x, y) = \begin{cases} \frac{y}{(1+x)^4} \exp\left(-\frac{y}{1+x}\right), & x \geq 0, y \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

QUESTION

What is the value of $E[Y \mid X = 10]$?

OPTIONS

- (a) 22
(b) 25
(c) 30
(d) 32

ANSWER (AI-derived — no official key exists for this paper)

(a) 22

EXAM SHORTCUT (AI-derived explanation)

For fixed x the y -part is proportional to $ye^{-y/(1+x)}$, $a\Gamma$ with shape 2 and scale $(1+x)$, whose mean is $2(1+x)$. At $x = 10$ this is 22.

TIPS & TRICKS (AI-derived explanation)

- Read the conditional law off the y -dependence: $y^{a-1}e^{-y/b}$ is $\Gamma(\text{shape } a, \text{scale } b)$, so here $a = 2$ and $b = 1 + x$.

- Γ mean = shape $\times$ scale = $2(1+x)$. Any function of x alone in the density (here $1/(1+x)^4$) is absorbed into the normalising constant and cannot affect the conditional mean.
- The power of y is 1, so the shape is 2, not 1 - mistaking it for an exponential gives $(1+x) = 11$ and misses every option.
- This is the same structural trick as 2018Q₂₆: identify the conditional family from the y -kernel and read off its mean.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The joint density is $f(x, y) = y(1+x)^{-4} \exp(-y/(1+x))$ for $x, y \geq 0$.
2. For a fixed value $X = x$, the conditional density of Y is proportional to $y \exp(-y/(1+x))$.
3. This is the kernel of a Γ distribution with shape parameter 2 and scale parameter $(1+x)$; normalising gives $f(y | x) = ye^{-y/(1+x)}/(1+x)^2$.
4. The mean of a $\Gamma(\text{shape } a, \text{scale } b)$ variable is ab , so $E(Y | X = x) = 2(1+x)$.
5. At $x = 10$: $E(Y | X = 10) = 2(11) = 22$.
6. Option (a) is correct.

12. 2023 · Q2 Normal Distribution · Numerical**QUESTION**

If $X \sim N(0, 1)$ and $Y | X = x \sim U(x-1, x+1)$, then what is the value of $E(E^2(Y | X))$?

OPTIONS

- (a) 1
(b) 2
(c) 3
(d) 4

ANSWER (AI-derived — no official key exists for this paper)

(a) 1

EXAM SHORTCUT (AI-derived explanation)

Y given $X = x$ is uniform on $(x-1, x+1)$, whose mean is the midpoint x . So $E(Y | X) = X$ and $E[E^2(Y | X)] = E(X^2) = 1$.

TIPS & TRICKS (AI-derived explanation)

- The mean of $U(a, b)$ is the midpoint $(a+b)/2$; here $((x-1) + (x+1))/2 = x$ exactly, so the conditional mean is X itself.
- Read the notation carefully: $E^2(Y | X)$ means the SQUARE of the conditional mean, not the second conditional moment.
- For $X \sim N(0, 1)$, $E(X^2) = \text{Var}(X) + [E(X)]^2 = 1 + 0 = 1$ - the whole answer once the conditional mean is identified.
- The half-width 1 of the uniform affects the conditional VARIANCE ($1/3$) but not the conditional mean, so it never enters this calculation.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Given $X = x$, the variable Y is uniform on the interval $(x-1, x+1)$.

2. The mean of a uniform distribution is the midpoint of its range, so $E(Y | X = x) = ((x - 1) + (x + 1))/2 = x$.
3. Hence $E(Y | X) = X$ as a random variable.
4. Therefore $E^2(Y | X) = X^2$, and $E[E^2(Y | X)] = E(X^2)$.
5. For $X \sim N(0, 1)$, $E(X^2) = \text{Var}(X) + [E(X)]^2 = 1 + 0 = 1$.
6. Hence the value is 1, option (a).

13. 2023 · Q10 Random Vectors & Joint Distributions · Numerical

COMMON DATA FOR THIS GROUP

Questions 8–10 are based on the following: let the joint pdf of two random variables X and Y be

$$f(x, y) = \frac{1}{\sqrt{2\pi y}} e^{-\frac{1}{2y}(x-12y)^2}, \quad -\infty < x < \infty, 0 < y < 1$$

QUESTION

What is the value of $\text{Cov}(X, Y)$?

OPTIONS

- (a) 1
- (b) 2
- (c) 2.5
- (d) 3

ANSWER (AI-derived — no official key exists for this paper)

(a) 1

EXAM SHORTCUT (AI-derived explanation)

$E(XY) = E[Y E(X | Y)] = E(12Y^2) = 12(1/3) = 4$, and $E(X) E(Y) = 6(1/2) = 3$, so $\text{Cov} = 4 - 3 = 1$.

TIPS & TRICKS (AI-derived explanation)

- Condition first: $E(XY) = E[E(XY | Y)] = E[Y E(X | Y)]$, because Y is fixed inside the inner expectation.
- For $U(0, 1)$, $E(Y^2) = 1/3$ - not $1/4$. Confusing $E(Y^2)$ with $[E(Y)]^2$ gives 3 and a covariance of 0.
- Reuse $E(X) = 6$ from the first part rather than recomputing; shared-stem items are built to chain.
- Consistency check: $\text{Cov} = 12 \text{Var}(Y) = 12/12 = 1$, since $X = 12Y$ plus conditional noise that is uncorrelated with Y - a neat one-line confirmation.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. From the joint density, $X | Y = y \sim N(12y, y)$ and $Y \sim \text{Uniform}(0, 1)$, with $E(X) = 6$ from the earlier part.
2. $E(XY) = E[E(XY | Y)] = E[Y E(X | Y)] = E[Y(12Y)] = 12 E(Y^2)$.
3. For $U(0, 1)$, $E(Y^2) = 1/3$, so $E(XY) = 12(1/3) = 4$.
4. $E(X) E(Y) = 6 \times (1/2) = 3$.
5. $\text{Cov}(X, Y) = E(XY) - E(X) E(Y) = 4 - 3 = 1$.
6. Equivalently $\text{Cov}(X, Y) = \text{Cov}(12Y, Y) = 12 \text{Var}(Y) = 12/12 = 1$. Option (a) is correct.

14. **2023 · Q14** Moment Generating Function · Numerical**QUESTION**

If a random variable assumes non-negative integral values and $E(X) = \mu$, $\text{Var}(X) = \sigma^2$, then what is the value of $2 \sum_{x=0}^{\infty} xP(X > x)$?

OPTIONS

- (a) $\sigma^2 + \mu^2$
 (b) $\sigma^2 + \mu^2 - \mu$
 (c) $\frac{\sigma^2 - \mu}{2}$
 (d) $\frac{\sigma^2 + \mu^2 - \mu}{2}$

ANSWER (AI-derived — no official key exists for this paper)**(b)** $\sigma^2 + \mu^2 - \mu$ **EXAM SHORTCUT** (AI-derived explanation)

Interchanging sums, $\sum x P(X > x) = \sum p_k [k(k-1)/2] = [E(X^2) - E(X)]/2$. Doubling gives $E(X^2) - E(X) = \sigma^2 + \mu^2 - \mu$.

TIPS & TRICKS (AI-derived explanation)

- Swap the summations: $\sum x \sum p_k = \sum p_k$ sum over $x = 0$ to $k-1$ of x , and the inner sum is $k(k-1)/2$.
- The factor 2 outside is placed exactly to cancel the $1/2$ from the triangular number - a strong hint that the identity is being tested.
- Finish with $E(X^2) = \sigma^2 + \mu^2$, so the answer is $\sigma^2 + \mu^2 - \mu$. Options (c) and (d) retain a stray $1/2$.
- Companion identity worth knowing: $\sum P(X > x) = E(X)$ for non-negative integer variables - the first-moment version of the same trick.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Write $S = \sum x P(X > x) = \sum x \sum p_k$, where $p_k = P(X = k)$.
2. Interchange the order of summation over the region $0 \leq x < k$: $S = \sum p_k$ sum over $x = 0$ to $k-1$ of x .
3. The inner sum is the triangular number $k(k-1)/2$.
4. So $S = \sum p_k k(k-1)/2 = [E(X^2) - E(X)]/2$.
5. Therefore $2S = E(X^2) - E(X)$.
6. Since $E(X^2) = \sigma^2 + \mu^2$ and $E(X) = \mu$, the value is $\sigma^2 + \mu^2 - \mu$, option (b).

15. **2024 · Q4** Uniform / Rectangular Distribution · Numerical**COMMON DATA FOR THIS GROUP**

Questions 4–6 are based on: let $X_1, X_2, X_3, \dots, X_n$ be iid $U(0, 1)$ variates and $T = -\ln(X_1 X_2 X_3 \cdots X_n)$.

QUESTION

What is $E(T)$ equal to?

OPTIONS

- (a) n

- (b) $2n$
- (c) $4n$
- (d) $n/2$

ANSWER (AI-derived — no official key exists for this paper)

(a) n

EXAM SHORTCUT (AI-derived explanation)

$-\ln X_i$ is $\text{Exp}(1)$ for X_i uniform, so T is the sum of n independent $\text{Exp}(1)$ variables, i.e. $\Gamma(n, 1)$, with mean n .

TIPS & TRICKS (AI-derived explanation)

- Convert the log of a product into a sum of logs FIRST: $T = \text{sum of } (-\ln X_i)$. The rest is the exponential distribution.
- For $U \sim U(0, 1)$, $P(-\ln U > t) = P(U < e^{-t}) = e^{-t}$, so $-\ln U \sim \text{Exponential}(1)$ with mean 1.
- Sum of n independent $\text{Exp}(1)$ variables is $\Gamma(\text{shape } n, \text{rate } 1)$, with mean n and variance n - both needed for the next parts.
- This is the inverse-transform construction used to simulate exponential and Γ variates, so the identification is worth knowing cold.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Since $\ln(X_1 X_2 \dots X_n) = \text{sum of } \ln X_i$, we can write $T = \sum (-\ln X_i)$.
2. For $X \sim U(0, 1)$ and $t > 0$, $P(-\ln X > t) = P(X < e^{-t}) = e^{-t}$, so $-\ln X \sim \text{Exponential}$ with rate 1.
3. Hence T is the sum of n independent $\text{Exponential}(1)$ variables, i.e. $T \sim \Gamma(\text{shape } n, \text{rate } 1)$.
4. The mean of $\text{Exponential}(1)$ is 1, so by linearity $E(T) = n(1) = n$.
5. Equivalently, the $\Gamma(n, 1)$ mean is $n/1 = n$.
6. Option (a) is correct.

16. **2024 · Q5** Uniform / Rectangular Distribution · Numerical

COMMON DATA FOR THIS GROUP

Questions 4–6 are based on: let $X_1, X_2, X_3, \dots, X_n$ be iid $U(0, 1)$ variates and $T = -\ln(X_1 X_2 X_3 \dots X_n)$.

QUESTION

What is $\text{Var}(T)$ equal to?

OPTIONS

- (a) $4n$
- (b) $3n$
- (c) $2n$
- (d) n

ANSWER (AI-derived — no official key exists for this paper)

(d) n

EXAM SHORTCUT (AI-derived explanation)

T is a sum of n independent $\text{Exp}(1)$ variables, each with variance 1, so $\text{Var}(T) = n$.

TIPS & TRICKS (AI-derived explanation)

- Variances of INDEPENDENT summands add, so $\text{Var}(T) = n \text{Var}(-\ln X_1) = n(1) = n$.
- Exponential(1) has mean 1 and variance 1 - both equal to 1, which is why $E(T)$ and $\text{Var}(T)$ coincide here at n .
- $\Gamma(\text{shape } a, \text{rate } b)$ has mean a/b and variance a/b^2 ; with $a = n$ and $b = 1$ both equal n .
- Options $2n$, $3n$ and $4n$ would correspond to Exp with rate $1/\sqrt{2}$ and so on - none arises from the uniform-to-exponential transformation.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. From the previous part, $T = \sum (-\ln X_i)$ where the $-\ln X_i$ are independent Exponential(1) variables.
2. For Exponential with rate 1, the variance is $1/1^2 = 1$.
3. Since the summands are independent, their variances add: $\text{Var}(T) = nx1 = n$.
4. Equivalently, $T \sim \Gamma(\text{shape } n, \text{rate } 1)$, whose variance is $n/1^2 = n$.
5. Note that here the mean and variance of T coincide, both equal to n .
6. Option (d) is correct.

17. 2024 · Q8 Normal Distribution · Numerical**COMMON DATA FOR THIS GROUP**

Questions 7–10 are based on: let $X_1, X_2, X_3, \dots, X_7$ be iid random variables, each having the $N(0, 1)$ distribution.

QUESTION

What is $E\left(\frac{1}{X_1^2 + X_2^2 + X_3^2 + X_4^2}\right)$ equal to?

OPTIONS

- (a) 0
- (b) 1
- (c) $1/2$
- (d) $1/9$

ANSWER (AI-derived — no official key exists for this paper)

(c) $1/2$

EXAM SHORTCUT (AI-derived explanation)

The sum of four squared standard normals is χ_4^2 , and $E(1/\chi_k^2) = 1/(k-2) = 1/2$.

TIPS & TRICKS (AI-derived explanation)

- Memorise $E(1/V) = 1/(k-2)$ for $V \sim \chi_k^2, k > 2$; it is the workhorse for every 'expected reciprocal' item in this chapter.
- Identify the degrees of freedom by COUNTING the squared standard normals: four here, so $k = 4$ and the answer is $1/2$.

- The expectation is finite only for $k > 2$ - with $k = 1$ or 2 the integral diverges, a useful caveat to state.
- Do not confuse with $E(1/X)$ for the F distribution or with $1/E(V) = 1/4$; the reciprocal of an expectation is not the expectation of a reciprocal.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. X_1, X_2, X_3, X_4 are independent $N(0, 1)$, so $V = X_1^2 + X_2^2 + X_3^2 + X_4^2 \sim \chi^2$ with 4 degrees of freedom.
2. For a χ^2 variable with k degrees of freedom, the density is proportional to $v^{k/2-1}e^{-v/2}$.
3. Computing $E(1/V)$ shifts the exponent down by one and yields the standard result $E(1/V) = 1/(k - 2)$, valid for $k > 2$.
4. With $k = 4$: $E(1/V) = 1/(4 - 2) = 1/2$.
5. Note that this differs from $1/E(V) = 1/4$, since the expectation of a reciprocal is not the reciprocal of the expectation.
6. Option (c) is correct.

18. 2024 · Q9 Normal Distribution · Numerical**COMMON DATA FOR THIS GROUP**

Questions 7–10 are based on: let $X_1, X_2, X_3, \dots, X_7$ be iid random variables, each having the $N(0, 1)$ distribution.

QUESTION

What is $E\left(\frac{X_1 + X_2 + X_3}{X_4^2 + X_5^2 + X_6^2 + X_7^2}\right)$ equal to?

OPTIONS

- (a) 0
(b) 1
(c) 2
(d) $1/2$

ANSWER (AI-derived — no official key exists for this paper)

(a) 0

EXAM SHORTCUT (AI-derived explanation)

The numerator and denominator involve disjoint variables and are therefore independent. Since $E(X_1 + X_2 + X_3) = 0$, the whole expectation is 0.

TIPS & TRICKS (AI-derived explanation)

- Check for DISJOINT index sets first: the numerator uses X_1, X_2, X_3 and the denominator X_4 to X_7 , so the two are independent and the expectation factorises.
- $E(\text{numerator}) = 0$ because each X_i has mean 0 - the answer follows without evaluating the denominator's expectation at all.
- The denominator's expectation $E(1/\chi_4^2) = 1/2$ is finite, which is what makes the product $0 \times 1/2$ legitimate rather than an indeterminate form.
- Contrast with a ratio built from the SAME variables (as in a t statistic), where independence fails and this shortcut is unavailable.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The numerator involves X_1, X_2, X_3 and the denominator involves X_4, X_5, X_6, X_7 - disjoint sets of independent variables.
2. Hence the numerator and the reciprocal of the denominator are independent, and the expectation factorises.
3. $E(X_1 + X_2 + X_3) = 0 + 0 + 0 = 0$.
4. The denominator $X_4^2 + \dots + X_7^2 \sim \chi_4^2$, so $E(1/(X_4^2 + \dots + X_7^2)) = 1/(4 - 2) = 1/2$, which is finite.
5. Therefore the expectation is $0 \times (1/2) = 0$.
6. Option (a) is correct.

19. 2024 · Q18 Conditional Expectation · Numerical**QUESTION**

If $4\sigma_x^2 = 9\sigma_y^2 = 25$, where x and y are independently distributed, then what is the value of $\text{Var}\left(\frac{2x + 3y}{\sqrt{2}}\right)$?

OPTIONS

- (a) 25
- (b) 50
- (c) 62.5
- (d) 75

ANSWER (AI-derived — no official key exists for this paper)

(a) 25

EXAM SHORTCUT (AI-derived explanation)

From $4\sigma_x^2 = 25$ and $9\sigma_y^2 = 25$ we get $\sigma_x^2 = 25/4$ and $\sigma_y^2 = 25/9$. Then
 $\text{Var}((2x + 3y)/\sqrt{2}) = (1/2)[4(25/4) + 9(25/9)] = (1/2)(50) = 25$.

TIPS & TRICKS (AI-derived explanation)

- Read the chain $4\sigma_x^2 = 9\sigma_y^2 = 25$ as TWO equations, then solve each for its variance separately.
- The design is deliberate: $4\sigma_x^2 = 25$ and $9\sigma_y^2 = 25$ make the two terms in the variance BOTH equal 25, so the bracket is simply 50.
- Independence means no covariance term; the $1/\sqrt{2}$ divisor contributes a factor $1/2$, not $1/\sqrt{2}$, because variance scales by the square.
- Forgetting to square the divisor gives $50/\sqrt{2} = 35.4$ and no matching option; forgetting it entirely gives 50, which is planted as option (b).

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. From $4\sigma_x^2 = 25$ we get $\sigma_x^2 = 25/4$, and from $9\sigma_y^2 = 25$ we get $\sigma_y^2 = 25/9$.
2. Since x and y are independent, $\text{Var}(2x + 3y) = 4\sigma_x^2 + 9\sigma_y^2 = 4(25/4) + 9(25/9) = 25 + 25 = 50$.
3. Dividing by $\sqrt{2}$ scales the variance by $(1/\sqrt{2})^2 = 1/2$.
4. $\text{Var}((2x + 3y)/\sqrt{2}) = 50/2 = 25$.
5. Option (a) is correct.

20. **2024 · Q19** Conditional Expectation · Numerical**QUESTION**

If $\text{Cov}(2x, y) = 4$, $\text{Var}(x/4) = 1$ and $\text{Var}(y/9) = 7$, then what is the value of $(18r(x, y))^2$ equal to?

OPTIONS

- (a) 7
(b) $1/7$
(c) 4
(d) $1/4$

ANSWER (AI-derived — no official key exists for this paper)

(b) $1/7$

EXAM SHORTCUT (AI-derived explanation)

$\text{Cov}(x, y) = 2$, $\text{Var}(x) = 16$ and $\text{Var}(y) = 567$, so $r = 2/(4\sqrt{567})$. Then $(18r)^2 = 324(4)/(16 \times 567) = 81/567 = 1/7$.

TIPS & TRICKS (AI-derived explanation)

- Undo each transformation first: $\text{Cov}(2x, y) = 2 \text{Cov}(x, y)$ gives $\text{Cov} = 2$; $\text{Var}(x/4) = \text{Var}(x)/16$ gives $\text{Var}(x) = 16$; $\text{Var}(y/9) = \text{Var}(y)/81$ gives $\text{Var}(y) = 567$.
- Keep $567 = 81 \times 7$ in factored form - it cancels beautifully at the end against the 81 from $18^2/4$.
- $(18r)^2 = 324r^2 = 324 \text{Cov}^2 / (\text{Var } x \text{ Var } y) = 324(4)/(16 \times 567) = 1296/9072 = 1/7$. Work with squares throughout to avoid surds.
- Sanity check: $r = 2/(4 \times 23.81) = 0.021$, a very weak correlation, so $(18r)^2 = 0.143 = 1/7$ is the right order of magnitude.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. From $\text{Cov}(2x, y) = 2 \text{Cov}(x, y) = 4$ we get $\text{Cov}(x, y) = 2$.
2. From $\text{Var}(x/4) = \text{Var}(x)/16 = 1$ we get $\text{Var}(x) = 16$, so $\text{sd}(x) = 4$.
3. From $\text{Var}(y/9) = \text{Var}(y)/81 = 7$ we get $\text{Var}(y) = 567$.
4. Hence $r(x, y) = \text{Cov} / (\text{sd}(x) \text{sd}(y)) = 2/(4\sqrt{567})$.
5. Then $(18r)^2 = 324r^2 = 324 \times 4/(16 \times 567) = 1296/9072$.
6. Since $567 = 81 \times 7$, this is $81/567 = 1/7$. Option (b) is correct.

21. **2024 · Q26** Moment Generating Function · Numerical**QUESTION**

Let $U(t)$ and $V(t)$ be the moment-generating functions of two random variables X and Y respectively, and $V(t) = U(\log U(t))$. If $E(X) = 2$, then what is the value of $E(Y)$?

OPTIONS

- (a) 1
(b) 2
(c) 4
(d) $\ln 2$

ANSWER (AI-derived — no official key exists for this paper)**(c)** 4**EXAM SHORTCUT (AI-derived explanation)**

Differentiate $V(t) = U(\log U(t))$ by the chain rule: $V'(t) = U'(\log U(t)) \cdot U'(t)/U(t)$. At $t = 0$, $U(0) = 1$ so $V'(0) = U'(0)^2 = 4$.

TIPS & TRICKS (AI-derived explanation)

- Every MGF satisfies $U(0) = 1$, so $\log U(0) = 0$ - that is the fact that makes the composite collapse at $t = 0$.
- Chain rule in two layers: the derivative of $\log U(t)$ is $U'(t)/U(t)$, which at $t = 0$ equals $U'(0)/1 = E(X)$.
- The result is $E(Y) = [E(X)]^2 = 4$, so the composition SQUARES the mean - a neat structural observation.
- Option (b) 2 is the trap for candidates who evaluate only the outer derivative and forget the inner factor.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. U and V are the moment generating functions of X and Y , so $E(X) = U'(0) = 2$ and $E(Y) = V'(0)$.
2. Every moment generating function satisfies $U(0) = E(e^0) = 1$.
3. Differentiating $V(t) = U(\log U(t))$ by the chain rule: $V'(t) = U'(\log U(t)) \cdot d/dt[\log U(t)] = U'(\log U(t)) \cdot U'(t)/U(t)$.
4. At $t = 0$: $\log U(0) = \log 1 = 0$, and $U(0) = 1$.
5. So $V'(0) = U'(0) \cdot U'(0)/1 = [U'(0)]^2 = 2^2$.
6. Hence $E(Y) = 4$, option (c).

22. **2024 · Q34** Mathematical Expectation · Numerical**QUESTION**

The probability mass function of a random variable X is given by $P(X = n) = 2^{-n}$; $n = 1, 2, 3, \dots$. Define a random variable

$$Y = \begin{cases} -1, & \text{if } X \text{ is odd} \\ +1, & \text{if } X \text{ is even} \end{cases}$$

With reference to the above, consider the following statements: I. $P(Y = 1) = 2/3$ II. $E(Y) = -1/3$ III. $V(Y) = 8/9$ Which of the statements given above are correct?

OPTIONS

- (a) I and II only
 (b) II and III only
 (c) I and III only
 (d) I, II and III

ANSWER (AI-derived — no official key exists for this paper)**(b)** II and III only**EXAM SHORTCUT (AI-derived explanation)**

$P(X \text{ even}) = 1/4 + 1/16 + \dots = 1/3$, so $P(Y = 1) = 1/3$ (not $2/3$) and statement I fails. Then $E(Y) = -2/3 + 1/3 = -1/3$ and $\text{Var}(Y) = 1 - 1/9 = 8/9$, so II and III hold.

TIPS & TRICKS (AI-derived explanation)

- Sum the even terms as a geometric series with first term $1/4$ and ratio $1/4$: total $(1/4)/(1 - 1/4) = 1/3$.
- The odd terms then account for $2/3$, so the distribution is weighted towards ODD X - and $Y = -1$ on odd X , which is why $E(Y)$ is negative.
- Statement I attaches $2/3$ to $Y = +1$ when it belongs to $Y = -1$ - a straight swap, and the only error in the set.
- Since $Y^2 = 1$ always, $E(Y^2) = 1$ and $\text{Var}(Y) = 1 - (EY)^2 = 1 - 1/9 = 8/9$ with no further computation.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. X has $P(X = n) = 2^{-n}$ for $n = 1, 2, 3, \dots$, which sums to 1.
2. $P(X \text{ even}) = 2^{-2} + 2^{-4} + 2^{-6} + \dots = (1/4)/(1 - 1/4) = 1/3$, so $P(X \text{ odd}) = 2/3$.
3. Statement I: $Y = +1$ when X is even, so $P(Y = 1) = 1/3$, not $2/3$. FALSE.
4. Statement II: $E(Y) = (-1)(2/3) + (1)(1/3) = -1/3$. TRUE.
5. Statement III: since Y takes only the values -1 and $+1$, $Y^2 = 1$ and $E(Y^2) = 1$, so $V(Y) = 1 - (1/3)^2 = 1 - 1/9 = 8/9$. TRUE.
6. Only statements II and III are correct, option (b).

23. 2025 · Q8 Mathematical Expectation · Numerical**QUESTION**

Let $k > 0$ be a constant such that pdf of X is $f(x) = \begin{cases} kx(1-x), & 0 < x < 1 \\ 0, & \text{elsewhere} \end{cases}$. What is $E(X)$?

OPTIONS

- (a) $\frac{1}{2}$
 (b) $\frac{1}{4}$
 (c) $\frac{1}{6}$
 (d) $\frac{1}{8}$

ANSWER (AI-derived — no official key exists for this paper)

(a) $\frac{1}{2}$

EXAM SHORTCUT (AI-derived explanation)

$kx(1-x)$ on $(0, 1)$ is the Beta(2, 2) density ($k = 6$), which is symmetric about $1/2$, so $E(X) = 1/2$.

TIPS & TRICKS (AI-derived explanation)

- Recognise the Beta kernel: $x^{a-1}(1-x)^{b-1}$ with $a = b = 2$, so the distribution is Beta(2, 2) and $k = 1/B(2, 2) = 6$.
- Beta mean $= a/(a+b) = 2/4 = 1/2$ - no integration needed once the family is identified.
- Symmetry shortcut: the density $kx(1-x)$ is symmetric about $x = 1/2$, so the mean must be $1/2$ whatever k is.
- The constant k never enters the answer, which is why the question can leave it unspecified - a hint that symmetry is the intended route.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The density $f(x) = kx(1-x)$ on $(0, 1)$ has the form $x^{a-1}(1-x)^{b-1}$ with $a = b = 2$, so $X \sim \text{Beta}(2, 2)$.
2. Normalisation: the $\int x(1-x)$ over $(0, 1)$ is $1/2 - 1/3 = 1/6$, so $k = 6$.

3. For $\text{Beta}(a, b)$ the mean is $a/(a + b) = 2/(2 + 2) = 1/2$.
4. Alternatively, the density is symmetric about $x = 1/2$ (since $x(1 - x)$ is unchanged by $x \rightarrow 1 - x$), so the mean is $1/2$ by symmetry.
5. Direct check: $E(X) = 6$ times the $\int x^2(1 - x) dx = 6(1/3 - 1/4) = 6(1/12) = 1/2$.
6. Option (a) is correct.

24. 2026 · Q15 Conditional Expectation · Numerical

QUESTION

X has pmf $p(x) = x/15$, for $x = 1, 2, 3, 4, 5$. What is $E(X \mid x > 1)$?

OPTIONS

- (a) 27/7
- (b) 45/14
- (c) 15/7
- (d) 15/14

ANSWER (AI-derived — no official key exists for this paper)

(a) 27/7

EXAM SHORTCUT (AI-derived explanation)

Drop $x = 1$ and renormalise. The 15 in the denominator cancels, so
 $E(X \mid X > 1) = (2^2 + 3^2 + 4^2 + 5^2)/(2 + 3 + 4 + 5) = 54/14 = 27/7$.

TIPS & TRICKS (AI-derived explanation)

- For a conditional expectation on a truncated support, take the ratio of unnormalised sums - the common denominator (15 here) cancels.
- Numerator terms are $x \cdot p(x)$ proportional to $x \cdot x = x^2$; denominator terms are $p(x)$ proportional to x . Recognising this turns the problem into sum of squares over sum.
- Check the pmf is valid: $(1 + 2 + 3 + 4 + 5)/15 = 1$, so $p(x) = x/15$ is proper.
- Bound check: conditioning on $X > 1$ must raise the mean above the unconditional $E(X) = 55/15 = 11/3 = 3.67$; $27/7 = 3.857$ is above it, while $15/7$ and $15/14$ are far too small.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The pmf is $p(x) = x/15$ for $x = 1, 2, 3, 4, 5$ (check: sum = $15/15 = 1$).
2. $P(X > 1) = (2 + 3 + 4 + 5)/15 = 14/15$.
3. $E(X \cdot \text{indicator}(X > 1)) = [2(2) + 3(3) + 4(4) + 5(5)]/15 = (4 + 9 + 16 + 25)/15 = 54/15$.
4. $E(X \mid X > 1) = (54/15)/(14/15) = 54/14 = 27/7$ (approximately 3.857).
5. Hence option (a) is correct.

P9 · PROBABILITY

Characteristic Function, MGF & PGF; Inversion, Uniqueness, Continuity

21 questions · 2018: 2 2019: 0 2020: 2 2021: 1 2022: 0 2023: 3 2024: 7 2025: 5 2026: 1

Weight. 21 questions (10.7% of Probability) · appeared in 7 of 9 papers · peak year 2024 (7)**Examiner's pattern.** Rising sharply — 7 items in 2024 and 5 in 2025. Recurring: identify a distribution from its MGF, MGF of a linear combination of normals, the pgf $P(s) = E(s^X)$, and the compound/random-sum result $M_{SN}(t) = M_N(\log M_X(t))$, which was asked in 2024, 2025 AND 2026. Characteristic functions of products and ratios also appear.**Must-know.** $M_{(aX+b)}(t) = e^{bt} M_X(at)$; uniqueness theorem; compound sums: $M_{SN}(t) = M_N(\log M_X(t))$; cumulants $k_1 = \mu_1'$, $k_2 = \mu_2'$, $k_3 = \mu_3'$, $k_4 = \mu_4' - 3\mu_2'^2$.**1.** **2018 · Q17** Moment Generating Function · Numerical

QUESTION

 X, Y independent normal, means 10, 5, variances 9, 4. $Z = 4X - 5Y$. The MGF $M_Z(t)$ is

OPTIONS

- (a) $\exp[15t + 22t^2]$
- (b) $\exp[15t - 22t^2]$
- (c) $\exp[15t + 122t^2]$
- (d) $\exp[15t - 122t^2]$

ANSWER (AI-derived — no official key exists for this paper)**(c)** $\exp[15t + 122t^2]$

EXAM SHORTCUT (AI-derived explanation)

 $Z = 4X - 5Y$ is normal with mean $4(10) - 5(5) = 15$ and variance $16(9) + 25(4) = 244$. The normal MGF is $\exp(mt + s^2 t^2 / 2) = \exp(15t + 122t^2)$. Note the variance ADDS even though the coefficient is negative.

TIPS & TRICKS (AI-derived explanation)

- $\text{Var}(aX - bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y)$ for independent X, Y - the minus sign disappears when squared. Options (b) and (d) with a negative t^2 coefficient are impossible for any MGF.
- The MGF exponent carries $\sigma^2/2$, not σ^2 . Here $244/2 = 122$, which separates (c) from the 22 in option (a).
- Option (a)'s 22 comes from $4(9) - 5(4) = 4$ i.e. from adding variances without squaring the coefficients - the single most common slip in this family.
- A linear combination of independent normals is always normal, so you only ever need its mean and variance; never attempt a convolution integral.

STEP-BY-STEP SOLUTION (AI-derived explanation)

- $X \sim N(10, 9)$ and $Y \sim N(5, 4)$ are independent, so $Z = 4X - 5Y$ is normally distributed.
- Mean: $E(Z) = 4E(X) - 5E(Y) = 4(10) - 5(5) = 40 - 25 = 15$.
- Variance: $\text{Var}(Z) = 4^2 \text{Var}(X) + (-5)^2 \text{Var}(Y) = 16(9) + 25(4) = 144 + 100 = 244$.
- The MGF of a $N(\mu, \sigma^2)$ variable is $M(t) = \exp(\mu t + \sigma^2 t^2 / 2)$.
- Substituting $\mu = 15$ and $\sigma^2 = 244$ gives $M_Z(t) = \exp(15t + 244t^2 / 2) = \exp(15t + 122t^2)$.

6. Hence option (c) is correct.

2. 2018 · Q18 Probability Generating Function · Numerical

QUESTION

A fair die is thrown repeatedly until 3 or 4 appears. X = number of throws. The p.g.f. of X for $|s| < 3/2$ is

OPTIONS

- (a) $P(s) = \frac{2s}{3-s}$
 (b) $P(s) = \frac{s}{3-2s}$
 (c) $P(s) = \frac{3-s}{2s}$
 (d) $P(s) = \frac{3-2s}{s}$

ANSWER (AI-derived — no official key exists for this paper)

(b) $P(s) = \frac{s}{3-2s}$

EXAM SHORTCUT (AI-derived explanation)

Success = 'a 3 or a 4' has $p = 2/6 = 1/3$, so X is geometric on $\{1, 2, 3, \dots\}$. Its PGF is $ps/(1 - qs) = (s/3)/(1 - 2s/3) = s/(3 - 2s)$.

TIPS & TRICKS (AI-derived explanation)

- Memorise the two geometric PGFs and keep them apart: counting TRIALS (support $1, 2, 3, \dots$) gives $ps/(1 - qs)$; counting FAILURES (support $0, 1, 2, \dots$) gives $p/(1 - qs)$.
- Radius-of-convergence check: $1 - qs = 0$ at $s = 1/q = 3/2$, which matches the stated $|s| < 3/2$. That single check confirms $q = 2/3$ and eliminates option (a), whose pole is at $s = 3$.
- Every PGF must satisfy $P(1) = 1$. Testing $s = 1$: option (b) gives $1/(3 - 2) = 1$ (valid); option (a) gives $2/2 = 1$ as well, so use the pole test to separate them.
- Options (c) and (d) are reciprocals of the first two and blow up at $s = 0$, so they are not PGFs at all - strike them immediately.

STEP-BY-STEP SOLUTION (AI-derived explanation)

- A 'success' is the appearance of a 3 or a 4, so $p = 2/6 = 1/3$ and $q = 1 - p = 2/3$.
- X is the number of throws required for the first success, so X takes values $1, 2, 3, \dots$ with $P(X = k) = q^{k-1}p$.
- The probability generating function is $P(s) = E(s^X) = \sum q^{k-1}ps^k = ps \sum (qs)^{k-1} = ps/(1 - qs)$, valid for $|qs| < 1$.
- Substituting $p = 1/3$ and $q = 2/3$: $P(s) = (s/3)/(1 - 2s/3)$.
- Multiplying numerator and denominator by 3: $P(s) = s/(3 - 2s)$.
- The condition $|qs| < 1$ becomes $|s| < 3/2$, exactly as stated. Hence option (b) is correct.

3. 2020 · Q28 Moment Generating Function · Numerical

QUESTION

Y has p.d.f. $f(y) = e^y$ if $y < 0$ (else 0). The MGF $M_Y(t)$ of Y is

OPTIONS

- (a) $\frac{1}{1-t}$ for $t < 1$

- (b) $\frac{1}{1-|t|}$ for all t
- (c) $\frac{1}{1+t}$ for $t > -1$
- (d) $\frac{1}{1+|t|}$ for all t

ANSWER (AI-derived — no official key exists for this paper)

(c) $\frac{1}{1+t}$ for $t > -1$

EXAM SHORTCUT (AI-derived explanation)

$M(t) = \text{the } \int e^{ty} e^y dy = \text{the } \int e^{(t+1)y} dy \text{ from minus } \infty \text{ to } 0 = 1/(t+1), \text{ converging when } t+1 > 0, \text{ i.e. } t > -1.$

TIPS & TRICKS (AI-derived explanation)

- Y here is the NEGATIVE of an exponential ($Y = -E$ with $E \sim \text{Exp}(1)$), so its MGF is the exponential MGF with t replaced by $-t$: $1/(1+t)$.
- Convergence direction flips with the sign of the support: for $\text{Exp}(1)$ the MGF exists for $t < 1$, so for $-\text{Exp}(1)$ it exists for $t > -1$.
- Always state the region of convergence - it is what distinguishes option (a) from option (c), since the formulas are mirror images.
- Options (b) and (d) use $|t|$ and are not differentiable at $t = 0$, so they cannot be MGFs (an MGF must be smooth at 0 to generate moments).

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The density is $f(y) = e^y$ for $y < 0$ and 0 otherwise. Check: the $\int e^y$ from minus ∞ to 0 is 1, so f is a valid density.
2. $M_Y(t) = E(e^{tY}) = \text{the integral from minus } \infty \text{ to } 0 \text{ of } e^{ty} e^y dy = \text{the } \int e^{(t+1)y} dy.$
3. The antiderivative is $e^{(t+1)y}/(t+1)$, evaluated from minus ∞ to 0.
4. For the lower limit to vanish we need $t+1 > 0$, i.e. $t > -1$; then the value is $1/(t+1) - 0 = 1/(1+t)$.
5. So $M_Y(t) = 1/(1+t)$ for $t > -1$.
6. Option (c) is correct.

4. 2020 · Q35 Probability Generating Function · Numerical

QUESTION

The p.g.f. of a Poisson distribution has value e^{-1} at $2/3$. The mean of the distribution is

OPTIONS

- (a) $e^{-2/3}$
- (b) $\frac{2}{3}$
- (c) $\frac{3}{2}$
- (d) 3

ANSWER (AI-derived — no official key exists for this paper)

(d) 3

EXAM SHORTCUT (AI-derived explanation)

Poisson PGF is $P(s) = e^{\lambda(s-1)}$. At $s = 2/3$ this is $e^{-\lambda/3} = e^{-1}$, so $\lambda/3 = 1$ and $\lambda = 3$. The Poisson mean is $\lambda = 3$.

TIPS & TRICKS (AI-derived explanation)

- Poisson PGF: $P(s) = e^{\lambda(s-1)}$; the MGF is $e^{\lambda(e^t-1)}$. Keep the two apart - substituting into the wrong one is the main trap.
- $s - 1 = 2/3 - 1 = -1/3$, so the exponent is $-\lambda/3$. Equating exponents is faster than taking logarithms.
- For the Poisson, mean = variance = λ , so once $\lambda = 3$ is found the answer is immediate.
- Check: $P(1) = e^0 = 1$ for every λ , as any PGF must satisfy - a quick validity test of the formula you are using.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The probability generating function of a Poisson variable with parameter λ is $P(s) = E(s^X) = e^{\lambda(s-1)}$.
2. We are told $P(2/3) = e^{-1}$.
3. Substituting $s = 2/3$: $P(2/3) = e^{\lambda(2/3-1)} = e^{-\lambda/3}$.
4. Setting $e^{-\lambda/3} = e^{-1}$ gives $\lambda/3 = 1$, so $\lambda = 3$.
5. For the Poisson distribution the mean equals λ .
6. Hence the mean is 3, option (d).

5. 2021 · Q8 Characteristic Function · Theoretical**QUESTION**

Consider the following statements in respect of the characteristic function of a random variable: 1. It always exists. 2. It is uniformly continuous on $\mathbb{R}$. 3. It is not independent of change of origin and scale. 4. If the characteristic function of the sum of two random variables equals the product of their individual characteristic functions, then the variables are independent. Which of the above statements are correct?

OPTIONS

- (a) 1, 2 and 3
- (b) 1, 2 and 4
- (c) 1, 3 and 4
- (d) 2, 3 and 4

ANSWER (AI-derived — no official key exists for this paper)

(a) 1, 2 and 3

EXAM SHORTCUT (AI-derived explanation)

Statements 1, 2 and 3 are standard properties of the characteristic function. Statement 4 is the FALSE converse: a factorising characteristic function of the sum does not imply independence.

TIPS & TRICKS (AI-derived explanation)

- Existence is guaranteed because $|e^{itx}| = 1$, so the integral always converges - unlike the MGF, which may fail to exist.
- Uniform continuity follows from $|\phi(t+h) - \phi(t)| \leq E |e^{i(t+h)X} - e^{itX}|$, a bound free of t . Learn this as a standard property.

- Change of origin and scale: $\phi_{aX+b}(t) = e^{ibt}\phi_X(at)$, so the characteristic function certainly DOES depend on both - statement 3 is correctly worded as a negative and is true.
- Statement 4 states the converse of a true theorem. Independence implies factorisation, but factorisation of the sum's characteristic function does not imply independence - counterexamples exist.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Statement 1: $\phi(t) = E(e^{itX})$ always exists because $|e^{itX}| = 1$, so the expectation is finite for every t and every distribution. TRUE.
2. Statement 2: $|\phi(t+h) - \phi(t)| = |E[e^{itX}(e^{ihX} - 1)]| \leq E|e^{ihX} - 1|$, which tends to 0 as h tends to 0 uniformly in t . Hence ϕ is uniformly continuous on $\mathbb{R}$. TRUE.
3. Statement 3: for $Y = aX + b$, $\phi_Y(t) = e^{ibt}\phi_X(at)$, so the characteristic function changes under a change of origin or scale - it is NOT independent of them. TRUE.
4. Statement 4: if X and Y are independent then $\phi_{X+Y} = \phi_X\phi_Y$, but the converse fails. There exist dependent pairs whose sum's characteristic function still factorises. FALSE.
5. The correct statements are 1, 2 and 3.
6. Hence option (a) is correct.

6. 2023 · Q11 Binomial Distribution · Numerical

QUESTION

If random variable X follows Binomial distribution with $n = 10$ and $p = \frac{1}{4}$, then what is the value of third cumulant of $Y = 10 - X$?

OPTIONS

- (a) $-\frac{15}{16}$
 (b) $\frac{15}{16}$
 (c) $\frac{5}{8}$
 (d) $\frac{5}{16}$

ANSWER (AI-derived — no official key exists for this paper)

(a) $-\frac{15}{16}$

EXAM SHORTCUT (AI-derived explanation)

$Y = 10 - X \sim \text{Binomial}(10, 3/4)$. The third cumulant of $B(n, p)$ is $npq(q-p) = 10(3/4)(1/4)(1/4 - 3/4) = -15/16$.

TIPS & TRICKS (AI-derived explanation)

- Reflection rule: $X \sim B(n, p)$ implies $n - X \sim B(n, 1 - p)$. Apply it FIRST and then use the standard cumulant formula.
- Third cumulant of the binomial: $k_3 = npq(q-p)$. It changes sign with the skewness, so $p > 1/2$ gives a negative value.
- Here $p = 3/4$ exceeds $q = 1/4$, so $q - p = -1/2$ and the answer must be NEGATIVE - only option (a) qualifies.
- Alternative route: cumulants of a reflected variable satisfy $k_3(c - X) = -k_3(X)$. Since $k_3(X) = 10(1/4)(3/4)(1/2) = 15/16$, the reflection gives $-15/16$.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. $X \sim \text{Binomial}(10, 1/4)$, and $Y = 10 - X$ counts the failures, so $Y \sim \text{Binomial}(10, 3/4)$.
2. For Y the parameters are $n = 10, p = 3/4$ and $q = 1/4$.
3. The third cumulant of a binomial variable is $k_3 = npq(q - p)$.
4. Substituting: $k_3 = 10(3/4)(1/4)(1/4 - 3/4) = 10(3/16)(-1/2) = -30/32 = -15/16$.
5. Equivalently, $k_3(10 - X) = -k_3(X) = -10(1/4)(3/4)(3/4 - 1/4) = -15/16$. Option (a) is correct.

7. 2023 · Q13 Mathematical Expectation · Numerical**QUESTION**

If random variable X assumes non-negative integral values and $P(s) = E(s^X)$, then what is the value of

$$\sum_{n=0}^{\infty} s^{2n} P(X \leq n)?$$

OPTIONS

- (a) $\frac{P(\sqrt{s})}{1 - s^2}$
 (b) $\frac{P(s)}{1 - s^2}$
 (c) $\frac{P(s^2)}{1 - s^2}$
 (d) $\frac{P(\sqrt{s})}{1 - s}$

ANSWER (AI-derived — no official key exists for this paper)

(c) $\frac{P(s^2)}{1 - s^2}$

EXAM SHORTCUT (AI-derived explanation)

Swap the order of summation: $\sum s^{2n}$ sum over $k \leq n$ of $p_k = \sum p_k \sum s^{2n} = \text{sum of } p_k s^{2k} / (1 - s^2) = P(s^2) / (1 - s^2)$.

TIPS & TRICKS (AI-derived explanation)

- Interchange the summations - the double sum over the region $k \leq n$ is the standard manoeuvre for turning a CDF generating function into a pmf one.
- The inner geometric sum over $n \geq k$ gives $s^{2k} / (1 - s^2)$, which supplies both the argument s^2 and the denominator.
- The surviving sum of $p_k (s^2)^k$ is by definition $P(s^2)$, since $P(t) = \sum p_k t^k$. Substituting the argument correctly is what separates the options.
- Sanity check at $s = 0$: the left side is $P(X \leq 0) = p_0$, and $P(0) / (1 - 0) = p_0$. Only options with argument s^2 or s pass, and the general identity settles it.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Write $P(X \leq n) = \text{sum over } k = 0 \text{ to } n \text{ of } p_k$, where $p_k = P(X = k)$.
2. Then the required sum is $S = \sum s^{2n}$ sum over $k = 0$ to n of p_k .
3. Interchange the order of summation over the region $0 \leq k \leq n$: $S = \sum p_k \sum s^{2n}$.
4. The inner geometric series is $s^{2k} + s^{2k+2} + \dots = s^{2k} / (1 - s^2)$ for $|s| < 1$.
5. So $S = [1 / (1 - s^2)] \sum p_k (s^2)^k = P(s^2) / (1 - s^2)$, since $P(t) = E(t^X) = \sum p_k t^k$.

6. Hence option (c) is correct.

8. 2023 · Q15 Moment Generating Function · Numerical

COMMON DATA FOR THIS GROUP

Questions 15–17 are based on the following joint pmf of X and Y ($0 < \theta < 1$):

$X \backslash Y$	-2	-1	0	1	2
-1	θ	2θ	3θ	θ	θ
0	2θ	3θ	θ	0	0
1	θ	θ	θ	θ	2θ

QUESTION

What is the moment generating function of $X \mid Y = 2$?

OPTIONS

- (a) $\frac{e^{-t} + 2e^t}{3}$
 (b) $\frac{1 + e^{-t} + e^t}{3}$
 (c) $\frac{2e^{-t} + e^t}{3}$
 (d) $(1 - 2t)^{-1}$

ANSWER (AI-derived — no official key exists for this paper)

(a) $\frac{e^{-t} + 2e^t}{3}$

EXAM SHORTCUT (AI-derived explanation)

The $Y = 2$ column has masses $\theta, 0, 2\theta$ for $X = -1, 0, 1$, so the conditional probabilities are $1/3, 0, 2/3$. The MGF is $(e^{-t} + 2e^t)/3$.

TIPS & TRICKS (AI-derived explanation)

- For a conditional distribution, take only the relevant COLUMN and renormalise by its total - the value of θ never matters.
- Column $Y = 2$ totals $\theta + 0 + 2\theta = 3\theta$, so the weights are $1/3$ and $2/3$ on $X = -1$ and $X = +1$.
- MGF = $\sum e^{tx} P(X = x \mid Y = 2) = (1/3)e^{-t} + (2/3)e^t$. Options (b) and (c) attach the weights to the wrong values.
- Validity check: every MGF must equal 1 at $t = 0$. Here $(1 + 2)/3 = 1$, as required; option (b) gives $3/3 = 1$ too, so use the weights, not this test, to choose.

STEP-BY-STEP SOLUTION (AI-derived explanation)

- Summing the whole table: row $X = -1$ gives 8θ , row $X = 0$ gives 6θ and row $X = 1$ gives 6θ , a total of $20\theta = 1$, so $\theta = 1/20$.
- The column $Y = 2$ has entries θ at $X = -1$, 0 at $X = 0$ and 2θ at $X = 1$, totalling 3θ .
- Conditional probabilities: $P(X = -1 \mid Y = 2) = \theta/3\theta = 1/3$, $P(X = 0 \mid Y = 2) = 0$, $P(X = 1 \mid Y = 2) = 2\theta/3\theta = 2/3$.
- The moment generating function is $M(t) = E(e^{tX} \mid Y = 2) = (1/3)e^{-t} + 0 + (2/3)e^t = (e^{-t} + 2e^t)/3$.

5. Check $M(0) = (1 + 2)/3 = 1$. Option (a) is correct.

9. **2024 · Q6** Moment Generating Function · Numerical

COMMON DATA FOR THIS GROUP

Questions 4–6 are based on: let $X_1, X_2, X_3, \dots, X_n$ be iid $U(0, 1)$ variates and $T = -\ln(X_1 X_2 X_3 \cdots X_n)$.

QUESTION

What is the moment-generating function of T ?

OPTIONS

- (a) $(1 - 2t)^{-2n}$
- (b) $(1 - t)^{-2n}$
- (c) $(1 - t)^{-n}$
- (d) $(1 - 2t)^{-n/2}$

ANSWER (AI-derived — no official key exists for this paper)

(c) $(1 - t)^{-n}$

EXAM SHORTCUT (AI-derived explanation)

Each $-\ln X_i$ is $\text{Exp}(1)$ with MGF $(1 - t)^{-1}$; n independent copies multiply, giving $(1 - t)^{-n}$.

TIPS & TRICKS (AI-derived explanation)

- MGF of $\text{Exp}(\text{rate } \lambda)$ is $\lambda/(\lambda - t) = (1 - t/\lambda)^{-1}$; with $\lambda = 1$ that is $(1 - t)^{-1}$, valid for $t < 1$.
- MGFs of INDEPENDENT variables multiply, so n copies give the n th power - the exponent equals the number of summands.
- Consistency check: differentiating $(1 - t)^{-n}$ at $t = 0$ gives n , matching $E(T) = n$; the second derivative gives $n(n + 1)$, so $\text{Var} = n(n + 1) - n^2 = n$, matching $\text{Var}(T)$.
- Note $(1 - 2t)^{-n}$ would be the MGF of χ^2_{2n} ; since $T = \chi^2_{2n}/2$ here, the factor 2 is absent - a common confusion.

STEP-BY-STEP SOLUTION (AI-derived explanation)

- $T = \sum(-\ln X_i)$, a sum of n independent $\text{Exponential}(1)$ variables.
- The moment generating function of an $\text{Exponential}(1)$ variable is $E(e^{tW}) = \int e^{tw} e^{-w} dw = 1/(1 - t)$ for $t < 1$.
- For independent summands the MGFs multiply, so $M_T(t) = [(1 - t)^{-1}]^n = (1 - t)^{-n}$.
- Check: $M'(0) = n$ gives $E(T) = n$, and $M''(0) = n(n + 1)$ gives $\text{Var}(T) = n(n + 1) - n^2 = n$, agreeing with the previous parts.
- This is the $\Gamma(\text{shape } n, \text{rate } 1)$ moment generating function, as expected.
- Option (c) is correct.

10. **2024 · Q16** Moment Generating Function · Factual

QUESTION

The moment-generating function of X , when X follows $\chi^2(10)$, is:

OPTIONS

- (a) $(1 - t)^{-5}$
- (b) $(1 - t)^{-10}$
- (c) $(1 - 2t)^{-5}$
- (d) $(1 - 2t)^{-10}$

ANSWER (AI-derived — no official key exists for this paper)**(c)** $(1 - 2t)^{-5}$ **EXAM SHORTCUT (AI-derived explanation)**

The χ^2 MGF is $(1 - 2t)^{-k/2}$; with $k = 10$ the exponent is -5, giving $(1 - 2t)^{-5}$.

TIPS & TRICKS (AI-derived explanation)

- Memorise $M(t) = (1 - 2t)^{-k/2}$ for χ^2 with k d.f., valid for $t < 1/2$. The 2 inside and the halved exponent are both essential.
- For $k = 10$ the exponent is $-10/2 = -5$. Options (b) and (d) keep the full 10 and options (a) and (c) drop the 2.
- Verify by differentiating: $M'(0) = k = 10$ gives the mean, and $M''(0) = k(k + 2) = 120$ gives $\text{Var} = 120 - 100 = 20 = 2k$. Both correct.
- The χ^2 is $\Gamma(\text{shape } k/2, \text{scale } 2)$, so its MGF is the Γ MGF $(1 - \text{scale} \times t)^{-(\text{shape})}$ - one formula covers both families.

STEP-BY-STEP SOLUTION (AI-derived explanation)

- A χ^2 variable with k degrees of freedom is Γ with shape $k/2$ and scale 2.
- The moment generating function of $\Gamma(\text{shape } a, \text{scale } b)$ is $(1 - bt)^{-a}$ for $t < 1/b$.
- With $a = k/2$ and $b = 2$, this gives $M(t) = (1 - 2t)^{-k/2}$, valid for $t < 1/2$.
- For $k = 10$ the exponent is $-10/2 = -5$.
- Hence $M(t) = (1 - 2t)^{-5}$.
- Check: $M'(0) = 10 = E(X)$ and $M''(0) = 120$, so $\text{Var}(X) = 120 - 100 = 20 = 2(10)$. Option (c) is correct.

11. **2024 · Q23** Characteristic Function · Numerical**QUESTION**

If X, Y and Z are iid $N(0, 1)$ variates, then the characteristic function of $\frac{X + YZ}{\sqrt{1 + Z^2}}$ is:

OPTIONS

- (a) $e^{-\frac{t^2}{2}}$
- (b) e^{-2t^2}
- (c) $\frac{1}{1 + t^2}$
- (d) $\frac{1}{\sqrt{1 + t^2}}$

ANSWER (AI-derived — no official key exists for this paper)

(a) $e^{-\frac{t^2}{2}}$

EXAM SHORTCUT (AI-derived explanation)

Condition on $Z = z$: $X + Yz$ is $N(0, 1 + z^2)$, so dividing by $\sqrt{1 + z^2}$ gives a standard normal for EVERY z . Hence the ratio is $N(0, 1)$ with characteristic function $e^{-t^2/2}$.

TIPS & TRICKS (AI-derived explanation)

- Condition on the awkward variable Z first - the conditional law turns out to be free of z , which is the whole design of the question.
- $\text{Var}(X + Yz) = 1 + z^2$ exactly matches the square of the divisor, so the standardisation is exact rather than approximate.
- When a conditional distribution does not depend on the conditioning variable, the unconditional distribution is the same - no averaging is needed.
- The characteristic function of $N(0, 1)$ is $e^{-t^2/2}$; options (c) and (d) are Laplace-type and cannot arise from a normal.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Condition on $Z = z$. Given this, X and Y are still independent $N(0, 1)$, so $X + Yz$ is normal with mean 0 and variance $1 + z^2$.
2. Dividing by the constant $\sqrt{1 + z^2}$ gives $(X + Yz)/\sqrt{1 + z^2} \sim N(0, 1)$.
3. This conditional distribution is the same for every value of z , so it does not depend on Z at all.
4. Hence the unconditional distribution of $(X + YZ)/\sqrt{1 + Z^2}$ is also $N(0, 1)$.
5. The characteristic function of a standard normal variable is $E(e^{itW}) = e^{-t^2/2}$.
6. Option (a) is correct.

12. 2024 · Q24 Characteristic Function · Numerical**QUESTION**

Two random variables X and Y are such that $P(X = \pm 1) = 0.5$ and $Y \sim N(0, 1)$. What is the characteristic function of $2XY$?

OPTIONS

- (a)** $e^{-\frac{t^2}{2}}$
- (b)** e^{-2t^2}
- (c)** $\frac{1}{4 + t^2}$
- (d)** $\frac{1}{\sqrt{4 + t^2}}$

ANSWER (AI-derived — no official key exists for this paper)

(b) e^{-2t^2}

EXAM SHORTCUT (AI-derived explanation)

Multiplying a standard normal by a symmetric random sign leaves it standard normal, so $XY \sim N(0, 1)$ and

$2XY \sim N(0, 4)$, with characteristic function $e^{-4t^2/2} = e^{-2t^2}$.

TIPS & TRICKS (AI-derived explanation)

- Random sign trick: if Y is symmetric about 0 and $X = \pm 1$ independently, then XY has the SAME distribution as Y . Condition on X to see it.
- $\text{Var}(2XY) = 4 \text{Var}(XY) = 4$, so the characteristic function exponent is $-\sigma^2 t^2/2 = -2t^2$.
- Do not halve twice: the formula is $e^{-\sigma^2 t^2/2}$ with $\sigma^2 = 4$, giving $-2t^2$, not $-t^2$.
- Options (c) and (d) are Laplace/Cauchy-type functions and cannot be the characteristic function of a normal variable.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. X takes the values $+1$ and -1 with probability $1/2$ each, independently of $Y \sim N(0, 1)$.
2. Condition on X : given $X = +1$, $XY = Y \sim N(0, 1)$; given $X = -1$, $XY = -Y$, which is also $N(0, 1)$ by the symmetry of the normal.
3. Since the conditional distribution is $N(0, 1)$ in both cases, $XY \sim N(0, 1)$ unconditionally.
4. Therefore $2XY \sim N(0, 4)$.
5. The characteristic function of $N(0, \sigma^2)$ is $e^{-\sigma^2 t^2/2}$, which with $\sigma^2 = 4$ gives e^{-2t^2} .
6. Option (b) is correct.

13. 2024 · Q25 Characteristic Function · Numerical

QUESTION

If X_1 and X_2 are iid $N(0, 1)$ variates, then the characteristic function of $X_1 X_2$ is:

OPTIONS

- (a) e^{-t^2}
- (b) $\frac{1}{1+t^2}$
- (c) $\frac{1}{\sqrt{1+t^2}}$
- (d) $e^{-2|t|}$

ANSWER (AI-derived — no official key exists for this paper)

(c) $\frac{1}{\sqrt{1+t^2}}$

EXAM SHORTCUT (AI-derived explanation)

Condition on $X_2 = z$: $X_1 X_2 \sim N(0, z^2)$ with characteristic function $e^{-t^2 z^2/2}$. Averaging over $Z^2 \sim \chi_1^2$ gives the MGF at $-t^2/2$, namely $(1+t^2)^{-1/2}$.

TIPS & TRICKS (AI-derived explanation)

- Condition and then average: $\mathbb{E}[e^{-t^2 Z^2/2}]$ is the χ_1^2 moment generating function evaluated at the NEGATIVE argument $-t^2/2$.
- χ_1^2 MGF is $(1-2s)^{-1/2}$; with $s = -t^2/2$ the bracket becomes $1+t^2$ and the exponent $-1/2$.
- The product of two independent standard normals has heavier tails than a normal but lighter than a Cauchy - a square-root singularity is exactly right.
- Option (b) $1/(1+t^2)$ is the Laplace characteristic function, which would correspond to the product being a difference of exponentials - close, but the exponent is wrong.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Condition on $X_2 = z$. Given this, $X_1|z$ is normal with mean 0 and variance z^2 , so its characteristic function is $e^{-t^2 z^2/2}$.
2. Averaging over the distribution of X_2 gives $\phi(t) = E[e^{-t^2 X_2^2/2}]$.
3. Now $X_2^2 \sim \chi^2$ with 1 degree of freedom, whose moment generating function is $M(s) = (1 - 2s)^{-1/2}$.
4. The required expectation is $M(s)$ evaluated at $s = -t^2/2$.
5. So $\phi(t) = (1 - 2(-t^2/2))^{-1/2} = (1 + t^2)^{-1/2}$.
6. Hence the characteristic function is $1/\sqrt{1+t^2}$, option (c).

14. 2024 · Q37 Moment Generating Function · Numerical**QUESTION**

Let X and Y be two independent random variables having moment-generating functions

$$M_X(t) = \left(\frac{3}{4} + \frac{1}{4}e^t\right)^3 \quad \text{and} \quad M_Y(t) = e^{2(e^t-1)}$$

What is $P[(X + Y) < 2]$ equal to?

OPTIONS

- (a) $\frac{27}{64}e^{-2}$
 (b) $\frac{27}{16}e^{-2}$
 (c) $\frac{81}{32}e^{-2}$
 (d) $\frac{27}{32}e^{-2}$

ANSWER (AI-derived — no official key exists for this paper)

(b) $\frac{27}{16}e^{-2}$

EXAM SHORTCUT (AI-derived explanation)

$X \sim B(3, 1/4)$ and $Y \sim \text{Poisson}(2)$.

$$P(X + Y < 2) = P(0, 0) + P(0, 1) + P(1, 0) = (27/64)e^{-2} + (27/64)(2e^{-2}) + (27/64)e^{-2} = (27/16)e^{-2}.$$

TIPS & TRICKS (AI-derived explanation)

- Identify the distributions from the MGFs: $(q + pe^t)^n$ is Binomial(n, p) and $e^{\lambda(e^t-1)}$ is Poisson(λ). Here $n = 3, p = 1/4$ and $\lambda = 2$.
- $X + Y < 2$ means the sum is 0 or 1, so only three (x,y) combinations arise - enumerate them rather than seeking a closed form.
- Both $P(X = 0)$ and $P(X = 1)$ equal $27/64$ for $B(3, 1/4)$, which is a pleasant coincidence: $(3/4)^3 = 27/64$ and $3(1/4)(9/16) = 27/64$.
- Factor $27/64$ out of all three terms: $(27/64)(1 + 2 + 1)e^{-2} = (27/64)(4)e^{-2} = (27/16)e^{-2}$.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. $M_X(t) = (3/4 + (1/4)e^t)^3$ is the MGF of a Binomial(3, 1/4) variable, and $M_Y(t) = e^{2(e^t-1)}$ is the MGF of a Poisson(2) variable; X and Y are independent.
2. $P(X + Y < 2) = P(X + Y = 0) + P(X + Y = 1)$.
3. $P(X = 0) = (3/4)^3 = 27/64$ and $P(X = 1) = 3(1/4)(3/4)^2 = 27/64$; $P(Y = 0) = e^{-2}$ and $P(Y = 1) = 2e^{-2}$.

4. $P(X + Y = 0) = P(X = 0)P(Y = 0) = (27/64)e^{-2}$.
5. $P(X + Y = 1) = P(X = 0)P(Y = 1) + P(X = 1)P(Y = 0) = (27/64)(2e^{-2}) + (27/64)e^{-2} = (81/64)e^{-2}$.
6. Total $= (27/64 + 81/64)e^{-2} = (108/64)e^{-2} = (27/16)e^{-2}$, option (b).

15. **2024 · Q40** Moment Generating Function · Numerical

QUESTION

If the moment-generating function of a distribution is $M_X(t) = \frac{\lambda}{\lambda - t}$ for $t < \lambda$, then what are mean and variance of the distribution respectively?

OPTIONS

- (a) $\frac{-1}{\lambda - t}, \frac{\lambda^2}{(\lambda - t)^2}$
- (b) $\frac{1}{\lambda}, \frac{1}{\lambda^2}$
- (c) λ, λ^2
- (d) $1, \lambda^2$

ANSWER (AI-derived — no official key exists for this paper)

(b) $\frac{1}{\lambda}, \frac{1}{\lambda^2}$

EXAM SHORTCUT (AI-derived explanation)

$M(t) = \lambda/(\lambda - t) = (1 - t/\lambda)^{-1}$ is the exponential MGF with rate λ , whose mean is $1/\lambda$ and variance $1/\lambda^2$.

TIPS & TRICKS (AI-derived explanation)

- Recognise the family: $\lambda/(\lambda - t)$ for $t < \lambda$ is the exponential MGF. The condition $t < \lambda$ in the stem is the giveaway.
- Exponential with RATE λ has mean $1/\lambda$ and variance $1/\lambda^2$ - reciprocals, not λ itself. Option (c) inverts this.
- If you prefer to differentiate: $M'(t) = \lambda/(\lambda - t)^2$ gives $M'(0) = 1/\lambda$, and $M''(t) = 2\lambda/(\lambda - t)^3$ gives $M''(0) = 2/\lambda^2$, so $\text{Var} = 2/\lambda^2 - (1/\lambda)^2 = 1/\lambda^2$.
- Option (a) leaves t in the answer, which is impossible - moments are numbers, obtained by evaluating derivatives AT $t = 0$.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The moment generating function $M(t) = \lambda/(\lambda - t)$, valid for $t < \lambda$, can be written as $(1 - t/\lambda)^{-1}$.
2. This is the moment generating function of an exponential distribution with rate parameter λ .
3. Differentiating: $M'(t) = \lambda/(\lambda - t)^2$, so $E(X) = M'(0) = \lambda/\lambda^2 = 1/\lambda$.
4. Differentiating again: $M''(t) = 2\lambda/(\lambda - t)^3$, so $E(X^2) = M''(0) = 2\lambda/\lambda^3 = 2/\lambda^2$.
5. Hence $\text{Var}(X) = 2/\lambda^2 - (1/\lambda)^2 = 1/\lambda^2$.
6. The mean and variance are $1/\lambda$ and $1/\lambda^2$, option (b).

16. **2025 · Q9** Moment Generating Function · Numerical

QUESTION

Let $M_X(t)$ and $M_Y(t)$ be moment generating functions of X and Y . If $E(X) = \frac{1}{2}$, $\text{Var}(X) = \frac{1}{4}$, and $M_Y(t) = (0.75 + 0.25e^{2t})M_X(t)$ for $t \in (-h, h)$, $h > 0$, what is $\text{Var}(Y)$?

OPTIONS

- (a) $\frac{1}{4}$
 (b) $\frac{1}{2}$
 (c) 1
 (d) 2

ANSWER (AI-derived — no official key exists for this paper)**(c)** 1**EXAM SHORTCUT** (AI-derived explanation)

The extra factor $0.75 + 0.25e^{2t}$ is the MGF of a variable Z taking 0 and 2 with probabilities 0.75 and 0.25, independent of X . $\text{Var}(Z) = 4(0.25) - 0.25 = 0.75$, so $\text{Var}(Y) = 0.25 + 0.75 = 1$.

TIPS & TRICKS (AI-derived explanation)

- A PRODUCT of MGFs means a SUM of independent variables - so Y is distributed as $X + Z$ with Z independent of X .
- Read Z off its MGF: coefficients are probabilities and the exponents are values, so $Z = 0$ with probability 0.75 and $Z = 2$ with probability 0.25.
- $\text{Var}(Z) = E(Z^2) - [E(Z)]^2 = 4(0.25) - (0.5)^2 = 1 - 0.25 = 0.75$. (Equivalently $Z = 2W$ with $W \sim \text{Bernoulli}(0.25)$, giving $4(0.25)(0.75)$.)
- Variances of independent summands ADD, so $\text{Var}(Y) = 0.25 + 0.75 = 1$ - no knowledge of X 's distribution is needed beyond its variance.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. $M_Y(t) = (0.75 + 0.25e^{2t})M_X(t)$ expresses M_Y as a product of two moment generating functions.
2. A product of MGFs corresponds to the sum of INDEPENDENT random variables, so Y has the distribution of $X + Z$ where Z is independent of X .
3. The factor $0.75 + 0.25e^{2t}$ is the MGF of a variable taking the value 0 with probability 0.75 and the value 2 with probability 0.25.
4. $E(Z) = 2(0.25) = 0.5$ and $E(Z^2) = 4(0.25) = 1$, so $\text{Var}(Z) = 1 - 0.25 = 0.75$.
5. By independence, $\text{Var}(Y) = \text{Var}(X) + \text{Var}(Z) = 1/4 + 3/4 = 1$, option (c).

17. **2025 · Q12** Mathematical Expectation · Theoretical**QUESTION**

If k_r and μ'_r denote the r^{th} cumulant and r^{th} moment about the origin, which statements are correct? I.

$$k_1 = \mu'_1 \quad \text{II.} \quad k_2 = \mu'_2 - \mu_1'^2 \quad \text{III.} \quad k_3 = \mu'_3 - 3\mu'_2\mu'_1 + 2\mu_1'^3$$

OPTIONS

- (a) I and II only
 (b) II and III only
 (c) I and III only
 (d) I, II and III

ANSWER (AI-derived — no official key exists for this paper)**(d)** I, II and III**EXAM SHORTCUT (AI-derived explanation)**

The first three cumulants are the mean, the variance and the third central moment: $k_1 = \mu'_1$, $k_2 = \mu'_2 - \mu_1'^2$ and $k_3 = \mu'_3 - 3\mu'_2\mu'_1 + 2\mu_1'^3$. All three statements are standard.

TIPS & TRICKS (AI-derived explanation)

- Cumulants come from $\log M(t)$: k_1 is the mean, k_2 the variance and k_3 the third CENTRAL moment - the first three have direct interpretations.
- Statement III is exactly the expansion of μ_3 (the third central moment) in terms of raw moments - a formula worth memorising in its own right.
- $k_4 = \mu'_4 - 4\mu'_3\mu'_1 - 3\mu_2'^2 + 12\mu'_2\mu_1'^2 - 6\mu_1'^4$ differs from μ_4 , which is why cumulants and central moments part company from order 4 onwards.
- All cumulants beyond the second vanish for the normal distribution - a memorable characterisation.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Cumulants are defined by the expansion of the cumulant generating function $K(t) = \log M(t) = \sum k_r t^r / r!$.
2. Statement I: differentiating K once at $t = 0$ gives $K'(0) = M'(0)/M(0) = \mu'_1$, so k_1 is the mean. TRUE.
3. Statement II: differentiating twice gives $K''(0) = M''(0)/M(0) - [M'(0)/M(0)]^2 = \mu'_2 - \mu_1'^2$, the variance. TRUE.
4. Statement III: the third derivative yields $k_3 = \mu'_3 - 3\mu'_2\mu'_1 + 2\mu_1'^3$, which is precisely the third CENTRAL moment μ_3 expressed in raw moments. TRUE.
5. (From the fourth order onwards cumulants and central moments differ, but the first three coincide as stated.)
6. All three statements are correct, option (d).

18. **2025 · Q45** Moment Generating Function · Numerical**COMMON DATA FOR THIS GROUP**

Shared data for Q45–Q47. X and Y are independently distributed as $\text{Poisson}(\lambda)$, and their MGFs together with random variable Z satisfy $M_Z(t) = M_Y(\log M_X(t))$ for $t \in (-h, h)$, $h > 0$.

QUESTIONWhat is $E(Z)$?**OPTIONS**

- (a)** $\lambda + 1$
- (b)** λ^2
- (c)** $\lambda^2 + \lambda$
- (d)** $\lambda^4 + 1$

ANSWER (AI-derived — no official key exists for this paper)**(b)** λ^2

EXAM SHORTCUT (AI-derived explanation)

The relation $M_Z(t) = M_Y(\log M_X(t))$ makes Z a random sum of X 's stopped at Y , so $E(Z) = E(Y) E(X) = \lambda \lambda = \lambda^2$.

TIPS & TRICKS (AI-derived explanation)

- Recognise the compound structure: $M_S(t) = G_N(M_X(t))$ for a random sum, and $G_N(s) = M_N(\log s)$, so $M_S(t) = M_N(\log M_X(t))$ - exactly the given relation.
- Wald's identity: $E(S) = E(N) E(X)$. With both $\text{Poisson}(\lambda)$, that is λ^2 .
- Direct differentiation also works: $M'_Z(0) = M'_Y(0) \times M'_X(0)/M_X(0) = \lambda \lambda$, since every MGF equals 1 at $t = 0$.
- Option (c) $\lambda^2 + \lambda$ is the VARIANCE-style answer; keep mean and variance apart across the three parts of this stem.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. For a random sum $S = X_1 + \dots + X_N$ with the X_i i.i.d. and independent of N , the moment generating function is $M_S(t) = M_N(\log M_X(t))$.
2. The given relation $M_Z(t) = M_Y(\log M_X(t))$ therefore identifies Z as the sum of Y independent copies of X .
3. Both X and Y are $\text{Poisson}(\lambda)$, so $E(X) = E(Y) = \lambda$.
4. By Wald's identity, $E(Z) = E(Y) E(X) = \lambda \cdot \lambda$.
5. Equivalently, differentiating $M_Z(t) = M_Y(\log M_X(t))$ at $t = 0$ and using $M_X(0) = 1$ gives $M'_Z(0) = M'_Y(0) M'_X(0) = \lambda^2$.
6. Hence $E(Z) = \lambda^2$, option (b).

19. 2025 · Q46 Moment Generating Function · Numerical**COMMON DATA FOR THIS GROUP**

Shared data for Q45–Q47. X and Y are independently distributed as $\text{Poisson}(\lambda)$, and their MGFs together with random variable Z satisfy $M_Z(t) = M_Y(\log M_X(t))$ for $t \in (-h, h)$, $h > 0$.

QUESTION

What is $\text{Var}(Z)$?

OPTIONS

- (a) $1 + \lambda + \lambda^3$
 (b) $1 + \lambda^2 + \lambda^3$
 (c) $\lambda^2 + \lambda^3$
 (d) $\lambda^2 + \lambda^3 + \lambda^4$

ANSWER (AI-derived — no official key exists for this paper)

(c) $\lambda^2 + \lambda^3$

EXAM SHORTCUT (AI-derived explanation)

For a random sum, $\text{Var}(Z) = E(Y) \text{Var}(X) + \text{Var}(Y)[E(X)]^2 = \lambda(\lambda) + \lambda(\lambda^2) = \lambda^2 + \lambda^3$.

TIPS & TRICKS (AI-derived explanation)

- Compound-sum variance formula: $\text{Var}(S) = E(N) \text{Var}(X) + \text{Var}(N)[E(X)]^2$. Both terms are needed - dropping either loses a power of λ .
- With N and X both $\text{Poisson}(\lambda)$, every mean and variance equals λ , so the two terms are λ^2 and λ^3 .
- The presence of λ^3 reflects the extra variability from the RANDOM number of summands - a pure sum of a fixed number would give only λ^2 .
- Options containing a constant 1 (options a and b) cannot be right: setting $\lambda = 0$ must give $\text{Var} = 0$, since Z is then identically zero.

STEP-BY-STEP SOLUTION (AI-derived explanation)

- From the previous part, Z is the sum of Y independent copies of X , where X and Y are independent $\text{Poisson}(\lambda)$.
- For such a random sum the variance is $\text{Var}(Z) = E(Y) \text{Var}(X) + \text{Var}(Y)[E(X)]^2$.
- For the Poisson, mean and variance both equal λ , so $E(X) = \text{Var}(X) = E(Y) = \text{Var}(Y) = \lambda$.
- First term: $E(Y) \text{Var}(X) = \lambda \cdot \lambda = \lambda^2$.
- Second term: $\text{Var}(Y)[E(X)]^2 = \lambda \cdot \lambda^2 = \lambda^3$.
- Hence $\text{Var}(Z) = \lambda^2 + \lambda^3$, option (c).

20. 2025 · Q47 Moment Generating Function · Numerical**COMMON DATA FOR THIS GROUP**

Shared data for Q45–Q47. X and Y are independently distributed as $\text{Poisson}(\lambda)$, and their MGFs together with random variable Z satisfy $M_Z(t) = M_Y(\log M_X(t))$ for $t \in (-h, h)$, $h > 0$.

QUESTION

What is the moment generating function of Z ?

OPTIONS

- (a) $e^{\lambda^2(e^t-1)}$
- (b) $e^{\lambda[e^{\lambda(e^t-1)}-1]}$
- (c) $e^{\lambda(e^{t^2}-1)}$
- (d) $e^{\lambda[e^{\lambda(e^t-1)}-1]}$

ANSWER (AI-derived — no official key exists for this paper)

(d) $e^{\lambda[e^{\lambda(e^t-1)}-1]}$

EXAM SHORTCUT (AI-derived explanation)

$M_X(t) = e^{\lambda(e^t-1)}$, so $\log M_X(t) = \lambda(e^t-1)$. Substituting into M_Y gives $M_Z(t) = e^{\lambda[e^{\lambda(e^t-1)}-1]}$.

TIPS & TRICKS (AI-derived explanation)

- Poisson MGF: $M(t) = e^{\lambda(e^t-1)}$. Taking the logarithm just strips the exponential, leaving $\lambda(e^t-1)$ - a clean inner argument.
- Substitute that argument wherever t appears in M_Y : the outer exponent becomes $\lambda[e^{\text{inner}}-1]$.
- The result is the compound-Poisson (Neyman type A) MGF - a nested double exponential, which is the structural signature to look for.

- Check at $t = 0$: the inner bracket is $\lambda(1 - 1) = 0$, so the outer is $\lambda(e^0 - 1) = 0$ and $M_Z(0) = 1$, as every MGF must satisfy.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. For $X \sim \text{Poisson}(\lambda)$, the moment generating function is $M_X(t) = \exp[\lambda(e^t - 1)]$.
2. Taking logarithms: $\log M_X(t) = \lambda(e^t - 1)$.
3. Y is also $\text{Poisson}(\lambda)$, so $M_Y(s) = \exp[\lambda(e^s - 1)]$.
4. Substituting $s = \log M_X(t) = \lambda(e^t - 1)$: $M_Z(t) = \exp\{\lambda[e^{\lambda(e^t - 1)} - 1]\}$.
5. Check at $t = 0$: the inner exponent is 0, so $M_Z(0) = e^{\lambda(1-1)} = 1$, as required.
6. Hence $M_Z(t) = e^{\lambda[e^{\lambda(e^t - 1)} - 1]}$, option (d).

21. 2026 · Q18 Moment Generating Function · Theoretical**QUESTION**

$X_1, X_2, \dots$ are iid, $S_N = X_1 + X_2 + \dots + X_N$ where N is a random variable independent of the X_i . If $M_{X_1}(t) = E(e^{tX_1})$, the MGF of S_N is:

OPTIONS

- (a) $\log(\log M_{X_1}(t))$
- (b) $M_N(\log M_{X_1}(t))$
- (c) $M_{X_1}(\log M_N(t))$
- (d) $M_N(e^{M_{X_1}(t)})$

ANSWER (AI-derived — no official key exists for this paper)

(b) $M_N(\log M_{X_1}(t))$

EXAM SHORTCUT (AI-derived explanation)

Condition on N : $E(e^{tS_N}) = E[(M_X(t))^N] = E[e^{N \log M_X(t)}] = M_N(\log M_X(t))$. Answer (b).

TIPS & TRICKS (AI-derived explanation)

- The compounding rule: for a random sum, the outer MGF is evaluated at $\log$ of the inner MGF - $M_{S_N}(t) = M_N(\log M_X(t))$.
- In pgf language it is cleaner still: $P_{S_N}(s) = P_N(P_X(s))$ - a straight composition, with no logarithm.
- Derivation in one line: $E[a^N] = M_N(\log a)$ with $a = M_X(t)$, by writing $a^N = e^{N \log a}$.
- Standard application: if N is $\text{Poisson}(\lambda)$ then $M_{S_N}(t) = \exp(\lambda[M_X(t) - 1])$, the compound Poisson MGF - a frequent follow-up question.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. By the law of total expectation, $M_{S_N}(t) = E[e^{tS_N}] = E[E(e^{tS_N} | N)]$.
2. Given $N = n$, S_N is the sum of n iid copies of X_1 , so $E(e^{tS_N} | N = n) = [M_{X_1}(t)]^n$ by independence.
3. Hence $M_{S_N}(t) = E[(M_{X_1}(t))^N]$.
4. Write $a = M_{X_1}(t) > 0$. Then $a^N = e^{N \log a}$, so $E[a^N] = E[e^{N \log a}] = M_N(\log a)$.
5. Therefore $M_{S_N}(t) = M_N(\log M_{X_1}(t))$.
6. Check with $N \text{Poisson}(\lambda)$: $M_N(u) = \exp(\lambda(e^u - 1))$, so $M_{S_N}(t) = \exp(\lambda(M_{X_1}(t) - 1))$, the standard compound Poisson result.

7. Hence option (b) is correct.

P10 · PROBABILITY

Modes of Convergence of Sequences of Random Variables

5 questions · 2018: 0 2019: 0 2020: 1 2021: 1 2022: 2 2023: 0 2024: 0 2025: 1 2026: 0

Weight. 5 questions (2.6% of Probability) · appeared in 4 of 9 papers · peak year 2022 (2)**Examiner's pattern.** Small but pure-theory. Always a constructed counterexample: a sequence that converges in probability but not in mean square, or a limiting CDF that must be checked at the boundaries. Know which implication fails in which direction.**Must-know.** a.s. $\Rightarrow$ in probability $\Rightarrow$ in distribution; mean-square $\Rightarrow$ in probability. None of the converses hold without extra conditions (uniform integrability, or a constant limit for dist $\Rightarrow$ prob).1. **2020 · Q32** Modes of Convergence · Theoretical

QUESTION

 $U \sim U(0, 1)$. The sequence $X_n = \sqrt{n}$ if $0 < U < 1/n^2$ (else 0), $n \geq 1$, converges

OPTIONS

- (a) both with probability one and in mean square
- (b) with probability one but not in mean square
- (c) in mean square but not with probability one
- (d) neither with probability one nor in mean square

ANSWER (AI-derived — no official key exists for this paper)**(a)** both with probability one and in mean square**EXAM SHORTCUT** (AI-derived explanation)

$P(X_n \text{ is not } 0) = 1/n^2$, and the series of these converges, so Borel-Cantelli gives $X_n \rightarrow 0$ with probability one. Also $E(X_n^2) = n(1/n^2) = 1/n \rightarrow 0$, so convergence in mean square holds too.

TIPS & TRICKS (AI-derived explanation)

- Two separate computations, one for each mode: $P(X_n \text{ not } 0)$ for almost-sure convergence, $E(X_n^2)$ for mean-square convergence. Do them independently.
- Borel-Cantelli I: if the sum of $P(A_n)$ converges then $P(A_n \text{ infinitely often}) = 0$. Here sum $1/n^2$ converges, so eventually X_n is 0 forever.
- $E(X_n^2) = (\sqrt{n})^2 x P(U < 1/n^2) = nx1/n^2 = 1/n$. The value taken grows but its probability shrinks faster, so the second moment still dies.
- Contrast with the classic counterexample $X_n = n$ on $(0, 1/n)$: there $P(X_n \text{ not } 0) = 1/n$ whose sum DIVERGES and $E(X_n^2) = n$, so neither mode holds. Changing the exponents changes the answer completely.

STEP-BY-STEP SOLUTION (AI-derived explanation)

- Let A_n be the event $\{0 < U < 1/n^2\}$, on which $X_n = \sqrt{n}$; elsewhere $X_n = 0$. Then $P(A_n) = 1/n^2$.
- The series sum of $P(A_n) = \text{sum of } 1/n^2$ converges (it equals $\pi^2/6$).
- By the first Borel-Cantelli lemma, $P(A_n \text{ occurs infinitely often}) = 0$, so with probability one only finitely many A_n occur and $X_n = 0$ for all large n .
- Hence X_n converges to 0 with probability one.
- For mean-square convergence, $E(X_n^2) = nP(A_n) + 0 = n(1/n^2) = 1/n$, which tends to 0 as n grows.

6. So X_n also converges to 0 in mean square. Both modes hold, option (a).

2. 2021 · Q25 Modes of Convergence · Numerical

QUESTION

Suppose $(X_1, \dots, X_n)$ are iid with finite second moment, and for some $x \in \mathbb{R}$, $P(X_1 + \dots + X_n \leq x\sqrt{n}) \rightarrow \Phi(2)$. If $\text{Var}(X_1) = 4$, then $P\left(X_1 + \dots + X_n \leq \frac{1}{x}\sqrt{n}\right)$

OPTIONS

- (a) does not converge
- (b) converges to 0
- (c) converges to $\Phi(1/2)$
- (d) converges to $\Phi(1/8)$

ANSWER (AI-derived — no official key exists for this paper)

(d) converges to $\Phi(1/8)$

EXAM SHORTCUT (AI-derived explanation)

A finite non-degenerate limit forces $\mu = 0$, and then $x/\sigma = 2$ with $\sigma = 2$ gives $x = 4$. Replacing x by $1/x = 1/4$ gives the limit $\Phi((1/4)/2) = \Phi(1/8)$.

TIPS & TRICKS (AI-derived explanation)

- Standardise first: $P(\text{sum} \leq x\sqrt{n}) = P(Z_n \leq (x - \sqrt{n}\mu)/\sigma)$. For the limit to be a finite constant, the $\sqrt{n}\mu$ term must vanish, forcing $\mu = 0$.
- With $\mu = 0$ the argument is simply x/σ , so the given limit $\Phi(2)$ pins $x = 2\sigma = 4$.
- The second probability replaces x by $1/x = 1/4$, giving the argument $(1/4)/2 = 1/8$ - hence $\Phi(1/8)$.
- $\text{Var} = 4$ means $\sigma = 2$, not 4. Forgetting the square root turns $\Phi(1/8)$ into $\Phi(1/16)$ and misses the option.

STEP-BY-STEP SOLUTION (AI-derived explanation)

- Let $\mu = E(X_1)$ and $\sigma^2 = \text{Var}(X_1) = 4$, so $\sigma = 2$. By the CLT, $(\text{sum } X_i - n\mu)/(\sigma\sqrt{n})$ converges to $N(0, 1)$.
- $P(\text{sum } X_i \leq x\sqrt{n}) = P[(\text{sum } X_i - n\mu)/(\sigma\sqrt{n}) \leq (x\sqrt{n} - n\mu)/(\sigma\sqrt{n})] = P[Z_n \leq (x - \sqrt{n}\mu)/\sigma]$.
- If μ were non-zero the bound $(x - \sqrt{n}\mu)/\sigma$ would tend to $+\infty$ or $-\infty$ and the probability would tend to 0 or 1, not to $\Phi(2)$. Hence $\mu = 0$.
- With $\mu = 0$ the limit is $\Phi(x/\sigma) = \Phi(x/2)$, and this is given to equal $\Phi(2)$, so $x/2 = 2$ and $x = 4$.
- Now consider $P(\text{sum } X_i \leq (1/x)\sqrt{n})$ with $1/x = 1/4$. By the same argument the limit is $\Phi((1/4)/2) = \Phi(1/8)$.
- Hence the probability converges to $\Phi(1/8)$, option (d).

3. 2022 · Q3 Modes of Convergence · Theoretical

QUESTION

Suppose $\{X_n : n \geq 1\}$ is a sequence of independent random variables, and for each $n \geq 1$, X takes values $-\frac{3n}{2}, -n, \frac{n}{2}, n$ with probability $\frac{1}{4}$ each. The sequence $Y_n = \frac{1}{n^\alpha} \sum_{i=1}^n \frac{x_i}{i}$ converges in probability to 0 if and only if:

OPTIONS

- (a) $\alpha > 1$
- (b) $\alpha = 1$
- (c) $\alpha \geq 1$
- (d) $\alpha > \frac{3}{2}$

ANSWER (AI-derived — no official key exists for this paper)**(a)** $\alpha > 1$ **EXAM SHORTCUT (AI-derived explanation)**

$E(X_i/i) = -1/4$, so $E(Y_n) = -n^{1-\alpha}/4$, which tends to 0 only if $\alpha > 1$. The variance term needs only $\alpha > 1/2$, so the binding condition is $\alpha > 1$.

TIPS & TRICKS (AI-derived explanation)

- Check the MEAN and the VARIANCE separately: convergence in probability to 0 needs both $E(Y_n) \rightarrow 0$ and $\text{Var}(Y_n) \rightarrow 0$. The stricter condition wins.
- The four values $-3n/2, -n, n/2, n$ do NOT average to zero: their mean is $-n/4$, so each X_i/i contributes a constant $-1/4$ to the sum. That non-zero drift is the crux.
- The drift term is n times $(-1/4)$ divided by n^α , i.e. of order $n^{1-\alpha}$ - it vanishes only for $\alpha > 1$, and equals $-1/4$ at $\alpha = 1$, which is why option (b) fails.
- $\text{Var}(X_i/i) = 17/16$ is free of i , so $\text{Var}(Y_n) = 17n/(16n^{2\alpha}) = O(n^{1-2\alpha})$, requiring only $\alpha > 1/2$ - the weaker requirement.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. For each n , X_n takes the values $-3n/2, -n, n/2, n$ with probability $1/4$ each.
2. $E(X_n) = (1/4)(-3n/2 - n + n/2 + n) = (1/4)(-n) = -n/4$, so $E(X_i/i) = -1/4$ for every i .
3. $E(X_n^2) = (1/4)(9n^2/4 + n^2 + n^2/4 + n^2) = 9n^2/8$, so $\text{Var}(X_n) = 9n^2/8 - n^2/16 = 17n^2/16$ and $\text{Var}(X_i/i) = 17/16$.
4. Therefore $E(Y_n) = (1/n^\alpha) \cdot n \cdot (-1/4) = -n^{1-\alpha}/4$, and $\text{Var}(Y_n) = (1/n^{2\alpha}) \cdot n \cdot (17/16) = 17n^{1-2\alpha}/16$.
5. $\text{Var}(Y_n)$ tends to 0 whenever $\alpha > 1/2$, but $E(Y_n)$ tends to 0 only when $1 - \alpha < 0$, i.e. $\alpha > 1$ (at $\alpha = 1$ the mean is the constant $-1/4$).
6. Both conditions hold precisely when $\alpha > 1$, option (a).

4. **2022 · Q45** Uniform / Rectangular Distribution · Theoretical**QUESTION**

What is the limiting distribution function of the sequence

$$F_n(x) = \begin{cases} 0, & x < -n \\ \frac{x+n}{2n}, & -n \leq x \leq n \\ 1, & x > n \end{cases}$$

OPTIONS

- (a) Distribution function of a random variable degenerating at 0
- (b) Distribution function of a random variable degenerating at 1
- (c) Distribution function of a $U(0, 1)$ random variable

(d) The limiting distribution function does not exist

ANSWER (AI-derived — no official key exists for this paper)

(d) The limiting distribution function does not exist

EXAM SHORTCUT (AI-derived explanation)

F_n is the $U(-n, n)$ distribution function. For any fixed x , $F_n(x) = (x + n)/(2n)$ tends to $1/2$, and the constant $1/2$ is not a distribution function - so no limiting distribution exists.

TIPS & TRICKS (AI-derived explanation)

- Test the limit POINTWISE at a fixed x , letting n grow: $(x + n)/(2n) = 1/2 + x/(2n) \rightarrow 1/2$ for every x .
- A valid distribution function must tend to 0 as $x \rightarrow -\infty$ and to 1 as $x \rightarrow +\infty$; the constant function $1/2$ does neither.
- This is the standard illustration that probability mass can 'escape to ∞ ' - the uniform spreads out without limit and no proper limit law survives.
- Contrast with a sequence like $U(-1/n, 1/n)$, where mass CONCENTRATES and the limit is degenerate at 0 - the opposite failure mode.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. F_n is the distribution function of a $\text{Uniform}(-n, n)$ random variable.
2. Fix any real x . For all n large enough that $-n \leq x \leq n$, $F_n(x) = (x + n)/(2n) = 1/2 + x/(2n)$.
3. As n tends to ∞ , $x/(2n)$ tends to 0, so $F_n(x)$ tends to $1/2$ for EVERY fixed x .
4. The pointwise limit is therefore the constant function $G(x) = 1/2$.
5. A distribution function must satisfy $G(x) \rightarrow 0$ as $x \rightarrow -\infty$ and $G(x) \rightarrow 1$ as $x \rightarrow +\infty$; the constant $1/2$ satisfies neither, so it is not a distribution function. The probability mass escapes to ∞ .
6. Hence the limiting distribution function does not exist, option (d).

5. **2025 · Q49** Modes of Convergence · Numerical

QUESTION

Let $X_1, X_2, \dots$ be an iid sequence with common pdf $f(x) = \begin{cases} e^{-x+\theta}, & x \geq \theta \\ 0, & x < \theta \end{cases}$. If $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$, then $\bar{X}_n$

converges in probability to:

OPTIONS

- (a)** $1 - \theta$
- (b)** $1 + \theta$
- (c)** $1 - \theta^2$
- (d)** $1 + \theta^2$

ANSWER (AI-derived — no official key exists for this paper)

(b) $1 + \theta$

EXAM SHORTCUT (AI-derived explanation)

The density is a shifted exponential with mean $\theta + 1$, so by the weak law of large numbers $\bar{X}$ converges in probability to $1 + \theta$.

TIPS & TRICKS (AI-derived explanation)

- Recognise the shift: writing $Y = X - \theta$ gives the standard $\text{Exp}(1)$ density e^{-y} on $y > 0$, so $X = \theta + Y$ and $E(X) = \theta + 1$.
- The WLLN says the sample mean converges in probability to the POPULATION MEAN - so the whole question reduces to computing $E(X)$.
- The answer must be LINEAR in θ , since shifting the data shifts the mean by the same amount. Options (c) and (d) with θ^2 are structurally impossible.
- Sign check: increasing θ shifts the distribution to the right, so the limit must increase with θ - eliminating option (a).

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The density $f(x) = e^{-(x-\theta)}$ for $x \geq \theta$ is the exponential distribution with rate 1 shifted to start at θ .
2. Substituting $Y = X - \theta$ gives the standard density e^{-y} on $y > 0$, so $Y \sim \text{Exponential}(1)$ with $E(Y) = 1$.
3. Hence $E(X) = \theta + E(Y) = \theta + 1$, which is finite.
4. The X_i are independent and identically distributed with this finite mean.
5. By the weak law of large numbers, $\bar{X}_{n}$ converges in probability to $E(X)$.
6. Therefore $\bar{X}_{n}$ converges in probability to $1 + \theta$, option (b).

P11 · PROBABILITY

Borel 0-1 Law, Kolmogorov 0-1 Law

3 questions · 2018: 0 2019: 0 2020: 0 2021: 2 2022: 1 2023: 0 2024: 0 2025: 0 2026: 0

Weight. 3 questions (1.5% of Probability) · appeared in 2 of 9 papers · peak year 2021 (2)**Examiner's pattern.** Only three appearances in nine years, all in 2021–2022, and all recognisable on sight: 'sum $P(A_n) < \infty$, what is $P(\limsup A_n)$?' and 'does the pattern SFSF occur infinitely often?'. Cheap marks if you know both halves of Borel-Cantelli.**Must-know.** BC-I: $\sum P(A_n) < \infty \Rightarrow P(\limsup A_n) = 0$. BC-II: if the A_n are independent and $\sum P(A_n) = \infty \Rightarrow P(\limsup A_n) = 1$.1. **2021 · Q7** Borel & Kolmogorov 0-1 Laws · Theoretical

QUESTION

Let $A_1, A_2, A_3, \dots$ be a sequence of events and let $E = \limsup A_n$. If $\sum_{n=1}^{\infty} P(A_n) < \infty$, then $P(E)$ is equal to

OPTIONS

- (a) 1
- (b) $\frac{1}{2}$
- (c) $\frac{1}{4}$
- (d) 0

ANSWER (AI-derived — no official key exists for this paper)

(d) 0

EXAM SHORTCUT (AI-derived explanation)

This is exactly the first Borel-Cantelli lemma: if the sum of $P(A_n)$ converges then $P(A_n \text{ infinitely often}) = P(\limsup A_n) = 0$.

TIPS & TRICKS (AI-derived explanation)

- $\limsup A_n$ is the event that infinitely many A_n occur - read the notation as ' A_n infinitely often' and the lemma is immediate.
- Borel-Cantelli I needs NO independence; Borel-Cantelli II (divergent sum implies probability 1) does require independence. Knowing which lemma needs what is a standard exam point.
- Convergent sum means the tail sums go to zero, and $P(\limsup A_n) \leq$ the tail sum for every n - that is the one-line proof.
- This lemma is the workhorse for proving almost-sure convergence: to show $X_n \rightarrow X$ a.s., show that $\sum P(|X_n - X| > e)$ converges for each e .

STEP-BY-STEP SOLUTION (AI-derived explanation)

- By definition, $\limsup A_n = \bigcap_{m=1}^{\infty} \bigcup_{n=m}^{\infty} A_n$, which is the event that infinitely many of the A_n occur.
- For every n , $P(\limsup A_n) \leq P(\bigcup_{m=n}^{\infty} A_m) \leq \sum_{m=n}^{\infty} P(A_m)$.
- Since the series of $P(A_n)$ converges, its tail sums tend to 0 as n grows.
- Hence $P(\limsup A_n)$ is bounded above by a quantity tending to 0, so $P(E) = 0$.
- This is the first Borel-Cantelli lemma, which requires no independence assumption.

6. Option (d) is correct.

2. **2021 · Q21** Borel & Kolmogorov 0-1 Laws · Theoretical

QUESTION

If $\{x_n : n \geq 1\}$ are iid with mean $\frac{1}{2}$, then $P(|x_n| > \frac{n}{2} \text{ for infinitely many } n)$

OPTIONS

- (a) is equal to 0
- (b) is equal to $\frac{1}{2}$
- (c) lies strictly between $\frac{1}{2}$ and 1
- (d) is equal to 1

ANSWER (AI-derived — no official key exists for this paper)

(a) is equal to 0

EXAM SHORTCUT (AI-derived explanation)

A finite mean forces the sum of $P(|X| > n/2)$ to converge (it is bounded by $2E|X|$), so Borel-Cantelli I gives probability 0.

TIPS & TRICKS (AI-derived explanation)

- Key inequality: the $\sum P(|X| > cn) \leq E|X|/c$ for $c > 0$. A finite mean therefore makes every such series converge.
- 'Infinitely many n ' is exactly $\limsup A_n$, so the first Borel-Cantelli lemma applies directly and needs no independence.
- This is the standard lemma behind the strong law: $E|X| < \infty$ is equivalent to the convergence of sum $P(|X| > n)$.
- If the mean were infinite the answer would flip to 1 by the second Borel-Cantelli lemma (the X_n being independent) - the two lemmas give a clean 0 – 1 dichotomy here.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The x_n are i.i.d. with mean $1/2$, so $E|x_1|$ is finite.
2. Let $A_n = \{|x_n| > n/2\}$. Since the x_n are identically distributed, $P(A_n) = P(|x_1| > n/2) = P(2 | x_1| > n)$.
3. The standard bound gives the $\sum P(2 | x_1| > n) \leq E(2 | x_1|) = 2E|x_1| < \infty$.
4. Hence the series of $P(A_n)$ converges.
5. By the first Borel-Cantelli lemma, $P(A_n \text{ occurs for infinitely many } n) = 0$.
6. Therefore the required probability is 0, option (a).

3. **2022 · Q46** Borel & Kolmogorov 0-1 Laws · Theoretical

QUESTION

The probability that in a sequence of Bernoulli trials, the pattern SFSF appears infinitely often, is equal to:

OPTIONS

- (a) 1
- (b) $\frac{1}{2}$
- (c) $\frac{1}{4}$
- (d) 0

ANSWER (AI-derived — no official key exists for this paper)

(a) 1

EXAM SHORTCUT (AI-derived explanation)

Split the sequence into disjoint blocks of four trials. Each block equals SFSF with the same positive probability independently, so the sum of these probabilities diverges and the second Borel-Cantelli lemma gives probability 1.

TIPS & TRICKS (AI-derived explanation)

- Use DISJOINT blocks to obtain independence: trials 1 – 4, 5 – 8, 9 – 12, ... are independent, which is exactly what Borel-Cantelli II requires.
- Each block matches SFSF with probability $p^2q^2 > 0$, a constant, so the sum over infinitely many blocks diverges immediately.
- Borel-Cantelli II: for INDEPENDENT events with divergent probability sum, $P(\text{infinitely often}) = 1$. Independence is essential and the blocking supplies it.
- General principle: in an infinite sequence of Bernoulli trials, ANY fixed finite pattern occurs infinitely often with probability 1 - the 'infinite monkey' result.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Let p be the probability of success and $q = 1 - p$ that of failure, with $0 < p < 1$ so that both are positive.
2. Partition the trials into disjoint consecutive blocks of four: trials 1 – 4, 5 – 8, 9 – 12, and so on. These blocks are independent of one another.
3. Let A_k be the event that the k th block reads exactly S, F, S, F. Then $P(A_k) = p^2q^2$, a fixed positive constant for every k .
4. The $\sum P(A_k)$ is therefore infinite.
5. Since the A_k are independent, the second Borel-Cantelli lemma gives $P(A_k \text{ occurs infinitely often}) = 1$.
6. The pattern SFSF must then appear infinitely often in the full sequence, so the probability is 1, option (a).

P12 · PROBABILITY

Tchebycheff's & Kolmogorov's Inequalities

7 questions · 2018: 1 2019: 2 2020: 1 2021: 2 2022: 1 2023: 0 2024: 0 2025: 0 2026: 0

Weight. 7 questions (3.6% of Probability) · appeared in 5 of 9 papers · peak year 2019 (2)**Examiner's pattern.** A fixed template: mean and variance are given, and the answer is a bound with the words 'at least' or 'at most'. It also appears as a sample-size question ('how large must n be so that the probability is at least 0.95'). Spotting 'at least' next to a mean and a variance is the whole skill.**Must-know.** $P(|X - \mu| \geq k\sigma) \leq 1/k^2$, so $P(|X - \mu| < k\sigma) \geq 1 - 1/k^2$. Markov: $P(X \geq a) \leq E(X)/a$. Sample size: $n \geq \sigma^2 / (\epsilon^2 \alpha)$.**1. 2018 · Q19** Classical & Axiomatic Definitions of Probability · Numerical**QUESTION**

Weekly automobile sales have $E = 16$, $\text{Var} = 9$. The probability that next week's sales are between 10 and 22 is at least

OPTIONS

- (a) 0.50
- (b) 0.56
- (c) 0.75
- (d) 0.95

ANSWER (AI-derived — no official key exists for this paper)**(c)** 0.75**EXAM SHORTCUT** (AI-derived explanation)

The interval 10 to 22 is centred at the mean 16 with half-width 6, and $\sigma = 3$, so the half-width is exactly $k = 2$ standard deviations. Chebyshev gives at least $1 - 1/k^2 = 1 - 1/4 = 0.75$.

TIPS & TRICKS (AI-derived explanation)

- Chebyshev's inequality: $P(|X - \mu| < k\sigma) \geq 1 - 1/k^2$. The words 'at least' in the stem are the signal to use it rather than a normal table.
- Always find k as (half-width)/ σ . Here $(22 - 16)/3 = 2$ and $(16 - 10)/3 = 2$ - the interval is symmetric, which is required for the standard form.
- 0.75 is the $k = 2$ bound and is worth memorising along with $k = 3$ giving $8/9 = 0.889$ and $k = 1.5$ giving $5/9 = 0.556$.
- Do NOT use the normal 95% figure: option (d) 0.95 is the planted trap for candidates who assume normality where none is given.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Let X be next week's sales, with $\mu = E(X) = 16$ and $\text{Var}(X) = 9$, so $\sigma = 3$.
2. The event is $10 < X < 22$, that is $|X - 16| < 6$.
3. Write the half-width as a multiple of σ : $6 = k\sigma = 3k$, so $k = 2$.
4. Chebyshev's inequality states $P(|X - \mu| < k\sigma) \geq 1 - 1/k^2$.
5. With $k = 2$ this gives $P(10 < X < 22) \geq 1 - 1/4 = 3/4 = 0.75$.

6. Hence the probability is at least 0.75, option (c).

2. **2019 · Q33** Discrete & Continuous Random Variables · Numerical

QUESTION

The number of items produced in a factory during a week is a random variable with mean 50. If the variance is known to be 25, the probability that this week's production will be between 40 and 60 is at least

OPTIONS

- (a) 0.50
- (b) 0.60
- (c) 0.75
- (d) 0.80

ANSWER (AI-derived — no official key exists for this paper)

(c) 0.75

EXAM SHORTCUT (AI-derived explanation)

$\sigma = 5$ and the interval 40 to 60 is the mean plus or minus $10 = 2\sigma$. Chebyshev gives at least $1 - 1/2^2 = 0.75$.

TIPS & TRICKS (AI-derived explanation)

- Chebyshev: $P(|X - \mu| < k\sigma) \geq 1 - 1/k^2$. The words 'at least' signal that no distributional assumption is available.
- Convert the half-width to σ units first: $10/5 = 2$, so $k = 2$ and the bound is the familiar 0.75.
- Memorise the small table: $k = 2$ gives 0.75, $k = 3$ gives 0.889, $k = 4$ gives 0.9375, $k = 5$ gives 0.96.
- Do not reach for the normal 95% figure - no normality is assumed, and the true probability could be anywhere from 0.75 to 1.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Let X be the weekly production, with $\mu = 50$ and $\text{Var}(X) = 25$, so $\sigma = 5$.
2. The event $40 < X < 60$ is $|X - 50| < 10$.
3. Express the half-width in standard deviations: $10 = k\sigma = 5k$, so $k = 2$.
4. Chebyshev's inequality gives $P(|X - \mu| < k\sigma) \geq 1 - 1/k^2$.
5. With $k = 2$ the bound is $1 - 1/4 = 0.75$.
6. Hence the probability is at least 0.75, option (c).

3. **2019 · Q34** Tchebycheff's Inequality · Numerical

QUESTION

The smallest value of k in Chebyshev's inequality, for which the probability that a random variable takes a value between $\mu - k\sigma$ and $\mu + k\sigma$ is at least 0.99, is given by

OPTIONS

- (a) 9.5

- (b) 10
(c) 10.5
(d) 11

ANSWER (AI-derived — no official key exists for this paper)

(b) 10

EXAM SHORTCUT (AI-derived explanation)

Require $1 - 1/k^2 \geq 0.99$, so $1/k^2 \leq 0.01$ and $k^2 \geq 100$. The smallest such k is 10.

TIPS & TRICKS (AI-derived explanation)

- Invert Chebyshev directly: $k = 1/\sqrt{1-p}$ for a required probability p . Here $1 - 0.99 = 0.01$ and $1/\sqrt{0.01} = 10$.
- Useful pairings: 0.75 needs $k = 2$, 0.90 needs $k = 3.16$, 0.95 needs $k = 4.47$, 0.99 needs $k = 10$.
- Note how conservative the bound is - the normal distribution reaches 99% at only $k = 2.58$, whereas Chebyshev demands 10. That contrast is itself examinable.
- 'Smallest value of k ' means take the boundary case, so $k = 10$ exactly rather than anything larger; options (a), (c) and (d) are eliminated by that word.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Chebyshev's inequality states $P(\mu - k \sigma < X < \mu + k \sigma) \geq 1 - 1/k^2$.
2. We need this lower bound to be at least 0.99: $1 - 1/k^2 \geq 0.99$.
3. Rearranging: $1/k^2 \leq 0.01$.
4. Hence $k^2 \geq 100$, so $k \geq 10$.
5. The smallest admissible value is $k = 10$.
6. Option (b) is correct.

4. 2020 · Q29 Discrete & Continuous Random Variables · Theoretical

QUESTION

X is a non-negative random variable with finite, nonzero mean μ . For every $t > 0$:

OPTIONS

- (a) $P(X \geq t\mu) \geq 1/(t\mu)$
(b) $P(X \geq t\mu) \leq 1/t$
(c) $P(X \leq t\mu) \leq 1/(t\mu)$
(d) $P(X \leq t\mu) \geq 1/t$

ANSWER (AI-derived — no official key exists for this paper)

(b) $P(X \geq t\mu) \leq 1/t$

EXAM SHORTCUT (AI-derived explanation)

Markov's inequality: for $X \geq 0$, $P(X \geq a) \leq E(X)/a$. Putting $a = t\mu$ gives $P(X \geq t\mu) \leq \mu/(t\mu) = 1/t$.

TIPS & TRICKS (AI-derived explanation)

- Markov's inequality needs only non-negativity and a finite mean - exactly the assumptions stated, which is the signal to use it rather than Chebyshev.
- The μ cancels completely, leaving the clean bound $1/t$. Any option retaining μ (options (a) and (c)) is therefore structurally wrong.
- Direction check: an upper tail probability must be BOUNDED ABOVE, so the inequality sign must be $\leq$. That eliminates options (a) and (d).
- Chebyshev's inequality is Markov applied to $(X - \mu)^2$; knowing that link lets you derive one from the other under exam pressure.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Markov's inequality states that for a non-negative random variable X with finite mean, and any $a > 0$, $P(X \geq a) \leq E(X)/a$.
2. Here $E(X) = \mu$, which is finite and non-zero, and $X \geq 0$ as required.
3. Take $a = t\mu$, which is positive for every $t > 0$.
4. Then $P(X \geq t\mu) \leq \mu/(t\mu) = 1/t$.
5. The bound is free of μ , as it must be, since scaling X does not change the ratio X/μ .
6. Hence $P(X \geq t\mu) \leq 1/t$, option (b).

5. 2021 · Q6 Tchebycheff's Inequality · Numerical**QUESTION**

How large a sample must be taken in order that the probability will be at least 0.95 that the sample mean will be within a 0.5-neighbourhood of the population mean, given population standard deviation 1?

OPTIONS

- (a) 80
- (b) 79
- (c) 74
- (d) Cannot be determined

ANSWER (AI-derived — no official key exists for this paper)

(a) 80

EXAM SHORTCUT (AI-derived explanation)

Chebyshev on the sample mean: $P(|\bar{X} - \mu| < e) \geq 1 - \sigma^2/(ne^2)$. Require $1 - 1/(0.25n) \geq 0.95$, so $0.25n \geq 20$ and $n \geq 80$.

TIPS & TRICKS (AI-derived explanation)

- Chebyshev applied to $\bar{X}$ uses $\text{Var}(\bar{X}) = \sigma^2/n$, so the bound is $1 - \sigma^2/(ne^2)$. Forgetting the n makes the problem unsolvable.
- Rearrange once: $n \geq \sigma^2/(e^2(1 - P)) = 1/(0.25 \times 0.05) = 80$. Memorise this compact form.
- $e^2 = 0.25$, not 0.5 - squaring the neighbourhood radius is the step most often skipped, and it changes the answer by a factor of 2.

- Under normality the requirement would be only $n = 16$ (from $1.96/\sqrt{n} \leq 0.5$); Chebyshev's 80 is far larger because it assumes nothing about the shape.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Let $\bar{X}$ be the mean of a sample of size n from a population with standard deviation $\sigma = 1$, so $\text{Var}(\bar{X}) = 1/n$.
2. Chebyshev's inequality gives $P(|\bar{X} - \mu| < e) \geq 1 - \text{Var}(\bar{X})/e^2 = 1 - 1/(ne^2)$.
3. Here the required neighbourhood is $e = 0.5$, so $e^2 = 0.25$ and the bound is $1 - 1/(0.25n) = 1 - 4/n$.
4. We need $1 - 4/n \geq 0.95$, i.e. $4/n \leq 0.05$.
5. Hence $n \geq 4/0.05 = 80$.
6. The smallest admissible sample size is 80, option (a).

6. 2021 · Q23 Tchebycheff's Inequality · Numerical**QUESTION**

A random variable X has mean 12, and Chebyshev's inequality gives $P(2 < X < 22) \geq \frac{3}{4}$. Then $E(X^2)$ is

OPTIONS

- (a) 17
(b) 37
(c) 112
(d) 169

ANSWER (AI-derived — no official key exists for this paper)

(d) 169

EXAM SHORTCUT (AI-derived explanation)

The interval $(2, 22)$ is 12 ± 10 , so $1 - \sigma^2/100 = 3/4$ gives $\sigma^2 = 25$. Then $E(X^2) = \sigma^2 + \mu^2 = 25 + 144 = 169$.

TIPS & TRICKS (AI-derived explanation)

- Work Chebyshev BACKWARDS here: the bound is given, so solve $1 - \sigma^2/e^2 = 3/4$ for σ^2 rather than for the probability.
- The half-width is $e = 10$, and the equality form of Chebyshev pins $\sigma^2 = e^2/4 = 25$ - equivalently $k = 2$ and $\sigma = 5$.
- $E(X^2) = \text{Var}(X) + [E(X)]^2$ is the step candidates forget; the question asks for the raw second moment, not the variance. Option (a) 17 and (b) 37 come from omitting μ^2 .
- Check: with $\mu = 12$ and $\sigma = 5$, the interval $(2, 22)$ is exactly $\mu \pm 2\sigma$, and Chebyshev's bound at $k = 2$ is $3/4$, matching the datum.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Chebyshev's inequality gives $P(|X - \mu| < e) \geq 1 - \sigma^2/e^2$.
2. The interval $2 < X < 22$ is $|X - 12| < 10$, so $e = 10$.
3. We are told the resulting bound is exactly $3/4$: $1 - \sigma^2/100 = 3/4$.
4. Hence $\sigma^2/100 = 1/4$, so $\sigma^2 = 25$ (i.e. $\sigma = 5$ and the interval is $\mu \pm 2\sigma$).
5. Then $E(X^2) = \text{Var}(X) + [E(X)]^2 = 25 + 12^2 = 25 + 144 = 169$.

6. Option (d) is correct.

7. **2022 · Q55** Mathematical Expectation · Theoretical

QUESTION

Let $\beta_n = E(|x|^n)$. For non-degenerate random values, consider: 1. $\beta_k^{1/k} > \beta_{k+1}^{1/(k+1)}$ for $k = 1, 2, 3, \dots$ 2. $\beta_1 \leq \beta_2^{1/2} \leq \beta_3^{1/3} \leq \dots \leq \beta_n^{1/n}$

OPTIONS

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

ANSWER (AI-derived — no official key exists for this paper)

(b) 2 only

EXAM SHORTCUT (AI-derived explanation)

Lyapunov's inequality says the sequence $\beta_k^{1/k}$ is NON-DECREASING in k. Statement 2 states exactly that; statement 1 reverses the inequality and is false.

TIPS & TRICKS (AI-derived explanation)

- Lyapunov's inequality: $[E|X|^r]^{1/r}$ is non-decreasing in r. Remember the DIRECTION - higher-order norms are larger, never smaller.
- It follows from Jensen's inequality applied to the convex function $t \rightarrow t^{(k+1)/k}$, or from Holder's inequality.
- Statements 1 and 2 assert opposite orderings, so exactly one can be true - decide the direction once and both are settled.
- Concrete check: for $X \sim U(0, 1)$, $\beta_1 = 1/2$ and $\beta_2^{1/2} = (1/3)^{1/2} = 0.577 > 0.5$, confirming the increasing direction.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Let $\beta_n = E(|X|^n)$ denote the nth absolute moment.
2. Lyapunov's inequality states that for $0 < r < s$, $[E|X|^r]^{1/r} \leq [E|X|^s]^{1/s}$, i.e. the sequence $\beta_k^{1/k}$ is non-decreasing in k.
3. It follows from Jensen's inequality applied to the convex function $t \rightarrow t^{s/r}$.
4. Statement 1 asserts $\beta_k^{1/k} > \beta_{k+1}^{1/(k+1)}$, the reverse ordering. FALSE.
5. Statement 2 asserts $\beta_1 \leq \beta_2^{1/2} \leq \beta_3^{1/3} \leq \dots \leq \beta_n^{1/n}$, which is exactly Lyapunov's inequality. TRUE.
6. Only statement 2 is correct, option (b).

P13 · PROBABILITY

Laws of Large Numbers & Central Limit Theorems

7 questions · 2018: 1 2019: 0 2020: 0 2021: 3 2022: 1 2023: 0 2024: 1 2025: 1 2026: 0

Weight. 7 questions (3.6% of Probability) · appeared in 5 of 9 papers · peak year 2021 (3)**Examiner's pattern.** CLT dressed up as a limit of a binomial sum, consistency of the sample proportion, the CLT applied to sums of squares, and the asymptotic distribution of the empirical CDF. Recognise a sum standardised by $\sqrt{n}$ and you are done.**Must-know.** CLT: $(\sum X_i - n\mu)/(\sigma\sqrt{n}) \rightarrow N(0,1)$. WLLN needs finite mean (Khinchine); SLLN gives almost-sure convergence.**1. 2018 · Q33** Normal Distribution · Conceptual

QUESTION

 $(X_1, \dots, X_n)$ is a random sample of size n from $U[-3, 3]$. The large-sample distribution of $\sqrt{n}\bar{X}$ is

OPTIONS

- (a) $N(0, 1)$
- (b) $N(0, 3)$
- (c) $U\left[-\frac{3}{\sqrt{n}}, \frac{3}{\sqrt{n}}\right]$
- (d) $U\left[-\frac{3}{n}, \frac{3}{n}\right]$

ANSWER (AI-derived — no official key exists for this paper)**(b)** $N(0, 3)$ **EXAM SHORTCUT (AI-derived explanation)**

For $U[-3, 3]$, mean = 0 and variance = $(6)^2/12 = 3$. By the CLT, $\sqrt{n}(\bar{X} - \mu)$ converges to $N(0, \sigma^2) = N(0, 3)$. No division by σ appears, so the variance stays 3.

TIPS & TRICKS (AI-derived explanation)

- Variance of $U[a, b]$ is $(b - a)^2/12$. Here $(3 - (-3))^2/12 = 36/12 = 3$ - compute this before anything else.
- Distinguish the two CLT forms: $\sqrt{n}\bar{X}/\sigma$ tends to $N(0, 1)$ whereas $\sqrt{n}\bar{X}$ tends to $N(0, \sigma^2)$. The stem has no σ in the denominator, so option (b) not (a).
- The limit is NORMAL, never uniform - the CLT destroys the parent shape. Options (c) and (d) are eliminated on principle without any arithmetic.
- The mean of $U[-a, a]$ is 0 by symmetry, which is why the limiting mean is 0 and no centring constant is needed.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. For the Uniform distribution on $[-3, 3]$, the mean is $\mu = (-3 + 3)/2 = 0$ and the variance is $\sigma^2 = (3 - (-3))^2/12 = 36/12 = 3$.
2. The Central Limit Theorem states that $\sqrt{n}(\bar{X} - \mu)/\sigma$ converges in distribution to $N(0, 1)$ as n grows.
3. Since $\mu = 0$, this reads $\sqrt{n}\bar{X}/\sigma$ tends to $N(0, 1)$.
4. Multiplying by the constant σ , $\sqrt{n}\bar{X}$ converges in distribution to $N(0, \sigma^2) = N(0, 3)$.

5. Hence the large-sample distribution is $N(0, 3)$, option (b).

2. 2021 · Q9 Central Limit Theorem · Numerical

QUESTION

What is the value of $\lim_{n \rightarrow \infty} \sum_{j=n}^{4n} \binom{4n}{j} \left(\frac{1}{4}\right)^j \left(\frac{3}{4}\right)^{4n-j}$?

OPTIONS

- (a) 0
- (b) $\frac{1}{4}$
- (c) $\frac{1}{2}$
- (d) $\frac{3}{4}$

ANSWER (AI-derived — no official key exists for this paper)

(c) $\frac{1}{2}$

EXAM SHORTCUT (AI-derived explanation)

The sum is $P(X \geq n)$ where $X \sim \text{Binomial}(4n, 1/4)$, whose mean is exactly n . By the CLT the probability of being at or above the mean tends to $1/2$.

TIPS & TRICKS (AI-derived explanation)

- Recognise the binomial tail first: $\binom{4n}{j} (1/4)^j (3/4)^{4n-j}$ summed from $j = n$ to $4n$ is $P(X \geq n)$ for $X \sim \text{Bin}(4n, 1/4)$.
- Compute the mean: $np = 4n(1/4) = n$, exactly the lower summation limit. That coincidence is the entire design of the question.
- By the CLT the standardised binomial tends to $N(0, 1)$, and $P(Z \geq 0) = 1/2$. The single-point mass $P(X = n)$ is of order $1/\sqrt{n}$ and vanishes.
- The de Moivre-Laplace theorem is the special case of the CLT being used here - worth naming in an answer for full credit.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Let $X \sim \text{Binomial}(4n, 1/4)$. Then the given sum is exactly $P(X \geq n)$.
2. The mean of X is $E(X) = 4n(1/4) = n$, and its variance is $4n(1/4)(3/4) = 3n/4$.
3. Standardising, $P(X \geq n) = P((X - n)/\sqrt{3n/4} \geq 0)$.
4. By the Central Limit Theorem (de Moivre-Laplace), the standardised variable converges in distribution to $N(0, 1)$.
5. Hence the probability tends to $P(Z \geq 0) = 1/2$; the atom at $X = n$ contributes $O(1/\sqrt{n})$, which vanishes.
6. Therefore the limit is $1/2$, option (c).

3. 2021 · Q10 Tchebycheff's Inequality · Conceptual

QUESTION

For a sequence of Bernoulli trials, the fraction of successes f/n is a consistent estimator of the probability of success p . This follows from

OPTIONS

- (a) Central limit theorem
- (b) De Moivre–Laplace theorem
- (c) Law of large numbers
- (d) Chebyshev's theorem

ANSWER (AI-derived — no official key exists for this paper)**(c)** Law of large numbers**EXAM SHORTCUT (AI-derived explanation)**

Convergence of the sample proportion to p in probability is precisely the (weak) law of large numbers, in its Bernoulli form.

TIPS & TRICKS (AI-derived explanation)

- Match the conclusion to the theorem: convergence of an AVERAGE to its expectation is the law of large numbers; convergence of a standardised sum to a normal law is the CLT.
- Bernoulli's theorem is the historical name for the LLN applied to binomial proportions - it is the same result the question describes.
- Chebyshev's inequality (option d) is the TOOL used to prove the weak LLN, not the theorem that states the conclusion.
- De Moivre-Laplace (option b) is the normal approximation to the binomial, a CLT statement about the distribution, not about consistency.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Write the number of successes as $f = X_1 + X_2 + \dots + X_n$ where the Ξ are independent Bernoulli(p) variables.
2. Then f/n is the sample mean of these Ξ , and $E(\Xi) = p$.
3. The weak law of large numbers states that the sample mean of i.i.d. variables with finite mean converges in probability to that mean.
4. Hence f/n converges in probability to p , which is exactly the definition of f/n being a consistent estimator of p .
5. In its binomial form this result is known as Bernoulli's theorem, a special case of the law of large numbers. Chebyshev's inequality is the tool used in its proof, and the CLT and de Moivre-Laplace theorem describe the limiting DISTRIBUTION rather than consistency.
6. Option (c) is correct.

4. 2021 · Q26 Normal Distribution · Numerical**QUESTION**

If $\{X_n : n \geq 1\}$ are iid with $E(X_1) = 1$, $E(X_1^4) < \infty$, and

$$\frac{X_1^2 + X_2^2 + \dots + X_n^2 - 5n}{\sqrt{12n}} \xrightarrow{d} N(0, 1),$$

then $V(X_1)$ equals

OPTIONS

- (a) 4

- (b) 5
- (c) $\sqrt{12}$
- (d) 12

ANSWER (AI-derived — no official key exists for this paper)

(a) 4

EXAM SHORTCUT (AI-derived explanation)

The CLT applied to $Y_i = X_i^2$ identifies $E(X_1^2) = 5$ and $\text{Var}(X_1^2) = 12$. Then $\text{Var}(X_1) = E(X_1^2) - [E(X_1)]^2 = 5 - 1 = 4$.

TIPS & TRICKS (AI-derived explanation)

- Read the centring and scaling constants directly: the '5n' is $nE(Y)$ and the '12n' under the root is $n\text{Var}(Y)$, where $Y = X^2$.
- Do not confuse $\text{Var}(X_1^2) = 12$ with $\text{Var}(X_1)$; the question deliberately supplies the variance of the SQUARED variable.
- The condition $E(X_1^4) < \infty$ is exactly what guarantees $\text{Var}(X_1^2)$ is finite so that the CLT applies to Y - a hint about which variable is being averaged.
- Final step is the standard identity $\text{Var}(X) = E(X^2) - [E(X)]^2 = 5 - 1^2 = 4$. Option (b) 5 is the raw second moment, the planted near-miss.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Set $Y_i = X_i^2$. These are i.i.d. with finite variance because $E(X_1^4)$ is finite.
2. The Central Limit Theorem gives $(\sum Y_i - nE(Y_1))/\sqrt{n\text{Var}(Y_1)}$ converging in distribution to $N(0, 1)$.
3. Comparing with the given expression $(\sum X_i^2 - 5n)/\sqrt{12n}$, we identify $E(Y_1) = E(X_1^2) = 5$ and $\text{Var}(Y_1) = 12$.
4. We are told $E(X_1) = 1$.
5. Therefore $\text{Var}(X_1) = E(X_1^2) - [E(X_1)]^2 = 5 - 1 = 4$.
6. Option (a) is correct.

5. 2022 · Q41 Laws of Large Numbers · Theoretical

QUESTION

A sequence of i.i.d. random variables $\{X_k\}$ obeys the strong law of large numbers if:

OPTIONS

- (a) $\text{Var}(X_k) < \infty$
- (b) $\text{Var}(X_k) = \infty$
- (c) $E(X_k) < \infty$
- (d) Information is insufficient to conclude

ANSWER (AI-derived — no official key exists for this paper)

(c) $E(X_k) < \infty$

EXAM SHORTCUT (AI-derived explanation)

Kolmogorov's strong law for i.i.d. variables needs only a finite mean, i.e. $E|X_k| < \infty$ - finite variance is sufficient but not necessary.

TIPS & TRICKS (AI-derived explanation)

- For i.i.d. sequences, $E|X| < \infty$ is NECESSARY AND SUFFICIENT for the strong law; the sample mean then converges almost surely to $E(X)$.
- Finite variance (option a) is a stronger condition than needed - it guarantees the SLLN but is not required, so it is not the correct characterisation.
- The Cauchy distribution is the standard counterexample: it has no finite mean and its sample means do not converge at all.
- Contrast with the weak law, which also needs only a finite mean for i.i.d. variables but gives convergence in probability rather than almost surely.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. Kolmogorov's strong law of large numbers states that for a sequence of independent and identically distributed random variables, the sample mean converges almost surely to the population mean if and only if $E|X_1|$ is finite.
2. That is, a finite first absolute moment is both necessary and sufficient in the i.i.d. case.
3. A finite variance implies a finite mean and therefore also implies the SLLN, but it is a stronger assumption than required.
4. An infinite variance does not prevent the SLLN, provided the mean exists - so option (b) is not a condition for it either.
5. The information given is sufficient to answer, so option (d) is wrong.
6. Hence the condition is $E(X_k)$ finite, option (c).

6. 2024 · Q13 Distributions of Functions of Random Variables · Numerical**QUESTION**

If X is a normally distributed random variable with $\mu = 1$ and $\sigma^2 = 2$, then the asymptotic distribution of sample median for $n = 20$ is:

OPTIONS

- (a) $N(1, 1/10)$
- (b) $N(1, \pi/20)$
- (c) $N(1, \pi/10)$
- (d) $N(1, \pi/2)$

ANSWER (AI-derived — no official key exists for this paper)

(b) $N(1, \pi/20)$

EXAM SHORTCUT (AI-derived explanation)

The sample median from $N(\mu, \sigma^2)$ is asymptotically $N(\mu, \pi\sigma^2/(2n))$. With $\sigma^2 = 2$ and $n = 20$ the variance is $2\pi/40 = \pi/20$.

TIPS & TRICKS (AI-derived explanation)

- Asymptotic variance of the median: $1/(4nf(m)^2)$. For the normal, $f(\mu) = 1/(\sigma\sqrt{2\pi})$, giving $\pi\sigma^2/(2n)$.
- The median is centred at the population median, which for a symmetric normal equals $\mu = 1$ - so all four options share that centre.
- Substitute carefully: $\pi(2)/(2 \times 20) = \pi/20$. Option (a) $1/10$ is the variance of the sample MEAN ($\sigma^2/n = 2/20$), the standard trap.
- The ratio $(\pi/20)/(1/10) = \pi/2 = 1.571$ is the reciprocal of the efficiency $2/\pi$ - a neat consistency check with the ARE result.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. For a random sample of size n from a distribution with density f and population median m , the sample median is asymptotically normal with mean m and variance $1/(4n[f(m)]^2)$.
2. For $N(\mu, \sigma^2)$ the median equals the mean μ , and the density at the mean is $f(\mu) = 1/(\sigma\sqrt{2\pi})$.
3. Hence $[f(\mu)]^2 = 1/(2\pi\sigma^2)$, and the asymptotic variance is $1/(4n) \times 2\pi\sigma^2 = \pi\sigma^2/(2n)$.
4. Substituting $\sigma^2 = 2$ and $n = 20$: $\pi(2)/(2 \times 20) = 2\pi/40 = \pi/20$.
5. The centre is $\mu = 1$.
6. So the asymptotic distribution is $N(1, \pi/20)$, option (b).

7. 2025 · Q51 Distributions of Functions of Random Variables · Theoretical**QUESTION**

Let $X_1, \dots, X_n$ be a random sample from $U[0, 1]$. The asymptotic distribution of the empirical cdf $F_n(x)$ is:

OPTIONS

- (a) Normal with mean zero and variance 1
- (b) Normal with mean zero and variance $x(1-x)/n$
- (c) Normal with mean x and variance $x(1-x)/n$
- (d) None of the above

ANSWER (AI-derived — no official key exists for this paper)

(c) Normal with mean x and variance $x(1-x)/n$

EXAM SHORTCUT (AI-derived explanation)

$nF_n(x)$ is Binomial($n, F(x)$) with $F(x) = x$ for $U[0, 1]$, so $F_n(x)$ is asymptotically normal with mean x and variance $x(1-x)/n$.

TIPS & TRICKS (AI-derived explanation)

- The empirical CDF at a fixed x is an AVERAGE of indicator variables, each Bernoulli($F(x)$) - so the CLT applies directly.
- $E[F_n(x)] = F(x)$ exactly (it is unbiased), and $\text{Var}[F_n(x)] = F(x)(1-F(x))/n$. For $U[0, 1]$, $F(x) = x$.
- Option (b) gets the variance right but centres at zero - the empirical CDF estimates x , so its mean must be x , not 0.
- The centred and scaled version $\sqrt{n}(F_n(x) - x)$ tends to $N(0, x(1-x))$; the option as stated keeps the $1/n$ inside, which is the finite-sample form.

STEP-BY-STEP SOLUTION (AI-derived explanation)

1. The empirical distribution function is $F_n(x) = (1/n)$ times the number of observations not exceeding x .
2. Each observation contributes an indicator $I(X_i \leq x)$, which is Bernoulli with success probability $F(x)$; for $U[0, 1]$ and $0 \leq x \leq 1$ this is $F(x) = x$.
3. Hence $nF_n(x) \sim \text{Binomial}(n, x)$, so $E[F_n(x)] = x$ and $\text{Var}[F_n(x)] = x(1 - x)/n$.
4. Since $F_n(x)$ is an average of i.i.d. bounded variables, the Central Limit Theorem applies.
5. Therefore $F_n(x)$ is asymptotically normal with mean x and variance $x(1 - x)/n$.
6. Option (c) is correct.